



**GREEN
CLIMATE
FUND**

Meeting of the Board

10 – 13 July 2023

Songdo, Incheon, Republic of Korea

Provisional agenda item 12

GCF/B.36/02/Add.03

19 June 2023

Consideration of funding proposals – Addendum III

Funding proposal package for FP208

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Enhanced climate resilience in the Trois-Rivières region of Haiti through Integrated Flood Management";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

Table of Contents

Funding proposal submitted by the accredited entity	3
No-objection letter issued by the national designated authority(ies) or focal point(s)	109
Environmental and social report(s) disclosure	110
Secretariat's assessment	114
Independent Technical Advisory Panel's assessment	135
Response from the accredited entity to the independent Technical Advisory Panel's assessment	149
Gender documentation	151

Funding Proposal

Project/Programme title:	<u>Enhanced climate resilience in the Trois-Rivières region of Haiti through Integrated Flood Management</u>
Country(ies):	<u>Haiti</u>
Accredited Entity:	<u>United Nations Development Programme, UNDP</u>
Date of first submission:	<u>2021/07/23</u>
Date of current submission	<u>2023/06/04</u>
Version number	<u>V.14</u>



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Contents

Section A	PROJECT / PROGRAMME SUMMARY
Section B	PROJECT / PROGRAMME INFORMATION
Section C	FINANCING INFORMATION
Section D	EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA
Section E	LOGICAL FRAMEWORK
Section F	RISK ASSESSMENT AND MANAGEMENT
Section G	GCF POLICIES AND STANDARDS
Section H	ANNEXES

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

PROJECT/PROGRAMME SUMMARY				
A.1. Project or programme	Project	A.2. Public or private sector	Public	
A.3. Request for Proposals (RFP)	If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+. <u>Not applicable</u>			
A.4. Result area(s)	Check the applicable GCF result area(s) that the <u>overall</u> proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.			
		GCF contribution	Co-financers' contribution¹	
	Mitigation total	<u>Enter number</u> %	<u>Enter number</u> %	
	☐ Energy generation and access	<u>Enter number</u> %	<u>Enter number</u> %	
	☐ Low-emission transport	<u>Enter number</u> %	<u>Enter number</u> %	
	☐ Buildings, cities, industries and appliances	<u>Enter number</u> %	<u>Enter number</u> %	
	☐ Forestry and land use	<u>Enter number</u> %	<u>Enter number</u> %	
	Adaptation total	<u>Enter number</u> %	<u>Enter number</u> %	
	☒ Most vulnerable people and communities	20 %	50 %	
	☒ Health and well-being, and food and water security	20 %	50 %	
	☐ Infrastructure and built environment	<u>Enter number</u> %	<u>Enter number</u> %	
	☒ Ecosystems and ecosystem services	60 %	<u>Enter number</u> %	
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)		A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	Total beneficiaries: 733,872	
			Direct beneficiaries: 292,600 people	Indirect beneficiaries: 441,272 people
			Approximately 3 % of total population ²	Approximately 4% of the total population
A.7. Total financing (GCF + co-finance³)	31,290,561.48 USD	A.9. Project size	Small (Upto USD 50 million)	
A.8. Total GCF funding requested	<u>22,427,206.48 USD</u> For multi-country proposals, please fill out annex 17.			

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal. 31,

A.10. Financial instrument(s) requested for the GCF funding	<i>Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.</i>		
	☒ Grant 22,427,206.48 ☐ Equity <u>Enter number</u> ☐ Loan <u>Enter number</u> ☐ Results-based payment <u>Enter number</u> ☐ Guarantee <u>Enter number</u>		
A.11. Implementation period	8 years	A.12. Total lifespan	25 Years
A.13. Expected date of AE internal approval	<i>This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/ programme, if available.</i> 7/23/2021	A.14. ESS category	<i>Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.</i> B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☒ No ☐		
A.20. Executing Entity information	The Executing Entity for the project is Haiti's Ministry of Environment (MoE). The Ministry is also the office designated as the NDA on behalf of the Government of Haiti to the GCF.		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			
- As a Small Island Developing State (SIDS) and the only Least Developed Country (LDC) in the Western Hemisphere, Haiti is one of the most vulnerable countries in the world to climate change. This vulnerability is compounded by high levels of poverty and longstanding macroeconomic challenges experienced in the country — with approximately 59% of Haitians living under the national poverty line of US\$2.41 per day and approximately 24% living under the national extreme poverty line of US\$1.23 per day. - Haiti is particularly affected by the climate change impacts of increasing frequency and intensity of floods, tropical storms and hurricanes. Between 1975 and 2012, such climate change-related events resulted in			

² Population estimated at 11,242,856 people per World Bank projections (based upon 2003 Census and taking into account population growth rates). Please note that 2003 is the latest official Census conducted by the Government of Haiti, hence population growth rates have been considered for the project

³ Refer to the Policy of Co-financing of the GCF.

damages and losses equivalent to 2% of the country's GDP on average per year. In 2004, one storm — Hurricane Jeanne — caused severe flooding in northern Haiti. The flooding caused by this storm resulted in approximately US\$41 million in damages, more than 2,800 deaths and the need to remove approximately 592,000 m³ of mud that clogged the drainage networks and infrastructure across the country. Since 2012, the impacts of climate change have continued to negatively affect the population of Haiti. For example, the Category 5 storm — Hurricane Matthew — that struck Haiti in 2016 caused widespread damages and economic losses and negatively affected more than 2 million people (20% of the country's population) primarily located in the poorest regions of the country. Specifically, Hurricane Matthew resulted in the deaths of 546 people, forced more than 175,500 people to seek refuge in shelters, disrupted schooling for more than 450,000 children, and led to more than 1.4 million people requiring immediate humanitarian assistance. The storm also inflicted economic damages and losses estimated at more than 22% of the country's GDP in that year (2016), including the losses of up to 90% of crops and livestock in some areas of the country. In addition, it caused a sharp increase in the number of cholera cases experienced throughout Haiti. The intense rainfall associated with extreme climate events such as Hurricane Matthew also leads to severe flooding, which exacerbates these associated negative impacts. Such high-impact flooding has been a regular occurrence in Haiti in recent decades. During the period 1993–2012, for example, Haiti experienced 31 floods, causing the deaths of approximately 2,870 people and adversely affecting approximately 9 million people through livelihood losses and personal damages. Of these flood events, 10 were directly attributed to hurricanes and a further 16 attributed to extreme rainfall events.

3. Projections under the RCP8.5 scenario simulate that climatic conditions contributing to flooding incidence are expected to become more prevalent in Haiti. Such conditions include more concentrated, intense rainfall as well as warming temperatures, which contribute to major storm development and exacerbate flood impacts. Furthermore, it is anticipated that a modest increase in the frequency of hurricanes and tropical storms affecting Haiti will be experienced in the future. A such watershed management actions have been prioritized within Haiti's National Adaptation Plans, prioritizing the Northwest and the Aribonite regions where Haiti's Trois-Rivieres (TR) watershed is located. The (TR) watershed is particularly vulnerable to the impacts of climate change-induced flooding and hence has been identified as the project's target area (see section 10.1 of the FS on its prioritization).
4. Although the TR watershed is not the largest watershed in the country, its population — located in six cities and three departments — is highly vulnerable to the impacts of climate change. TR watershed is prone to frequent flooding in the municipalities it crosses, with adverse effects particularly on poor communities who do not have sufficient resources to adequately respond. The rural areas of Trois-Rivières — such as Marmelade, Plaisance, Pilate, Gros-Morne, Bassin Bleu, Chansolme, and Port-de-Paix — are home to 80% of Haiti's population of extremely poor people. These areas have an extreme poverty rate of 40%, which is almost twice the national average of 23.9%, and much higher than any other area (for comparison, Port-au-Prince has an extreme poverty rate of 4.9%). In addition to the high rate of poverty, which inhibit communities from adapting and responding to climate impacts, the impacts of extreme climate events are being compounded by ecosystem degradation caused by unsustainable agricultural practices such as the increasing area under crop cultivation, and particularly the practice of leaving soil bare and exposed to extreme weather for most of the year.
5. Current agricultural practices, such as the increasing prevalence of annual crop cultivation, leave the soil bare and exposed to extreme weather for most of the year. During intense rainfall and storm events, the limited ground cover leads to increased surface runoff and consequently exacerbates soil erosion in the watershed. This erosion degrades soils, reducing agricultural productivity and necessitating the need for the expansion of agricultural land. The negative feedback loop created is further compounded by extensive deforestation in the mountainous parts of the watershed. This deforestation is primarily linked to charcoal production, with many farmers in the TR watershed resorting to charcoal production as an alternative livelihood in response to low yields, crop failure or unexpected expenses — particularly in extremely wet or dry years. As a result, most of the native forests have been harvested, with minimal effort towards reforestation activities. Given that the demand for charcoal in Haiti is high and is unlikely to decrease in the foreseeable future, charcoal production will likely continue to drive deforestation in the absence of alternative livelihood options.
6. Unsustainable land-use practices have radically changed the TR watershed's natural landscape, where the large-scale removal of tree cover for farming and charcoal production has decreased infiltration capacity of the soil and contributed to increased surface runoff during heavy rainfall events. This, in turn, has considerably increased flood frequency and intensity in the TR watershed, where floods wash away fertile soils, cause sedimentation of riverbeds and block drainage infrastructure. Given the near-complete absence of embankments and levees in the watershed, this cycle intensifies future flooding, leading to further destruction of crops, farmland and agricultural infrastructure, as well as the loss of livestock. Moreover, the erosion of

riverbanks and sedimentation build-up in rivers caused by flooding decreases water quality in the watershed considerably, leading to an increased risk of water- and vector-borne diseases.

7. At present, the Government of Haiti (GoH) has limited technical and financial capacity to fully address the impacts of climate change-induced flooding on the population of the TR watershed through domestic financing. This is highlighted by the fact that more than half of Haiti's national budget is accounted for by foreign aid. The existing high levels of public debt, combined with the urgent need to address poverty in the country, further constrains the GoH's capacity to invest in climate change adaptation. Consequently, the GoH is requesting GCF grant finance to initiate a paradigm shift towards climate-resilient flood management across seven target communes in the TR watershed (Port-de-Paix, Chansolme, Bassin Bleu, Plaisance, Pilate, Marmelade, Gros Morne). GCF finance will be combined with co-financing from Heifer International Haiti to transform the GoH's approach to addressing climate change-induced flood impacts on the watershed and population living nearby. This approach will involve shifting the current uncoordinated decision-making processes on flood management at the national and local levels towards an integrated approach to land and water resources management in the watershed. The proposed transformation of land and water resources management will be achieved by: i) implementing agroforestry systems and reforesting degraded landscapes at priority intervention sites in the TR watershed; ii) enhancing technical and institutional capacity for productive climate-resilient land management at the national and local levels; and iii) establishing the required governance framework for integrated water resources management (IWRM) to support the climate-resilient land management systems and facilitate sustainable use and management of water resources over the long term. This transformative approach the combined transformative effect of project interventions will incentivise sustainable land and water resources management in Haiti, while simultaneously promoting a shift towards the adoption and implementation of a climate-resilient, integrated approach to flood management that can be readily scaled up and replicated nationally, and across the Caribbean region.
8. Overall, the project will confer direct and indirect adaptation benefits to approximately 292,600 and 441,272 vulnerable people in Haiti, respectively. By implementing 17,740 ha of agroforestry systems and rehabilitating 7,700 ha of forests in priority areas in the TR watershed to address the impacts of climate change-induced flooding, the project will result in a more than 50% reduction in the number of households affected by 100-year flood events by restoring the ecosystems' capacity for water infiltration hence reducing land degradation. This reduction will be from 1,342 households at risk of flooding under the baseline scenario down to approximately 640 households with project interventions. Similarly, agroforestry and reforestation interventions under this project are expected to result in a 35% reduction in the number of households at risk to 20-year flood events in the TR watershed while providing the benefit of mitigating an estimated 3,636,750 tCO₂-e during the project's total 25-year lifespan.

PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Context

9. Haiti forms part of the Caribbean island Hispaniola, sharing a border with the Dominican Republic to the east. Hispaniola is the second-largest island in the Greater Antilles, and at 27,750 km², Haiti is the third-largest country in the Caribbean. As a Small Island Developing State (SIDS) and the only Least Developed Country (LDC) in the Western Hemisphere⁴, Haiti is exposed to multiple natural hazards, including hurricanes, floods, earthquakes, landslides and droughts⁵. Many of the country's primary economic activities⁶ (including agricultural production and industry), as well as more than 96% of the country's population of 11.1 million people⁷, are situated in areas that are at risk of being impacted by two or more of the hazards mentioned above. The vulnerability of the population is compounded by high levels of poverty and longstanding macroeconomic challenges experienced in the country⁸ — with approximately 59% of Haitians living under the national poverty line of US\$2.41 per day and approximately 24% living under the national extreme poverty line of US\$1.23 per day⁹.
10. The existing exposure and vulnerability to natural hazards are being exacerbated by the impacts of climate change, with Haiti ranking first on the Global Climate Risk Index and third on the Long-Term Climate Risk Index¹⁰. Moreover, climate change impacts — particularly regarding floods, hurricanes and tropical storms — have considerable implications for health and livelihoods, as well as the local economy. The Government of Haiti (GoH) has prioritised addressing the baseline socio-economic challenges facing the nation as well as funding disaster recovery, which has resulted in limited financial resources being available for long-term solutions to enhance climate resilience in the country.

Baseline scenario

11. Haiti is very mountainous, with more than 50% of the island at a steep slope of 40%¹¹ or more. While cities are mostly located in the coastal plains, much of the most productive land is located along steep mountainsides. As a result, agricultural production mainly occurs in these mountainous areas. Soil conditions in the country are intrinsically fragile, comprised of newly formed shallow soils that can support forests and grasslands but that are also susceptible to erosion and leaching of water-soluble nutrients. The location of arable land, combined with the fragile soil conditions in the country, increases erosion along the mountainous slopes of Haiti. For example, in 1999, soil erosion in Haiti was estimated at a total of 36 million tonnes, or 1,319 tonnes per km²/year.^{12,13} This erosion, combined with the steep topography of the country, has severe consequences for some of the country's most populous cities which are situated in valleys along the coast. When it rains, the steep hills in the upper reaches of the valleys channel rainwater towards urban areas. Given the widespread degradation of the watersheds and the associated surface runoff and erosion, the resulting floodwaters contain substantial sediment loads which are deposited downstream. These sediments block already limited drainage infrastructure in the urban areas, which, coupled with the intensity of rainfall events, creates an environment conducive to intense flooding.
12. The conditions mentioned above are particularly evident in the Trois-Rivières (TR) watershed — one of the largest and most productive watersheds in the country. The TR watershed, which includes the Aribonite, Nord (hereafter "North") and Nord Ouest (hereafter "Northwest") Departments, is the third-largest in the country, covering 896 km². Originating in the Massif du Nord mountain range and Artibonite Departments, the TR watershed has steep topography, with its highest elevation at 1,167 m above sea level and its lowest elevation at the watershed's discharge site into the Atlantic Ocean. The TR watershed is particularly important to the country in that it provides critical ecosystem goods and services (including aquifer recharge), as well as contributing to food security through

⁴ UN Office of the High Representative for The Least Developed Countries, Landlocked Developing Countries and Small Island Developing States. Available at: <http://unohrrls.org/about-lidcs/about-lidcs/>

⁵ Global Facility for Disaster Reduction and Recovery (GFDRR). Available at: <https://www.gfdr.org/en/haiti>.

⁶ which contribute approximately 56% of the country's GDP

⁷ World Bank data. 2018.

⁸ Further information on the country's financial challenges are presented in Section B.5 of the Funding Proposal, and specific details on these challenges can be found at the following reference: International Monetary Fund. 2019. Haiti. IMF Country Report number 20/121.

⁹ Haiti Overview: World Bank. Available at: <http://www.worldbank.org/en/country/haiti/overview>.

¹⁰ GermanWatch, Global Climate Risk Index 2016. Available at: <https://germanwatch.org/fr/download/13503.pdf>.

¹¹ Slope is expressed as a percentage, whereby the percentage indicates the vertical distance divided by the horizontal distance of a land area.

¹² Kulkarni C: The Implications of Composting Toilets for Nitrogen Flows in Haiti. Final Report. 2009, Providence: Brown University Centre for Environmental Studies.

¹³ Jolly CM, Shannon DA, Bannister M, Laurentin G, Dale JZ, Binns A, Lindo P: Income efficiency of soil conservation techniques in Haiti. Proceeding of the 26th West Indies Agricultural Economics Conference: July 2007; St. Augustine, Trinidad and Tobago. Edited by: Neela Badrie. 2007, 156-163.: Caribbean Agro-Economic Society.

its high potential for agricultural production and opportunities for economic development^{14,15}. These factors have led to the watershed becoming one of the most densely populated parts of Haiti, with an estimated population of approximately 628,800 people — most of whom depend on agriculture for their livelihoods. While the TR watershed has numerous opportunities for livelihood security and economic growth, the watershed is also among the most vulnerable to the impacts of flooding caused by intense, concentrated rainfall and storms.

Climate Rationale

13. As a Caribbean SIDS and LDC, Haiti is one of the most vulnerable countries in the world to climate change, including increases in the frequency and intensity of flooding, tropical storms and hurricanes. The impact of climate change-induced flooding, in particular, is already adversely affecting Haiti's population that is already highly exposed. These impacts are particularly notable for the agricultural sector, which is highly vulnerable to flooding and which makes a considerable contribution to GDP (approximately 21.5%) and employment (approximately 40% of the country's population is dependent on crops, livestock, and fisheries for income and livelihoods) in the country.
14. Limited data is available on the historical extent and frequency of floods in Haiti. However, given the nature of watersheds in Haiti – including the targeted TR watershed – major hurricane events generally result in extensive flooding. Over the last 30 years (1990-2020), the number of categories 3, 4 and 5 hurricanes have increased significantly compared to the previous 30 years (period 1959-1989)^{16,17}. This illustrates an increase in the frequency of extreme events in the Atlantic basin and therefore the corresponding increasing trend in flooding in Haiti over the last 30 years.
15. Impacts from flooding after storms and hurricanes have already been experienced by much of Haiti's population. In the aftermath of Hurricane Jeanne in September 2004, for example, hillside erosion caused by flooding led to widespread destruction of crops as gravel was deposited onto the farmers' fields. As a result, approximately 80% of the land area in Gonaïves was flooded for three days, along with approximately 30% of land in Port-de-Paix¹⁸. The deposited gravel also made cultivation difficult for all affected areas in the aftermath of the storm. Thousands of people died or went missing during the resulting floods. Under future climate change conditions, it is expected that both small-scale subsistence farming and larger, commercial agricultural operations that contribute considerably to Haiti's overall GDP are likely to be increasingly impacted by climate change-induced flooding^{19,20,21}.
16. Such adverse impacts of climate change on the livelihoods of vulnerable communities are already being experienced frequently in Haiti. During the period 1993–2012, for example, Haiti experienced 31 floods²², 10 of which were directly attributed to hurricanes and a further 16 attributed to extreme rainfall events since 2000²³. Combined, these flood events caused the deaths of approximately 2,870 people and adversely affected approximately 9 million people through livelihood losses and personal damages. In 2004, for example, Hurricane Jeanne caused flooding in the northern part of Haiti that resulted in more than 2,800 deaths, approximately US\$41 million in damages and the need to remove approximately 592,000 m³ of mud that clogged drainage networks and

¹⁴ through investment in agroforestry systems, such as planting fruit trees, which have proven long-term economic benefits in the country

¹⁵ Priority watersheds: Trois-Rivières (code 3); Bassin du Limbé (5); Cap Haïtien (6); Trou du Nord (8a); Jassa (8c); La Quinte (9); Saint Marc (12a); Montrouis (12b); Cabaret (12c); Cul de Sac (13); Fond Verrettes (14); Rivière Froide (15a); Momance (15b); Belle Anse (16b); Marigot (16c); Gde. Rivière de Jacmel (17); Cavaillon (20); Les Cayes (21); Grande Anse (24); Gde. Rivière de Nippes (27); Petit Goave (28b). Smith Scot E. and Hersey Daniel. 2008. Analysis of Watershed Vulnerability to Flooding in Haiti.

<http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.388.1772&rep=rep1&type=pdf>.

¹⁶ By analyzing the data from coast.NOAA.gov/hurricanes and www.wunderground.com/hurricane/archive for category 3 to 5 hurricanes originating in the Atlantic basin.

¹⁷ Over the period 1959–1989, the following numbers of hurricanes occurred:

Category 5: 10

Category 4: 24

Category 3: 26

For the period 1990–220, the following occurred:

Category 5: 17

Category 4: 44

Category 3: 36

¹⁸ Dartmouth Flood Observatory. 2005. 2004 global register of major flood events. Available at: http://www.dartmouth.edu/approximately_floods/Archives/2004sum.htm

¹⁹ World Bank. 2011. Vulnerability, Risk Reduction, and Adaptation to Climate Change: Haiti. Available at: http://sdwebx.worldbank.org/climateportalb/doc/GFDRRCountryProfiles/wb_gfdr climate change country profile for HTI.pdf

²⁰ USAID. 2017. Climate Risk Profile: Haiti. Available at: https://www.climateinsights.org/sites/default/files/asset/document/2017_Cadmus_Climate-Risk-Profile_Haiti.pdf

²¹ Centro Internacional de Agricultura Tropical. 2013. Prediction of the impact of climate change on coffee and mango growing areas in Haiti. Available at: http://dapa.ciat.cgiar.org/wp-content/uploads/2014/03/CC_impact_coffee-mango_Haiti_CRS-CIAT_final.pdf

²² The World Bank. 2018. Resilient Productive Landscapes in Haiti (P162908).

²³ which are directly attributed to hurricanes and tropical storms.

infrastructure across the country²⁴. The impacts of Hurricane Jeanne exacerbated the effects of torrential rains experienced earlier in May of the same year, which caused the death or disappearance of more than 4,000 people in Haiti.

17. In 2008, four tropical storms – including two hurricane level storms – hit Haiti in a period of less than 30 days, with significant impacts on the TR watershed. These storms included: i) Tropical Storm Fay, August 15-16; ii) Hurricane Gustav, August 25-28; iii) Tropical Storm Hanna, September 1 to 4; and iv) Hurricane Ike, September 5 to 8). The combined impact of these four successive storms was immense, with 7.8% of the population, or nearly 800,000 people directly affected including 40,000 people living in temporary shelters 2 months after the last storm. Approximately 159,950 hectares of agricultural land was impacted, of which ~104,310 ha was considered damaged, with the department of Artibonite most affected. These damages were exacerbated by the lasting impacts of the 2007 hurricane season, which resulted in breaches of waterways that facilitated the entry of water into the cities of the departments of Artibonite, Centre and South. The cost damages from the storms, combined with the resultant loss of agricultural production, has been estimated at UD\$125,600,000. In 2016, Matthew, Category 5 storm caused intense flooding across the country and resulted in economic damages and losses estimated at more than 22% of the country's GDP in that year, including the losses of up to 90% of crops and livestock in some areas of the country, and caused a sharp increase in the number of cholera cases²⁵. There have been more than 820,000 cases of cholera²⁶ recorded in Haiti since 2010. Given that approximately 35% of the population lack access to safe drinking water, and approximately 65% have limited to no access to sanitation services²⁷, the population of Haiti is extremely vulnerable to these diseases and their spread which is aggravated by intense rainfall and ensuing flooding conditions²⁸.
18. The impacts of intense rainfall, both from tropical storms and torrential rainfall, have had numerous knock-on effects on the country's ecosystems by compounding the already high rate of soil erosion and degradation, and contributing to a general decline in soil health. At least 90% of Haiti's soils are considered severely degraded due to deforestation and inappropriate cultivation, causing an increase in run-off, a decline in soil water infiltration, and therefore the surface and groundwater needed to support agriculture. The declining soil quality caused by intense rainfall and the loss of forest systems — in combination with increasing average temperatures — are consequently contributing to a reduction in soil moisture, thereby exacerbating the impacts of climate change on food security by increasing desertification potential. In the absence of soil conservation structures or protective vegetative cover on cropped land and grazed fallow land, the impacts of increasingly intense rainfall will continue to decrease soil health and increase the rate of desertification. Given that over 60% of the country has hillsides, of which more than 20% of slopes are degraded, the conservation of soils on these slopes are directly linked to the food security of Haiti's population.
19. While most of Haiti is at risk to the impacts of climate change-induced flooding, the country's TR watershed has been identified as particularly high-risk²⁹. This area has seen an increased occurrence in extreme rainfall events since 1981. Further details on the projected changes in storm frequency and intensity are contained in Section 3 of Annex 2: Feasibility Study, with summaries of the projected changes in temperature and rainfall patterns under the RCP4.5 and 8.5 scenarios, presented below.

Climate change risks

20. Flooding in Haiti is closely linked with intense, concentrated seasonal rainfall (April–June and August–October), storm events and warm temperatures. The high intensity of rainfall during an extreme storm event results in a particularly notable flood risk. Projections under the RCP8.5 scenario simulate that these climatic conditions that contribute to flooding, and hurricane development are expected to become more prevalent in Haiti (FS pg 40 and Section 3.3). Furthermore, it is anticipated that a modest increase in the frequency of hurricanes and tropical storms affecting Haiti will be experienced in the future³⁰. While detailed data on historical flood impacts and trends is not

²⁴ Gaspard G. 2013. Flood Loss Estimate Model: Recasting Flood Disaster Assessment and Mitigation for Haiti; The Case of Gonaives. Available at: <https://pdfs.semanticscholar.org/b32b/d0ccdcce15f58b07a70f5b1bd5d570baf7f79.pdf>.

²⁵ World Bank. 2017. Rapidly Assessing the Impact of Hurricane Matthew in Haiti. Available at: <https://www.worldbank.org/en/results/2017/10/20/rapidly-assessing-the-impact-of-hurricane-matthew-in-haiti#:~:text=The%20hurricane%20resulted%20in%20flooding,and%20bridges%20were%20washed%20away>.

²⁶ Cholera is a diarrheal disease, and the spread thereof is directly related to limited access to adequate water, sanitation, hygiene and health (WASH) facilities.

²⁷ Etienne CF. 2019. Cholera in Haiti: Gone but not forgotten. Available at: <https://reliefweb.int/report/haiti/cholera-haiti-gone-not-forgotten>. Accessed 15/06/2020.

²⁸ Eisenberg, M. 2013: Examining rainfall and cholera dynamics in Haiti using statistical and dynamic modeling approaches. *Epidemics*. Vol 5 Issue 4. p 197- 204. Available at <https://www.sciencedirect.com/science/article/pii/S1755436513000431>

²⁹ according to four different studies and assessments (including Oxfam's).

³⁰ Further details on the projected changes in storm intensity and frequency are presented in Section 3 of Annex 2: Feasibility Study.

available for the TR watershed alone, and no formal record and quantification of damages could be found for this watershed, flooding is known, anecdotally, to have considerable impacts on communities. The best proxy that is available, given the sparse historic data available for Haiti, is to consider historical satellite data observations as well the modelled impacts for different flooding return periods under the baseline scenario (without climate change). Under a 100-year flooding scenario, 1,342 households in the target watershed are at risk of flooding. Under 20-year flooding scenario, 463 households in the target watershed are at risk of flooding.

Temperature

21. The annual mean daytime temperatures for the Caribbean region show an increase of 0.19°C per decade during the period 1961–2010³¹. This is smaller than the increase in mean night-time temperatures of 0.28°C per decade during the same period, with the resultant change being a decrease in the mean annual daily temperature range. Under the RCP4.5 scenario, maximum increases in temperature of 1.02°C and 1.87°C are expected for 2025 and 2055, respectively, and maximum increases of 1.18°C for 2025 and 2.57°C are expected for 2055 under the RCP8.5 scenario³². Additionally, an analysis of temperature trends for Haiti for the period 1901–2013 reveals that minimum temperatures increased at a slightly higher rate (approximately 0.12°C per decade) than maximum temperatures (approximately 0.10°C per decade), which is consistent with the decrease in daily temperature range observed in the Caribbean region. Under the RCP8.5 scenario for the middle and end of century time periods, daily maximum temperatures are projected to increase by 1.5°C and 3.3°C, respectively. Furthermore, under this scenario, the temperature profile will shift towards warmer temperatures, and the extreme warm events will become more extreme, with current extreme temperatures likely to become the new normal (See FS Section 3.1).
22. Reliable satellite observations (Reynolds et al. 2002) also demonstrate a trend towards increased sea surface temperature (SST) with the ocean surrounding Haiti warming significantly. The observed trend in SST averaged over 75°W–72°W, 17°N–21°N was 0.17°C per decade, for a total warming of 0.70°C over the period 1982–2021.

Rainfall

23. Haiti's First National Communication to the United Nations Convention on Climate Change (UNFCCC) also reports that, under the RCP8.5 scenario, the period 2030–2040 is expected to be up to 6% drier in Haiti, the period 2050–2060 up to 17% drier, and the end-of-century time period expected to be 20% drier. Under the RCP4.5 and 6.0 scenarios, Haiti is expected to be between 9 and 12% drier, respectively, for the end-of-century time period.³³
24. Historical satellite observations from 1981 to the present have demonstrated an increase in annual rainfall in the TR region by 152mm (15%), a trend that is statistically significant at the 90% confidence level which is at odds with these future projections. A projected shift in the precipitation character is also observed, with a projected decrease in frequency of lowrainfall amounts (approximately 10–20 mm per day), and a noted increase in the occurrence of more extreme rainfall events (+30 mm per day) of approximately 10–17% indicating that daily rainfall extremes are becoming more extreme (See FS Section 2.4.2).

Hydrological changes

25. Compared with baseline (1991–2010) flood flow simulations, future projections using four climate models (CCSM4, CMCC, CNRM, HadGEM4) indicate a decrease in mid-century (2041–2060) daily maximum flood flow followed by a significant increase in daily maximum flood flow in the late-century (2081–2100). These projections were analysed under the RCP8.5 scenario. The future projections of 100-year flow events and 20-year flow events³⁴ mimic the projected changes for daily maximum flow, also exhibiting a mid-century decrease and late-century increase. Similar changes in flood extent are also expected, with river discharge, depth and flood extent projections also indicating decreasing trends through the mid-century, before increasing in the late century. The exception to this is the forecasted maximum for 20-year flows, where projections indicate an increase from the historical baseline into the mid- and late-century³⁵. Furthermore, for the 100-year flow event, the following changes in the maximum extent of flooding in the TR watershed are projected: i) a decrease of 17% from hist100 to mid100; and ii) an increase by 12% from mid100 to late100. For the 20-year flow event, the maximum extent of flooding is projected to: i) increase

³¹ Campbell JD, MA Taylor, TS Stephenson, RA Watson & FS Whyte. 2011: Future climate of the Caribbean from a regional climate model. *Int. J. Climatol.* 31: 1866–1878.

³² Haiti. First National Communication to the United Nations Framework Convention on Climate Change (UNFCCC).

³⁴ This terminology pertains to the theoretical return period for an event, which is the inverse of the average frequency of occurrence. For example, a 100-year event has a 1% chance of occurring while a 20-year event has a 5% chance of occurring.

³⁵ Minear JT & Water Science International LLC. No date. Summary Report of Hydrodynamic Model Results: Assessment of Past and Future Flood Potential, Trois-Rivieres Watershed, Haiti.

by 7% from hist20 to mid20; and ii) increase by 7% from mid20 to late20^{36,37}. Further details on projected hydrological changes expected in Haiti and the TR watershed, in particular, are presented in Section 3.5 of Annex 2: Feasibility Study.

26. Given the overall steepness of the gradient of the landscape in the TR watershed, even small flows are enough to mobilise large volumes of sediment. However, in the flood flow model results summarised above, the bed and banks of the river are assumed to be immobile. The above results are therefore likely to be conservative, as a small volume of sediment added from tributaries to the mainstream river would be sufficient to reduce local channel capacities and subsequently increase flood risk^{38,39}.

Climate change impacts and vulnerability

27. The impacts of climate change-induced flooding in the TR watershed are exacerbated by the negative feedback loop of increasingly intense soil erosion that has been created in the watershed. Current agricultural practices, such as the increasing prevalence of annual crop cultivation, leave the soil bare and exposed to extreme weather for most of the year. During intense rainfall and storm events that cause flooding, the limited ground cover leads to increased surface runoff and consequently exacerbates soil erosion in the watershed. This erosion is compounded by extensive deforestation in the mountainous parts of the watershed, primarily linked to charcoal production. The demand for charcoal in Haiti is high and is unlikely to decrease in the foreseeable future, with many farmers in the TR watershed resorting to charcoal production as an alternative livelihood in response to low yields, crop failure or unexpected expenses (particularly in extremely wet or dry years). Given this high demand, charcoal production will likely continue to drive deforestation in the absence of alternative livelihood options. Most of the native forests on communal lands have already been harvested, with minimal effort towards reforestation activities⁴⁰. These practices have radically changed the TR watershed's natural landscape, where the large-scale removal of tree cover has decreased infiltration capacity of the soil and contributed to increased surface runoff during heavy rainfall events. This, in turn, has considerably increased flood frequency and intensity in the TR watershed, with flooding washing away fertile soils and causing sedimentation of riverbeds and blocking drainage infrastructure. Given the near-complete absence of embankments and levees in the watershed, this cycle then intensifies the next round of flooding, leading to the destruction of crops, farmland, and agricultural infrastructure, as well as the loss of livestock and human lives⁴¹. Vulnerable communities that depend on agriculture for their livelihoods will be particularly affected by the increased flooding, exacerbating the adverse economic, environmental and social conditions already being experienced in the area.
28. While frequent and intense flooding, soil erosion and sedimentation of rivers have considerable adverse impacts on the landscape and agriculture in the TR watershed, they also have severe consequences for water resources and human health. The erosion of riverbanks and sedimentation build-up in rivers caused by flooding decreases water quality in the watershed considerably, with multiple impacts on freshwater and marine biodiversity as well as increasing the risk of waterborne diseases in surrounding communities. Women and girls are particularly at risk as their household duties — including cooking and cleaning — involve the highest level of exposure to potentially contaminated food and water⁴². The adverse health risks of diarrheal diseases are exacerbated by flooding, which further extends the spread of water pollutants, disrupts water supply and increases the potential of exposure to contaminated water. Flooding also causes blockages in drainage channels with sediment and debris which results in flood waters lingering for longer periods of time. This exacerbates the risk of increased spread of water- and vector-borne diseases. The TR watershed is particularly vulnerable to such diseases given the area's exposure to flooding. For example, approximately 19,100 diarrhoea cases were reported in the watershed in 2018, accounting for approximately 35% of all cases of diarrhoea in the country at that time.
29. Without urgent intervention in the management of land and water resources, the vulnerable population of the TR region in Haiti will continue to become increasingly at risk to flooding and the associated negative impacts on the environment and the population's livelihoods. An integrated approach to flood management under future climate

³⁶ Hist100, mid100 and late100 are the historical, mid-century and late-century 100-year ensemble maximum flow, respectively. Hist20, mid20 and late20 are the historical, mid-century and late-century 20-year ensemble maximum flow, respectively.

³⁷ Minear JT & Water Science International LLC. No date. Summary Report of Hydrodynamic Model Results: Assessment of Past and Future Flood Potential, Trois-Rivieres Watershed, Haiti.

³⁸ Ibid.

³⁹ Further details on the methodology for the hydrological modelling undertaken in support of the proposed project are included in Sections 2.4 and 3.4 of Annex 2: Feasibility Study.

⁴⁰ World Bank. 2017. Charcoal in Haiti. A National Assessment of Charcoal Production and Consumption Trends. World Bank, Washington. Available at: <http://documents.worldbank.org/curated/en/697221548446232632/pdf/134058-CharcoalHaitiWeb.pdf>.

⁴¹ OXFAM, Welt Hunger Life, NATHAT, and Cap-Haitien.

⁴² GBV Sub-Sector Nigeria. 2017. Briefing Note: Integrating Gender in Cholera Prevention and Control Interventions in North East Nigeria. Available at: https://reliefweb.int/sites/reliefweb.int/files/resources/briefing_note-gender_in_cholera_response.pdf. Accessed 15/06/2020.

change conditions is consequently required to adequately address the diverse impacts of climate change-induced flooding on the TR watershed in Haiti. An integrated approach to flood management must take into account the environmental vulnerability within the region and the role that ecosystems, through sustainable land management can play as an integral adaptation solution that allows for community appropriation and will result in enhancing the capacity of the watershed to manage flood conditions and restore the infiltration capacity of the watershed.

Baseline projects, programmes and initiatives

30. Considerable baseline investments are being made through public expenditure and donor-funded initiatives to increase the resilience of Haiti's watersheds to climate change impacts, particularly flooding. Lessons learned and best practices from these investments, have been incorporated into the design of the proposed project to replicate successful adaptation and flood management techniques and ensure that there is complementarity between the project and existing actions. The proposed project interventions are designed to complement several of the most relevant baseline investments. An analysis of each of these projects was conducted to ensure that the proposed project will not duplicate or adversely affect these ongoing initiatives (refer to Section 6 in Annex 2 for a full list). Instead, GCF resources will be used to build on and complement existing efforts to achieve maximum adaptation impact in the TR watershed. The most relevant baseline investments, including a description of how the proposed project will complement these investments, are presented below⁴³.
31. USAID has committed to providing US\$40 million for the *Haiti Reforestation Project (2018–2022)* to reduce the threat of deforestation and increase tree cover in and around targeted areas in the northern parts of the country. The USAID project is aligned with the mutual goals of the GoH and USAID of restoring the provision of ecosystem goods and services in targeted forest areas in Haiti to enhance food security of surrounding communities. Specific interventions under the USAID project include: i) addressing the threat of deforestation in Haiti through reducing charcoal use by 80,000 metric tonnes in five priority watersheds; ii) increasing tree cover in selected areas, with the aim of planting a total of 5 million trees by project closure; iii) developing a total of 15,000 hectares under agroforestry systems to improve the resilience of those people employed in key economic sectors (e.g. agriculture) to economic shocks caused by climate and non-climate hazards; and iv) improving environmental governance and coordination in the country. The proposed GCF project will build on this ongoing baseline initiative by promoting an integrated approach to land and water resources management in the TR watershed and demonstrating the benefits of ecosystem-based adaptation (EbA) solutions such as land rehabilitation and agroforestry as effective flood management strategies. By promoting the adoption of these interventions as appropriate adaptation measures not only for reducing the impacts of climate change-induced flooding, but for contributing to food security and health in vulnerable Haitian communities, the project will catalyse the uptake of these measures across the TR region and throughout the country. The project proposal has also taken into account the project's evaluation into its design by ensuring that project interventions are folded into national and subnational government plans to ensure not only local but also national appropriation and long term sustainability. The project will also look to incorporate community best practices in reforestation and agroforestry to enhance a sense of local ownership in the techniques that will be promoted by the project to reduce the perception of a top down approach in project solutions.
32. The Natural Disaster Mitigation Program II (2016–2021) is a US\$42 million programme financed by the Inter-American Development Bank (IDB). The overarching objective of the programme is to improve climate-resilient watershed management by the agricultural sector in Haiti's Artibonite Department⁴⁴. Specific interventions under the project include: i) increasing climate resilience and disaster risk management in the agricultural sector; ii) improving water and sediment conservation in selected gullies of priority watersheds; iii) reducing the risk of rural economic losses as a result of flooding in targeted watersheds; and iv) protecting the institutional knowledge generated and housed within the country's Faculty of Agronomy and Veterinary Medicine, which is a central institution responsible for innovation and capacity development in Haiti's agricultural sector as well as in natural resources management and rural engineering. These interventions provide valuable lessons on climate-resilient, sustainable land management approaches for the agricultural sector, focusing primarily on implementing EbA approaches for disaster risk reduction. However, the interventions do not form part of an integrated and adaptive framework that can be used to inform climate-responsive planning at the watershed level. The proposed GCF project will address this gap by implementing an integrated approach to managing flooding in Haiti's TR watershed. Such an approach comprises: i) enhancing climate-resilient flood management in the watershed will be enhanced by establishing agroforestry systems and restoring degraded landscapes at priority intervention sites; ii) strengthening the institutional and technical capacity of decision-makers involved in agriculture at national and local levels to facilitate the uptake of productive land-use management techniques; iii) implementing a social protection system to support climate-vulnerable households affected by chronic food insecurity to recover their nutritional and

⁴³ A comprehensive list of baseline projects, programmes and initiatives as well as details of their focus and alignment with the proposed project is presented in Section 6 of Annex 2: Feasibility Study.

⁴⁴ Supporting documentation and project monitoring reports are available at: <https://www.iadb.org/en/project/HA-L1097>.

financial autonomy immediately after being impacted by climate change-related events such as flooding; and iv) establishing the required governance framework for integrated water resources management (IWRM) in Haiti to support the climate-resilient land management systems and facilitate sustainable use and management of water resources over the long term.

33. The World Bank-funded *Resilient Productive Landscapes Project (RPLP) (2018–2024)* is a US\$26 million project that focuses on climate-resilient agriculture and landscape management in Haiti. This project has two main objectives, namely to: i) improve the adoption of climate-resilient agricultural and landscape management practices in selected sub-catchments in the country; and ii) enable the GoH to respond promptly and effectively to extreme climate events such as floods, hurricanes and tropical storms. These objectives will be achieved through four interlinked components, focusing on: i) strengthening the institutional and organisational capacity for sustainable, climate-resilient agricultural and land-use management interventions at the landscape level; ii) strengthening climate-resilient agricultural production and practices in the project areas; iii) project coordination and undertaking regular monitoring and evaluation (M&E) of project interventions; and iv) establishing contingency processes and enhancing emergency responses. The proposed GCF project will complement and build on these four components by strengthening the capacity of farmers and community members in the TR watershed to implement climate-resilient agricultural and land-use practices. This will include: i) promoting a shift from unsustainable charcoal production to agroforestry systems as an alternative livelihood strategy; ii) improving the knowledge of these targeted user groups for implementing an integrated approach to land and water resources management; and iii) demonstrating the considerable food security (e.g. reduced agricultural losses as a result of implementing climate-resilient farming techniques) and health benefits (e.g. reduced spread of water- and vector-borne diseases because of the improved water quality resulting from reduced erosion and river sedimentation) associated with adopting such an integrated approach.
34. The Inter-American Development Bank (IDB), Global Agriculture and Food Security Program (GAFSP), International Fund For Agricultural Development (IFAD) and the GoH (through the Ministry of Agriculture, Natural Resources and Rural Development (MARNDR⁴⁵) in the country) have partnered to design and implement *Haiti's Technological Innovation for Agriculture and Agroforestry Program (PITAG) (2018–2022)*, valued at US\$77 million⁴⁶. This programme focuses on enhancing small agricultural producers' revenues and food security in the country. Interventions under the programme aim to increase on-farm productivity and improve the sustainable use of natural resources by implementing climate-resilient agricultural and agroforestry technologies that have long-term livelihood benefits. The proposed GCF project will build on this programme by implementing an integrated, climate-resilient approach to land and water resources management in Haiti. Specifically, agroforestry and reforestation activities under the PITAG programme will be scaled up and replicated at different sites under the proposed project to maximise the restoration of degraded ecosystems in the country and serve as an adaptation response to climate change-induced flooding in the TR watershed. Results from the PITAG programme's terminal evaluation, particularly those pertaining to the implementation of agroforestry on steep slopes, combined with lessons generated under the programme, will be used to inform the implementation of agroforestry systems under the proposed project. Lessons learned and best practices for implementing sustainable land management in Haiti generated under the PITAG programme will be used to inform the implementation of the adaptation interventions under the proposed GCF project to maximise the adaptation impact of these interventions and demonstrate the associated benefits to a wider beneficiary group. This will be extremely valuable for ensuring that these systems deliver maximum adaptation benefits to the communities living adjacent to project sites. The implementation of EbA interventions under the proposed project will be complemented by strengthening the capacity of national and local decision-makers, as well as local communities, involved in agricultural production in the TR watershed to implement productive climate-resilient land management practices. These capacity-building activities will be supported by the establishment of a finance mechanism for farmers in the TR watershed. Given that access to finance for climate-resilient agricultural inputs was not a focus of the PITAG programme, this intervention will bridge the gap of access to finance for these inputs and contribute to further safeguarding vulnerable people involved in agriculture against the impacts of climate change, particularly flooding. As a result of these interventions, decision-makers at the national and local levels will be better equipped to manage flood impacts under future conditions of climate change. Furthermore, local community members who depend on agriculture for their livelihoods will have enhanced technical and institutional capacity to implement productive land-use practices that are sustainable over the long term.
35. The GCF Readiness Project *Integrating climate change risks into national development planning processes in Haiti (NAP)* aims to strengthen institutional and technical capacities for the iterative development of a National

⁴⁵ MARNDR is the French acronym for this ministry. This acronym has been included at all mentions of the Ministry of Agriculture, Natural Resources and Rural Development throughout this Funding Proposal.

⁴⁶ Documentation on project including monitoring reports is available at <https://www.iadb.org/en/project/HA-G1038>.

Adaptation Plan (NAP) to effectively integrate climate change adaptation (CCA) into national and sub-national coordination, planning and budgeting processes. The project intends to achieve this objective through three outputs, namely: i) strengthening coordination mechanisms for multi-sectoral adaptation planning and implementation at different levels; ii) ensuring the NAP is compiled with a strong evidence base for adaptation planning and subsequently integrating priorities into the Strategic Development Plan (PSDH) and Disaster Risk Reduction Plan and Strategy (PNGRD); and iii) establishing a financing framework for CCA in the medium- to long-term. The proposed GCF project complement this initiative by enhancing institutional capacity for sustainable land management, as well as strengthening institutional capacity for the implementation of the Water Act⁴⁷ and developing an integrated, climate-resilient water management governance framework.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Problem statement

36. Haiti is already experiencing the impacts of extreme climate events, which contribute to its ranking as the most vulnerable country in the world to climate change, according to the 2016 Global Climate Risk Index⁴⁸. Climate change is causing an increase in temperatures and creating more favourable conditions for increased frequency and intensity of hurricanes in the Caribbean region. Moreover, climate change is resulting in a decrease in total annual rainfall in Haiti, compounded by a shortening of the wet season and a resulting increase in concentration and intensity. These changing climatic conditions are contributing to an increase in the frequency and intensity of flooding in the country, particularly in the Trois-Rivières (TR) watershed. The impacts of this climate change-induced flooding are exacerbated in the watershed by the steep topography, degraded landscapes caused by unsustainable agricultural practices and deforestation because of large-scale tree clearing for charcoal production. During heavy rainfall and flood events, these degraded areas experience increased surface runoff, compounded by decreased infiltration capacity of the soil because of reduced vegetation cover. These factors, combined with the steep topography of the TR watershed and near-complete absence of embankments along the river, increase the flow of water into the low-lying plains, increasing the flood risk of the population living near the river's edge. Furthermore, increased water flow intensifies erosion of the watershed and results in sediment deposits on the riverbed and drainage infrastructure, further exacerbating the impacts of flooding. The resultant pollution of freshwater resources also increases the risk of water- and vector-borne diseases (e.g. cholera) spreading throughout local communities. Without the implementation of adaptation measures in the TR watershed, these impacts of flooding on land and water resources, as well as the health and livelihoods of the population living in the watershed, will continue to become increasingly severe as rainfall patterns shift to become more concentrated and intense under future climate change conditions⁴⁹.

Preferred adaptation solution

37. For Haiti to adapt to the impacts of climate change-induced flooding on the TR watershed, it is necessary to implement an integrated approach to flood management to adequately address the diverse impacts of climate change-induced flooding on the watershed and population living within it. This approach will be implemented under the project and in particular focus on: i) improving land-use planning and management at national and local levels; ii) implementing Ecosystem-based Adaptation (EbA) solutions in the form of restoration and agroforestry activities to address soil erosion caused by flooding, reduce health risks associated with water- and vector-borne diseases, and provide alternative climate-resilient livelihood options to target communities; and iii) enhancing the capacity of national- and local-level actors responsible for decision-making around the use of water resources to implement climate-resilient integrated water resources management (IWRM) in Haiti. This approach is required to shift decision-making on flood management away from the existing uncoordinated and compartmentalised approach, towards an integrated approach that is underpinned by the uptake and implementation of climate-resilient land-use practices, strong governance, as well as the sustainable use and management of land and water resources.

38. Specifically, sustainable flood mitigation solutions are required to reduce the peak surface runoff and flooding extent in the watershed, while simultaneously securing the livelihoods of vulnerable communities. Solutions need to take into account long terms sustainability and local capacity for their maintenance as well as their own capacity to deliver on flooding benefits (see Annex 2 Section 4). Within the context of the TR watershed, Ecosystem-based Adaptation (EbA) solutions — in the form of reforestation and agroforestry activities — are the preferred option because of their cost-effectiveness, flood-reduction potential, and provision of valuable ecosystem goods and services to climate-vulnerable communities living in the watershed. Hydrological and economic studies commissioned by UNDP to support the development of the project indicate that actions related to forest

⁴⁷ The Water Act acts as regulatory policy towards water integrated management that may take the form of a National Regulatory Decree.

⁴⁸ GermanWatch, Global Climate Risk Index 2016. Available at: <https://germanwatch.org/fr/download/13503.pdf>.

⁴⁹ Further details on the baseline and projected changes in temperature, rainfall and flood profiles are presented in Section B.1 of this Funding Proposal.

management including the implementing agroforestry systems and forest rehabilitation will reduce the impacts of climate change-induced flooding through a 40% reduction in peak runoff in the watershed.

39. The establishment of agroforestry systems will also provide an alternative livelihood option to unsustainable charcoal production, further contributing to environmental sustainability in the TR watershed. These interventions should be supported by the creation of an enabling environment in Haiti for the adoption of sustainable, climate-resilient land-use practices with multiple benefits including food and water security. The technical and institutional capacity of national- and local-level representatives involved in agriculture, in particular, should be enhanced for productive climate-resilient land management. This capacity-building should be supported by the establishment of a governance framework that promotes integrated water resources management (IWRM) to facilitate a shift towards the sustainable use and management of water resources over the long term.

Barriers to the adaptation solution

40. Several barriers to the preferred adaptation solution have been identified for Haiti. These are summarised below and explained in detail in Section 8 of Annex 2: Feasibility Study. The barriers fall broadly in the themes of technical and institutional barriers (Barriers 1, 2 and 3), financial barriers (Barrier 4) and social/psychological barriers (Barriers 5 and 6).

Barrier 1. Limited technologies, resources and knowledge for climate-resilient flood management

41. Communities in Haiti, particularly farmers, are not informed about current and projected climate change impacts, particularly flooding, on agricultural production, other land-use practices and water resources. There is also limited technical knowledge amongst communities on how to adapt to these changes and increase climate resilience, particularly using EbA solutions. The limited understanding of communities and farmers of the impacts of climate change and how to address them (e.g. through adopting climate-resilient agricultural practices) increases their vulnerability over the long term. This vulnerability is compounded by existing limitations in practical and technical experience to establish and manage local, financially sustainable small-scale tree propagation operations that serve reforestation and agroforestry needs. Community members generally prefer to migrate to urban areas or engage in alternative rural occupations, as these options are perceived to offer more financial stability than tree propagation operations and agroforestry.

Barrier 2. Absence of integrated, climate-sensitive water and land management policy and governance frameworks

42. Currently, there are no integrated, climate-sensitive water or sustainable land management policies and governance frameworks in Haiti. The country lacks overarching water management legislation⁵⁰, a formal cadaster⁵¹, a standard practice for land tenure arrangements, or relevant strategies to promote sustainable landscape management. Absence of a regulatory framework and enforcement of sanctions and property rights also reinforces an open-access interpretation of community assets, which is compounded by social barriers (see Barriers 5 and 6) such as the lack of private incentives for reforestation. In terms of governance, there are no administrative structures at the catchment or sub-catchment levels, and there are currently no local water resources user associations or other community committees that have the skill and capacity for IWRM. This lack of vertical coordination and oversight is evident in line ministries providing weak or minimal support to local communities, which leads to limited support for farmer associations to implement changes in conservation or to manage water resources. Constraints in these policy and governance frameworks — as well as the limited integration of lessons learned — has produced three major concerns, namely: i) existing legal instruments that do not consider the projected threats from climate change; ii) lack of land use planning implementation and enforcement has led to deforestation and degradation; and iii) a lack of coordination in planning, implementation and accountability caused by undefined roles and responsibilities of institutions dealing with water management

Barrier 3. Limited coordination and capacity to implement/execute

43. Implementing/executing entities, government and non-governmental stakeholders in Haiti have limited opportunity and incentive to coordinate efforts to address climate risks. This leads to: i) projects by different organisations often being designed independently, without consideration of how they affect each other or address the broader resilience of the country; and ii) limited coordination among donors or between GoH ministries during project implementation. Local authorities and stakeholders at the forefront of managing climate change impacts are also not equipped and trained to incorporate climate change considerations into decision-making or to design and implement climate-resilient water management. This is evident in the lack of catchment and sub-catchment administration structures and capacity to oversee participatory, equitable and adaptation-focused decision-making on water and land use, which further amplifies households prioritising private gains over catchment-wide benefits.

⁵⁰ Although the establishment of a Water Act has been identified as one of the Ministry of Environment's priorities in its 2019-2022 Triennial Plan.

⁵¹ Or 'cadastre', an official register of the ownership, extent, and value of real property in a given area, used as a basis of taxation.

Barrier 4. Limited access to credit, savings and financial instruments or incentives among farmers

44. There are financial barriers to farmers wanting to implement sustainable land management practices in Haiti. These include a lack of access to capital to invest in start-up costs, maintenance materials and tools, as well as limited “bridging” household reserves to cope with crises⁵². Alternative sources of funding and credit⁵³ are not available or are difficult to access, which leads to short planning horizons. These planning horizons drive short-term relief activities, such as the production of charcoal at unsustainable rates to meet pressing short-term financial needs without recognizing its potential impact on valuable ecosystem that are providing flood management benefits.
45. For agroforestry and climate-resilient agricultural practices to be sustainable, incentives for farmers to implement these systems are needed to enable farmers to overcome this short-termism and to allow time for these systems to produce returns. Alternative financial mechanisms that are appropriate for the agroforestry and allow for long term continuity (are sustainable) and increased scale of action is also required to go beyond a pilot mentality to allow for transformative change. These mechanisms must take into account community practices and risk analysis to address existing systemic barriers to provide much needed capital for the success of these systems.

Barrier 5. Limited incentives for communities to change land management and farming practices or contribute to communal resilience building

46. Limited understanding of climate change impacts compounds prevailing circumstances, — such as poverty — that emphasise the prevalence of household or individual short-term decision-making over long-term communal resilience and well-being. Households (which have a high discount rate⁵⁴) make decisions in their own immediate interest, without necessarily considering the impact of those decisions on the sub-catchment and the long-term impacts on their own wellbeing. Additionally, charcoal production is entrenched in communities as the major source of energy for cooking, having significant economic importance and few available alternatives to households. There is also no experience or incentive among water users to consider sub-catchment scale water management in their household decision-making. Finally, the familiarity of established, unsustainable farming practices is a barrier to better alternatives as households are reluctant to change to new crops or practices without the development of local markets and social safety nets.

Barrier 6. Absence of inclusive and participatory community consultation mechanisms for climate change-induced flood management planning and decision-making

47. The absence of participatory and inclusive planning, which is a fundamental principle of an integrated flood management approach for climate resilience, means that communities are not currently incentivised to take up adaptation interventions in Haiti. Underlying factors for this include technical and financial capacity constraints at the municipal level. These have historically limited the ability of public officials to effectively engage with appropriate communities and stakeholders during decision-making on land and water resource management. Given the strong influence of social factors such as gender inequality on vulnerability to flood impacts, it is necessary to promote inclusive decision-making processes to ensure buy-in and effectiveness in interventions. Past interventions have been criticised for failing to take gender and other considerations into account by focusing on general agricultural incomes and outputs, access to which is shaped by prevailing systems of privilege.

Project approach to overcoming the barriers to adaptation

48. The project will overcome the abovementioned barriers to build the climate resilience of Haiti’s TR watershed by implementing a sustainable land and water management approach in the TR watershed as an adaptive measure to the impacts of climate induced flooding. These will be implemented through community based governance and planning mechanisms and will be linked to tangible benefits to address immediate food insecurity while providing flood protection to those most vulnerable. Sustainable land management as implemented by the project will result in enhanced food and water security in the TR area hence creating the incentives to sustain sustainable productive practices underpinning the ecosystem based interventions. The project’s strong focus on communication, monitoring and capacity building will allow for these linkages to become evident to project beneficiaries hence creating a shift in their approach to unsustainable business as usual practices. Barriers to sustainability will be addressed by the project in linking sustainable land management to enhanced productive capacity and markets as well as by mobilizing community based financing for continued investment. More importantly project investments

⁵² In this context, “limited bridging of household reserves” refers to the absence of emergency savings or other financial support for farmers so that, if crops fail because of climate change impacts and/or other disasters, they would be able to support their livelihoods immediately after these events until they are able to start generating their own income again.

⁵³ World Bank (2019). Agricultural Financing in Haiti: Diagnosis and Recommendations. Available at <https://openknowledge.worldbank.org/handle/10986/33178>

⁵⁴ A discount rate is the interest rate used to determine the present value of future cash flows in a discounted cash flow analysis. This helps determine if the future cash flows from an investment will be worth more than the capital investment in the present. Within this context, the high discount rate of households is as a result of their general financial insecurity, which is a barrier to applying for, and receiving, financial support (for example, in the form of bank loans).

will create a practical evidence base for the communities as well as the governance structures required to promote their continuity and upscale project actions into more ambitious adaptation measures.

49. The project's theory of change can be evidenced through its various interrelated Outputs. For example, Output 1 will enhance the resilience in the TR watershed to the impacts of climate change-induced flooding through a combination of on-the-ground ecosystem-based adaptation (EbA) interventions (agroforestry and reforestation activities) at strategic sites in communal sections within the target communes, while establishing the local governance structures required to implement and manage climate-resilient land-use practices, and building the capacity of community groups for climate-responsive planning. Doing so will not only reduce climate change-induced flooding in the TR watershed but also address the problem of unsustainable charcoal production through that is exacerbating degradation pressures in the watershed while aggravating flooding vulnerability to adjacent communities. This will be done through the identification of species that are focused on addressing community needs such as sustainable charcoal (through species that allow for sustainable charcoal through correct pruning measures) as well as food and livelihood security (fruit species). Output 2 will in turn further support the EbA Interventions through sustainable land management in agricultural systems while creating a social safety net that encourages agricultural development and desensitizes the need for unsustainable charcoal production within the ecosystem that in turn reduces its protective cover to manage flooding. It will also develop the financial mechanism through credit associations to ensure that these investments can be maintained and replicated even beyond the project's implementation scope. Finally, Output 3 will ensure the sustainability of these measures by creating the local (catchment and sub-catchment) and national structural capacities for Integrated Water Management in the form of improved land use management for water security and management while linking improved water management to public health outcomes.
50. In this sense, barrier 1 will be addressed by investing directly in implementing an integrated approach to climate-resilient flood management in Haiti. This approach will involve: i) strengthening governance for and national- and local-level decision-makers' and communities' capacity to implement Integrated Land Management (ILM) and Integrated Water Resources Management (IWRM) in Haiti, using a participatory approach; ii) restoring 7,700 ha of degraded forest and implementing 17,740 ha of agroforestry systems in the watershed to reduce the impacts of climate change-induced flooding on ecosystems and communities; and iii) implementing a social protection system, coupled with enhanced technical capacity and access to finance of national and local representatives to facilitate the adoption of climate-resilient sustainable land-use practices and livelihoods.
51. Under the climate-resilient Integrated Land and Water Resources Management (ILWRM) approach proposed by the project, community groups (including farmer and women's associations) will be encouraged to participate in the development of community land-use plans for each of the 33 communal sections within the seven target communes in the TR watershed. The development and implementation of these community land-use plans will be supported by the establishment of the necessary national governance framework for climate-resilient IWRM. This will involve strengthening national capacities for the implementation of the Water Act and developing an inclusive and equitable water management governance framework for the TR catchment, namely the Catchment Integrated Water Resources Management Committee (CIWRMC). Furthermore, lessons generated through the implementation of all project interventions will be shared through the relevant committees and associations — at the catchment and sub-catchment levels — to inform adaptive management of these interventions under future climate change conditions. Combined, these key activities will overcome Barriers 2 and 3, with the participatory decision-making around land and water resources management contributing to addressing Barrier 6.
52. Barrier 4 will be addressed through a combination of long- and short-term interventions to address the immediate impacts of a flood event in Haiti, including food security and health, as well as facilitate access to finance to support the uptake of climate change adaptation solutions in response to the projected increasing impacts of climate change in Haiti. Specifically, the technical capacity and access to finance of national and local representatives involved in agriculture in the TR watershed will be enhanced to facilitate their adoption of climate-resilient sustainable land-use practices and provide an alternative livelihood option to unsustainable charcoal production in the watershed. This will be supported by providing immediate financial relief to vulnerable households in the TR watershed to enable these households to sustain themselves immediately after a flooding event. These efforts will be underpinned by the Agricultural Village Savings and Credit Associations (VSCA) that will be created through the project to finance ongoing productive investments (by harnessing savings). The VSCA will act as an appropriate financial mechanism for community farmers to allow for the long term sustainability of the project's approach to sustainable land management.
53. Barrier 5 will be addressed by creating incentives in the TR watershed for the adoption of sustainable land and water resources management techniques that are resilient to climate change-induced flooding. This will be

achieved under each element of the ILWRM approach proposed under the project by: i) demonstrating the wide variety of adaptation benefits and alternative livelihood options associated with the proposed EbA solutions such as reforestation and agroforestry (Output 1); ii) optimising the agricultural value chain and increasing the knowledge of vulnerable communities of the benefits of implementing climate-resilient agricultural techniques (Output 2); and iii) creating an enabling policy environment and strengthening governance in Haiti at the national and local levels for integrated, climate-resilient land and water resources management (Output 3).

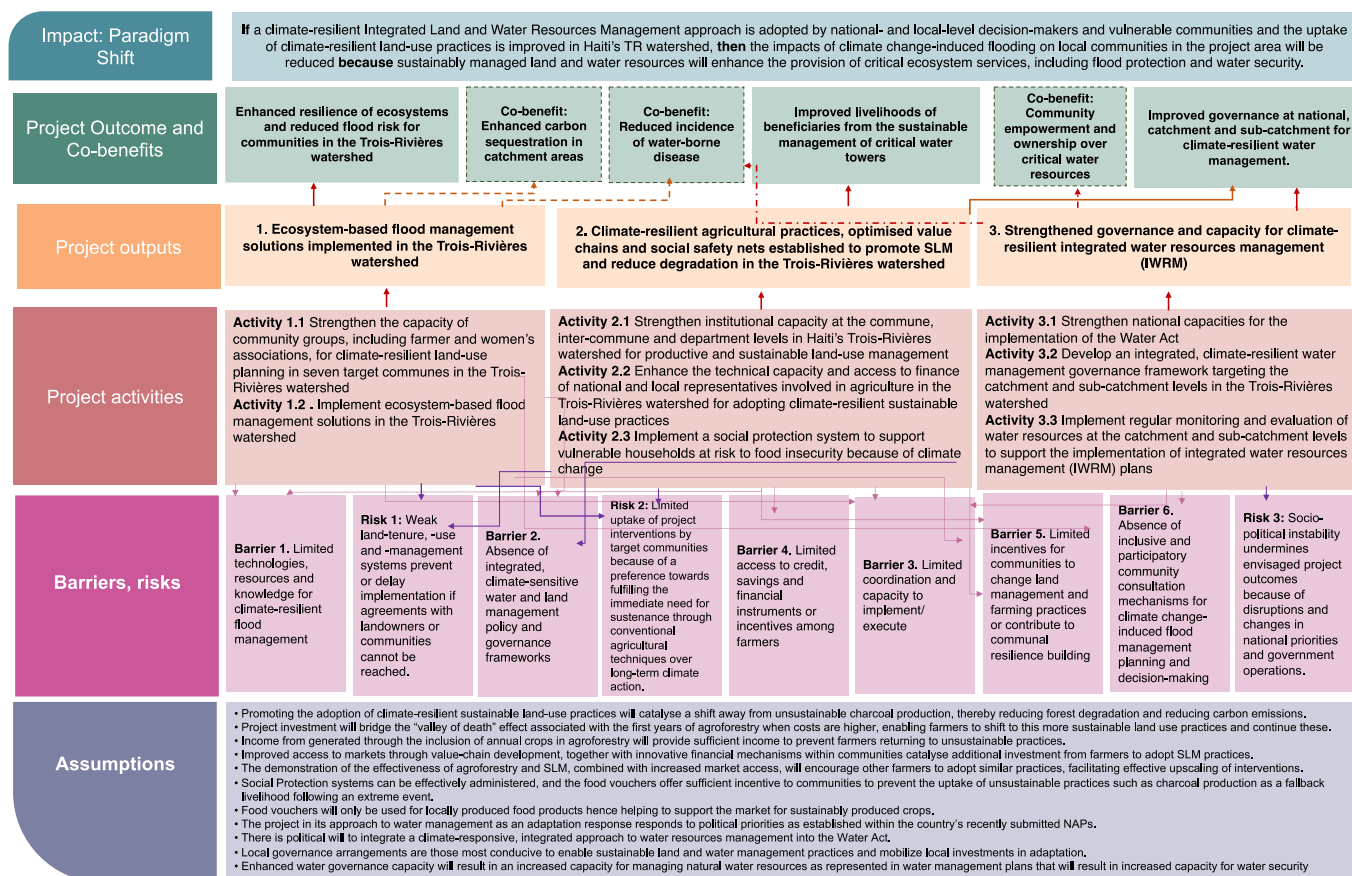


Figure 1. Project Theory of Change diagram.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Fill in the GCF results area table below to map each project/programme outcome identified in section B.2(a) to the contributing GCF results area(s) by referring to the description of eight results areas provided in the guidance note.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services

1. Enhanced resilience of ecosystems and reduced flood risk for communities in the Trois-Rivières watershed	☐	☐	☐	☐	☒	☐	☐	☒
2. Improved livelihoods of beneficiaries from the sustainable management of critical water towers	☐	☐	☐	☐	☐	☒	☐	☐
3. Improved governance at national, catchment and sub-catchment for climate-resilient water management	☐	☐	☐	☐	☐	☒	☐	☐

If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Enhanced carbon sequestration in catchment areas	☐	☐	☐	☐	☐	☒
Co-benefit 2: Reduced incidence of water-borne disease	☐	☒	☐	☐	☐	☐
Co-benefit 3: Community empowerment and ownership over critical water resources	☐	☒	☐	☐	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Project objective

54. The proposed project will contribute to GCF's paradigm shift objective of increased climate-resilient sustainable development by enhancing the resilience of Haiti's Trois-Rivières (TR) watershed to the impacts of climate change-induced flooding. This shift will be achieved by implementing a transformative approach to the way in which the GoH addresses flood impacts. Specifically, direct investments from the GCF combined with co-financing from the Heifer International Haiti-funded proposal⁵⁵ will be used to: i) implement agroforestry systems and rehabilitate 'water towers'⁵⁶ through reforestation⁵⁷ of degraded landscapes at priority intervention sites⁵⁸; ii) enhance technical and institutional capacity for productive climate-resilient land management at the national and local levels; and iii) establish the required governance framework for integrated water resources management (IWRM) to support the climate-resilient land management systems and facilitate sustainable use and management of water resources over the long term. The combined effect of project interventions will result in the adoption and implementation of a climate-resilient, integrated approach to flood management that can be readily scaled up and replicated nationally and across the Caribbean region. Proposed project outputs and activities are described in detail below with summaries of the project activities presented in Section E.6 of this Funding Proposal.

Output 1. Ecosystem-based flood management solutions implemented in 25,440 hectares of the Trois-Rivières watershed

55. At present, climate change-induced flooding is exacerbated by degraded landscapes and forests because of large-scale tree clearing for agricultural and charcoal production. During heavy rainfall events and flooding, the limited surface cover combined with the near-complete absence of embankments along the river intensifies soil erosion and river sedimentation. Implementing agroforestry systems and rehabilitating degraded forests in the TR watershed will address these problems by significantly reducing peak runoff and flooding extent in the area. These flood-reduction benefits will be realised through increasing vegetative cover in the watershed, which will attenuate flow by considerably reducing surface runoff and increasing the infiltration capacity of the soil. Reductions in water flow will reduce the risk of damage to public infrastructure and private property (e.g. homes and agricultural equipment), which will likely result in a decline in the observed annual damages and economic losses caused by flooding in Haiti. Furthermore, the implementation of these EbA interventions will have numerous environmental and social benefits for vulnerable households living in the target communes in the TR watershed. Rehabilitating degraded forests and establishing agroforestry systems will improve natural filtration of overland flows, removing water pollutants and, consequently, contribute towards avoiding sedimentation of key freshwater sources in the TR watershed. This will lead to improved water quality of these sources which, in turn, will contribute to reducing the spread of water- and vector-borne diseases (e.g. cholera), therefore safeguarding the health of communities living near to the priority intervention sites. The EbA interventions will also increase the ecosystem goods and services they provide (e.g. an improved quality and quantity of water supply, increased prevalence of agricultural pollinators and enhanced soil fertility⁵⁹), which will contribute positively to improving these communities' livelihoods.

56. Further livelihood benefits will be realised through the implementation of agroforestry systems in the target communes, including: i) Agri silvicultural systems — which are a combination of crops and trees, such as alley cropping; ii) silvopastoral systems — which combine forestry and grazing of domesticated animals⁶⁰; and iii) agrosilvopastoral systems that combine trees, crops and animals⁶¹. At present, most of the vulnerable communities living in the TR watershed depend on agriculture to sustain their livelihoods, with an observed increase in unsustainable charcoal production in years where the impacts of climate change have had particularly severe effects on agricultural yields and other livelihood options. Under this project output, vulnerable communities will be encouraged to adopt alternative livelihood options (such as beekeeping and mushroom cultivation). It is feasible for these livelihood options to be implemented alongside or independently of agroforestry services to catalyse a shift away from unsustainable charcoal production towards land use practices that will continue to provide long-term benefits under future climate change conditions.

⁵⁵ entitled *Project for improved food and nutrition security and access to social protection in the Trois Rivières watershed (also referred to as the "Pasan-Apros project")*

⁵⁶ "Water towers" refers to mountainous areas in Haiti where water for agricultural production and drinking is obtained.

⁵⁷ Restoration of arboreal land cover will be achieved through both agroforestry and reforestation. By promoting agroforestry in areas closer to households, beneficiaries will be able to integrate tree cover into their plots and obtain financial benefits from it. In areas further from households (i.e. water towers) the agricultural component of agroforestry will not be implemented, with only trees being planted in these areas. Reforestation will consist of reforesting the perimeters upstream of water points (using forest species present in the intervention zone) and also planting forest species in degraded mountainous slopes.

⁵⁸ Further details on the selected priority intervention sites are presented in Section 10 of Annex 2: Feasibility Study.

⁵⁹ Thompson, I.D. et al. 2011. "Forest Biodiversity and the Delivery of Ecosystem Goods and Services: Translating Science into Policy", *BioScience*, 61 (12).

⁶⁰ It should be noted that the project will not promote additional animal husbandry or breeding, but will instead focus on ensuring that existing livestock are managed sustainably using the principles of silvopastoralism.

⁶¹ FAO. 2015. Forestry: Agroforestry definition. Available at: <http://www.fao.org/forestry/agroforestry/80338/en/>

57. The implementation of agroforestry systems and reforestation of arboreal land cover on degraded mountainous slopes to manage flooding in Haiti's TR watershed will be supported under this project output by strengthening the technical and institutional capacity of community groups, including farmer and women's associations, for climate-resilient land-use planning. Specifically, this will include developing community land-use plans for use within 33 communal sections situated within the 7 target communes in the watershed. The development of these plans will facilitate the identification of climate change impacts, particularly flooding, on the TR watershed and the design of appropriate actions for addressing these impacts. Such actions include the implementation of EbA interventions such as flood reduction measures and the promotion of sustainable, climate-resilient land-use practices and techniques (e.g. agroforestry) to reduce the risk of increased land degradation under future climate change conditions. These plans will then be used to inform training on climate-resilient land-use management that will be delivered by the Responsible Parties to communities and farmer and women's associations in all seven communes. The training will focus on the practical application of climate-resilient land-use management and agricultural techniques and improving communities' understanding of the linkages between land use and hydrology of the TR watershed, particularly the implications of unsustainable land-use practices on the frequency and intensity of flooding in the watershed. The local and central authorities of the and MARNDR will be fully involved in the development of the community land use plans and accompanying trainings. These ministries already have systems in place for sharing information and organizational structures exist at the local level that will ensure the sustainability of these activities beyond the project lifetime. Exchange visits within the framework of other projects implemented by the and MARNDR will also be carried out to ensure information-sharing and coordination between the proposed project and ongoing national initiatives. Furthermore, lessons learned will be systematised under Output 2 of the project to be scaled up and replicated nationally and regionally.
58. To support this training, awareness-raising campaigns will be designed and implemented in each of the seven target communes in the watershed. These campaigns will focus on: i) increasing the knowledge of people living in the TR watershed on sustainable land-use practices and promoting the uptake of these practices; ii) encouraging community members to participate in the development of community land-use plans; and iii) raising awareness among people living in the target communes on the impacts of climate change-induced flooding and effective adaptation and landscape management solutions for addressing these impacts. These capacity-building initiatives combined with the implementation of EbA interventions will contribute towards a shift in the TR watershed towards the adoption of sustainable land-use management techniques and practices that are resilient to climate change-induced flooding.
59. Project activities that will be implemented under Output 1 are summarised below. Further details on these activities, including on the associated sub-activities and deliverables, are presented in Section E.6 of this Funding Proposal and Section 9 of Annex 2: Feasibility Study.

Activity 1.1. Strengthen the capacity of community groups, including farmer and women's associations, for climate-resilient land-use planning in seven target communes in the Trois-Rivières watershed

60. Under this activity, 33 communal sections⁶² within the 7 target communes in the TR watershed (Port-de-Paix, Chansolme, Bassin Bleu, Plaisance, Pilate, Marmelade, Gros Morne) will be validated through the following criteria vulnerability of the areas and populations, level of environmental degradation (including forest cover), proximity of the areas to the river and potential impact of flooding and their agricultural potential. Community members and farmer and women's associations within these 33 communal sections will be trained by the project's Responsible Parties on climate-resilient land-use management that will result in the development of 33 community land use plans that will guide on the ground interventions for community land use management.
61. The 33 community land use plans (Sub activity 1.1.1) will be shaped through community engagement in a consultative process and will focus on identifying the impacts of climate change, particularly flooding, on the TR watershed and designing appropriate actions for addressing these impacts. Such actions include the implementation of EbA solutions to reduce flood risk in the watershed as well as the promotion of sustainable, climate-resilient land-use practices and techniques (e.g. agroforestry) to reduce the risk of increased land degradation under future climate change conditions. The 33 land use plans developed will in turn the training that will be provided by Responsible Parties that will be provided in each specific area to the various land users (sub activity 1.1.2).

⁶² Aubert, La Pointe, Paulin Lacorne, Beaudin (Port de Paix); Chansolme, Source Beauvoir (chansolme), Haut des Monstiques, Carreay Datty, La Plate (Bassing Bleu), Pendu, L'Acul, Savane Carre, Moulin, Ravine Gros-Morne, Riviere Blanche, Riviere Mancelle, Boucan Richard (Gros-Morne), Platana (Marmelade), Riviere Laporte, Ravine Trompette, Joly, Piment, Ballon, Dubourgs, Margot, Beaudin (Pilate), Colline Gobert, Haut Martineau, Champagne, Bassin, Mapou, La Trouble, La Ville (Plaisance)

62. Hence, farmers in the target areas will also receive training by the Responsible Parties on the development of sustainable agroforestry business models that will contribute to both improving livelihood security and enhancing the use of sustainable land-use practices in the TR watershed (sub activity 1.1.2). The focuses of the trainings will be on:

- the expected impacts of climate change, particularly flooding, in Haiti and the associated consequences of these impacts on the population's livelihoods;
- linkages between land-use and hydrology of the TR watershed, particularly the implications of unsustainable land-use practices on the frequency and intensity of flooding in the watershed;
- promoting sustainable, climate-resilient agricultural practices such as slope stabilisation, riparian protection, and contour line farming, as well as the importance of protecting forest areas, as viable options for improving hydrological function of the watershed and reducing flood impacts;
- the role of the community in sustainable charcoal production and reforestation, as a strategy for increasing climate resilience in the watershed; and
- practical aspects of agroforestry as a climate-resilient livelihood alternative.

63. The above approach will create buy-in from beneficiary groups for the sustainable implementation of climate-resilient land-use practices such as agroforestry and securing commitment from beneficiary communities to maintain the adaptation interventions that will be implemented under Activity 1.2 during and after the project period. This approach will be enhanced by capacity-building initiatives and awareness-raising campaigns (sub activity 1.1.3) that will be designed and implemented throughout the project's seven target communes (going beyond just the 33 communal sections) through city hall meetings and community radio broadcasts. These campaigns will focus on: i) increasing the knowledge of people living in and in direct proximity to the TR watershed on sustainable land-use practices and promoting the uptake of these practices; ii) encouraging community members to participate in the development of community land-use plans; and iii) raising awareness among people living in the target communes on the impacts of climate change-induced flooding and effective adaptation and landscape management solutions for addressing these impacts.

Activity 1.2. Implement ecosystem-based flood management solutions in the Trois-Rivières watershed

64. This project activity will focus on reducing the impacts of climate change-induced flooding in the TR catchment and Port-de-Paix, specifically targeting degraded communal land that has been deforested in the past and that has resulted in increased flooding risk particularly for people living near the Trois-Rivières river (calculated at an estimated 292,600 people). An ecosystem based adaptation approach (EBA) will be taken by this activity through the implementation of 17,740 ha of agroforestry systems and the rehabilitation of 7,700ha of forest areas across targeted areas⁶³ within the 33 communal sections identified and validated through Activity 1.1. that will result in reducing the impacts of climate induced flooding by reducing peak runoff during extreme events.

65. Hydrological and economic studies commissioned by UNDP to support the development of the project indicate that implementing 17,740 ha of agroforestry systems and rehabilitating 7,700 ha of forests in priority areas in the watershed will considerably reduce the impacts of climate change-induced flooding. Specifically, implementing these EbA interventions will result in a 40% reduction in peak runoff, with rehabilitation interventions, in particular, expected to reduce floodplain extents in the target areas by 20% and 26% under the 20-year and 100-year flood scenarios, respectively. The EbA interventions are projected to result in a more than 50% reduction in the number of households affected by 100-year flood events (declining from 1,342 households at risk of flooding under the baseline scenario to 638 households with project interventions). Similarly, agroforestry and reforestation interventions under this project activity are expected to result in a 35% reduction in the number of households at risk to 20-year flood events in the TR watershed. This is representative of a decline from 463 under the baseline scenario to 300 houses at the end of the century under a 20-year flooding scenario. Further details on the modelling, calculations and methodologies used to determine these flood reduction benefits are presented in Sections 4.2 of Annex 2: Feasibility Study.

66. Specifically under Activity 1.2, the establishment of 17,740 ha of agroforestry systems (sub activity 1.2.1) and the rehabilitation of 7,700 ha of forest (sub activity 1.2.2) will be implemented and will include establishing nurseries, planting seedlings (e.g. for fruit production and reforestation) and providing ongoing technical assistance to end users including on the establishment and monitoring of nurseries, on the planning of seedlings, use of technical packages for agroforestry and monitoring of the agroforestry systems. To account for the growing times of many

⁶³ The total forest area in the watershed is 278 km², representing 31% the total watershed area, and the total urban area is 125.6 km², representing 14% of the total watershed area. Additionally, the total non-forest area (which will be targeted for rehabilitation with agroforestry or reforestation under this project activity) is 493.4 km², representing 55% of the watershed.

agroforestry species, the spaces between the agroforestry species will be filled with perennial food crops (e.g. plantains, sugar cane, cassava, pigeon peas and pineapple), in combination with annual crops (e.g. sweet potato, yams, and other annuals) (Tables 3 and 4). The implementation of agroforestry systems has a particularly high adaptation impact potential because of the provision of ecosystem goods and services beyond the target intervention areas (e.g. improved quality and quantity of water, more productive and stable soils, and crop pollination) combined with the creation of alternative livelihood options to unsustainable land uses (e.g. unregulated charcoal production, through the growing of species and shrubs that allow for regeneration through the use of correct pruning practices).

67. All agroforestry activities will also involve extensive engagements⁶⁴ — through workshops — with farmer and women's associations to encourage a progressive and gender-responsive transition towards sustainable agroforestry businesses that are sustainable over the long term. In addition the project will look to collect local and traditional best practices to enhance local ownership and appropriation. This includes the use of *jadden lakowu* and *ban manje* practices as adapted to local conditions and climates.

68. The rehabilitation of water towers through reforestation (sub activity 1.2.2) will also be undertaken in a participatory way and include: i) community engagement; ii) the provision of ongoing technical assistance for reforestation activities; iii) the development of long-term sustainable management plans for reforested areas at the sub-catchment level; and iv) encouraging the involvement of public institutions in the development and implementation of sustainable land management and reforestation strategies. Technical assistance and the development of management plan will be promoted through engagement workshops with community leaders, farmers and women's associations (sub activity 1.2.3)

Table 1. Perennial species suited for cultivation within the TR catchment.

Perennial species for the implementation of agroforestry systems in the TR catchment													
	Scientific name	Type of plant	Food	Shadow	Fodder	Fuel	Medicinal	Income	Timber	Hedge	Erosion control	Soil improver	Native (N), Currently Grown (G), or Newly Introduced (I) in Haiti
Fruit production													
Avocado (Avocatier)	<i>Persea americana</i>	Tree	X					X					G
Cocoa (Cacaoyer)	<i>Theobroma cacao</i>	Tree	X		X	X	X	X				X	G
Custard-apple (Cachiman)	<i>Annona reticulata</i>	Tree	X				X	X					G
Coffee Caféier	<i>Coffea arabica</i>	Shrub	X					X					G
Kassod tree (Cassia)	<i>Senna siamea</i>	Tree			X	X	X		X	X	X	X	G
Jamaica cherry (Cerise)	<i>Muntingia calabura</i>	Tree	X				X	X				X	N
Grapefruit (Chadequier)	<i>Citrus maxima</i>	Shrub	X				X	X					G
Lemon (Citronier)	<i>Citrus limon</i>	Shrub	X				X	X					G
Coconut tree (Cocotier)		Tree	X					X					G
Soursop (Corossol)	<i>Annona muricata</i>	Tree	X				X	X	X				N
Breadfruit (Lamveritable (arbre à pain))	<i>Artocarpus altilis</i>	Tree	X	X	X	X	X	X				X	G

⁶⁴ These engagements will not be undertaken using a cash-for-work approach, but rather encourage targeted groups to participate in the establishment of agroforestry systems and enhance the uptake of these systems as viable alternatives to unsustainable land uses (e.g. unsustainable charcoal production).

Leucaena (Lisina)	<i>Leucaena leucocephala</i>	Shrub			X	X			X		X	X	G
Mango (Manguier)	<i>Mangifera indica</i>	Tree	X					X					G
Orange (Orange doux/sure)	<i>Citrus sinensis</i>	Shrub	X					X					G
Forest													
American mahogany (Acajou)	<i>Swietenia mahagoni</i>	Tree		X		X	X	X	X				N
Mesquite (Bayawonn)	<i>Prosopis juliflora</i>	Shrub			X	X	X		X		X	X	G
Moringa tree (Benzolive)	<i>Moringa oleifera</i>	Tree	X		X	X	X		X		X	X	G
Cedar (Cèdre)	<i>Cedrella odorata</i>	Tree				X	X	X	X				N
Oak (Chene)	<i>Quercus petraea</i>	Tree		X				X	X				G
Bitterwood (Frene)	<i>Simaruba glauca</i>	Tree					X	X					N
Guaiaac (Gaiac)	<i>Guaiacum officinale</i>	Tree		X			X	X	X				N
West Indian Laurel (Laurier)	<i>Ocotea leucoxylon</i>	Tree						X	X				G
Hispaniolan pine (Pin)	<i>Pinus occidentalis</i>	Tree		X		X		X	X				G
Ice-cream bean (Poix doux ou Sucrin)	<i>Inga edulis</i>	Shrub		X	X	X						X	G
Saman or Rain tree (Saman)	<i>Albizia saman</i>	Tree		X	X	X	X		X			X	G

Table 2. Examples of agroforestry systems *with* perennial species presented above deployed in, and suited for, the TR catchment. Before perennial species reach full tree/shrub size, the spaces between them can be filled with perennial food crops (e.g. plantains, sugar cane, cassava, pigeon peas and pineapple), in combination with annual crops (e.g. sweet potato, yams, and other annuals).

Example 1		
Stage (year)	Species	Ecosystem goods and services ⁶⁵
1	Ice-cream bean	- Ice-cream bean seeds germinate at a rapid rate. - As they contain nitrogen-fixing properties, ice-cream beans are planted in improved fallows. - Ice-cream bean litter is high in nitrogen, lignins and polyphenols, which resultantly improves soil quality. Although the litter decomposes slowly, it provides for long-term accumulation of organic nitrogen and serves as effective weed control.
2	West Indian Laurels and/or <i>Saman</i> trees	West Indian Laurels and <i>Saman</i> trees provide fodder for livestock and shade for coffee and cacao plantations.
3	Cacao and Coffee	Cacao and coffee beans are sold to generate an income.
Example 2		
1	Moringa tree	- Moringa trees grow at a rapid rate. - Given their nutritional content and drought-resistance, moringa trees are an appropriate food source for ensuring food security. - Moringa trees are suited to environments which experience strong winds and droughts and resultantly provide soil erosion control. - The seed cake, which is a by-product of oil extraction, can be used as soil conditioner or fertiliser. - Moringa leaves are a good source of protein, Vitamins A, B and C and minerals such as calcium and iron. In addition, the leaves make a useful mulch.

⁶⁵ Available at: <http://www.worldagroforestry.org>

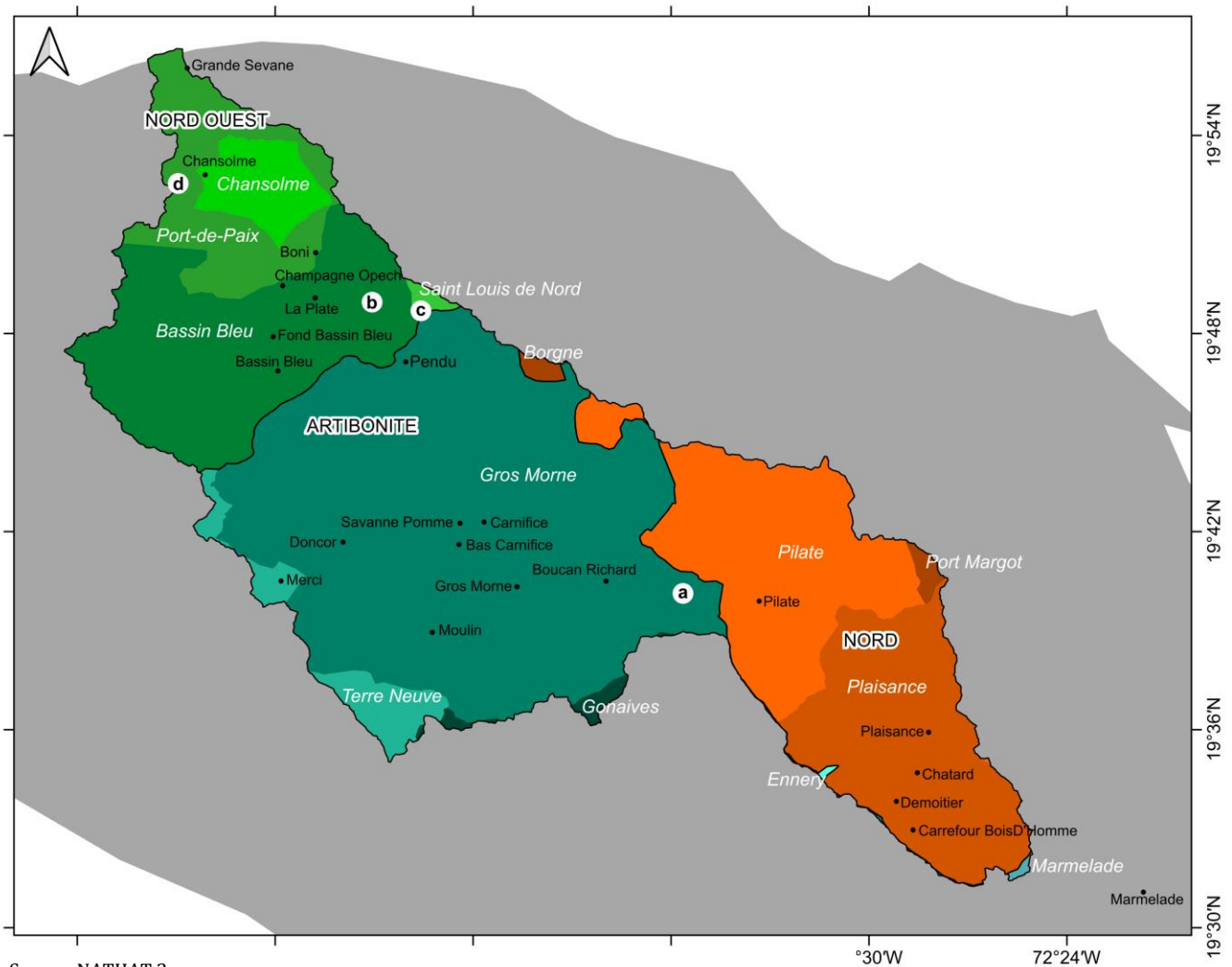
2	Soursop	- Soursop is native to Haiti and is a source of food for humans. - While small and early-bearing, soursop trees are planted as intercrops between larger fruit trees (such as mango, avocado and santol) and later uprooted when the fruit tree requires more space.
3	Mango or avocado	Mangoes and avocados are sold to generate income.
Example 3		
1	Bitterwood	- Bitterwood is native in Haiti. - The press cake produced from the milling operation is used as an organic fertiliser, particularly in coffee plantations and sugarcane, cotton and maize fields. - In addition to their rapid and linear growth, bitterwood trees are adaptable to shallow soils, making them a suitable species for agroforestry.
2	<i>Saman</i> and /or American mahogany	<i>Saman</i> trees provide shade as well as a cool microclimate for plants that grow beneath their canopies. - The seed pods contain a sweet-flavoured pulp which is often consumed by children. The pulp can also be used in making fruit beverages. <i>Saman</i> tree pods have a crude protein content of 12–18% (dry matter) with 41% digestibility for goats. The pods are also suitable fodder for cattle and horses. - The trees are suitable for sustainable non-destructive wood harvesting methods, such as pollarding. <i>Saman</i> wood produces 5,200–5,600 kcal/kg of biomass fuel when burned and regrows rapidly after lopping or pollarding, making it a valuable source of high-quality firewood and charcoal. - The <i>Saman</i> tree is highly prized for carvings, furniture and panelling because of the light and dark pattern the sapwood and the heartwood create.
3	Cacao and Coffee	Cacao and coffee beans are sold to generate an income.
Example 4		
1	Leucaena	- Leucaena leaves have a high nutritive value (high palatability, digestibility, intake and crude protein content) and is suitable fodder. - As it is in bloom almost throughout the year, Leucaena provides constant forage to honeybees. - It is a suitable firewood species. - Leucaena has a forceful taproot system which helps break-up compacted subsoil layers, improving the penetration of moisture into the soil and subsequently decreasing surface runoff. - Leucaena is used as a shade tree on cocoa, coffee and tea plantations. It often serves as a shelterbelt, providing shade and wind protection for a variety of crops, particularly during early growth. - The tree has high nitrogen-fixing potential (100–300 kg N/ha a year) related to its abundant root nodulation. - Leucaena was one of the first species to be used to produce green manure in alley-cropping systems. The leaves, even with moderate yields, contain enough nitrogen to sustain a maize crop.
2	Cedar	- Cedar flowers are a source of nectar for honey production. - Cedar trees are suitable for firewood. - Cedar wood is lightweight and comparatively soft. It is suitable timber for furniture, decorative veneer, musical instruments, wooden novelties and doors. - As the trees have many low branches and a spreading crown, they provide shade and serve as windbreakers in courtyard gardens and cocoa and coffee plantations. - Cedar trees can be planted as ornamentals along roads and in parks. They are planted for this purpose in, for example, peninsular Malaysia, Papua New Guinea and Singapore. - Cedar trees are highly susceptible to <i>Hypsipyla</i> infestations and as a result, it is recommended that the trees are planted in mixed plantations, for example with <i>Leucaena leucocephala</i>.
3	Cacao and Coffee	Cacao and coffee beans are sold to generate an income.

69. The implementation of agroforestry activities will be primarily focused for farmers with agricultural plots that are suitable for farming and have a certain level of degradation. The implementation of reforestation activities (including EBA activities), will be focused for farmers/landowners with land that has an accelerated level of degradation and

is not suitable for agricultural practices (high slopes, inaccessibility). In both cases, the properties of the public domain of the State will also be considered.

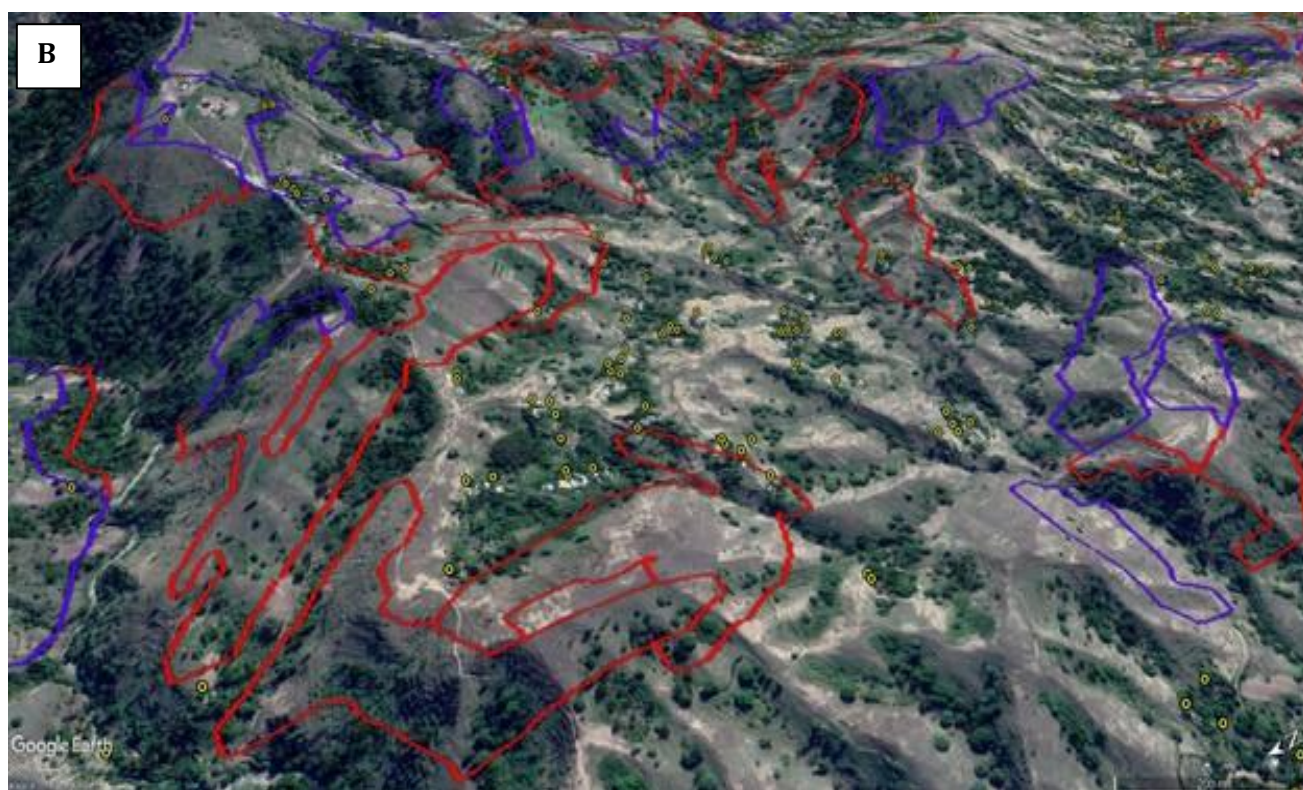
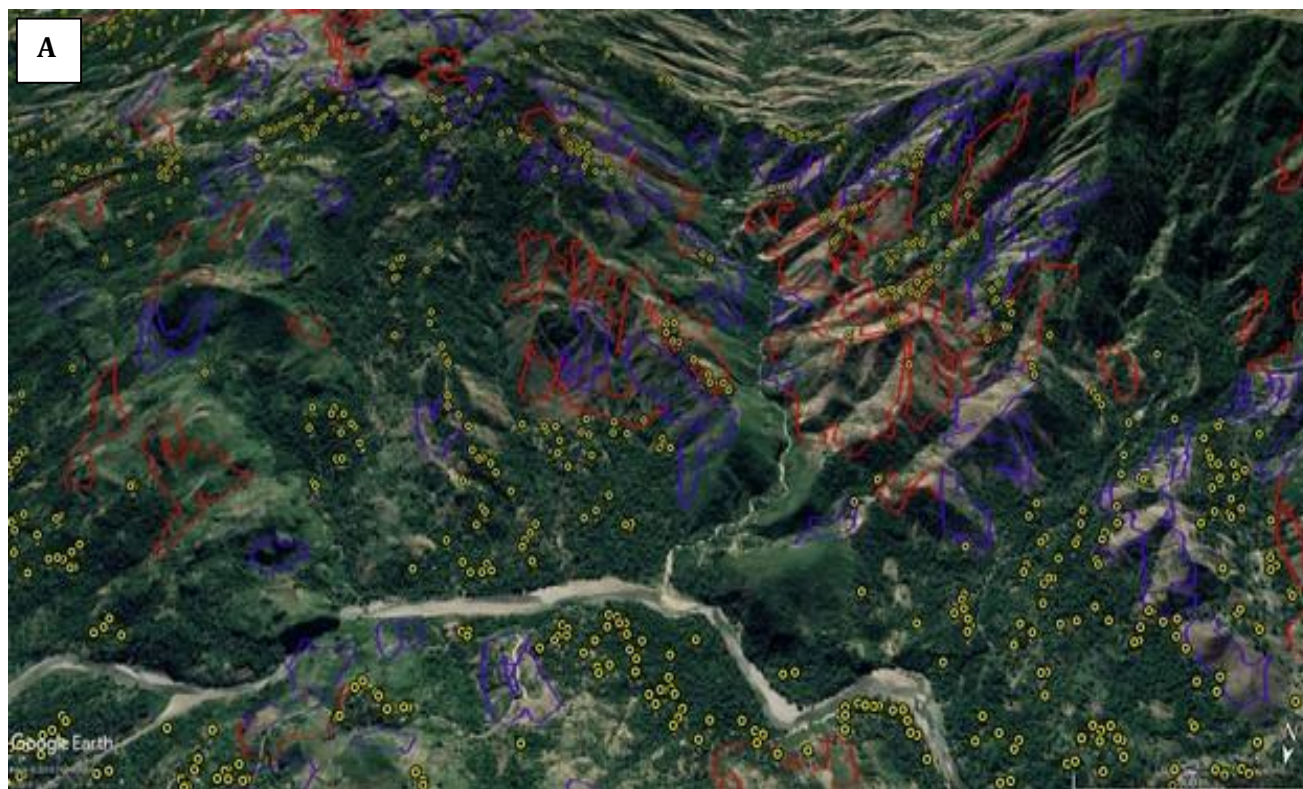
Priority intervention areas for agroforestry and reforestation interventions

70. The priority areas for the implementation of agroforestry and reforestation interventions in the TR watershed were defined using a multicriteria model that considered three variables. These were the slope of the terrain, distance from river channels and a maximum natural vegetation cover of 50%. Data from the ALOS DSM model — a global digital surface model with a 30-metre spatial resolution pixel, generated from the Panchromatic Remote-Sensing Instrument for Stereo Mapping (PRISM) on the ALOS satellite — was used to run the multicriteria model. Once the polygons identified as priority areas were established, the multicriteria model was then built to classify the optimal land use of each polygon into two intervention options, agroforestry or reforestation. The key variables used to classify polygons were density of houses/buildings and proximity to rivers and/or water sources. Finally, an assessment of available agricultural land was undertaken, informed by engagements with farmers to determine the expected participation levels in the implementation of agroforestry systems under the project. Since not all agricultural land where agroforestry will be introduced is classed as a priority area for reforestation, the actual area reforested (through the introduction of both agroforestry and reforestation) under the project will be larger than the prioritised areas. This will extend beneficiary farmers within the TR catchment beyond target communities whose land falls within priority zones. Based on an assessment of all these factors, 77 km² of land will be prioritised for reforestation and a total of 177.4 km² for agroforestry, considering participation rates. Of this 177.4 km² for agroforestry, 75.2 km² is classified as a priority area for agroforestry systems, and a further 102.2 km² is classified as a non-priority area. Further details on the site selection process undertaken for the proposed project are presented in Section 10 of Annex 2: Feasibility Study.
71. The locations of priority sites for reforestation and agroforestry interventions are presented in Figure 2Figure 3a and 3b below.



Source: NATHAT 2

Figure 2. Map showing the northern part of Haiti where the project intervention sites are located.



Source: Project Feasibility Study (Multicriteria model analysis developed for the project through Google Earth)

Figure 3a and b. Priority areas for reforestation (blue polygons) and priority areas for agroforestry (red polygons) interventions for various areas within the TR catchment, demonstrating the results from the classification method. a) Department Artibonite, Gros Morne Commune, Boucan Richard Section; and b) Bassin Bleu Commune, Nord Ouest Department, La Plate Section.

Output 2. Climate-resilient agricultural practices, optimised value chains and social safety nets established to promote SLM and reduce degradation in the Trois-Rivières watershed

72. Output 2 will be fully co-financed by Heifer. The key objectives of the Heifer support are to: i) strengthen food security governance; ii) encourage sustainable development in the agricultural sector; iii) safeguard natural resources in the TR watershed; and iv) promote access to a sustainable social safety net that encourages local agricultural development. While most of the Heifer-funded interventions can be implemented in the absence of the proposed GCF project, its impact will be minimised without the key adaptation interventions that are to be implemented under Output 1 of the proposed project. Therefore, the co-finance has been made available conditional on the approval of the GCF project. Likewise, the proposed project is dependent on the co-financed interventions to complete the integrated approach to climate-resilient flood management in Haiti (Output 1) and maximise the paradigm shift potential of the project. This will be achieved by facilitating a transformative shift in Haiti's agricultural sector from unsustainable practices (e.g. the cultivation of annual crop types and clearing of forests for charcoal production) towards sustainable options that are resilient to the impacts of climate change-induced flooding.
73. To facilitate this shift, a suite of complementary interventions will be implemented to promote the adoption and implementation of sustainable land-use practices in Haiti. This will be achieved by: i) strengthening the technical and institutional capacity of people involved in agriculture at the commune, inter-commune and department levels for implementing productive climate-resilient agricultural practices (such as agroforestry systems with alley cropping and contour bunds, as well as local Gad Manjé⁶⁶ systems); ii) enhancing smallholder farmers' access to financial resources to facilitate the uptake and implementation of climate-resilient agricultural practices over the long term; and iii) establishing a social protection system for chronically food insecure households in vulnerable areas affected by climate change impacts, particularly flooding.
74. Together, interventions under this output will stabilise access to markets for farmers and producers, incentivising the uptake of climate-resilient, sustainable land-use practices. By diversifying agricultural produce and agroforestry system varieties to prevent seasonal production peaks and providing technical support to farmers to adapt supply to demand while providing inputs that will facilitate their processing and their climate resiliency. This, in turn, will contribute to regulating price volatility of agricultural produce, attracting further investment into sustainable products and making the adoption of alternative livelihood options more feasible for the vulnerable people living in the TR watershed. As a result of these interventions, the project will enable vulnerable farmers and households living in the watershed to invest in climate resilient productive practices through a sustainable land management perspective in keeping with the EBA interventions and land planning established through Output 1. This will in turn reduce the adverse social and economic impacts associated with these events and promote livelihood security amongst these people.
75. These interventions will further complement the activities that will be implemented under the GCF-funded project currently under development, entitled *Increasing resilience of vulnerable farmers in Southern Haiti* being implemented with the support of the Food and Agriculture Organization (FAO) on the South Department of Haiti. While the geographic scope of each project will prevent from in overlap of any on-the-ground interventions the outcomes of these projects related to institutional capacity building at national level will be strongly complementary securing an integrated approach to land and water resources management that will contribute to a paradigm in Haiti towards climate-resilient sustainable development.

Activity 2.1. Strengthen institutional capacity at the commune, inter-commune and department levels in Haiti's Trois-Rivières watershed for productive and sustainable land-use management

76. Under this activity, a capacity needs assessment will be conducted for the agricultural sector in the 7 targeted communes to identify the technical and institutional requirements for transitioning the sector from unsustainable land-use practices towards a sustainable landscape management (SLM) approach to land and water resources management that is climate resilient (Sub activity 2.1.1). Capacity will be assessed in the seven target communes in the TR watershed, as part of this assessment the project will consult with farmers associations at a commune and sub commune level. From the capacity needs assessments, opportunities will be identified, and plans made to enhance the management of natural resources and promote the uptake of climate-resilient agricultural practices to increase productivity.

⁶⁶ Gad Manjé translates to "a band of food", hedgerows that comprise perennial crops (e.g. plantains and cassava), in combination with annual crops (e.g. sweet potato and yam). The hedgerows provide structural integrity to stabilise soils to prevent erosion, whilst the crops provide a source of food security and income generation.

77. To facilitate the uptake of these new sustainable practices, governance mechanisms and frameworks for SLM will be developed in the TR watershed through the establishment of communal development councils (Sub activity 2.1.2). These councils will represent all agricultural sector stakeholders, including smallholder farmers in the seven communes both during and after the project implementation with their primary function being to enhance the management of natural resources in combination with improving the climate-resilience of agricultural practices and enhancing sustainable productivity. Members to these councils will be appointed by Ministry of Agriculture in collaboration with the Ministry of Environment and the local authorities. Multi-sectoral governance structures will also be established to oversee and manage the implementation of interventions designed to address the priority areas identified through the capacity needs assessment developed under Sub activity 2.1.1.

78. Support for the governance structures developed under Sub Activity 2.1.2 will be provided through strengthening the capacity of relevant government entities, including the Ministry of Agriculture, Natural Resources and Rural Development (MARNDR), National Coordination of Food Security (NCFS) and Ministry of Social Affairs and Labour (MSAL), to implement SLM at the department level (Sub activity 2.1.3). These sectoral entities and ministries have a primary role to play in the implementation of sustainable land management hence training and information workshops to members of these ministries will allow for increased ownership and sensitization of their role and mandates for sustainable land management. Training and information workshops under this subactivity will focus on the content of management plans, implementation strategies, and the role and implications of departmental stakeholders. This will enable the adoption of SLM practices to be guided in a manner that is compatible with the effective functioning of the entire watershed and that contributes to enhancing flood management across the TR watershed.

Activity 2.2. Enhance the technical capacity and access to finance of national and local representatives involved in agriculture in the Trois-Rivières watershed for adopting climate-resilient sustainable land-use practices

79. This project activity will increase the capacity of farmers in the 7 target communes to implement climate-resilient agricultural techniques and land-use practices, using the findings from the capacity needs assessments conducted as part of Activity 2.1. This will be achieved through implementing targeted adaptation interventions in Haiti's TR watershed to optimise the agricultural value chain and promote the use of climate-resilient agricultural techniques across the sector (Sub activity 2.2.1). Such interventions will include enhancing the transportation of agricultural produce (e.g. by developing agricultural tracks), as well infrastructure upgrades to address decreasing water security caused by the impacts of climate change-induced flooding. Infrastructural upgrades will also be undertaken in the form of increasing the storage capacity and quality of post-harvest facilities, as well as establishing local processing facilities for agricultural produce to optimise agricultural production in the watershed. These investments will be communal in nature and will be prioritized with the support of local farmers association to benefit all members.

80. Technical capacity provided will also include the delivering of technical skills training on climate-resilient agricultural techniques and SLM to farmers. This will be done through a farmer field school approach where approximately 30 farmers will work in groups (estimated 180 groups along the 33 subcommunes) to put in practice the training provided. These members will then form their own group and provide training to 10 of their neighbors. Capacity building will involve: i) enhancing their ability to effectively manage and organise irrigation and other associations; ii) facilitating participatory access to agricultural inputs, farming equipment, maintenance facilities; and iii) upskilling of maintenance staff. These interventions will be supported by the provision of climate-resilient agricultural inputs (e.g. climate-resilient seed varieties) to vulnerable households that depend on subsistence agriculture for their livelihoods (Sub activity 2.2.2).

81. Strengthening the above key elements of the agricultural value chain in Haiti's TR watershed will not only enhance the climate resilience of agricultural production in the country, but also contribute to increased food security in the watershed. Sustainability of these investments will be supported by looking to improve the access to finance to producers in a manner that is appropriate for communities in line with the sustainable productive management capacities that are being created. Hence this activity will also develop social feasibility assessments, including risk assessments and financial mapping analysis (Sub activity 2.2.3) to inform the establishment and enhancement of Agricultural Village Savings and Credit Associations (VSCA) that will be led by the communities and supported through technical expertise by the project as part of Activity 2.3. Within the framework of the proposed project, VSCAs are levers for households benefiting from the social safety net to develop small economic activities with the credits or the amount of "remunerated" savings obtained at the end of the year, in order to increase their income in a sustainable way and establish a means to enhance sustainable productive investments and enhancing local markets.

Activity 2.3. Implement a social protection system to support vulnerable households at risk to food insecurity because of climate change

82. This project activity will support households affected by chronic food insecurity to recover their nutritional and financial autonomy being those most vulnerable to climate change-induced flooding and resorting to unsustainable land management practices. Subsistence farmers living near floodplains along the TR watershed are particularly vulnerable to these events, which often result in the near-complete destruction of their agricultural yields grown to support their livelihoods. When this happens, most of these farmers take up unsustainable charcoal production as a last resort to support their livelihoods, which in turn results in increased degradation of the landscape, exacerbating soil erosion and sedimentation of the TR watershed and its tributaries. While Activities 1.2 and 2.1 provide a long-term solution to addressing these negative impacts, this activity will provide the required immediate relief to vulnerable people affected by these events, concomitantly reducing further pressure on the environment by providing an alternative option to resorting to unsustainable charcoal production.
83. Specifically, Activity 2.3 will facilitate the development of a social safety net whereby food coupons will be provided to vulnerable households affected by food insecurity. The food coupons will be designed for the project and will be provided by Heifer. To determine eligible households for these coupons, vulnerable households in the 33 communal sections will be identified through a site survey and/or assessment, with their details captured in a database for future reference (Sub Activity 2.3.1). Recipients of these coupons will be able to redeem them in exchange for local produce (food basket) at selected local markets (Sub activity 2.3.2). An estimated 934⁶⁷ farmers (households) will be selected to participate in the operation of the food coupons mechanism. Participating farmers will receive training on *inter alia* hygiene, modalities of exchange and personal security (Sub activity 2.3.3).
84. This activity will also support in the the creation of 150 new VSCAs in the TR watershed (Sub Activity 2.3.4). VSCAs are relevant as they provide their members with small, flexible financial services of savings, microcredit and micro-insurance (see footnote 63 above) for capital investments. Hence, VSCAs will become a means to mobilize local financing even after the project is completed. It is important to note that GCF grants nor cofinancing will be provided in the capitalization of the VSCAs rather these will be capatilized by the initial investment provided by each of the VSCA members as a way to ensure full community ownership of the VSCA and grant flexibility in the bylaws that are created. The VSCAs will be created with the support of a network of an estimated 150 village agents (Sub activity 2.3.5) who will act as extension officers to the VSCAs and that will be formed by the project. The village agents will have as their primary responsibilities the monitoring of existing VSCAs and creation of new ones. Moreover, to optimise selected farming markets contributing to the implementation of the food coupons mechanism, market performance will be monitored regularly, and the relevant adjustments made to the system where required. This process will be community-driven, led by the selected network of village agents. Through these actions an exit strategy for the food coupon mechanism is created that is linked to the establishment of agroforestry systems (long term productive means) as well as in the creation of community appropriate financial mechanisms (VSCAs) and capacities (village agents) within the target areas to mobilize saving for continued capital investment.

Output 3 Strengthened governance and capacity for climate-resilient integrated water resources management (IWRM)

85. At present, the government is organised along departmental and communal lines. As a result, there is an overlap between existing water governance structures, particularly at the catchment and sub-catchment levels. This leads to challenges in coordination and planning, as well the effective and efficient implementation of an integrated approach to water resources management in the TR watershed. While the GoH has made good progress in drafting the Water Act⁶⁸ to address these challenges, there is a need for technical assistance for the 'last mile' of its drafting. Under Output 3, the capacity of the GoH will be strengthened to develop and implement the Water Act, which is an essential legal foundation for enabling participatory, integrated water resources management in Haiti's TR watershed. This will involve: i) drafting provisions to Haiti's Water Act for the adoption of an integrated approach to water resources management (Sub activity 3.1.1); ii) establishing Catchment IWRM Committees (Comités de GIRE des Bassins Versants; CGBVs) within 31 catchments in Haiti (Sub activity 3.1.2); and iii) establishing Sub-Catchment Water Resource Users Associations (Associations d'Usagers des Ressources en Eau du Sous-bassin versant; AssURES) within the sub-catchments of the 31 catchments (Sub activity 3.1.3). These committees will

⁶⁷ Averaging 133 farmer households per commune (7) based on an estimated prioritization taken from Haiti's PROCSIMAST (Programme de Renforcement Opérationnel du Ciblage et Système d'Information du Ministère des Affaires Sociales et du Travail) which is an official master list of vulnerable people by community provided by the Ministry of Social Affairs. From this list the following people will be selected: Women head of household in charge of kids and other elderly and/or disabled with no income or low income (living under poverty level); 60 + years of age; Disabled and Residing permanently in the community. This will allow the programme to focus on the most vulnerable and with least adaptive capacity.

⁶⁸ The Water Act acts as regulatory policy towards water integrated management that may take the form of a National Regulatory Decree.

work with different stakeholders in the 31 catchments and sub-catchments to manage the efficient use of land and water resources. Furthermore, the committees will be responsible for fostering community empowerment and ownership over these critical resources under future climate change conditions. Together, these activities will ensure that the Water Act is developed and implemented to respond not only to national level priorities, but also to the adaptation needs of vulnerable communities living in the TR watershed.

86. To support the implementation of the Water Act and the transition towards an integrated management structure for land and water resources in the TR watershed, a Catchment Integrated Water Resources Management Committee (CIWRMC) will be established (Sub activity 3.2.1). This committee will work with representatives from productive sectors (e.g. agriculture), land users at sub-catchment level and Sub-Catchment Water Resources User Associations (SCWRUAs) to develop, implement and manage an inclusive, integrated water management governance framework in the watershed . This will involve these committees working together to develop climate-resilient IWRM plans (Sub activity 3.2.2), which define roles and responsibilities for end users that promote sustainable water resources management and an equitable distribution of the available resources at the catchment and sub-catchment levels. The CIWRMC, together with the relevant land users, SCWRUAs, CGBVs and AssURES will ensure that the management of water resources in the catchment considers climate change impacts, particularly the expected impacts at landscape level (Sub activity 3.2.3) as well as on upstream and downstream flow dynamics .
87. Ensuring that all adaptation interventions are monitored throughout the implementation phase is critical for facilitating adaptive management and assessing the effectiveness of these interventions in reducing climate change-induced flooding in the TR watershed. The implementation of EbA interventions under Output 1, combined with the improved governance and capacity of relevant decision-makers and actors involved in land and water resources management (Outputs 2 and 3), is expected to reduce the impact of climate change-induced flooding on the TR watershed and communities living nearby, while concomitantly catalysing a shift from segregated land and water resources management towards the use of an integrated governance framework that is resilient to the impacts of such flooding. Moreover, these interventions will generate benefits for the health of vulnerable for individuals, households and communities living in the TR watershed, reducing the spread of water- and vector-borne diseases as contamination of freshwater sources (e.g. rivers) and food is reduced.
88. To ensure that these adaptation benefits continue to be realised by all beneficiary groups throughout the project period, continuous health monitoring and impact evaluations will be undertaken under Activity 3.3. This Monitoring and Evaluation (M&E) will commence at the end of Year 1 and will continue throughout the implementation phase until project closure. UNDP will commission an expert to undertake the health monitoring and impact evaluations of the EbA flood management solutions (Sub activity 3.3.1). The World Health Organisation (WHO) will provide technical support to this expert, as they are currently monitoring cases of water- and vector-borne diseases in Haiti and therefore have a pre-defined methodology that would enable comparison studies to be undertaken across watersheds. The contracted expert will be responsible for: i) undertaking a baseline assessment of the prevalence of water- and vector-borne diseases within the TR watershed (Sub activity 3.3.2); ii) analysing the impact of climate change-induced flooding on the spread of these diseases; and iii) assessing and reporting on the health benefits of the adaptation interventions implemented under the project, particularly the reduced spread of water- and vector-borne diseases as a result of implementing EbA interventions under Output 1 (Sub activity 3.3.3). Finally, lessons generated through the implementation of all project interventions will be shared through the relevant committees and associations at the catchment and sub-catchment levels to inform adaptive management of these interventions under future climate change conditions (Sub activity 3.3.4).
89. By strengthening national capacities for the implementation of the Water Act, promoting the adoption of an integrated, climate-resilient water management governance framework at the catchment and sub-catchment levels in the watershed, and monitoring the impact of IWRM plans and EbA interventions (Output 1) on flooding and water resources management in the TR watershed, the interventions under Output 3 will contribute to establishing the governance framework required to facilitate the adoption of a climate-resilient integrated water resources management (IWRM) in Haiti's TR watershed. The first step that is required for this process is to strengthen the capacity of the GoH to develop and implement the Water Act, which is an essential legal foundation for enabling participatory, integrated water resources management in Haiti's TR watershed. Once this is done, governance around water resources at the local level will then be improved in line with the Water Act to promote the more sustainable management of these resources, through establishing the necessary local-level regulatory structures. Finally, to complete the cycle of understanding and improving water resources management, adequate monitoring of these resources — that also considers flooding and its impacts on health — is important to feed back into the regular updating of the Water Act and facilitate adaptive management through the local-level regulatory structures.

90. Project activities that will be implemented under Output 3 are summarised below. Further details on these activities, including on the associated sub-activities and deliverables, are presented in Section E.6 of this Funding Proposal and Section 9 of Annex 2: Feasibility Study.

Activity 3.1 Strengthen national capacities for the implementation of the Water Act

91. Under this activity, the capacity of the GoH will be strengthened to develop and implement the Water Act, which is an essential legal foundation for enabling participatory, integrated water resources management in Haiti's TR watershed. At present, the government is organised along departmental and communal lines. As a result, there is an overlap between existing water governance structures, particularly at the catchment and sub-catchment levels. This leads to challenges in coordination and planning, as well the effective and efficient implementation of an integrated approach to water resources management in the TR watershed. While the GoH has made good progress in drafting the Water Act to address these challenges, there is a need for technical assistance for the 'last mile' of its drafting. Under Activity 3.1, provisions will be drafted to the Water Act to ensure that it promotes an integrated approach to water resources management, including considerations for the coordinated development of water, land and related resources (Sub activity 3.1.1). This will contribute towards: i) maximising economic and social welfare in an equitable manner without compromising the sustainability of vital ecosystems under future climate change conditions; and ii) facilitating efficient, integrated adaptation responses to the impacts of climate change in Haiti. Provisions to the Water Act will also recommend the establishment of Catchment IWRM Committees (Committees de GIRE des Bassins Versants, CGBVs) within each of the 31 catchments in the country (Sub activity 3.1.2), and the establishment of Sub-Catchment Water Resource Users Associations (Associations d'Usagers des Ressources en Eau du Sous-bassin versant, or AssURES) (Sub Activity 3.1.3).

Activity 3.2. Develop an integrated, climate-resilient water management governance framework targeting the catchment and sub-catchment levels in the Trois-Rivières watershed

92. Given that the TR watershed is one of the most vulnerable watersheds in Haiti to climate change-induced flooding, the implementation of an integrated water resources management strategy is a priority action for increasing the climate resilience of its population. Under this activity, an inclusive and equitable water management governance framework will be developed for the TR catchment, namely the Catchment Integrated Water Resources Management Committee (CIWRMC) (Sub activity 3.2.1). This committee will be made up of actors directly or indirectly involved in the agricultural and water sector (state institutions including the MoE and Haiti's Hydrometeorological Unit (UHM), local elected officials, farmers, schools (in the integrated development approach), representatives of civil society (economic or social organizations, producers' organizations), associations and grassroots groups) and its members will be defined during the consultative workshops.

93. Together they will develop climate-resilient Integrated water resource management plans for end users at the sub-catchment and catchment levels that promote sustainable water resources management and equitable distribution of the available resources among all users (Sub activity 3.2.2). The gender-responsive IWRM plans will define roles and responsibilities of institutions and civil society stakeholders for water resources management (complementary to the decentralised management of water resources to be established in the Water Act under Activity 3.1). It will also promote a landscape management approach that will be linked to the implementation of community land-use plans developed under Activity 1.1 considering gender dynamics (Sub activity 3.2.3). The CIWRMC, will also work with the relevant land users and SCWRUAs, to ensure that the management of water resources in the catchment considers climate change impacts, particularly the expected impacts at landscape level as well as on upstream and downstream flow dynamics. Training on the climate-resilient IWRM plans will also be delivered to these stakeholder groups, focusing on the application of these plans in decision-making (Sub activity 3.2.4).

94. The combined result of these actions will be the establishment and operation of an integrated governance framework that will contribute to preserving climate-resilient, sustainable land and water resources management beyond the lifetime of the project. During the implementation of Activity 3.2, considered efforts will be made to securing equitable representation, participation and leadership of women.

Activity 3.3. Implement regular monitoring and evaluation of water resources at the catchment and sub-catchment levels to support the implementation of integrated water resources management (IWRM) plans

95. From project inception (Year 1), continuous health monitoring and impact evaluations will be undertaken and will continue throughout the implementation phase. The implementation of EbA solutions under Output 1 of the project are expected to significantly reduce the extent of climate change-induced flooding in the TR watershed and reduce erosion caused by extreme rainfall events. Such reductions in flood extent are also likely to reduce the spread of water- and vector-borne diseases, as contamination of freshwater sources (e.g. rivers) and food is reduced. By monitoring the evolution, frequency and location of flood-related illnesses from the start of the project, the impact of the project can be assessed once a sufficient time series of observations is completed. UNDP will commission

an expert to undertake the health monitoring and develop impact evaluations of the EbA flood management solutions (Sub activity 3.3.1). The World Health Organisation (WHO) will provide technical support to the health monitoring expert, as they are currently monitoring cases of water- and vector-borne diseases in Haiti and therefore have a pre-defined methodology that would allow comparison studies to be undertaken across watersheds. The contracted expert will undertake a baseline assessment of the prevalence of water- and vector-borne diseases within the TR watershed and analyse the impact of climate change-induced flooding on the spread of these diseases (Sub activity 3.3.2). Following this assessment, the expert will report on her/his findings and provide the required health impact data to assess the health benefits of the EbA interventions implemented under the project.

96. Simultaneously, the project will engage an integrated water resource management expert that assisted by representatives from SCWRUAs and in consultation with the UHM, will monitor dry season baseflow at four locations along the main stem of the Trois-Rivières (TR) river. This will be done to provide a stable indicator of the increased infiltration capacity of the soil because of forest rehabilitation and restoration activities to be implemented under Activity 1.2 of the project (Sub activity 3.3.3). The monitoring will facilitate an evidence-based assessment of these EbA solutions as effective measures for reducing the impacts of climate change-induced flooding in the TR watershed and for improving the health of the watershed's population. Finally, lessons generated through the implementation of all project interventions will be shared through the relevant committees and associations at the catchment and sub-catchment levels as well as to relevant national ministries including the UHM to inform adaptive management of these interventions under future climate change conditions (Sub activity 3.3.4).

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

97. The project will be implemented following UNDP's Full Support to National Implementation Modality (NIM), according to the Standard Basic Assistance Agreement (SBAA) between UNDP and the Government of Haiti signed on 28 June 1973, the Country Programme Action Plan (CPAP), and as policies and procedures outlined in the [UNDP - POPP - Chapter](#).
98. The national executing entity - also referred to as the national 'Implementing Partner' in UNDP terminology is required to implement the project in compliance with UNDP rules and regulations, policies, and procedures (POPP), including the NIM Guidelines ([POPP NIM](#)). These include relevant requirements on fiduciary, procurement, environmental and social safeguards, and other performance standards. The (national) Implementing Partner for this project is the Ministry of Environment (MOE).
99. Implementing partner (IP) is accountable to UNDP for managing the project, including the monitoring and evaluation of project interventions achieving project outcomes and for the effective use of UNDP resources. A subsidiary agreement (project document) will be established between UNDP and MoE as per Article 3 of the SBAA. All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner".

Executing Entity (IP per UNDP terminology)

100. The Ministry of Environment of Haiti (MoE) will act as an Executing Entity (IP) for the project and as such is accountable to UNDP for managing the project, including the monitoring and evaluation of project interventions achieving project outcomes and for the effective use of GCF resources. The MoE as Executing Entity will designate a National Project Director, responsible for the overall direction, strategic guidance and timely delivery of the Project. The project, through the National Project Director, will inform the Technical Secretariat of the National Committee on Climate Change of the progress of the project's implementation to ensure alignment and coordination with the GoH's national climate strategies and policies.
101. The MoE is a national Ministry for the Government of Haiti. It was created by the Haitian law of January 28, 1995 with its mission, attributions and intervention modes being defined in decrees of January 26, 2006 (Décret portant sur la gestion de l'environnement et de la régulation de la conduite des citoyens) and the decree of August 16, 2020 (décret organisant le Ministère de l'environnement). It is the National Ministry responsible for overseeing the environment and natural resources for Haiti as mandated by law. Its attributions includes: (1) to formulate, guide and enforce the GoH's policy on environmental matters; (2) to regulate any public or private action falling within the domain of the environment and; (3) to guarantee sovereign measures conducive to the protection of the environment, the conservation of biodiversity, the fight against climate change and a better living environment for the benefit and well-being of the population. The MoE acts as the National Designated Authority on behalf of the GoH to the GCF.
102. The MoE delegates its various functions to numerous agencies and institutions related to its management area, among these is the Change Directorate which acts as the lead environmental body in the system. As a public institution, it receives an estimated USD 12million annual funding from the GoH's national budget. It has expertise in implementing the enabling activities such as national communications, Biennial reports. The MoE

is currently executing the CBIT project with UNDP oversight. The MoE also collaborates with partners such as civil society organizations in the execution of this work.

103. A HACT microassessment has been developed for the MoE identifying its capacity to manage projects with UNDP and partner support. The HACT assessment identified the MoE's overall risk level as moderate taking into account a low risk rating in for partnerships and collaborations, moderate risks in governance, project management and human resources and a significant risk in monitoring and evaluation and procurement and financial management.
104. The Government of Haiti has requested UNDP to provide direct project services for this project, including in the procurement of goods and services through UNDP processes, providing direct payments to suppliers and Responsible Parties, hiring of project staff, as well as in the delivery of training services. Under these terms UNDP will make payments and provide related services at the instruction of the MoE who will be responsible for the preparation of terms of references and selection criteria. The UNDP and the Government of Haiti acknowledge and agree that those services are not mandatory and will be provided only upon Government request and specified in the Letter of Agreement (LoA). If requested, the direct project services would follow UNDP policies on the recovery of direct project costs relating to GCF funded projects. The project has been designed to enable UNDP to be able to provide support as the government needs for execution under its support to NIM Modality that has been reflected within the project's budget, in addition UNDP is providing co financing support to enhance project management and will provide implementation support in the delivery of activities under Output 3.
105. UNDP will provide implementation support in the delivery of activities included within the project's Output 3. A firewall will be implemented to separate the work being provided for the implementation of Output 3 and UNDP's oversight responsibilities in its role as an AE. A Letter of Agreement (as specified in paragraph 86) will be signed between the Government of Haiti and UNDP specifying UNDP's role in Output 3

Accredited Entity

106. UNDP provides a three – tier oversight and quality assurance role involving UNDP staff in Country Offices and at regional and headquarters levels. The quality assurance role supports the Project Board by carrying out objective and independent project oversight and monitoring functions. This role ensures appropriate project management milestones are managed and completed. Project Assurance must be independent of the Project Management function; the Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. The project assurance role is covered by the accredited entity fee provided by the GCF. As an Accredited Entity to the GCF, UNDP is required to deliver GCF-specific oversight and quality assurance services including: (i) Day-to-day oversight supervision, (ii) Oversight of project completion, (iii) Oversight of project reporting."
107. The justification to engage UNDP support services responds to HACT evaluation assessment of the MoE in particular in regards to procurement and financial management processes. UNDP will be the best entity to engage various partners from both public and private sector, either following competitive procurement process or other partnership arrangement. Such support services and the cost involved, shall be detailed in a Letter of Agreement (LoA) for UNDP support services. As the support services will be provided by UNDP, usual UNDP regulations, rules and procedures apply. A procurement and administrative analyst will also be hired to support the project operations.

Responsible Parties

108. Responsible parties will be overseen by and report to the MoE in its role as Executing Entity. As such they will enter upon written agreements with the MoE specifying their role in project implementation including deliverables, budgets and workplan. Responsible parties will report on the usage of project funds and delivery of the workplan to the MoE.
109. Heifer International Haiti will act as Responsible Party for Output 1. Heifer International Haiti will also execute all activities under Output 2 through its own co-finance. Once the FAA is signed between UNDP and the GCF and upon indication by the MoE, a Responsible Party Agreement will be established with Heifer International Haiti on behalf of the Executing Entity for the provision of services for the execution of these outputs.
110. Heifer International Haiti is recognized in Haiti as a non-profit corporation working in Haiti since 1999 by supporting rural communities through sustainable and integrated crop-livestock projects as a means of poverty reduction. Heifer International Haiti has managed more than 20 million in Haiti in the past 8 years and has plenty of experience working in the northern part of the country with community organizations, farmers organizations, local authorities, and government institutions. Its work places an emphasis on community mobilization, resilience, social capital development, environmental protection, personal leadership, the promotion of gender equity, agricultural

production and food security, agribusiness development, and the mobilization of young people and women in agriculture. It is currently implementing a portfolio in Haiti of over USD12 million.

111. Heifer International Haiti holds offices in the Trois Rivières (TR) Watershed and has demonstrated implementation capacity (low-risk rating) as evaluated through UNDP's Harmonized Approach to Cash Transfers (HACT) assessments. UNDP's HACT assessment noted Heifer's high capacity for monitoring (including the use of a QR system for project beneficiaries) and reporting, the presence of sufficient technical staff that receives continuous training to address donor and program needs and its appropriate financial accounting measures and procedures.

Project Board

112. The Project Board will ensure that activities under all three outputs of the project comply with the strategies, principles and overall approach outlined in this proposal. The Project Board is responsible for making decisions by consensus. Its responsibilities including recommendation for UNDP and Implementing Partner approval of project plans, substantive revisions and acceptance of the project progress reports. In order to ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development results, the best value for money, fairness, integrity, transparency and effective international competition. In the case where a consensus cannot be reached, the final decision shall rest with UNDP. Project reviews by the Board will take place at least twice a year, or as necessary when raised by the Project Manager or one of the Board members. The Board must be consulted by the Project Manager for decisions where there are substantial changes required to the Annual Workplan as per deviation tolerances set by a donor or UNDP, whichever is lower.
113. The specific roles of the Project Board are to:
- set strategic direction, reinforce government leadership of the program and coordinate all interventions;
 - provide guidance and agree on possible countermeasures/management actions to address specific risks;
 - authorise the Annual Work Plan (prior to approval by UNDP);
 - conduct bi-annual meetings to review the Project Progress and provide direction and recommendations to ensure that the agreed deliverables are produced satisfactorily according to the approved Annual Work Plan;
 - provide ad-hoc guidance when project manager's is unable to come to a decision;
 - review and approve all activities that are supported by the program based on the program objectives, work plan and availability of funding;
 - provide technical advice to create synergy and uniformity between program supported activities and policy;
 - guide and support program delivery at sectoral level;
 - provide support in resource mobilization to support program funding gaps;
 - monitor and evaluate project activities through periodic meetings and occasional site visits; and
 - receive reports on all activities supported by the program to serve as an additional basis to assess and monitor the program performance and delivery.
114. The composition of the Project Board, along with each participant's roles and responsibilities are listed below.
- The Executive will preside over the Project Board. This role will be fulfilled by the MoE.
 - As the "Development Partner" for the project, UNDP's role is to represent the interests of the parties, which provide funding and/or technical expertise to the project (designing, developing, facilitating, procuring, implementing), including providing guidance regarding the technical feasibility of the project. Within this role, UNDP will have the authority to commit or acquire the necessary resources to carry out its function to the full extent.
 - The Ministry of Environment (MoE)'s role is to ensure that investment decisions at the project level are relevant to national policies and ensure continued involvement and strengthening of national capacities.
 - The Ministry of Agriculture, Natural Resources and Rural Development (MARND) will fulfil the 'Main Beneficiary' role for the project. Within this role, the MARND is mandated with representing the interests of project beneficiaries to ensure that project outputs serve the needs of the beneficiaries.
 - Representatives from the beneficiaries will assist to annual Project Board meetings by providing necessary information on the execution of activities and act as non-voting observers.

Project Management Unit (PMU)

115. The Project Manager (PM) will provide strategic orientation to the proposed project and ensure technical and operational coordination of the Project Management Unit (PMU) that will receive training from UNDP. Specifically, the PM will undertake day-to-day management and decision-making for the project. In line with this mandate, the PM will be primarily responsible for guiding the implementation of adaptation interventions to achieve the targets specified in the Logical Framework (Section E of this Funding Proposal) and ensuring that these interventions are

implemented to the required standards, within the specified time periods and according to the project budget. The PM will be supported by the technical staff under the PMU, including: i) national and international consultants and support staff; ii) an Agroforestry Expert who will coordinate activities under Outputs 1.1 and 2.1; iii) a Governance and Policy Expert who will coordinate activities under Output 3; iv) a full-time Gender and Safeguards Specialist, who will be responsible for tracking project interventions against the Environmental and Social Impact Assessment and Management Plan (Annex 6) and Gender Assessment and Action Plan (Annex 8); and v) a full-time Monitoring and Evaluation (M&E) Expert, who will be responsible for detailed monitoring of the progress of project activities against the targets specified in the Logical Framework. The PM's responsibilities will conclude when the project Terminal Evaluation report and other relevant documentation required by the GCF and UNDP have been completed and submitted to UNDP for clearance. While UNDP's process will be used for the hiring of the PMU, it is important to highlight that the MoE will elaborate the terms of reference for project staff including the selection criteria.

Experience and track record of UNDP as the Accredited Entity and Support services provider:

116. Within its central role in the UN system, UNDP facilitates a multi-sectoral approach in assisting developing countries to respond to the complex challenges associated with climate change. Furthermore, through its focus on promoting a long-term, people-centred approach to capacity development, UNDP is recognised as an experienced agency in institutional development which will add considerable value to the implementation of an integrated approach to land and water resources management in Haiti.
117. To date, UNDP has led numerous environmental protection and climate change responses in Haiti through *inter alia* policy formulation, strategic planning, coordination and information sharing. Through these actions, UNDP has built strong relationships with decision-makers at all levels and proven its strengths as an impartial provider of technical guidance and support to communities in the country. Key resilience projects using an ecosystems-based adaptation (EbA) approach that were managed and implemented by UNDP in Haiti in recent years are listed below.
 - Increasing resilience to climate change and anthropic threats through a ridge to reef approach — GEF Trust Fund; approximately US\$ 9.3 million.
 - The Post-Matthew Recovery Programme — Government of Canada, Korea International Cooperation Agency (KOICA), World Food Programme; approximately US\$ 15 million.
 - Support for the Preparation and Management of Disasters — Government of Japan; approximately US\$ 2.1 million.
 - Natural Disaster Prevention and Territorial Governance — United States Agency for International Development (USAID), Norwegian Ministry of Foreign Affairs; approximately US\$ 1.7 million.
 - National Risk and Disaster Management System — Department for International Development (DFID), USAID, European Commission, The World Bank; approximately US\$ 10.5 million.
 - Resilience to Disasters in *Département Nord* — Government of Japan; approximately US\$ 4.6 million.
118. UNDP in Haiti acts as co chair to various donor and partner groups for Haiti that are relevant to the project, including the Technical and Financial Partner Group for the Environment Sector and the International Community Support Group (GACI) for Disaster Risk Management. UNDP will provide general information on the progress of the project and lessons learned to the aforementioned partner groups to allow for enhanced coordination amongst partners as well as allowing for building on best practices and lessons learned.
119. The management arrangements for this project are summarised in the chart below (Figure 4).

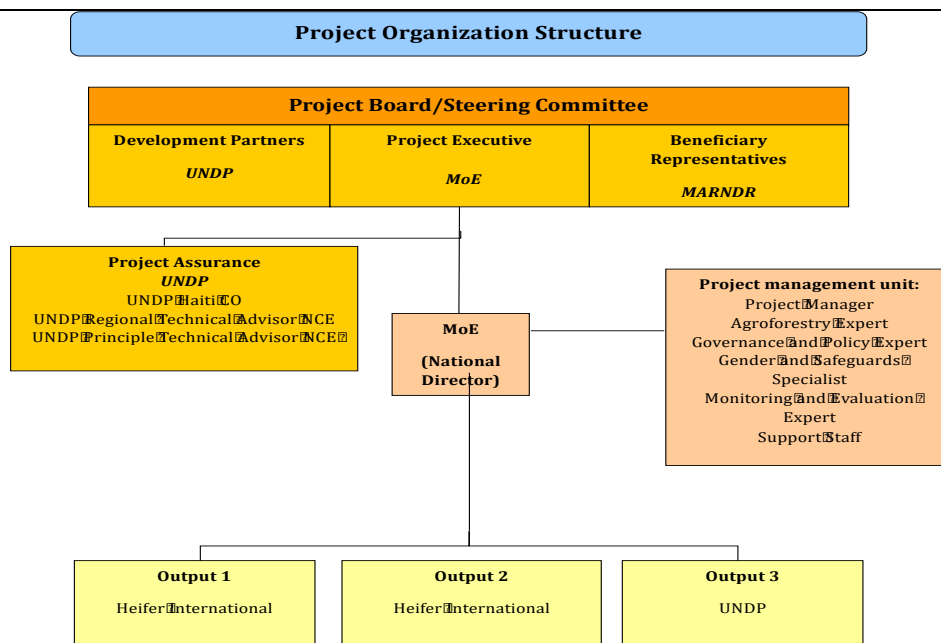


Figure 4. Institutional arrangements.

120. GCF funding will be received by the administrative agent (UNDP) as the AE based on a standard letter of agreement (FAA) signed between the GCF and UNDP (Figure 5). As funds will be channelled through UNDP, a 'Direct Payments' option will be utilised, whereby transfers will be made directly from UNDP to the goods and service providers based on annual workplans submitted by the MoE. Heifer International Haiti's co-financing resources (see red line on Figure 5), which are conditional on the approval of this project by the GCF, will be used by Heifer International Haiti to implement the project interventions outlined in Output 2.

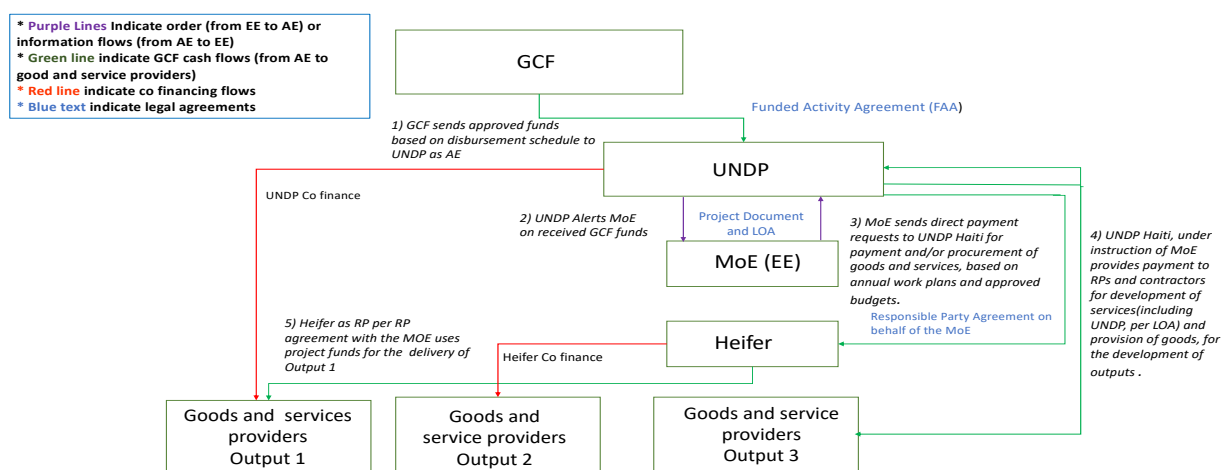


Figure 5. Simplified infographic of the flow of funds/payments and contractual arrangements.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

121. As a Caribbean SIDS and LDC, Haiti is one of the most vulnerable countries to the impacts of climate change and extreme events, particularly flooding, hurricanes and tropical storms. Climate change-related events are already adversely affecting the country's economy, with negative impacts extending across several of Haiti's major economic sectors. Between 1975 and 2012, such events resulted in damages and losses equivalent to 2% of the

country's GDP on average, per year. Since then, the impacts of climate change have continued to negatively affect the population of Haiti. For example, Hurricane Matthew (Category 5) that struck Haiti in 2016 resulted in the deaths of 546 people, forced more than 175,500 people to seek refuge in shelters and led to more than 1.4 million people requiring immediate humanitarian assistance. The storm also inflicted economic damages and losses estimated at more than 22% of the country's GDP in that year (2016), including the losses of up to 90% of crops and livestock in some areas of the country, and caused a sharp increase in the number of cholera cases. The intense rainfall associated with these extreme events also leads to severe flooding, which exacerbates the associated negative impacts. For example, in 2004, Hurricane Jeanne caused severe flooding in northern Haiti, resulting in more than 2,800 deaths, approximately US\$41 million in damages and required removing approximately 592,000 m³ of mud that clogged drainage networks and infrastructure across the country⁶⁹.

122. The Trois-Rivières (TR) watershed in the north of the country is particularly vulnerable to the impacts of climate change-induced flooding. Given the near-complete absence of embankments and levees in the watershed, such flooding causes severe damage to crops and farmland, as well as agricultural infrastructure and equipment. Flooding in the region also has severe consequences for both human and livestock health, resulting in an increased risk of loss of human lives and livestock⁷⁰. Climate change projections simulate an increase in rainfall intensity in the TR region, which is expected to result in further adverse impacts on the country's economy as well as on the most vulnerable communities and their livelihoods. Addressing the impact of climate change-induced flooding on vulnerable communities in the region, requires adopting an integrated approach to flood management that responds to the impacts of climate change. The Government of Haiti (GoH) acknowledges this requirement and is consequently requesting US\$25 million in grant financing to facilitate the adoption of this management approach. At present, the GoH does not have the capacity to fully finance integrated climate-resilient flood management interventions through domestic finance in both the private and public sector. This is largely because of the persistent occurrence and impact of extreme climate events such as hurricanes and tropical storms on the country, which has caused the GoH to channel funding towards reactive disaster response interventions. Consequently, the amount of domestic financing available for implementing climate change adaptation solutions in the country is limited.
123. Currently, foreign aid accounts for more than half of Haiti's national budget. The International Monetary Fund (IMF) estimates that public debt in Haiti will increase from approximately 29% of the country's GDP in 2016 (equivalent to US\$2.2 billion in the same year) to approximately 38% in 2021 and approximately 49% in 2026. Such considerable increases in public debt will place the country at a high risk of debt distress. The majority of Haiti's approximately US\$2 billion debt as of 2015/16 is attributed to external loans on concessional terms. Moreover, domestic public debt amounts to approximately US\$112 million, mostly in the form of treasury bills held by commercial banks, with the account deficit (including international grants) estimated at approximately 4% for 2016/17⁷¹. High levels of debt as a financial barrier to adapting to climate change is exacerbated by the need to address poverty, with approximately 59% and approximately 24% of the population living below the national and extreme poverty lines, respectively. This is compounded by the negative effects of climate change-induced flooding on public health, as inundations facilitate the increased spread of waterborne diseases such as cholera. The resultant cost of no-action in addressing these health risks places further burdens on the country's already constrained healthcare system. Consequently, poverty combined with the impacts of climate change-induced flooding and extreme climate events are locking the GoH into a negative cycle of short-term reactivity, with insufficient funding being available for climate-responsive planning in Haiti.
124. Several alternative financing options were considered during project development, as fully financing the project through government funds is not possible. This is because the country's current public debt burden limits the GoH from accessing loans from both regional and international development banks. Given that the GoH does not have the financial capacity to finance the project, the government has secured co-financing of US\$8.3 million from the Heifer International Haiti to support the implementation of interventions under Output 2. While this contribution is significant, external support is required to fully finance a transition in Haiti towards implementing and adopting an integrated approach to land and water resources management that will reduce the impacts of climate change-induced flooding on the TR watershed and vulnerable communities living nearby.
125. External support in the form of GCF grant financing will be necessary to catalyse the adoption of such an approach to flood management in Haiti's TR watershed. With the support of GCF funding, the proposed project will deliver several adaptation benefits that will contribute to achieving the GCF paradigm shift objective of climate-resilient sustainable development in the country. Specifically, this will be achieved by: i) implementing agroforestry

⁶⁹ Gaspard, G. 2013. Flood loss estimate model: Recasting flood disaster assessment and mitigation for Haiti, the case of Gonaives. Available at: <https://pdfs.semanticscholar.org/b32b/d0ccdcce15f58b07a70f5b1bd5d570baf7f79.pdf>.

⁷⁰ OXFAM, Welt Hunger Life, NATHAT, and Cap-Haitian.

⁷¹ Further details on Haiti's fiscal situation are presented in Section 2.2 of Annex 2: Feasibility Study.

17,740 ha of agroforestry systems and reforestation 7,700 ha in priority sites⁷² in the TR watershed; ii) enhancing national and local technical and institutional capacity for productive climate-resilient land management; and iii) strengthening national and sub-catchment governance and capacity for climate-resilient integrated water resources management (IWRM). Implementing this suite of adaptation interventions in the TR watershed will reduce the impacts of climate change-induced flooding on the vulnerable communities living in the TR watershed and increase the resilience of the agricultural sector — a major economic sector in the country — to these impacts. Moreover, adaptation benefits conferred by IWRM will be realised in public health improvements, as the prevalence of waterborne diseases will decrease alongside improved flood management.

126. Without GCF support, the GoH will continue to operate under the existing paradigm of reactive recovery from the impacts of climate change. In this scenario, local-level decision-makers in the TR watershed will remain limited in their capacity to plan for and respond to the impacts of climate change — particularly those related to increasingly severe flooding. Moreover, as agriculture will be severely affected by climate change — through damages to crops and ecosystems as a result of flooding, as well as projected reductions in annual rainfall and increases in maximum temperatures contributing to increased droughts — both the country's livelihoods and economy will be negatively impacted. Under future climate change conditions, this will result in *inter alia*: i) continued declines in agricultural productivity and food security; ii) public health concerns, including diarrhoeal diseases; and iii) increased economic damages and loss of life.

B.6. Exit strategy (max. 500 words, approximately 1 page)

127. The proposed project interventions have been designed to deliver maximum adaptation benefits to vulnerable communities in Haiti's Trois Rivières (TR) watershed beyond the project lifetime. Such benefits include reducing flood impacts on vulnerable communities living in the watershed and improving environmental sustainability under future climate change conditions through implementing sustainable land and water resources management practices. To secure these benefits, the principles of institutional, technical and financial sustainability have been incorporated into the design of all project interventions, as described below.

Participatory approach to climate-resilient sustainable development

128. The country-driven project has been developed using a participatory and consultative process, enabling all relevant stakeholders to contribute to the design of interventions. These interventions are centred around enhancing the resilience of the TR watershed to the impacts of climate change-induced flooding by implementing an integrated approach to land and water resources management. Uptake of such an approach across the watershed requires buy-in and commitment from national- and local-level stakeholders to ensure that the adaptation interventions implemented under the project respond directly to the needs of all beneficiary groups and continue to deliver adaptation benefits to these groups over the long term.
129. Given that the project promotes an integrated approach to land and water resources management, participation from national and local stakeholders, particularly communities and the most vulnerable groups (e.g. women and people with disabilities), is critical for securing project ownership and long-term adaptation benefits in Haiti. As a result, all project stakeholders will continue to be engaged in the implementation of relevant interventions throughout the implementation phase. This engagement will be supported by awareness-raising campaigns (Activity 1.1), which will be designed and implemented in target communes in the TR watershed to increase local knowledge of sustainable land-use practices and promote their adoption. These campaigns will also encourage community members to participate in the development of community land-use plans, fostering ownership of these plans to secure the buy-in of stakeholders at the community level, which will ensure the sustainability of on-the-ground interventions. Community ownership will also be fostered for the implementation of agroforestry systems and rehabilitation of water towers (Activity 1.2) by promoting community involvement in related activities, such as the establishment of nurseries and planting of seedlings. Demonstrating these activities as viable alternative livelihood options under the project will contribute to maintaining their social and environmental sustainability beyond the project lifetime.
130. An extensive stakeholder engagement plan (Annex 7) has also been developed to ensure that this participatory approach is continued during implementation. The impact and sustainability of project interventions will be further enhanced through the project's focus on ensuring gender-responsive benefits are provided to project beneficiaries. Gender-sensitive targets for all project activities have therefore been set in the project's Gender Action Plan (Annex 8) to ensure that project interventions are not gender-biased. This will contribute to the ownership of project activities by both women and men which, in turn, will result in gender-sensitive adaptation benefits being conferred beyond the project lifetime.

⁷² Detailed information on the identification and selection of sites for the EbA interventions is presented in Section B.3 of the Funding Proposal and Section 10 of Annex 2: Feasibility Study.

Strengthened technical and institutional capacity for sustainable climate-resilient land and water resources management

131. Under the project, the technical and institutional capacity of relevant stakeholders at national, department and commune levels will be strengthened for the implementation of an integrated approach to land and water resources management that is centred around sustainability and climate resilience. This will be achieved by training relevant government entities, including the Ministry of Agriculture, Natural Resources and Rural Development (MARNDR), National Coordination of Food Security (NCFS) and Ministry of Social Affairs and Labour (MSAL), to implement sustainable land management (SLM) at the department level. Doing so, will enable the compatibility of SLM with the effective functioning of the entire watershed which, in turn, will contribute to enhancing flood management across the TR watershed. By building the capacity of the targeted institutions, the transfer of knowledge and skills related to sustainable watershed and flood management will extend beyond the project. This will result in: i) a sound understanding of the impacts of climate change-induced flooding on land use, water resources, food security, health and governance in Haiti; and ii) strengthened decision-making around the use of land and water resources under future conditions of climate change. The local and central authorities of the MoE and MARNDR will be strongly involved in the capacity-building interventions under the project to secure the sustainability of these interventions beyond project closure. Exchange visits within the framework of other projects implemented by the MoE and MARNDR will also be carried out to ensure information-sharing and coordination between the proposed project and ongoing national initiatives.
132. Such capacity-building activities will also extend beyond government departments and ministries to demonstrate the effectiveness of improved climate-resilient land-use planning and water resources management as a viable climate change adaptation option to be implemented in Haiti. Specifically, this will involve training community members, and farmers' and women's associations in 33 communities within the 7 target communes in the TR watershed (Activity 1.1). This training will focus on climate-resilient land-use management to enhance participant's awareness of the need for the continued prevention of land degradation and implementation of sustainable agricultural practices. In addition, technical capacity will be strengthened among farmers in the TR watershed for the implementation of climate-resilient agricultural practices (Activity 2.2). This activity will include the implementation of interventions that will provide households with increased market access in the long term for produce grown from agroforestry systems established under Activity 1.2, contributing to project sustainability. Long-term sustainability of the project will be further reinforced by providing training to these households on the development of sustainable agroforestry business models that will contribute to both improving livelihood security and enhancing the implementation of sustainable land-use practices. These models will be designed to promote a progressive and gender-responsive transition from unsustainable agricultural and charcoal production towards the development of sustainable agroforestry businesses (Activity 2.2). To support this training, institutional capacity for sustainable land-use and management will be developed in the TR watershed. This will be achieved through the establishment of an integrated governance framework for SLM under newly established communal development councils and multi-sectoral management structures (Output2). These councils and management structures will represent all agricultural sector stakeholders, including smallholder farmers, both during and after the project implementation. Their primary function will be to enhance the management of natural resources in combination with improving the climate-resilience of agricultural practices, to increase productivity across agricultural subsectors. Consequently, the governance structures will complement activities aimed at community members' capacity building by providing oversight of the implementation of EbA interventions (agroforestry and reforestation) and assisting with the management and maintenance of these interventions.
133. In addition to strengthening governance for SLM in the TR watershed, capacity will be built for the climate-resilient management of water resources in Haiti (Output2). This will be achieved by drafting provisions to the Water Act (under Activity 3.1) to ensure that the act promotes an integrated approach to water resources management in the country. This will establish a legal foundation for optimising the use of these resources without compromising the sustainability of vital ecosystems under future climate change conditions, consequently, securing equitable economic and social welfare. The Water Act will also provide a framework for the institutionalisation of integrated watershed management to empower regional administrations across Haiti — through establishing Catchment Water Resources Management Committees (CIWRMCs) and Sub-Catchment Water Resource Users Associations (SCWRUAs). The primary functions of these committees will be to coordinate planning of water and land use that both maximises benefits to local communities and increases resilience to climate change. This will facilitate a participatory, bottom-up process for the development and implementation of Integrated Water Resources Management (IWRM) in Haiti through local governance processes. Long-term sustainability of governance frameworks will be further ensured through the generation of lessons and under Activity 3.3 and the dissemination of these lessons through the committees and councils established under Outputs 1 and 2. This will inform and improve future adaptation efforts as well as adaptive management of the project, thereby enhancing the success of climate-resilient sustainable development in Haiti.

Strengthening agricultural value chains and unlocking public and private sector investment into sustainable climate-resilient land-use practices

134. Financial sustainability will be ensured through the GoH's commitment to prioritise climate change adaptation, including in national budgeting. This commitment is evidenced in Activity 3.2 of the project, in particular, where the CIWRMCs will work with representatives from productive sectors (for example agriculture), land users at the sub-catchment level and SCWRUAs to develop climate-resilient IWRM plans for end users. These plans will promote sustainable water resources management and an equitable distribution of available resources among all users. The financial sustainability of the project will be further secured by establishing an adaptation finance mechanism under Activity 2.3 through the VSCAs. This mechanism will increase access to finance (through peer to peer loans) for smallholder farmers, providing much needed financial services for continued investment in the maintenance and enhancement of the project supported productive measures, therefore, encouraging a shift in land use away from unsustainable farming practices such as charcoal production by establishing financially sustainable ecosystem compatible livelihoods. To support this shift, a social protection system will be created (Activity 2.3) to support households affected by chronic food insecurity, enabling them to recover their nutritional and financial autonomy immediately after being impacted by climate change-induced flooding. Specifically, this will involve the establishment and operationalisation of a food coupons mechanism targeting these vulnerable households. To support the efficient and effective implementation of the food coupons mechanism, existing VSCAs will receive operational improvements supplemented by the creation of 150 new VSCAs in the TR watershed. Furthermore, a network of village agents will be established, with their primary responsibilities being: i) monitoring existing VSCAs and creating new ones; ii) monitoring market performance and implementing adjustments to the system where required. The community-driven nature of these interventions will contribute to securing the financial sustainability of interventions beyond the project lifetime.

Operation and maintenance (O&M) beyond the project lifetime

135. Sustainable operation and management of all adaptation interventions under the project has been secured by identifying appropriate government and beneficiaries to oversee specific project activities. The GoH is committed to: i) financing all operations, maintenance and insurance activities required under the project; and ii) supporting the governance structures established under Outputs 1 and 2, as well as the community land-use plans developed under Activity 1.1 To support the GoH and ensure that adaptation interventions are implemented efficiently and effectively under the project the Responsible Parties will be beneficiaries for the project. In this role, these institutions will provide specific technical inputs and support to MOE on the implementation of all interventions. The RP (Heifer International Haiti) will implement Outputs 1 and, 2. UNDP will provide support in the implementation of Output 3. The GoH is committed to undertaking the required O&M of these interventions to ensure that they continue to confer adaptation benefits to project beneficiaries beyond project closure, this is outlined in the letter attached to this FP. O&M will be further secured through incentivising community groups (e.g. farmer associations, ROSCAs) to maintain the project interventions because of livelihood support and alternative income streams provided by these interventions.

136. During the implementation of UNDP-supported projects in the field, close attention is paid to institutional capacity building to facilitate the sustainability and durability of projects. UNDP, as accredited entity, will work in close collaboration with local, departmental and national elected officials to ensure maximum effectiveness of the project even after its lifetime. In this capacity building, UNDP will first train the institutional managers to facilitate the transfer of skills, then we will equip the different institutions to better implement the project.

137. Two recent examples of projects that implemented similar activities (agroforestry and supply chain) that have proven to be sustainable beyond the project implementation are "Increasing resilience of ecosystems and vulnerable communities to CC and anthropic threats through a ridge to reef approach to BD conservation and watershed management" (funded by the LDCF) and "Strengthening adaptive capacities to address climate change threats on sustainable development strategies for coastal communities in Haiti" (funded by the GEF).

138. The table below provides a *per hectare* estimate of implementation and O&M costs to be covered by GCF and agroforestry beneficiaries. Implementation is considered to be the investments incurred in year 1 and O&M any investments thereafter.

	Implementation costs (USD/ha)	O&M (USD/ha)	
		during implementation	after implementation
GCF	350.62	348.19	
beneficiaries	702.48	2505.52	2563.10

FINANCING INFORMATION

C.1. Total financing						
(a) Request ed GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	22,427,206.48			million USD (\$)		
GCF financial instru ment	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter %</u>		
(ii) Subordin ated loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter %</u>		
(iii) Equity	<u>Enter amount</u>			<u>Enter % equity return</u>		
(iv) Guarante es	<u>Enter amount</u>	<u>Enter years</u>				
(v) Reimbur sable grants	<u>Enter amount</u>					
(vi) Grants	22,427,206.48					
(vii) Results- based payment s	<u>Enter amount</u>					
(b) Co- financin g informat ion	Total amount		Currency			
	<u>Enter amount</u>		Options			
Name of institutio n	Financ ial instru ment	Amount	Currency	Tenor & grace	Pricing	Seniori ty
Heifer Internatio nal Haiti	<u>Gr ant</u>	8,303,355	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Option s</u>
UNDP	<u>Gr ant</u>	560,000	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Option s</u>
Click here to enter text.	<u>Opti ons</u>	<u>Enter amount</u>	<u>Options</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Option s</u>

Click here to enter text.	<u>Options</u>	<u>Enter amount</u>	<u>Options</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Options</u> <u>s</u>
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	31,290,561.48			million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	This is not applicable for the proposed project.					

C.2. Financing by component

Output	Activity	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
1. Ecosystem-based flood management solutions implemented in 25,440 hectares of the Trois-Rivières watershed	1.1.Strengthen the capacity of community groups, including farmer and women's associations, for climate-resilient land-use planning in seven target communes in the Trois-Rivières watershed	1,702,000.00	1,702,000.00	<u>Grants</u>	0	<u>Grants</u>	
	1.2 Implement ecosystem-based flood management solutions in the Trois-Rivières watershed	16,396,800.00	16,396,800.00	<u>Grants</u>		<u>Grants</u>	
	M&E	300,800.00	100,800.00	<u>Grants</u>	200,000.00	<u>Grants</u>	<u>UNDP</u>

2. Climate-resilient agricultural practices, optimised value chains and social safety nets established to promote SLM and reduce degradation in the Trois-Rivières watershed	2.1. Strengthen institutional capacity at the commune, inter-commune and department levels in Haiti's Trois-Rivières watershed for productive and sustainable land-use management	1,484,597	0		1,484,597	<u>Grants</u>	Heifer International Haiti
	2.2 Enhance the technical capacity and access to finance of national and local representatives involved in agriculture in the Trois-Rivières watershed for adopting climate-resilient sustainable land-use practices	5,686,057	0		5,686,057	<u>Grants</u>	
	2.3 Implement a social protection system to support vulnerable households at risk to food insecurity because of climate change	1,072,701	0		1,072,701	<u>Grants</u>	
	M&E	60,000			60,000	<u>Grants</u>	
3. Strengthened governance and capacity for climate-resilient integrated water resources management (IWRM)	3.1 Strengthen national capacities for the implementation of the Water Act	1,015,140	1,015,140.00	<u>Grants</u>	0		
	3.2 Develop an integrated, climate-resilient water management governance framework targeting the catchment and sub-catchment levels in the Trois-Rivières watershed	1,090,066.96	1,090,066.96	<u>Grants</u>	0		

	3.3 Implement regular monitoring and evaluation of water resources at the catchment and sub-catchment levels to support the implementation of integrated water resources management (IWRM) plans	650,952.08	650,952.08	<u>Grants</u>	<u>0</u>		
	M&E	517,360.00	517,360.00	<u>Grants</u>			
Project management cost (USD)		1,314,087.44	954,087.44	<u>Grants</u>	<u>360,000</u>	<u>Grants</u>	<u>UNDP</u>
Indicative total cost (USD)		31,290,561.48	22,427,206.48		8,863,355.00		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☒ No ☐

Technology transfer

139. Output 1 of the proposed project will facilitate technology transfer by implementing EbA interventions in Haiti's TR watershed, leading to a reduction of the impacts of climate change-induced flooding. These interventions will include establishing 17,740 ha of agroforestry systems and rehabilitating 7,700 ha of water towers⁷³ through reforestation at priority sites⁷⁴ in the watershed (Activity 1.2). The technologies to be implemented under the project are not entirely new to Haiti, having all been implemented in various projects. For example, under the project, entitled "*Applying ecosystem-based disaster risk reduction*" (2016), 36,300 mangroves and sea grape trees were planted on shores and in river mouths to reduce the impacts of flooding in coastal regions in Haiti. What is novel under the proposed project, is the use of EbA interventions as part of an integrated approach to flood management in the country. Such an approach will not only demonstrate the flood-reduction benefits of EbA interventions but also create the required governance frameworks and technical capacities at all levels for climate-resilient, sustainable land and water resources management in Haiti.

Capacity-building interventions

140. To support the implementation of EbA interventions under Activity 1.2, the technical and institutional capacity of decision-makers and local communities will be strengthened under Activity 1.1 as well as Outputs 1 and 2 of the project. These capacity-building activities will focus on creating the enabling environment for the adoption and application of an integrated approach to land and water resources management at all levels in Haiti's TR watershed including capacity for carbon monitoring. In particular, capacity-building activities under Output 2 include: i) strengthening institutional capacity at the commune, inter-commune and department levels in Haiti's TR watershed for productive and sustainable land-use management and ii) enhancing the technical capacity and access to finance of national and local representatives involved in agriculture in the TR watershed for adopting climate-resilient sustainable land-use practices (Activity 2.2).

141. Under Outputs 2 and 3 of the project, capacity-building initiatives will focus on improving water resources management in Haiti's TR watershed. Specifically, these initiatives include: i) strengthening national capacities for the implementation of the Water Act; ii) developing an integrated, climate-resilient water

⁷³ 'Water towers' refers to mountainous areas in Haiti where water for agricultural production and drinking is obtained.

⁷⁴ Further details on the priority intervention sites are presented in Section B.3 of this Funding Proposal and Section 10 of Annex 2: Feasibility Study.

management governance framework targeting the catchment and sub-catchment levels in the TR watershed; and iii) implementing regular monitoring and evaluation of water resources at the catchment and sub-catchment levels to support the implementation of integrated water resources management (IWRM) plans. The complementarity between the respective capacity-building activities focused on SLM (Activity 1.1 and Output 2) and IWRM (Output 3) to enable the integrated, sustainable and climate-resilient management of land and water resources highlights the transformative potential of the project in managing climate change-induced flooding in the TR watershed. From the project total budget USD 5,760,272 has been allocated for capacity building including USD 3.9 million in co financing.

EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

142. The proposed project will contribute to the GCF's paradigm shift adaptation objective of increased climate-resilient sustainable development by enhancing the capacity of vulnerable communities in the Trois Rivières (TR) region of Haiti to adopt and implement an integrated approach to climate change-induced flood management. To accomplish this objective, project interventions have been designed to achieve three GCF fund-level impacts and two fund-level outcomes, as presented below⁷⁵.

Fund-level Outcomes under IRMF

143. The proposed project will enhance the sustainability of Haiti's approach to flood management in the TR region by achieving the following fund-level impacts:

- ARA1.0 — Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions.
- ARA2.0 — Improved health, wellbeing, food and water security.
- ARA4.0 — Improved resilience of ecosystems and ecosystem services.

Project Outcomes

144. The proposed project will be evaluated against the relevant factors specified in the GCF Integrated Results Management Framework. It is anticipated that the proposed project will achieve three Fund-level enabling environment Outcomes, as described below⁷⁶.

- Core indicator 5 Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner.
- Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation.
- Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level

145. The proposed project outputs include activities for enhancing climate-responsive planning and development — in particular, through the sustainable improvement of land and water resources management in Haiti. First, the collaborative development of community land-use plans will provide an integrated framework for sustainable landscape management (SLM) at the local level (Activity 1.1). This will be supported by the establishment of communal development councils and the provision of training to appropriate public stakeholders, ensuring the effective implementation of SLM (Activity 2.1). Second, climate-resilient IWRM will be integrated into Haiti's Water Act by including provisions for Catchment IWRM Committees and Sub-Catchment Water Resource User Associations. These committees and associations will provide a platform for cross-sectoral, vertical and horizontal collaboration between stakeholders (Activity 3.1). This collaboration will be supported by the development of climate-resilient IWRM plans as well as providing training to members of the aforementioned institutions on the implementation of IWRM in decision-making activities (Activity 3.2). Finally, the implementation of consistent monitoring and evaluation of all project activities at the catchment and sub-catchment levels will provide an important regulatory function in optimising the uptake and effectiveness of climate-resilient flood management solutions (Activity 3.3).

146. Implementing 17,740 ha of agroforestry systems and rehabilitating 7,700 ha of forests under Activity 1.2 will restore degraded ecosystems and considerably reduce the impacts of climate change-induced flooding in the TR watershed. The combined effect of these EbA interventions will result in a 40% reduction in peak runoff, with rehabilitation interventions expected to reduce floodplain extents in the target areas by 20% and 26% under the 20-year and 100-year flood scenarios, respectively. By the end of this century, these EbA interventions are projected to result in a more than 50% reduction in the number of households affected by 100-year flood events (declining from 1,342 households at risk of flooding under the baseline scenario to 638 households with project interventions). Similarly, agroforestry and reforestation interventions under this project activity are expected to result in a 35% reduction in the number of households at risk to 20-year flood events in the TR watershed. This

⁷⁵ Detailed information on the assumptions and estimates of the impact potential of the proposed project are provided in the project's results framework and economic analysis (Section E of this document and Annex 3, respectively).

⁷⁶ A detailed description of the project activities that will be implemented under each output to achieve the GCF project outcomes is presented in Section B.3 of this Funding Proposal.

represents a decline from 463 under the baseline scenario to approximately 300 houses at the end of the century under a 20-year flooding scenario.

147. In addition to their significant flood-reduction potential, the EbA interventions will also deliver numerous social benefits for vulnerable households in the target communes in the TR watershed. Rehabilitating degraded forests will increase the ecosystem goods and services they provide (e.g. improved quality and quantity of water supply, increased prevalence of agricultural pollinators and enhanced soil fertility⁷⁷), while new livelihood benefits will be generated through the implementation of agroforestry systems in the seven target communes. This will incentivise target communities to shift away from unsustainable practices — such as unregulated charcoal production — towards the adoption and implementation of sustainable, climate-resilient land uses. Moreover, these interventions will generate benefits for the health of vulnerable individuals, households and communities living in the TR watershed, reducing the spread of water- and vector-borne diseases as contamination of freshwater sources (e.g. rivers) and food is reduced. Specifically, agroforestry and reforestation interventions will: i) reduce the transmission of communicable diseases; ii) safeguard agricultural livelihoods; and iii) improve basic water quantity and quality, as well as sanitation and hygiene (WASH) services⁷⁸. Access to WASH, in particular, reduces the spread of water- and vector-borne diseases including, *inter alia*, cholera, diarrhoea and typhoid — the World Health Organisation has estimated that improving access to WASH facilities could reduce the global burden of disease by approximately 10%⁷⁹. Finally, the development of rigorous monitoring and evaluation of project activities and implementation will allow the iterative adjustment of project implementation practices, to ensure the effectiveness of, *inter alia*, EbA interventions in reducing the spread of water- and vector-borne diseases.
148. Demonstrating the adaptation and livelihood benefits of EbA interventions will be supported by awareness-raising activities under Output 1. These activities will strengthen the adaptive capacity of communities by increasing their knowledge of, *inter alia*, the benefits of climate-resilient agricultural practices and agroforestry as well as the alternative livelihood options that these activities provide (Activity 1.1.). As a result, project beneficiaries will have increased technical and institutional capacity to improve their long-term food and income security. Moreover, the dissemination of information on sustainable charcoal production and reforestation will enhance the energy security of these beneficiaries and enable them to sustainably utilise forest resources. The adaptive capacity of communities will be further strengthened by the: i) provision of climate-resilient agricultural inputs to communities (Activity 2.1); ii) creation of an agricultural finance mechanism (Activity 2. 2.); and iii) establishment of a social protection system for food insecure households (Activity 2.3). These activities will provide a combination of short- and long-term support to vulnerable communities in response to climate change-induced flooding, ensuring that the most vulnerable smallholder farmers are included in the agricultural value chain and are able to improve their food and income security.
149. The proposed project will be implemented in the TR watershed area, covering 33 communal sections and 118 communities within the seven target communes. As a result of direct investments from the GCF and Heifer International Haiti, the project will confer direct and indirect climate change adaptation benefits to approximately 292,600⁸⁰ (approximately 152,152 women and 140,448 men) and approximately 441,272⁸¹ (approximately 229,461 women and 211,811 men) vulnerable people in Haiti, respectively. These investments will transform Haiti's approach to land and water resources management and create an enabling environment for the uptake of climate-resilient, sustainable flood management practices within the public and private sector. It will then become possible to scale up and replicate these practices nationally, as well as across the Caribbean region.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

⁷⁷ Thompson, I.D. *et al.* 2011. "Forest Biodiversity and the Delivery of Ecosystem Goods and Services: Translating Science into Policy", *BioScience*, 61 (12).

⁷⁸ APFM. 2015. Health and Sanitation Aspects of Flood Management.

⁷⁹ *Ibid.*

⁸⁰ Direct beneficiaries include the total population living in proximity to waterways in the target communes. This was calculated by mapping all infrastructure and calculating their distance to the river. All houses within 30m of the river are considered direct beneficiaries.

⁸¹ Indirect beneficiaries include the total population of the target communes, minus direct beneficiaries based on 2003 census data adjusted for population growth. See Annex 3 – Economic Analysis for a full description of the adjustment methodology.

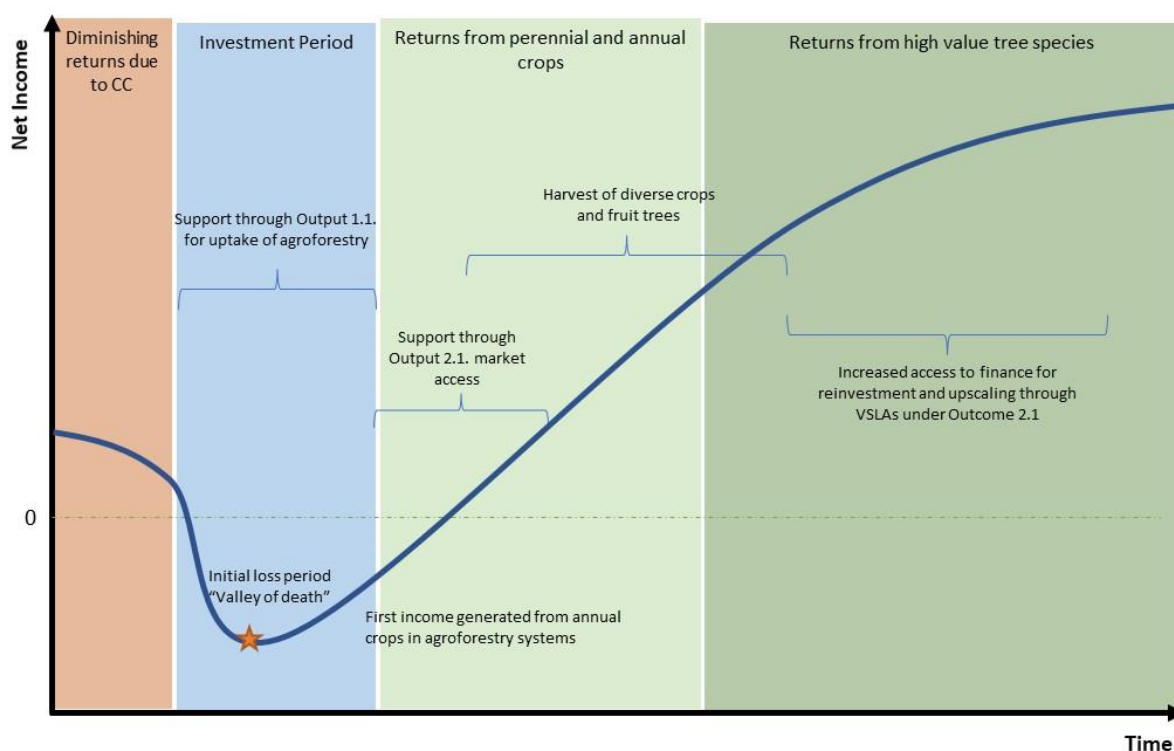
150. The proposed project will catalyse a paradigm shift in Haiti's approach to flood management in the TR watershed by initiating a shift away from uncoordinated decision-making and unsustainable land-use practices towards an integrated, climate-resilient approach to land and water resources management. Details on how the project will contribute to GCF's Paradigm Shift Objective of climate-resilient sustainable development are presented below.

Potential for scaling up and replication, as well as strengthening Haiti's institutional and regulatory framework

151. Under the proposed project, an integrated climate-resilient approach to flood management will be implemented to address the severe impacts of climate change-induced flooding on soil erosion, well-being and livelihoods of vulnerable communities in Haiti's TR watershed. This approach is built around the principles of climate-resilient, sustainable development and underpinned by three main interventions. First, agroforestry systems will be established and degraded forests restored at priority intervention sites across the watershed to: i) reduce peak surface runoff and flood extent; and ii) provide alternative livelihood options to the unsustainable land-use practices on which most vulnerable communities in the TR watershed depend. As a result, these agroforestry systems will contribute to reducing the spread of water- and vector-borne diseases such as cholera by improving water quality and reducing flood impacts in the watershed. The implementation of agroforestry systems will be supported by establishing the local governance structures required to implement and manage climate-resilient land-use practices and providing training to community groups — including farmer and women's associations — for strengthened climate-responsive planning. Second, complementary interventions focused on enhancing technical and institutional capacity at the commune, inter-commune and department levels will be implemented to promote productive climate-resilient agricultural practices and sustainable land management. Last, the necessary governance framework for integrated water resources management (IWRM) will be established in Haiti to support the uptake of climate-resilient land management systems and facilitate sustainable use and management of water resources over the long term. Together, project interventions will transform flood management in Haiti by catalysing a shift away from uncoordinated, compartmentalised decision-making around flood management, towards an integrated approach that can be scaled up and replicated nationally and across the Caribbean region.

152. To enhance the adaptation impact of these interventions, Haiti's governance framework for the sustainable management of natural resources will be strengthened. This will be done primarily through adapting the draft Water Act to include, *inter alia*, provisions for the adoption of IWRM Committees in departments across the country. As a result of these changes, a new strategy will be developed for IWRM in Haiti that addresses both the existing challenges of a fragmented and uncoordinated system of water management, particularly at a catchment and sub-catchment level, as well as the current and projected impacts of climate change-induced flooding. Regulation around the sustainable use and management of land and water resources will also be strengthened under the project through the development of community land-use plans, which will facilitate the implementation of a landscape approach to land and water resource management. This will contribute to the creation of a more coherent land and water resources management framework in Haiti that considers climate change.

153. A transition model underpinning the paradigm shift for the agroforestry interventions is depicted in the figure below, demonstrating how the project will support farmers in making the transition to sustainable agroforestry possible. This includes the role of social protection and technical support to assist farmers in transitioning through the initial investment period.



The diagram shows that GCF finance is needed to allow farmers to bridge the “valley of death” of the first years of investment in agroforestry, when costs are higher than revenues. By providing inputs to farmers, the costs of implementing agroforestry become lower, enabling the farmer to shift to this more sustainable land use and guarantee income from year 1 (through harvests of annual crops within the agroforestry system).

Knowledge sharing, awareness and the creation of an enabling environment

154. The absence of a platform for coordination between the multiple stakeholders involved in land and water management in Haiti is a barrier to effective flood management and has resulted in fragmented and ineffective decision-making as well as the duplication of activities⁸². In this context, horizontal coordination will be enabled through the establishment of IWRM Committees (Activity 3.1), which will promote cross-sectoral decision-making between all stakeholders regarding resource use and allocation. The project will also strengthen vertical coordination between local water resource users and public sector officials at both municipal and national levels. For example, both community members and public institutions (such as municipalities) will be involved in the creation of community land-use management plans (Activity 1.1 and 1.2). This will ensure a rigorous balance of bottom-up and top-down integration through participatory processes. The establishment of Sub-Catchment Water Resource Users Associations (Activity 3.1) will also promote a freer flow of information and communication between stakeholders at different levels, thereby promoting vertical coordination on the sustainable management and use of land and water resources.

155. To further support the shift away from uncoordinated decision-making and unsustainable land-use practices, agricultural productivity in the TR watershed will be enhanced. This will be achieved by: i) strengthening the technical and institutional capacity of people involved in agriculture at the commune, inter-commune and department levels for implementing productive climate-resilient agricultural practices; ii) enhancing smallholder farmers’ access to financial resources to facilitate the long-term implementation of climate-resilient agricultural practices; and iii) establishing a social protection system to provide immediate support to vulnerable households affected by climate change impacts, particularly flooding. Furthermore, public sector stakeholders will receive training on how to implement SLM and IWRM to mainstream sustainable land and water resource management into the public sector. The provision of training on climate-resilient agricultural practices will improve the climate resilience of beneficiary communities, resulting in improved food security for subsistence farmers as well as higher incomes for smallholder farmers. The incorporation of these income- and livelihood-generating activities has been previously demonstrated as an important factor that will ensure the long-term sustainability of adaptation and development projects⁸³. Knowledge sharing on sustainable charcoal production methodologies will help improve

⁸² Widmer *et al.* 2018. A Vision for Water in Haiti, 2018.

⁸³ Gibbons, A. 2010. Greening Haiti, Tree by Tree. *Science*, 327, pp. 640–641.

the efficiency of these activities and reduce wood loss during the production process — presently, approximately 70% of wood energy is lost during the production of charcoal. This will reduce deforestation and contribute to improved climate-resilient flood management. In addition, awareness-raising campaigns will be promoted in city hall meetings as well as community radio broadcasts, focusing on: i) promoting the benefits sustainable land-use practices; ii) encouraging community participation in land-use plans; and iii) creating awareness of adaptation and landscape management solutions to address climate change-induced flooding. These capacity-building activities, combined with the monitoring and evaluation systems included in the proposed project design, will produce an evidence base detailing the success of project interventions. This will be an important incentive for the creation of an enabling environment for future upscaling and replication of project activities in other areas of Haiti and the Caribbean region.

D.3. Sustainable development (max. 500 words, approximately 1 page)

156. The proposed project will contribute towards the achievement of at least 5 of the 17 United Nation Sustainable Development Goals (SDGs), including: i) SDG 2 — Zero Hunger; ii) SDG 5 — Gender Equality; iii) SDG 6 — Clean Water and Sanitation; iv) SDG 11 — Sustainable Cities and Communities; and v) SDG 15 — Life on Land. In addition, project interventions will generate several environmental, social, economic and gender-responsive co-benefits, as described below.

Environmental co-benefits

157. The restoration of degraded forests through targeted reforestation interventions and expanded agroforestry practices, as well as the implementation of integrated water resource management (IWRM) practices (Outputs 1.1 and 3.1, respectively) will generate several environmental co-benefits. Such benefits include reduced surface runoff and soil erosion, carbon sequestration and improvements in air quality.

158. Historically, deforestation in Haiti has increased erosion rates by six-fold and river flows by 25%⁸⁴. This problem has also been experienced in the country's Trois-Rivières (TR) watershed, where soil erosion is exacerbated by climate change-induced flooding. To address this climate change problem, 17,740 ha of agroforestry systems will be implemented, and 7,700 ha of priority forests restored in the TR watershed (Activity 1. 2). These interventions will not only have direct flood-reduction benefits for vulnerable communities in the watershed⁸⁵, but they will also contribute to soil conservation. This will lead to *inter alia*: i) reduced surface runoff and soil erosion caused by flooding; ii) increased water infiltration and storage capacity of the soil; iii) improved water quality, particularly in near-shore marine environments because of reduced sedimentation; improved air quality; and iv) improved soil quality and productivity. Furthermore, the stabilisation of degraded watersheds will result in greater biodiversity, increased provisioning of ecosystem goods and services to communities living adjacent to intervention sites, and improved conservation of indigenous plant and animal species.

159. While most of the environmental co-benefits that will be conferred by agroforestry and reforestation interventions are adaptation-focused, there is also the potential for mitigation co-benefits to be provided through the project. Implementing agroforestry and reforestation activities will contribute to enhancing the carbon sequestration potential of the restored landscapes, while promoting the adoption of climate-resilient sustainable land-use practices (Activity 2. 2) will catalyse a shift away from unsustainable charcoal production that contributes to increasing CO₂ emissions in the country. It should be noted that the potential mitigation benefits associated with reforestation interventions have not been calculated for this project as the potential benefits will be peripheral to the adaptation impact of these interventions. Instead, the carbon sequestration potential of reforestation interventions is supported by evidence across the United States, where reforestation interventions have demonstrated that “the establishment of woody vegetation delivers immediate to multi-decadal carbon sequestration benefits in biomass and woody debris pools”⁸⁶.

Social co-benefits

160. Implementing targeted landscape restoration and IWRM, as well as enhancing the technical capacity of smallholder farmers and communities to utilise climate-resilient agricultural practices will generate several social co-benefits. These benefits are described in further detail below and include improved food security, social empowerment, greater energy security, as well as health benefits.

⁸⁴ Inter-American Development Bank (IADB). 2014. Haiti: Natural Disaster Mitigation Program II – Proposal for Operation Development. [online] Available: https://www.climateinvestmentfunds.org/sites/cif_enc/files/meeting-documents/haiti_climate_proofing_of_agriculture_in_the_centre-artibonite_loop_idb-project_document_0.pdf

⁸⁵ as described in Sections B.2 and B.3 of this Funding Proposal

⁸⁶ Nave, L.E. *et al.* 2019. The role of reforestation in carbon sequestration. *New Forests*, 50: pp. 115-137.

161. Training communities and individual farmers in climate-resilient agroforestry practices (Activity 1.1.1), as well as expanding access to agricultural markets and credit facilities (Activity 2.1.2) will improve food security at the household level. These benefits will be realised through improved food production and increased income. In addition, a reduction in the incidence and intensity of flooding achieved through reforestation and agroforestry activities (Activity 1.1.2) will improve food security, as flooding often undermines agricultural production. In particular, productive floodplains are impacted by increased rates of erosion, which contributes to food and water being contaminated by heavy chemicals and/or pesticides transported by flooding⁸⁷. Enhanced food security will also increase the likelihood of project beneficiaries continuing agroforestry practices and the conservation of restored forests after project closure.
162. Project interventions will also generate positive social co-benefits for beneficiary communities in the form of greater social empowerment. First, communities will have improved stewardship over forests and other natural resources and will receive long-term benefits from the sustainable use of non-timber forest products and other forest resources. This will be achieved through training provided on, *inter alia*, community land-use plans and sustainable charcoal production and reforestation (Activity 1. 1). Second, the active engagement of communities in the development of these community land-use plans (Activity 1.1) will also improve community cohesion. Finally, the development of community land-use plans (Activity 1.1), as well as the establishment of communal development councils (Activity 2.1), Catchment IWRM Committees and Sub-Catchment Resource Users Associations (Activity 3.1) will improve the decision-making power of traditionally marginalised groups — including women and local communities — on natural resource management. These benefits will be supported by the diversified and improved climate resilience of incomes amongst smallholder farmers through the promotion of agroforestry systems (Activity 1. 2) and the provision of training on climate-resilient agricultural practices (Activity 2.2).

Economic co-benefits

163. Project interventions — particularly reforestation, IWRM, and the expansion of agricultural markets — will generate several economic co-benefits. These are described in detail below and include reduced damage to infrastructure, employment creation, enhanced climate resilience of the agricultural sector and the adoption of climate resilient livelihoods amongst smallholder farmers and other community beneficiaries.
164. At present, approximately 50% of built-up areas in Haiti are vulnerable to climate change-induced flooding. This is evidenced by the densely populated cities of Port-de-Paix and Gonaïves, which saw 30 and 80% of their areas flooded after Hurricane Jeanne made landfall in northern Haiti, a tropical storm which caused approximately US\$41 million in total damages in 2004. Flood management interventions under the project are expected to reduce peak surface runoff and floodplain extents in Haiti's TR watershed. This, in turn, will contribute to reducing the economic damages to existing infrastructure, generating public savings and indirectly increasing the fiscal space available for other social and developmental investments⁸⁸. These economic co-benefits will be supported by the creation of additional employment opportunities and through the adoption of agroforestry systems and reforestation activities. Such opportunities will include the involvement of farmers and other community members in the establishment of nurseries, physical planting of seedlings, and promotion of regeneration activities. Encouraging farmers, in particular, to participate in planting specific tree seedlings (e.g. coffee, cocoa and cassava) will provide them with an important economic safety net to reduce the economic impacts of climate change-induced flooding on their production. These impacts will be further reduced by expanding agricultural markets (Activity 2.2), which will increase the resilience of the agricultural sector to external climate change shocks, such as flood and hurricane impacts. This, in turn, will contribute to reducing fluctuations of agricultural employment and improved food price stability.
165. Smallholder farmers and beneficiary communities will also secure climate-resilient livelihoods as a result of several project interventions including: i) establishing 17,740 ha of agroforestry systems (Activity 1.2), which will diversify farmers product base, generate higher levels of income and improve climate resilience as beneficiaries will have a wider resource base to draw from; ii) training community members who depend on agriculture for the livelihoods on climate-resilient agricultural practices and sustainable land management (Activity 2.2), which will improve agricultural productivity under future climate change conditions; and iii) increasing access to agricultural finance (Activity 2. 2), which will allow beneficiaries to purchase necessary agricultural inputs and improve their yields. In addition, the reforestation of 7,700 ha of forest in priority areas in the TR watershed will result in, *inter alia*, reduced erosion and improved soil fertility. This will bolster yields, resulting in higher income levels and contributing to more resilient livelihoods.

⁸⁷ Associated Programme on Flood Management (APFM). 2015. Health and Sanitation Aspects of Flood Management. *Integrated Flood Management Tools Series*, 23.

⁸⁸ Ibid.

Gender-sensitive development impact

166. Approximately 52% of direct beneficiaries of the proposed project are women. In addition, project interventions will result in several gender-specific co-benefits described below, including gender-responsive health and socio-economic benefits.
167. Implementing an integrated approach to flood management in the TR watershed will have several positive gender-based co-benefits. First, this will reduce the rate of displacement, and in turn reduce the amount of violence — particularly sexual violence — experienced by women. Previous studies have documented increased incidence of gender-based violence when displacement has occurred. In the aftermath of Hurricane Matthew, field assessments reported an increased incidence of sexual violence against displaced women and girls. Increased incidences of gender-based violence in the aftermath of the 2010 earthquake further support this finding, and it has been partially attributed to the breakdown of traditional social networks between neighbours and communities that occur alongside displacement⁸⁹. Second, reduced flooding will result in a decline in the spread of water-borne diseases as the contamination of water supplies will decline. This will generate a disproportionately positive impact on women, who are the most exposed to contaminated water because of gender roles in the home⁹⁰.
168. Additional gender-responsive health co-benefits that will be achieved in the proposed project include a reduction in the horizontal transmission of HIV/AIDS and other STIs amongst women because of improved food security. This is because women are more likely than men to engage in high-risk strategies to mitigate food insecurity — including sex work — which exposes them to the disease⁹¹. The project will also generate gender-responsive economic co-benefits in the form of resilient livelihoods. Haiti has a high percentage of female-headed rural households (approximately 39%), the majority of whom are smallholder farmers. These women, in particular, will benefit from Activity 2. 2, which will include the provision of training on climate-resilient agricultural practices and sustainable land management as well as the extension of agricultural finance and provision of climate-resilient agricultural inputs. Studies have found that closing the gender gap in agriculture⁹² results in improved agricultural production levels overall, as well as increased levels of food availability and a decline in average food prices⁹³.
169. The promotion of gender equality has been mainstreamed throughout proposed project interventions wherever possible, based on the gender assessment and action plan (Annex 8). This gender mainstreaming will contribute to the promotion of gender-sensitive development beyond the conclusion of the project.

commentD.4. Needs of recipient (max. 500 words, approximately 1 page)

Vulnerability of the country

170. Ranking 163rd out of 188 countries on the 2016 UN Human Development Index, Haiti faces multiple development challenges related to food and economic security, health, education, political violence, and environmental degradation. These challenges are compounded by extreme climate events such as hurricanes and tropical storms that cause flooding and landslides and the frequent occurrence of natural disasters (e.g. earthquakes) in the country. As the inhabitants of a SIDS and LDC, most Haitians are particularly vulnerable to impacts of these events. Indeed, with more than 96% of its population living at risk of multiple climate and natural hazards⁹⁴, Haiti is considered one of the world's most vulnerable countries to the acute shocks and chronic stresses of climate change. This is supported by the findings of the 2020 Climate Risk Index⁹⁵, where Haiti was listed as the third most affected country by hydrometeorological disasters that occurred during the period 1998–2018. Additionally, with a vulnerability index of 0.58 and a sensitivity index of 0.22, the Development Bank of Latin America listed Haiti as the most vulnerable country to climate change in Latin America and the Caribbean⁹⁶.

⁸⁹ International Federation of Red Cross and Red Crescent Societies (IFRC). 2015. *Unseen, Unheard: Gender-based violence in disasters*. Geneva, IFRC.

⁹⁰ APFM. 2015. Health and Sanitation Aspects of Flood Management.

⁹¹ Anema, J., Vogenthaler, N., Frongillo, E.A., Kadiyala, S., and Weiser, S.D. 2009. Food Insecurity and HIV/AIDS: Current Knowledge, Gaps, and Research Priorities. *Current HIV/AIDS Reports*, 6(4): 224-231.

⁹² The "gender gap in agriculture" is an observed difference in yields per ha between male and female farmers, which has predominantly been attributed to differences in input levels. On average, female farmers are less productive than their male counterparts because of lower levels of access to agricultural inputs including, *inter alia*, seeds, fertilisers and pesticides. From: FAO. 2011. *The State of Food and Agriculture: Women in Agriculture*. Office of Knowledge Exchange, Research and Extension, Rome. Retrieved from: <http://www.fao.org/3/i2050e/i2050e.pdf>

⁹³ FAO. 2011. *The State of Food and Agriculture: Women in Agriculture*.

⁹⁴ Global Facility for Disaster Reduction and Recovery (GFDRR). Available at: <https://www.gfdr.org/en/haiti>.

⁹⁵ D. Eckstein, V. Künzel, and L. Schäfer (2020). Global Climate Risk Index 2020. Germanwatch, Bonn. Available at: https://germanwatch.org/sites/germanwatch.org/files/20-2-01e%20Global%20Climate%20Risk%20Index%202020_14.pdf

⁹⁶ CAF (2014). Vulnerability index to climate change in the Latin American and Caribbean Region. CAF, Caracas. 206 pp. Available at: <http://scioteca.caf.com/handle/123456789/509>

171. In the past two decades, Haiti has experienced 26 floods (12 of which were catastrophic), 8 hurricanes, and numerous other extreme climate events such as tropical storms and earthquakes. With a total of 55 extreme climate events, 2.81 related deaths per 100,000 inhabitants and approximately US\$390 million in losses during the period 1999–2018, the country ranks in the top three at risk countries to climate change impacts⁹⁷. Of these impacts, the damages and losses caused by climate change-induced flooding are particularly severe. For example, in 2004, Hurricane Jeanne caused severe flooding in the northern part of Haiti, resulting in more than 2,800 deaths and approximately US\$41 million in damages. More recently, Hurricane Matthew struck Haiti in 2016, causing 546 deaths and the loss of up to 90% of crops and livestock in some areas⁹⁸. The frequency and intensity of flood events is already increasing in Haiti. This trend is expected to continue under future climate change conditions as a result of projected higher average annual temperatures and increased intensity and concentration of precipitation events across the country⁹⁹. The impacts of these climatic changes will be particularly severe for the country, which is already extremely vulnerable to external shocks caused by climate change. This is because Haiti is currently locked into a reactive cycle of costly disaster response and rebuilding that is preventing the adoption and implementation of climate-resilient approaches to managing the cross-sectoral impacts of flooding in the country. These impacts are further exacerbated by the country's fiscal challenges and dependency on neighbouring countries for importing critical supplies. For example, approximately 50% of the national budget comes from foreign aid, 100% of the hydrocarbons used for energy production are imported, and 60% of its food requirements are imported, despite agriculture being the primary economic activity. Extreme climate events are already affecting these imports, resulting in shipping routes becoming unsafe and insurance premiums for shipping companies becoming unaffordable. The cumulative impact of these factors inhibits the capacity of national and local decision-makers to implement long-term strategies for responding to the adverse impacts of climate change on the country's population.

Vulnerability of beneficiary groups

172. While the TR watershed covers three departments in Haiti (namely the Artibonite, North and Northwest departments), the Northwest department has the joint highest number of inhabitants living under poverty, with an extreme poverty rate exceeding 40% (representing approximately 20% of the extreme poor)¹⁰⁰. Poverty is particularly prevalent in the Trois Rivières (TR) region, which is the most populous of the three watersheds in the Northwest department. This region also encompasses Port-de-Paix, a major city ranked as the third-most food-insecure area in Haiti¹⁰¹. High levels of poverty and food insecurity experienced in the TR region contribute directly to increasing the vulnerability of inhabitants to the impacts of climate change.

173. Within the Northwest department, the TR region is particularly vulnerable to the impacts of climate change-induced flooding. The reason for this is two-fold. First, the region's steep topography contributes to increased runoff during extreme precipitation events which consequently exacerbates flood impacts. Second, the TR region is located near Haiti's north coast which increases the exposure of the region's population to the impacts of hurricanes and tropical storms coming from the north-east. The cumulative negative impacts of climate change-induced flooding and extreme events are compounded by land degradation in the TR region, resulting from deforestation and unsustainable crop farming on the surrounding steep slopes. Current agricultural practices — particularly the increasing prevalence of annual crop cultivation which leaves the soil bare during most of the year — and charcoal production in the mountainous areas of the catchment cause widespread deforestation. Consequently, these practices have resulted in large-scale changes in the natural landscape of the TR region, with the clearing of tress, in particular, contributing to decreased infiltration and increased surface runoff in the catchment. This, in turn, has resulted in increased erosion and soil degradation within the catchment area, further exacerbating flood impacts.

Socioeconomic status

174. Haiti is the only LDC in the Western Hemisphere and the least wealthy country in the Americas with an annual GDP per capita of approximately US\$1,850. Poverty is endemic with approximately 59% of the population living below the national poverty line of US\$2.41 per day, and approximately 24% living below the national extreme poverty line of US\$1.23 per day. In rural areas, poverty is highest, where 52% of the population and 63% of extremely poor households reside. Inhabitants of rural areas also have limited capacity to adopt alternative livelihoods, resulting in an increased dependence on agriculture. Approximately 40% of the population is dependent

⁹⁷ D. Eckstein, V. Künzel, and L. Schäfer (2020). Global Climate Risk Index 2020. Germanwatch, Bonn. Available at: https://germanwatch.org/sites/germanwatch.org/files/20-2-01e%20Global%20Climate%20Risk%20Index%202020_14.pdf.

⁹⁸ World Bank. 2017. Rapidly Assessing the Impact of Hurricane Matthew in Haiti. Available at: <https://www.worldbank.org/en/results/2017/10/20/rapidly-assessing-the-impact-of-hurricane-matthew-in-haiti>

⁹⁹ BRGM (France) & CIAT. "Atlas des menaces naturelles en Haiti."

¹⁰⁰ World Bank. 2014. Poverty and Inclusion in Haiti: Social gains at timid pace. Available at: <http://documents.worldbank.org/curated/en/643771468257721618/pdf/895220BRI00pau00Box385284B00PUBLIC0.pdf>

¹⁰¹ 35% of households are exposed to food insecurity. WFP. 2016. Evaluation de la sécurité alimentaire en milieu urbain.

on the agricultural sector for their livelihoods which consists mostly of small-scale subsistence farming that is highly vulnerable to the impacts of climate change and natural disasters. Poverty, corruption, vulnerability to climate change and natural disasters, as well as low levels of education across much of the population, represent some of the most serious impediments to Haiti's economic growth.

175. The country's small tax and market base as well as its high public debt — outstanding external debt was cancelled by donor countries following the 2010 earthquake, but has since risen to approximately US\$2.8 billion as of December 2017¹⁰² — constrain the government's ability to allocate funding from the national budget for climate change adaptation. Moreover, limited opportunities exist for the public and private sector to access financial resources for addressing climate change impacts, which is estimated to cost the country approximately US\$20 million per annum¹⁰³. Given that this cost is likely to increase significantly under future climate change conditions, external investment is critical to increase the climate resilience of Haiti's population.

Institutional capacity needs

176. Governance of land and water resources, as well as improving the sustainability of agricultural practices, are major challenges that need to be addressed to strengthen the resilience of vulnerable communities in the TR region to the impacts of climate change. Critical conditions needed for the implementation of an effective and efficient resilience strategy for vulnerable populations are the strengthening of governance and technical capacities of institutions and the establishment of appropriate legal frameworks in relation to land and water resources management. A multi-sectoral approach is therefore required to establish an integrated water resource management (IWRM) approach in the TR region. The adoption of such an approach will contribute to the implementation of sustainable agricultural practices, which will concomitantly increase food security and reduce the adverse impacts of climate change-induced flooding on the most vulnerable communities in the region.
177. Most national and local decision-makers, as well as farmers, are aware of the adverse impacts of climate change-induced flooding on land and water resources in the TR catchment. However, there is limited knowledge on adaptation options for communities, as well as limited awareness surrounding the urgent need for an integrated approach to flood management in the region. Consequently, people in the TR region that rely on farming for their livelihoods are unable to understand the implications of conventional, unplanned agricultural practices on the landscape under climate change conditions. For example, the cultivation of annual crops leaves soil bare during most of the year, resulting in its degradation and erosion caused by extreme weather. This leads to decreased surface cover and reduced infiltration capacity of the soil, subsequently exacerbating the impacts of flooding on the landscape and surrounding communities. Furthermore, profit margins for the farmers within the catchment are small and government budgets are extremely limited. As a result, on-farm decision-making is aimed at sustaining livelihoods in the short term, while government investments do not currently prioritise the long-term sustainability of climate-resilient activities within the catchment. GCF resources supported by government co-financing will, therefore, be used to: i) demonstrate the benefits of implementing an integrated, climate-resilient flood management approach in the region; and ii) strengthen the technical and institutional capacities of national and local decision-makers, farmers, community members, CSOs and association representatives to implement coordinated and planned long-term adaptation interventions. The combined effect of these adaptation interventions will contribute enhanced climate resilience of vulnerable communities in the TR region.

D.5. Country ownership (max. 500 words, approximately 1 page)

178. The proposed project is strongly aligned with Haiti's national priorities, having been designed in collaboration with project stakeholders at all levels. These priorities are summarised below.
- The revised **National Adaptation Programme of Action (NAPA)**¹⁰⁴, developed in 2017, identified four priority areas for addressing the impacts of climate change — namely, soils, agriculture, coastal zones and water resources. Focus was placed on these areas because they are among the most vulnerable to the impacts of climate change such as flooding, hurricanes and tropical storms, and addressing these negative impacts will contribute considerably to climate-resilient sustainable development in Haiti. The NAPA is of relevance because of its linkages with, and integration into, other national-level plans and priorities centred around climate-resilient land and water resources management. Specifically, national policies that have been integrated into the revised NAPA include the Interim Cooperation Framework¹⁰⁵ and the Poverty Reduction

¹⁰² CIA World Factbook. 2019. Field listing: Debt – external. Available at: <https://www.cia.gov/library/publications/the-world-factbook/fields/246.html>

¹⁰³ D. Eckstein, V. Künzel, and L. Schäfer (2020). Global Climate Risk Index 2020. Germanwatch, Bonn. Available at: https://germanwatch.org/sites/germanwatch.org/files/20-2-01e%20Global%20Climate%20Risk%20Index%202020_14.pdf

¹⁰⁴ In French, referred to as *Programme d'Action National d'Adaptation* or PANA. 2006. <https://unfccc.int/resource/docs/napa/hti01f.pdf>

¹⁰⁵ Available at: <https://reliefweb.int/report/haiti/haiti-interim-cooperation-framework-2004-2006>

Strategy Paper (PRSP)¹⁰⁶. These policies share common goals relevant to the proposed project such as ramping up agricultural production in a sustainable way, preventing land degradation and improving access to clean water. Haiti's NAPA and PRSP also share mutual objectives towards implementing activities that reduce the vulnerability of communities to climate change alongside the National Disaster Risk and Management Plan¹⁰⁷. Another relevant plan integrated into the NAPA is the National Environmental Action Plan (NEAP), released in 1999, which prioritises addressing the impacts of climate change, particularly flooding, on land and water resources in the country. The proposed project strongly aligns with the NAPA, as it will complement the response to priority needs outlined within this national plan, as well as other plans that have been integrated into it. This alignment is evidenced in the project's focus on promoting the adoption of an integrated approach to land and water resources management to increase the climate resilience of the most vulnerable communities in the Trois Rivières (TR) region of Haiti.

- The proposed project aims to build on achieved results of the national adaptation planning (NAP) process, supported by a GCF Readiness project, through the strengthening of institutional and technical capacities and the strengthening of existing frameworks and systems. The NAPA called for increased local and sub-regional capacity to support adaptation initiatives and NAP implementation, in the form of an Inter-sectoral Regional Programme Approach (APRIS). This approach aims to better focus the efforts of government and non-government actors at the regional level by making it possible to adequately implement programs and projects, and to better meet the needs of populations in the regions of the country. APRIS opens the possibility for broader inclusion of civil society groups at a decentralized level and to build their capacities for influencing and implementing localized actions on climate change adaptation. It also aims to develop regions considering the environmental and economic potential of each, while seeking coherence and complementarity in actions, in line with land-use planning approach and socio-economic national development strategy. The GCF NAP project builds on these existing processes and cooperates with ongoing initiatives to strengthen the Government of Haiti's capacity to effectively respond to climate change threats while avoiding duplication of efforts.
- The **Strategic Development Plan for Haiti (SDPH)**¹⁰⁸ (*Plan Stratégique de Développement d'Haïti*) of 2012 presents a plan for long-term development centred around four core themes. Specifically, this plan focuses on building resilience to climate change impacts by: i) mainstreaming climate change considerations into planning and awareness-raising initiatives at the national and local level; ii) establishing a network of protected areas; iii) restoring degraded ecosystems; and iv) managing the use of watersheds and forests in a sustainable way. The proposed project strongly aligns with the country's adaptation priorities in that project interventions will increase the climate resilience of the TR region through ecosystem restoration, as well as improved watershed and landscape management.
- Haiti's **Nationally Determined Contribution (NDC)**¹⁰⁹ has the following adaptation priorities: i) enhancing the resilience and sustainable management of 15 watersheds; ii) undertaking afforestation and/or reforestation activities of 137,500 ha of land; iii) strengthening the protection of coastal areas; and iv) preserving marine protected areas across the country's coastal zone. The focus of the proposed project on enhancing climate-resilient land and water resources management supported by agroforestry interventions is, therefore, strongly aligned with the priorities and objectives set out in the NDC.
- The **Initial (First) and Second National Communication on Climate Change to the UNFCCC (INC and SNC)**^{110,111} highlight the importance of managing watersheds to address the impacts of climate change. Both communications emphasise the need to: i) implement climate-resilient solutions that will address the impacts of climate-induced flooding on vulnerable communities and the main economic sectors (e.g. agriculture); ii) promote the adoption of an integrated approach to land and water resources management in the country; and iii) strengthen the cohesion between existing and planned initiatives focused on landscape and watershed management, informed by best practices generated from past projects and/or programmes across the country and Caribbean region. The proposed project is strongly aligned with the abovementioned adaptation needs. Specifically, project interventions will promote the adoption of an integrated approach to land and water resources management in the TR region and, in so doing, will initiate a shift away from existing decentralised management of natural resources towards climate-resilient sustainable development.

¹⁰⁶ Available at: <https://www.imf.org/external/pubs/ft/scr/2008/cr08115.pdf>

¹⁰⁷ In French, referred as *Plan National de Gestion des Risques et des Désastres* — PNGRD.

¹⁰⁸ In French, *Plan Stratégique de Développement*— PSDH. Available at:

http://www.ht.undp.org/content/dam/haïti/docs/Gouvernance%20démocratique%20et%20état%20de%20droit/UNDP_HT_PLAN%20STRATÉGIQUE%20de%20développement%20Haïti_tome1.pdf

¹⁰⁹ MdE, *Contribution Prévue Déterminée au Niveau National*. 2015.

¹¹⁰ Available at: <https://www.ldc-climate.org/country/haïti/>

¹¹¹ UNEP. Enabling Activities for the Preparation of the Second National Communication under the UNFCCC. Available at: http://www.un-gsp.org/sites/default/files/documents/haïti_prod0c_29.09.2005.pdf

- The **Action Plan for Water Resources Management in Haiti** was developed in 1999. Outlined in this document are the needs for: i) reforms; ii) legal frameworks; iii) capacity building among policymakers and water users; and iv) recognition of the economic importance of water resources and the need for integrated management of these resources. In a 2018 review of the 1999 Action Plan where the progress of implementing integrated water resources management (IWRM) initiatives in Haiti was assessed, the country obtained a score of only 29 out of 100¹¹². The Action Plan noted gaps in the country's capacity for implementing IWRM. These gaps were largely because of: i) a limited enabling environment; ii) the unavailability of management instruments; and iii) limited finance for the efficient implementation of interventions. The gaps were also highlighted in the review of Action Plan, showing limited progress towards IWRM in Haiti since 1999. Project interventions will support the current three-year work plan (2019–2021) through prioritising addressing the capacity gaps noted in the 2018 review of the action plan.

Stakeholder engagement

179. The proposed project has been designed following extensive consultations with members of the Haitian government, public authorities, civil societies, and the private sector¹¹³. Haiti's NDA, the MoE, has been fully involved in the development of this project since inception through its valuable contributions to the consultation process and inputs during field missions. UNDP has facilitated dialogue between the MARNDR, other relevant institutions, as well as the MoE, to confirm that the project responds to the adaptation priorities of the country. Furthermore, the proposed adaptation interventions under this project have been validated by all the relevant project stakeholders, including the NDA.
180. Consultations were organised by UNDP in the North-West, North and Artibonite regions of Haiti. Given the observed impacts of flooding on the communities in these regions, developing climate-resilient solutions that are supported by technical assessments to address this problem were prioritised by the GoH and other stakeholders, including women's and farmers' associations. Discussions — with the regional branches of the MoE, Civil Protection Agency (DPC), Ministry of Economy and Finance (MEF), the National Directorate for Water Supply and Sanitation (DINEPA) in the Ministry of Public Works, Transport and Communications (MTPTC) and the Regional Water Supply and Sanitation Office (OREPA) and irrigation associations — also underlined the need for a coordinated effort towards improved flood risk management as the preferred solution for increasing resilience to climate change-induced flooding. Additional adaptation priorities that were discussed during stakeholder consultations included the need for: i) improved policies on natural resource management; ii) strengthened coordination between adaptation projects, donors, and their executing entities; iii) further information on climate change and adaptation solutions to facilitate decision-making around flood management; and iv) the implementation of climate-resilient solutions to reduce the impact of climate change and natural hazards on vulnerable communities in the regions.
181. During the implementation phase, all target beneficiary groups will remain engaged to ensure that the benefits conferred by project interventions respond directly to the needs of these groups. This will simultaneously encourage stronger inputs from stakeholders, while allowing for challenges that may be encountered to be addressed early on, which will reduce adverse effects on project implementation. Further details on the stakeholder engagement process relevant to the project development and implementation phases are presented in Annex 7: Stakeholder engagement plan, with specific details related to the grievance mechanism and environmental and social risk mitigation measures presented in Section 10 of Annex 6: Environmental and Social Impact Assessment and Management Plan.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

182. The Government of Haiti (GoH) is requesting US\$22.7 million in GCF grant finance to implement climate-resilient integrated water resources management (IWRM) to respond to the impacts of flooding on the Trois Rivières (TR) watershed. This GCF investment will be supported by the Heifer International Haiti, which has committed US\$8.86 million¹¹⁴ in co-financing for the implementation of project interventions. Funding from the GCF will only be used to address the additional impacts of climate change, with co-finance from the Heifer being used to cover all baseline costs — such as business as usual operations and maintenance necessary to ensure sustainable land and water resources management. With the support of GCF grant funding, the proposed project will deliver several adaptation benefits using an integrated approach to land and water resources management that will contribute to achieving a paradigm shift in the country's approach to addressing the impacts of climate change-induced flooding¹¹⁵. A GCF

¹¹² UN Environment. 2018. Progress on integrated water resources management. Global baseline for SDG 6 Indicator 6.5.1: degree of IWRM implementation.

¹¹³ Further details on the engagements conducted for the development of the proposed project are presented in Annex 7: Stakeholder engagement plan.

¹¹⁴ the equivalent of €7.06 million

¹¹⁵ A detailed description of project activities is presented in Sections B.3 and E.6 of the Funding Proposal.

grant is deemed the most appropriate level of concessionality given: i) the public good nature of the project's activities and lack of financial reflows; ii) Haiti's status as a SIDS and LDC; and iii) the level of debt of the country, with its limited capacity to repay loans and iv) the project beneficiaries are small poor farmers that need grants to make these investments. Indeed, the absence of financial resources required to implement the necessary adaptation investments has been identified as a considerable barrier to implementing climate-resilient IWRM in Haiti.

183. The combined investment of nearly US\$31.58 million will be used efficiently and effectively to achieve three main project objectives. First, agroforestry and ecosystem restoration interventions will be implemented at priority sites in the TR watershed to improve the infiltration of water into soils and increase surface roughness. This will contribute to attenuating overland flows from heavy rainfall events and consequently reduce the severity of associated flood events. Second, a shift from unsustainable land-use practices, including charcoal production, towards productive land-use management will be initiated in the TR watershed. This will be achieved by enhancing the technical and institutional capacity of decision-makers involved in agriculture and charcoal production at the commune, inter-commune and department levels in the watershed to implement climate-resilient and sustainable land-use management techniques. Specifically, communal development councils and multi-sectoral management structures will be established to oversee and manage the implementation of Sustainable Land Management (SLM) interventions. To support this, training will be provided to representatives from relevant government entities, including MARNDR, NCFS and MSAL, on the implementation of SLM in the TR watershed. Additionally, an agricultural finance mechanism will be established under the VSCA. Under this mechanism, financing will be made available to smallholder farmers to facilitate their adoption and implementation of sustainable, climate-resilient agricultural practices. This long-term solution will be supplemented by providing immediate relief, in the form of food coupons, to vulnerable people affected by climate change. This will not only safeguard the lives of these people against the immediate climate change shocks but also contribute to reducing further pressure on the environment by providing an alternative option to resorting to unsustainable charcoal production.
184. Last, the required governance framework for integrated land and water resources management will be established and implemented at catchment and sub-catchment levels in the TR watershed to optimise the sustainable use of these resources under future climate change conditions. The cost-effectiveness of proposed project interventions as well as a summary of the international best practices and lessons learned used to inform project development are presented below.

Cost-effectiveness of project interventions

185. Climate change-induced flooding in Haiti severely impacts vulnerable communities living in the TR region through inter alia the degradation of land and water resources, reduced resilience of agricultural livelihoods and increased prevalence of waterborne diseases. The projected reductions in total average annual rainfall combined with an increase in intensity of concentrated precipitation events under RCP4.5 and RCP8.5 scenarios are expected to exacerbate these flood impacts. In order to take decisions regarding the type of climate change adaptation options that the Government of Haiti (GoH) should implement to respond to this, it is necessary to understand the adaptation expenditure options and expected internal rate of return (IRR) on these options. To do this, a financial and economic analysis (FEA) was conducted to inform this decision-making¹¹⁶. This analysis focused on determining costs and benefits associated with: i) implementing EbA interventions, including agroforestry and reforestation activities, to enhance the resilience of the TR watershed to climate change-induced flooding; ii) facilitating sustainable catchment management practices with the strengthening of agricultural value chains; and iii) supporting the GoH in the development of a national-level policy framework to facilitate IWRM in Haiti.
186. The economic analysis estimated the net present value (NPV) of interventions to be US\$376 million with an internal rate of return (IRR) of 40.7%, therefore representing a highly cost-effective spend for GoH. The estimated health benefits under the FEA considered water-borne diseases (such as diarrhoea and cholera) that would, without project interventions, likely increase in prevalence because of increased flooding and related impacts. Relevant literature on the link between these diseases and floods was used to estimate the likely number of additional cases,

¹¹⁶ The financial and economic analysis undertaken to support the proposed project is presented in Annex 3: Financial and Economic Analysis.

morbidity, and mortality^{117,118,119,120,121,122}. Benefits from agroforestry interventions were estimated based on field data in the financial analysis. Benefits of avoided infrastructure damage due to floods was calculated based on Gaspard (2016) which is limited to calculating direct damages to households.

187. The economic analysis of the proposed project was carried out in accordance with the *Guidelines for the Economic Analysis of Projects of United Nations Development Program*.¹²³ The economic efficiency of the investment was determined by computing the economic net present value (NPV) with an assumed 10% discount rate, and the economic internal rate of return (EIRR). For consistency purposes, all proposals developed with the support of UNDP have thus far opted to use a 10% discount rate, in line with the existing practice of multilateral development banks.
188. Economic values (costs and benefits) are all measured in real terms of 2019. Economic costs of the project are net of taxes, duties, and price contingencies. Furthermore, the analysis assumes a shadow wage rate of 1.00 for unskilled and semi-skilled labour in Haiti. Given the official unemployment rate of approximately 14%,¹²⁴ the economic cost of unskilled labour in Haiti is expected to be considerably lower than the market wage rate (financial cost). As a result, this assumption leads to significantly *over-estimating* the economic cost of the project and thereby *under-estimating* the true net economic value of the project. The above assumption allows the use of financial cost as a measure of the economic cost of the project (once again noting that in doing so, the economic cost of the project is over-estimated, and the net present value of the investment is then under-estimated).
189. As is common when undertaking the economic analysis of investment projects, numerous assumptions were used to delineate the “with project scenario” from the “without project scenario”. These assumptions are presented and discussed in Annex III where the detailed economic analysis is presented. Assumptions were made so as to under-estimate the true net economic value of the proposed investment project. The analysis covers a period of 25 years, from 2019 to 2043.
190. *Economic costs.* Economic costs were obtained from the project budget. They were available for each of the 3 outputs in addition to project management. With the exception of the co-financing, project costs were also available for each of the 8 years of project implementation. Solely for purpose of the economic analysis, it is assumed that the co-financing is disbursed annually at the same as GCF funding. This provided a flow of project cost as presented in the table below.

Total Cost of the Project per Output per Year

Output	Total	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Output 1	18,399,600	301,200	1,079,340	1,148,990	4,446,050	4,316,700	4,231,050	1,865,130	1,011,140
Output 2	8,303,355	1,088,628	1,416,536	1,335,623	1,373,123	1,144,106	1,046,831	898,508	0
Output 3	3,273,519	481,880	395,720	527,666	432,866	428,666	360,220	299,820	346,680
Mgmt	1,314,087	152,867	171,573	158,280	177,564	172,559	158,118	154,163	168,965
Total	31,290,561	2,024,575	3,063,169	3,170,559	6,429,603	6,062,031	5,796,219	3,217,621	1,526,785

¹¹⁷ Eisenberg, M.C. *et al.* 2013. Examining Rainfall and Cholera Dynamics in Haiti Using Statistical and Dynamic Modelling Approaches. *Epidemics* 5 (4).

¹¹⁸ Barzilay, E.J. *et al.* 2013. Cholera Surveillance During the Haiti Epidemic — The First 2 Years. *New England Journal of Medicine* 368 (7).

¹¹⁹ WHO. 2010. Public health risk assessment and interventions. Earthquake: Haiti. Available at: https://www.preventionweb.net/files/12945_haitiearthquake201001181.pdf

¹²⁰ WHO. 2004. Cholera outbreak: Assessing the outbreak response and improving preparedness. Available at: https://apps.who.int/iris/bitstream/handle/10665/43017/WHO_CDS_CPE_ZFk_2004.4_eng.pdf;jsessionid=79C09A3DE1F0B1E71168734BACFD790D?sequence=1

¹²¹ Gaudart, J. *et al.* 2013. Spatio-Temporal Dynamics of Cholera during the First Year of the Epidemic in Haiti. *PLoS neglected tropical diseases*. 7.

¹²² CDC. 2011. Lessons Learned during Public Health Response to Cholera Epidemic in Haiti and the Dominican Republic. Available at: https://wwwnc.cdc.gov/eid/article/17/11/11-0827_article

¹²³ UNDP. 2015. *Guidance on the conduct and reporting of the Economic and Financial Analysis of Climate Change Adaptation and Mitigation Projects and Programmes*. UNDP.

¹²⁴ <https://data.worldbank.org/indicator/SL.UEM.TOTL.ZS?locations=HT>

191. Project benefits included in the economic analysis are health benefits of the project and improvement in rural livelihoods. Both of these benefits are estimated on an annual basis, comparing with and without project scenarios in presence of climate change.

192. *Health benefits.* Health benefits were limited to a reduction in the number of cases of cholera and mortality. These reductions were estimated by constructing a scenario of the potential number of such cases (of cholera and mortality) with climate change in the absence of the project, and another scenario of the potential number of such cases with climate change with the project. Three scenarios are developed in terms of the possible impacts of the project on reducing the number of cases morbidity and mortality: Scenario 1: a 10% reduction; Scenario 2: a 20% reduction; and Scenario 3: a 30% reduction. Based on best available information, a cost per case of cholera in Haiti was estimated to be \$35. For purpose of assessing the economic value of mitigating the death toll, a value of statistical life (VSL) of \$223,500 is used in the analysis (a sensitivity analysis on this estimate was also calculated). Details are available in Annex III.

193. *Rural livelihood benefits.* A financial analysis of agroforestry was carried out based on field data. The table below provides a summary of these results with and without GCF support:

Perspective	NPV		IRR		Payback period	
	Without GCF	With GCF	Without GCF	With GCF	Without GCF	With GCF
Smallholder farmer - Agroforestry	USD 1.818,33	USD 3.618,41	25%	42%	4	3

194. Additional assumptions for the economic analysis are:

- As reported in the Feasibility Study, the average farm size within the TR catchment ranges between 0.8 and 1.2 hectares (ha). For purpose of the economic analysis, it is assumed that average farm size is 1.0 hectare.¹²⁵ It is also assumed that the number of household farms benefiting from the agro-forestry activities of the project is 17,740 farms, thus reaching a total of 17,740 ha.
- Profit margins in the sector are estimated to be approximately 10%.
- It is important to note that the net incremental benefits from agroforestry are deducted by \$9.15 per ha which is an estimate of the foregone net revenues from charcoal production.¹²⁶

195. *Infrastructure benefits.* The proposed investment project shows a significant reduction of the projected extent of flood zones and impacts of the project on the number of houses and infrastructure exposed to projected floods (relative to a no project scenario). There is however limited information currently available on *historic* losses and damages to the stock of housing and infrastructure. The only estimate we could find is Gaspard (2016) which estimated the direct losses due to flooding in 2004 by interviewing households throughout the Gonaïves region (located adjacent to our project area). He estimated direct damages (by asking how much it would cost to repair the houses due to flood damages) at between USD960.85 and USD1489.5 per household. It is important to note that these benefits only consider direct damages from flooding (e.g. the cost to repair the damage) and only consider household costs (and not costs to public infrastructure, schools, hospitals, public administration. Therefore, we present these numbers with a strong word of caution that they do not represent the extent of flooding damages the project will reduce. These costs were included in the economic analysis without changing results significantly.

196. *Economic efficiency.* In the base case, the NPV of the project is estimated to reach between \$375 and \$412 million depending of the scenario of flood reduction benefits developed earlier (as presented above). The IRR ranges between 41% and 43%.

Base Case: Estimated NPV and IRR

¹²⁵ In a World Bank recently funded project of a somewhat similar nature but in the Southern departments of Haiti, a similar assumption of an average farm size of 1.0 hectare is also made. See: World Bank. 2018. *Project Appraisal Document on a Proposed Grant to the Republic of Haiti for a Resilient Productive Landscapes Project*. World Bank. Washington, D.C. (page 74)

¹²⁶ Kennedy, N.S. 2015. Id.

	NPV	IRR
Scenario 1	375.516.111	41%
Scenario 2	393.838.100	42%
Scenario 3	412.160.089	43%

197. The above estimates already include a sensitivity analysis pertaining to the expected flood benefits of the project (from the construction of the 3 scenarios). In addition, a further sensitivity analysis was included: a reduction of 20% in total benefits; an increase of 20% in total costs; and a combination of both of these scenarios. Results are presented in the table below. Results show the project's economic efficiency to be robust to these changes.

Sensitivity Analysis (1): Changes in Costs and Benefits

		NPV	IRR
Benefits: - 20%	Scenario 1	297,547,350	38%
	Scenario 2	312,204,941	39%
	Scenario 3	326,862,533	40%
Costs: +20%	Scenario 1	372,961,434	38%
	Scenario 2	391,283,423	40%
	Scenario 3	409,605,412	41%
Benefits: - 20% and Costs: +20%	Scenario 1	293,438,365	36%
	Scenario 2	308,095,957	37%
	Scenario 3	322,753,548	38%

Application of best practices in land and water resources management

198. Best practices from similar projects¹²⁷ were used to inform the design of IWRM interventions to be implemented under the proposed project. Lessons learned from numerous baseline investments into climate change adaptation in Haiti were also used to guide the design of all project interventions. Such lessons include appropriate mechanisms for ensuring that project activities are implemented in a participatory, gender-inclusive and sustainable way. The specific international best practices and lessons learned that were applied to the design of the proposed project and that will contribute to the efficiency and effectiveness of project interventions are summarised below. Additionally, details on all best practices and lessons learned relevant to the project are presented in Section 7 of Annex 2: Feasibility Study.

199. International best practices and lessons learned identified for the project include:

- ensuring that landscape planning, implementation of policies and strategies and monitoring processes are implemented using an integrated and iterative approach to account for the dynamic nature of landscape processes;
- acknowledging the need to regularly engage stakeholders in an equitable manner to ensure that adaptation benefits and incentives of integrated land and water resources management are responsive to the needs of the target groups;
- ensuring that water resource planning and management is linked to Haiti's overall sustainable development strategy and public administration framework;
- acknowledging that approaches to land and water resources management will evolve as the pressures on natural resources change;
- facilitating participatory monitoring and adaptive management by ensuring that information is widely accessible and not controlled by any single stakeholder;
- incorporating traditional knowledge into training for farmers and local communities on climate-resilient agricultural practices as well as land and water resources management strategies;

¹²⁷ as outlined in Section B.1 of this Funding Proposal and Section 6 of Annex 2: Feasibility Study

- ensuring the provision of high-quality seedlings and starter kits for nurseries to create a sustainable agroforestry production cycle;
- drawing on regional, national and local experiences and resources to increase capacity to respond to the impacts of extreme events;
- developing appropriate strategies for securing financial resources for project development and implementation;
- effectively managing the available financial and human resources to ensure the efficient and cost-effective implementation of project interventions;
- undertaking regular monitoring and evaluation (M&E) of climate-adaptive interventions to ensure that the most effective and appropriate solutions are being implemented under future conditions of climate change;
- engaging and collaborating extensively with all relevant project stakeholders to encourage buy-in from national- and local-level decision-makers, which will contribute to the sustainability of proposed adaptation interventions over the long term; and
- implementing effective financial and project management strategies to ensure the efficient use of financial resources and avoid delays during the implementation phase.

LOGICAL FRAMEWORK

*This section refers to the project/programme's logical framework in accordance with the **GCF's Integrated Results Management Framework** to which the project/programme contributes as a whole, including in respect of any co-financing.*

E.1. Project/Programme Focus

- ☐ Reduced emissions (mitigation)
☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.

Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/ transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Various initiatives in Haiti have been developed to increase its capacity for ecosystem management as well as to enhance its capacity for disaster risk reduction. However, while these initiatives have mobilized important resources, many of these have been unsuccessful in reaching the territorial scale that is required to	<u>Low</u>	The paradigm shift will be two fold. First it will take on a user end approach in its ecosystem based interventions that will allow the project to create the enabling conditions that will allow for these interventions to achieve the scale that is required to result in adaptation benefits in the form of flood reduction. This will include a strong community based focus that is linked to livelihood and health benefits but is also informed by the hydrological models developed by the project, thus ensuring that ecosystem based interventions are prioritized in areas that will effectively reduce climate induced flood risk.	The scale of the project is built around the principles of climate-resilient, sustainable development and underpinned by three main interventions. First, agroforestry systems will be established and degraded forests restored at priority intervention sites across the watershed to: i) reduce peak surface runoff and flood extent; and ii) provide alternative livelihood options to the unsustainable land-use practices on which most vulnerable communities in the TR watershed depend. Second, complementary interventions focused on enhancing technical and institutional capacity at the commune, inter-commune and

	achieve meaningful adaptation benefits. Often these projects are focused at a sub catchment rather than a catchment level, or invest in limited actions at an ecosystem level that do not allow them to reach their disaster reduction objectives that are originally expected. The lack of scale in results by these initiatives comes as a result of initiatives that fail to include an end user approach to the reforestation initiatives that is required to ensure the level of impact (in the form of survival rates and sustainability of ecosystem based investments) to produce the adaptation benefits that are required. Failure to produce these or other tangible benefits also limit the scale and reach of these projects to a meaningful part of the population (as ecosystem investments are often not guided on producing adaptation benefits). As these interventions are often implemented without linkages to national policies they also fail to reach the transformative		At a community level, the paradigm shift will involve shifting away from current degrading practices – resulting from poor land-use planning and a lack of viable livelihood alternatives – that are increasing flood risk in Haiti, towards an integrated approach to flood management to adequately address the diverse impacts of climate change-induced flooding on the watershed and population living within it. This will include shifting decision-making on flood management away from the existing uncoordinated and compartmentalised approach, towards an integrated approach that is underpinned by the uptake and implementation of climate-resilient land-use practices that will result in enhanced ecosystem based flood protection, strong governance, as well as the sustainable use and management of land and water resources. Together these actions will create a concept of proof that will be observed at a catchment level scale (rather than sub catchment) and will be observed in the project's capacity to deliver on benefits to over 700,000 people both directly and indirectly. The scale of these results will in turn be measured to carbon sequestration, health monitoring, increased water supply and reduced risk to allow for it to become a relevant to policy makers and development practitioners working in the country	department levels will be implemented to promote productive climate-resilient agricultural practices and sustainable land management. In addition, the necessary governance framework for IWRM will be established in Haiti to support the uptake of climate-resilient land management systems and facilitate sustainable use and management of water resources over the long term. Finally the project has integrated within it's the design the monitoring of the projects benefits as felt by the communities to create linkages between the tangible results seen as a result of the project in the form of improved livelihoods, food security, health and flood protection thus creating a connection between the scale of the project's investments (at a catchment level) to the concrete benefits that are being felt by the community within the project's target areas.
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	effect that is needed to evidence the role of EBA as a viable adaptation strategy in the country that is linked to multiple co benefits as these fail to materialize at the level that is required or are not monitored.			
Replicability	While several initiatives have attempted to introduce sustainable land-use systems in Haiti, the lack of a coordinated and integrated decision making processes by land planners, as well as the knowledge and technical capacity of communities to adopt sustainable practices, and immediate economic vulnerability is constraining the uptake of agroforestry and restoration activities in Haiti. Lack of information on the economic returns of agroforestry at the community level for example have made this transition appear risky in light of immediate economic needs and there are limited financial mechanisms to help enable these.	<u>Low</u>	Together, project interventions will transform flood management in Haiti by catalysing a shift away from uncoordinated, compartmentalised decision-making around flood management, towards an integrated approach that can be scaled up and replicated nationally. This will be enhanced by demonstrating the viability of eco-businesses linked to agroforestry (reducing the risk assessment constraint), will promote the uptake of project interventions across other watersheds in the country by other farmers. From a development perspective it will also demonstrate the importance of generating social protection to communities during the transition period between unsustainable land management practices to encourage a just transition, thus informing future approaches linking EBA to green growth opportunities. It will also generate financial mechanisms through VSCA to help not only sustain these practices but encourage its replicability through access to savings.	The replicability of the project interventions will be underpinned by the drafting of revisions to the Water Act for the adoption of an integrated approach to water resources management in Haiti as well as the establishment of Catchment IWRM Committees, as well as Sub-Catchment Water Resource Users Associations in each of the country's watersheds (Output 3). As a result of these changes, a new strategy will be developed for IWRM in Haiti that addresses both the existing challenges of a fragmented and uncoordinated system of water management, particularly at a catchment and sub-catchment level, as well as the current and projected impacts of climate change-induced flooding. Output 2 will also train representatives from appropriate government entities, including MARNDR, NCFS and MSAL, on the implementation of climate-resilient agricultural techniques and SLM, which can be transferred across other regions in the country. More significantly the tangible benefits from the EBA approach will motivate replicability of the sustainable management of water and land resources by farmers in the region as these will be linked to increased food security and incomes. The project will work to create

				VSCAs as financial mechanisms to enable upscale and continued investment.
Sustainability	Although the establishment of a Water Act has been identified as one of the Ministry of Environment's priorities in its 2019-2022 Triennial Plan, there are currently no integrated, climate-sensitive water or sustainable land management policies and governance frameworks in Haiti to allow for its practical implementation. The country lacks overarching water management legislation, a formal cadaster, a standard practice for land tenure arrangements, or relevant strategies to promote sustainable landscape management. Households make decisions in their own immediate interest, without necessarily considering the impact of those decisions on the sub-catchment and the long-term impacts on their own wellbeing thus increasing their vulnerability to climate impacts such as flooding. There is no experience or incentive among water	<u>Low</u>	Relevant stakeholders at national, department and commune levels have the technical and institutional capacity for the implementation of an integrated approach to land and water resources management that is centered around sustainability and climate resilience. This will enable the compatibility of SLM with the effective functioning of the entire watershed which, in turn, will contribute to enhancing flood management across the TR watershed. By building the capacity of the targeted institutions, the transfer of knowledge and skills related to sustainable watershed and flood management will extend beyond the project. These results are enhanced through monitoring (such as health) that will provide relevant information regarding the benefits of IWRM for local communities. Vulnerable farmers are equipped with the necessary knowledge and support to transition from unsustainable agricultural and charcoal production practices towards the development of sustainable agroforestry businesses that are progressive and gender-responsive. These in turn provide not only economic benefits but also increase resilience through EBA induced flood protection at a territorial level thereby reducing the potential impacts from climate induced flooding to lives and livelihoods. Positive impacts of EBA seen in water and food security as well as increased income and improved health will sustain the EBA interventions by creating an added value to	Under the project, the technical and institutional capacity of relevant stakeholders at national, department and commune levels will be strengthened for the implementation of an integrated approach to land and water resources management that is centered around sustainability and climate resilience. By building the capacity of the targeted institutions, the transfer of knowledge and skills related to sustainable watershed and flood management will extend beyond the project. Strengthening of the Water Act will establish a legal foundation for optimising the use of water resources without compromising the sustainability of vital ecosystems under future climate change conditions, consequently, securing equitable economic and social welfare. The Water Act will also provide a framework for the institutionalization of integrated watershed management to empower regional administrations across Haiti — through establishing Catchment Water Resources Management Committees (CIWRMCs) and Sub-Catchment Water Resource Users Associations (SCWRUAs). The country-driven project has been developed using a participatory and consultative process, enabling all relevant stakeholders to contribute to the design of interventions, ensuring buy-in and commitment from national- and local-level stakeholders. More importantly to communities, the social protection system provided through the project as well as the EBA interventions that will be installed will provide tangible benefits to communities

	users to consider sub-catchment scale water management in their household decision-making to enable long term strategies to reduce their own vulnerability by protecting critical water sources and ecosystems on which they depend.		communities through sustainable land and water management	linking the IWRM approach to reduced vulnerability.
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E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ¹²⁸	
<u>Total number of project beneficiaries</u> Sum of all project beneficiaries identified under ARA1, ARA2, and ARA 4 accounting for overlaps	<u>Core 2: Direct and indirect beneficiaries reached</u>	Primary MoV Baseline and Completion Surveys and project participation lists¹²⁹ Simulation of future flooding impact on households taking into account historical, no action scenarios and conditions derived from project	<u>Direct: 0 (Male: 0, Female: 0)</u> <u>Indirect: 0 (Male: 0, Female: 0)</u>	<u>Direct: At least 54,714 people (Male: 40,561 Female: 14,153)</u> <u>Indirect: At least 237,886 people in Haiti (Male: 114,185, Female: 123,701)</u>	<u>Direct: At least 292,600 people in Haiti (Male: 140,448, Female: 152,152)</u> <u>Indirect: At least 441,272 people in Haiti (Male: 211,811 Female: 229,461)</u>	Projections for population growth taken by the project for the TR area are accurate (see Annex 3 for numbers) Direct beneficiaries calculated by project mid term do not account for EBA benefits that are expected to be seen only by the project's Y7. Identified farmers within the 33 communal section are willing to

¹²⁸ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

¹²⁹ Means of Verification for Fund-level Impact and Outcome indicators will be triangulated with the baseline surveys that will be undertaken by project consultants, and at project mid-term and project end with the latest available national census data.

		implementation (increased forest cover). Increased forest coverage within the project areas measured by GIS mapping (see above MoV) Statutes of Creation of the committees that specify their coverage area.				participate in land planning and land management capacity building exercises by the project's mid term accounting for direct beneficiary calculation at mid term Direct beneficiary calculation at project end includes all households identified that are located less than 30m to the waterways in the target communes (See Section 10.3 Site Selection Section in FS and Annex 3) per mapping by the project. Project investments in agroforestry and forest rehabilitation in priority areas are completed by the project's Y7 with 50% being achieved by Y5. Water management system provides effective governance to the benefit of all water users within the project's 33 communal sections
<u>ARA1 Most vulnerable people and communities</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Baseline and Completion Surveys ¹³⁰	<u>Direct: 0 (Male: 0, Female: 0)</u>	<u>Direct: At least 54,714 people</u>	<u>Direct: At least 292,600 people in Haiti (Male: 140,448,</u>	Projections for population growth taken by the project for the TR area are accurate (see Annex 3 for numbers)

¹³⁰ Means of Verification for Fund-level Impact and Outcome indicators will be triangulated with the baseline surveys that will be undertaken by project consultants, and at project mid-term and project end with the latest available national census data.

		Simulation of future flooding impact on households taking into account historical, no action scenarios and conditions derived from project implementation (increased forest cover). Increased forest coverage within the project areas measured by GIS mapping (see above MoV) Community land use plans developed Capacity assessment of farmers that have been provided with training	<u>Indirect: 0 (Male: 0, Female: 0)</u>	<u>(Male: 40,561 Female: 14,153)</u> <u>Indirect: At least 237,886 people in Haiti (Male:114,185, Female: 123,701)</u>	<u>Female: 152,152)</u> <u>Indirect: At least 441,272 people in Haiti (Male: 212,064 Female: 229,736)</u>	Direct beneficiaries calculated by project mid term do not account for EBA benefits that are expected to be seen only by the project's Y7. Identified farmers within the 33 communal section are willing to participate in land planning and land management capacity building exercises by the project's midterm accounting for direct beneficiary calculation at mid term Direct beneficiary calculation at project end includes all households identified that are located less than 30m to the waterways (those most vulnerable) in the target communes (See Section 10.3 Site Selection Section in FS and Annex 3) per mapping by the project. Project investments in agroforestry and forest rehabilitation in priority areas are completed by the project's Y7 with 50% being achieved by Y5.
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<u>ARA2 Health, well-being, food and water security</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Baseline and Completion Surveys and project participation lists¹³¹ Capacity needs assessment of farmers that have been provided with training Statutes of Creation of the water governance committees that specify their coverage area. Statutes of Creation/Reports of farmer field schools created by the project Adaptation interventions developed by the project and report on their usage Monitoring reports of beneficiaries receiving food coupons Statutes of Creation/Reports of VSCAs	<u>Total: 0</u>	<u>Direct:</u> At least 10,000 people <u>Indirect:</u> At least 226,400 people in Haiti (Male:176,112, Female: 190,788)	<u>Direct:</u> 58,328 (male:42,362/ female: 15,966) <u>Indirect:</u> 441,272 <u>Male: 211,811</u> <u>Female: 229,461</u>	Direct beneficiaries under ARA2 include farmers with enhanced productive capacity through training provided by the project's Output 2 & members of households receiving food coupons as well as 340 technical staff from local governments (Refer to Annex 2 Section 10). Indirect beneficiaries under ARA2 include all water users within the 33 sub communes (total 733,872). To avoid double counting the number of direct beneficiaries under ARA 1&4 which are also water users (292,600) has been subtracted. These will benefit from improved water governance systems and their enhanced capacity in monitoring water quality and implementing improved (climate smart) water management practices Projections for population growth used for the indirect total beneficiary number taken by the project for the TR area are accurate (see Annex 3 for numbers)
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¹³¹ Means of Verification for Fund-level Impact and Outcome indicators will be triangulated with the baseline surveys that will be undertaken by project consultants, and at project mid-term and project end with the latest available national census data.

		enhanced/formed by the project				Information on amount of farmers in the 33 sub communes (representing direct beneficiary numbers) responds to the 2008 the Agricultural Census and does not take into account population growth as it relates to associated farmers Representatives from farmers' association are willing to participate in the project through training for productive purposes as well as in the water management committees
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Baseline and Completion Surveys¹³² Simulation of future flooding impact on households taking into account historical, no action scenarios and conditions derived from project implementation (increased forest cover). Increased forest coverage within the project areas measured by GIS mapping (see above MoV) Community land use plans developed	<u>Direct: 0 (Male: 0, Female: 0)</u> <u>Indirect: 0 (Male: 0, Female: 0)</u>	<u>Direct: At least 54,714 people (Male: 40,561 Female: 14,153)</u> <u>Indirect: At least 237,886 people in Haiti (Male: 114,185, Female: 123,701)</u>	<u>Direct: At least 292,600 people in Haiti (Male: 140,448, Female: 152,152)</u> <u>Indirect: At least 441,272 people in Haiti (Male: 211,811 Female: 229,461)</u>	Projections for population growth taken by the project for the TR area are accurate (see Annex 3 for numbers) Direct beneficiaries calculated by project mid term represent farmers that will participate in the implementation of agroforestry model and do not account for EBA benefits that are expected to be seen only by the project's Y7. Identified farmers within the 33 communal section are willing to participate in land planning and land management capacity building exercises by the project's midterm accounting

¹³² Means of Verification for Fund-level Impact and Outcome indicators will be triangulated with the baseline surveys that will be undertaken by project consultants, and at project mid-term and project end with the latest available national census data.

		Capacity assessment of farmers that have been provided with training for improved land use management in keeping with land use plans				for direct beneficiary calculation at mid term Please refer to Annex 2 Section 11 for more details on calculations for total direct and indirect beneficiaries Project investments in agroforestry and forest rehabilitation in priority areas are completed by the project's Y7 with 50% being achieved by Y5.
<u>ARA1 Most vulnerable people and communities</u>	<u>Supplementary 3.1: Change in expected losses of economic assets due to the impact of extreme climate-related disasters in the geographic area of the GCF intervention</u>	Baseline Survey and Year 7 monitoring if there is a flood event¹³³ Simulation of future flooding impact on households taking into account historical, no action scenarios and conditions derived from project implementation (increased forest cover).	463 households affected by 20 year floods under the baseline scenario 1,342 houses affected by a 100-year floods under the baseline scenario Assets in USD TBD¹³⁴	0 USD No immediate impact as agroforestry and reforestation actions will require increased time span to provide significant flood mitigation impacts .	At least USD 134,100 of savings on household losses assuming 35% fewer houses affected by the 20-year floods (163 households) at a minimum damage cost of USD 822.7 per household	Flooding simulations prepared during project development are able to project flooding impact based on climate change scenarios Population numbers for this output are those taken from the existing 2003 census information and do not take into account population growth.

¹³³ Methodologically it is challenging to anticipate the change in losses of lives and set targets, so the indicator to be reported would focus on change in losses of economic assets US\$, noting that this can only be measured at project end point if there is a flood event at that time. In the absence of these, the project will be able to simulate impact based on the climate modelling used in the development of the project (See FS Annex II). The project in its economic analysis (Annex III) was able to calculate assets based on Gaspard (2016) that estimated household damage cost between USD 822.7 to USD 1275.35 due to flooding in the region in 2004. It is important to note that these benefits only consider direct damages from flooding (e.g. the cost to repair the damage) and only consider household costs (and not costs to public infrastructure, schools, hospitals, public administration). Therefore, these numbers while available are presented with a strong word of caution that they do not represent the extent of flooding damages the project will reduce and as such are not included within the log frame. A survey will be able to better present these amounts and will also account the cost of health impacts derived from extreme climate related disasters in the area. Annex III includes information re a reduction rate of 10-30% of diarrhea and its avoided costs (between USD 2.1 million to 6.4 million) by 2030 and demonstrating increased reduction amount over time (USD 3 million- 9 million by 2043) .

¹³⁴ Baseline Survey to be conducted in Year 1 of project implementation and reported in the first two annual progress report.

					based on floodings in 2004 At least USD 579,181 of savings on household losses assuming 50% fewer houses affected by a 100 year flood (704 households) at a minimum damage cost of USD 822.7 per household based on floodings in 2004 Final US\$ amount will be determined based on survey information.	If there is a flood event in Year 7, appropriate tracking of the final target will be undertaken.
<u>ARA2 Health, well-being, food and water security</u>	<u>Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options</u>	Agricultural and beneficiary surveys to measure farmers' capacity for sustainable and climate resilient production, income and satisfaction Site visits	<u>Total:</u> 0	10,000 farmers adopting sustainable productive land management measures as a result of the project	54,252 farmers (13,921 women and 40,331 men) adopting sustainable productive land management measures as a	Agricultural census information (2008) used for beneficiary calculation remains accurate (these numbers do not account for population growth amongst associated farmers) The project's farmer field school approach for providing capacity building to agricultural

					result of the project	associations allows for project derived training to extend to all associated farmers within the 33 sub communes Adaptation interventions invested by the project are accessible to all associated farmers within the 33 communal sections
<u>ARA2 Health, well-being, food and water security</u>	<u>Supplementary 2.2: Beneficiaries (female/male) with improved food security</u>	Baseline and completion survey Household surveys ¹³⁵	<u>Total: 0</u>	At least 25,245 beneficiaries (Male: 12,118, Female: 13,127) decrease in food insecurity. Accounting for an at least 8% households decreasing in food insecurity	At least 88,359 beneficiaries (Male: 42,412, Female: 45,947) decrease in food insecurity. Accounting for at least 28% of households that decrease in food insecurity	Calculations were made on the assumption that at the beginning of the project 43% of households in the TR watershed are food insecure. Calculation based on direct beneficiaries (292,600) and indirect beneficiaries (441,272) with an average of 5 persons per household, so based on the number of households in TR watershed: 146,774.4 a total of 63,113 households in the TR watershed are food insecure at the start of the project. Food vouchers provided through the project are

¹³⁵ The change in food security will be measured using the Household Economy Approach — HEA. The HEA methodology is based on the analysis of family/household livelihoods. It highlights how households usually support themselves, assess the impact of external factors on their livelihoods and provides recommendations to support coping strategies. It is a dynamic approach that divides households into four socio-economic groups, namely the "very poor", "poor", "better-off" and "haves". Evidence for Development is available at: <http://efd.org/methods/the-household-economy-approach-hea>.

						effective in addressing immediate food security needs allowing for a reduction in unsustainable charcoal production. Households in the seven target communes adopt climate-resilient agriculture and sustainable land-use practices. Farmers in the project target area are willing to adopt new market opportunities for their produce. Communities participate in the VSCAs to mobilize savings for capital investments that increase their own productivity rate At end of project at least 28% decrease in food insecurity for farmers, communities and households in target communes.
<u>ARA2 Health, well-being, food and water security</u>	<u>Supplementary 2.3: Beneficiaries (female/male) with more climate-resilient water security</u>	Baseline and Completion Surveys¹³⁶ Statutes of Creation of the water governance committees that specify their coverage area. Capacity assessment of the water governance	<u>Total:</u> 0	Direct: 340 (102 women and 238 men) <u>Indirect:</u> At least 226,400 people in Haiti (Male:176,112, Female: 190,788)	Direct: At least 340 (102 women and 238 men) Indirect: At least 441,272 people in Haiti (<u>Male: 211,811 Female: 229,461</u>)	Beneficiaries are water users that will fall under the jurisdiction of the newly created water governance committees (Output 3) and will have increased water security as a result of improved water management capacities created through the project

¹³⁶ Means of Verification for Fund-level Impact and Outcome indicators will be triangulated with the baseline surveys that will be undertaken by project consultants, and at project mid-term and project end with the latest available national census data.

		committees created under Output 3 Minutes of meetings taking place in water governance committees at all levels specifying interventions made for enhanced water management Site visits				Improved capacity for water management planning and governance combined with improved land management practices will result in improved water security amongst water users. A governance water scorecard adapted to assess water catchment and sub catchment levels will be used based on UNDPs guide on assessing water governance.¹³⁷ For Indirect Beneficiaries, the total number takes into account the total population of the TR watershed that are water users (733,872) minus the number of direct beneficiaries of water users under ARA 1& 4. The 340 are assumed to be part of the 292,600 direct beneficiaries.
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice</u>	Baseline Survey; Site surveys; and GIS mapping	0	At least 8,870 ha of agroforestry systems established At least 3,850 ha of water towers rehabilitated	17,740 ha of agroforestry systems established 7,700 ha of water towers rehabilitated	Sustainable land management practices will improve the provision of ecosystem services, including flood management (reduced run off hence reducing losses), increased water quality and agricultural productivity

¹³⁷ UNDPs User's Guide on Assessing Water Governance available at:

<https://www.undp.org/sites/g/files/zskgke326/files/publications/Users%20Guide%20on%20Assessing%20Water%20Governance1.pdf> will be adapted, using Box 6.2 as reference.

<u>ARA4 Ecosystems and ecosystem services</u>	<u>Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal marine areas brought under resoration and/or improved ecosystems</u>	Baseline Survey; Site surveys; and GIS mapping	0	At least 8,870 ha of agroforestry systems established At least 3,850 ha of water towers rehabilitated	17,740 ha of agroforestry systems established 7,700 ha of water towers rehabilitated	Sustainable land management practices will improve the provision of ecosystem services, including increased water quality and agricultural productivity.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner</u>	At present, the government is organised along departmental and communal lines. As a result, there is an overlap between existing water governance structures, particularly at the catchment and sub-catchment levels. This leads to challenges in coordination and planning, as well the effective and efficient implementation of an integrated approach to water resources management in the TR watershed.	<u>low</u>	The Water Act is developed and implemented to respond not only to national level priorities, but also to the adaptation needs of vulnerable communities living in vulnerable watersheds. Vulnerable communities are in turn empowered to act in support of IWRM through coordinated and more inclusive governance mechanisms.	The project will fund the establishment of the governance framework required to facilitate the adoption of a climate-resilient integrated water resources management (IWRM) approach in Haiti's TR watershed. Under Output 3, the capacity of the GoH will be strengthened to develop and implement the Water Act, which is an essential legal foundation for enabling participatory, integrated water resources	<u>Multiple sub-national areas within a country</u>

	More importantly it has excluded the role of communities in the correct use of these critical resources particularly in consideration of future climate change conditions. It has also failed to create awareness of the linkages of IWRM to development. While the GoH has made good progress in drafting the Water Act to address these challenges, there is a need for technical assistance for the 'last mile' of its drafting, meaning its linkage at the territorial basin level and to the role of water users. This last mile is required to ensure the effectiveness of the Water Act in practice.		By strengthening national capacities for the implementation of the Water Act, promoting the adoption of an integrated, climate-resilient water management governance framework at the catchment and sub-catchment levels, and monitoring the impact of IWRM plans while linking these to concrete benefits, the governance framework will be establishing to facilitate the adoption of IWRM in Haiti both in a legal framework as well as in practice at a basin level.	management in Haiti's vulnerable watersheds. A Catchment Integrated Water Resources Management Committee (CIWRMC) will also be established to support the implementation of the Water Act. Together the project will create local capacities that will be able to enable the framework that is being promoted to the Water Act ensuring that once it is enacted it (and even if it is not) will detonate the change that is required, it will also create a local framework for action.	
<u>Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation</u>	Local communities have limited practical and technical experience to establish and manage local, financially sustainable small-scale operations for tree propagation, reforestation and agroforestry – compounding their climate vulnerability.	<u>low</u>	Local communities have the technical capacity to implement EbA interventions as flood reduction measures, including sustainable, climate-resilient land-use practices and techniques (e.g. agroforestry) to reduce the risk of increased land degradation under future climate change conditions. Enabling communities to adopt these activities as viable alternative livelihood	The project will strengthen the capacity of community groups, including farmer and women's associations, for climate-resilient land-use planning in seven target communes in the Trois-Rivières watershed. This will be complemented by direct support for the uptake agroforestry and restoration activities and the establishment of alternative livelihood practices and sustainable agroforestry businesses.	<u>Single sub-national area within a country</u>

			options under the project will also incentivise their uptake and contribute to maintaining their social and environmental sustainability beyond the project lifetime.		
<u>Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level</u>	Farmers currently lack access to capital to invest in start-up costs, maintenance materials and tools, as well as limited “bridging” household reserves to cope with crises . Alternative sources of funding and credit are not available or are difficult to access, which leads to short planning horizons. These planning horizons drive short-term relief activities, such as the production of charcoal at unsustainable rates to meet pressing short-term financial needs.	<u>low</u>	Farmers have access to sustainable and affordable finance options that enable private investment in agroforestry, climate-resilient agriculture and agricultural value-chains.	The project will establish an adaptation finance mechanism for the agricultural sector. This mechanism will increase access to finance for smallholder farmers, therefore, encouraging a shift in land use away from unsustainable farming practices such as charcoal production.	<u>Single sub-national area within a country</u>
<u>Choose an item.</u>		<u>Choose an item.</u>			<u>Choose an item.</u>

E.5. Project/programme specific indicators (project outcomes and outputs)

This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the “Project/programme co-benefit indicators” section in table below. Add rows as needed.

Please number each outcome and output as shown below to indicate association of outputs to the contributing outcome. The numbering for outputs under this section should correspond to the output numbering in annex 4 (detailed budget plan).

			Baseline	Target	Assumptions / Note
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[illegible]

Co-benefit 1: Social Empowerment	Number of community land-use plans giving stewardship over forests and other natural resources	Land use plans created and validated by communities	0 community land-use plans	33 community land-use plans	33 community land-use plans	Participatory design of community land use plans provides local communities with stewardship over natural resources and promotes social empowerment.
Co-benefit 2: Carbon mitigation	GHG emissions reduced, avoided or removed/sequestered	Baseline Survey; Site surveys; and GIS mapping	0	436,410 tCO ₂ -e	1,018,290 tCO ₂ -e	Rate of sequestration is evenly distributed across the 25 year lifespan of project interventions. See ME Annex for mitigation calculation methodology
Co-benefit 3: Reduced incidence of water borne diseases	Reduced incidence of water borne diseases	Baseline Survey; Health Monitoring by WHO ¹³⁸ Government reports	Average incidence rate of cholera in the TR basin at 11.1 per 1,000 people ¹³⁹	0	At least a 10% reduction	No further aggravating conditions other than flooding occur that may increase the rate of water borne diseases (i.e. cancellation of preventive action campaigns) See Annex 3a Section C.3.2.1 for methodological assumptions for target and calculations.

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 1.1 Strengthen the capacity of community groups, including farmer and women's associations, for climate-resilient land-use planning in seven target communes in the Trois-Rivières watershed	Under this activity, community members and farmer and women's associations in 33 communal sections within the 7 target communes in the TR watershed (Port-de-Paix, Chansolme, Bassin Bleu, Plaisance,	1.1.1 Develop 33 community land-use plans. 1.1.2 Train communities, farmer and women's associations in the	33 community land-use plans 20 Trainings delivered in the seven target communes for communities, farmer and women's associations

¹³⁸ The World Health Organisation (WHO) will provide technical support to the project in the development of methodology that would enable comparison studies to be undertaken across watersheds similar to the methodologies being used for the monitoring of HIV and cholera in the country. This line of cooperation is backed up by the technical collaboration already established through the NAP process.

¹³⁹ Please refer to Annex III for methodological notes on calculation. The baseline will be updated as part of the project's scope and reported upon on the project's Y1 to include not just cholera but all.

	Pilate, Marmelade, Gros Morne) will be trained on climate-resilient land-use management. 118 communities will be reached by these plans and the associated training, which will focus on: - the expected impacts of climate change, particularly flooding, in Haiti and the associated consequences of these impacts on the population's livelihoods; - linkages between land-use and hydrology of the TR watershed, particularly the implications of unsustainable land-use practices on the frequency and intensity of flooding in the watershed; - promoting sustainable, climate-resilient agricultural practices such as slope stabilisation, riparian protection, and contour line farming, as well as the importance of protecting forest areas, as viable options for improving hydrological function of the watershed and reducing flood impacts; - the role of the community in sustainable charcoal production and reforestation, as a strategy for increasing climate resilience in the watershed; and - practical aspects of agroforestry as a climate-resilient livelihood alternative. The groups of people targeted for training will also be encouraged to participate in the development of the community land-use plans, which will inform this training. The development of these plans will focus on identifying the impacts of climate change, particularly flooding, on the TR	seven target communes on the application of the community land-use plans and implementation of climate-resilient land-use practices. 1.1.3 Conduct one awareness campaign in each of the seven target communes in Haiti.	Awareness campaigns designed and implemented in the seven target communes
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	watershed and designing appropriate actions for addressing these impacts. Furthermore, farmers will be targeted to receive training on the development of sustainable agroforestry business models that will contribute to both improving livelihood security and enhancing the use of sustainable land-use practices in the TR watershed. To support these capacity-building initiatives, awareness-raising campaigns will be designed and implemented in the seven target communes through city hall meetings and community radio broadcasts. These campaigns will focus on: i) increasing the knowledge of people living in the TR watershed on sustainable land-use practices and promoting the uptake of these practices; ii) encouraging community members to participate in the development of community land-use plans; and iii) raising awareness among people living in the target communes on the impacts of climate change-induced flooding and effective adaptation and landscape management solutions for addressing these impacts.		
Activity 1.2. Implement ecosystem-based flood management solutions in the Trois-Rivières watershed	This project activity will focus on reducing the impacts of climate change-induced flooding in the TR catchment and Port-de-Paix. This will be achieved by implementing 17,740 ha of agroforestry systems and rehabilitating 7,700 ha of forests in priority areas in the watershed. Implementing these EbA interventions will result in a 40% reduction in peak runoff, with rehabilitation interventions, in particular, expected to reduce floodplain extents in the target	1.2.1. Establish 17,740 ha of agroforestry systems in priority areas the TR watershed, including: i) agrisilvicultural systems — which are a combination of crops and trees, such as alley cropping; ii) silvopastoral systems — which combine forestry and grazing of domesticated animals; and iii) agrosilvopastoral systems that combine trees, crops and animals.	17,740 ha of agroforestry systems established in priority areas the TR watershed, including: i) agrisilvicultural systems — which are a combination of crops and trees, such as alley cropping; ii) silvopastoral systems — which combine forestry and grazing of domesticated animals; and iii) agrosilvopastoral systems that combine trees, crops and animals 7,700 ha of forests in priority areas the TR watershed reforested

	areas by 20% and 26% under the 20-year and 100-year flood scenarios, respectively. The implementation of agroforestry systems has a particularly high adaptation impact potential because of the provision of ecosystem goods and services (e.g. improved quality and quantity of water, more productive and stable soils, and crop pollination) combined with the creation of alternative livelihood options to unsustainable land uses, such as unregulated charcoal production. All agroforestry activities will also involve extensive engagements with farmer and women's associations, as well as relevant community representatives and groups, to encourage a progressive and gender-responsive transition towards sustainable agroforestry businesses that are sustainable over the long term.	1.2.2. Restore 7,700 ha of forests in priority areas the TR watershed through planting indigenous tree species in degraded areas within the seven target communes. 1.2.3. Hold 8 engagement workshops with community leaders, farmer and women's associations to facilitate the implementation of Sub-activities 1.1.2 and 1.1.3.	8 engagement workshops with community leaders, farmer and women's associations
Activity 2.1. Strengthen institutional capacity at the commune, inter-commune and department levels in Haiti's Trois-Rivières watershed for productive and sustainable land-use management	Under this activity, a capacity needs assessment will be conducted for the agricultural sector to identify the technical and institutional requirements for transitioning the sector from unsustainable land-use practices towards a sustainable landscape management (SLM) approach to land and water resources management that is climate resilient. Capacity will be assessed in six of the seven target communes in the TR watershed. From the capacity needs assessments, opportunities will be identified, and plans made to enhance the management of natural resources and promote the uptake of climate-resilient agricultural practices to increase productivity. To facilitate the uptake of these new sustainable practices, governance mechanisms and frameworks	2.1.1 Conduct capacity gap/needs assessments for the agricultural sector in seven priority communes in the TR region. 2.1.2 Establish communal development councils and multi-sectoral management structures to oversee and manage the implementation of interventions designed to address priority areas identified in Sub-activity 2.1.1 2.1.3 Train representatives from appropriate government entities, including MARNDR, NCFS and MSAL, on the implementation of climate-resilient agricultural techniques and SLM.	- Capacity needs assessment - Communal development councils established - Trainings for government entities, including MARNDR, NCFS and MSAL, on SLM in the TR watershed

	for SLM will be developed in the TR watershed through the establishment of communal development councils. These councils will represent all agricultural sector stakeholders, including smallholder farmers, in the seven communes. Additionally, multi-sectoral governance structures will be established to oversee and manage the implementation of interventions designed to address priority areas identified under the capacity needs assessment. Support for the governance structures established will be provided through strengthening the capacity of relevant government entities, including the Ministry of Agriculture, Natural Resources and Rural Development (MARNDR), National Coordination of Food Security (NCFS) and Ministry of Social Affairs and Labour (MSAL), to implement SLM at the department level. This will enable the adoption of SLM practices to be guided in a manner that is compatible with the effective functioning of the entire watershed and that contributes to enhancing flood management across the TR watershed.		
Activity 2.2. Enhance the technical capacity and access to finance of national and local representatives involved in agriculture in the Trois-Rivières watershed for adopting climate-resilient sustainable land-use practices	This project activity will increase the technical capacity of farmers to implement climate-resilient agricultural techniques and land-use practices, using the findings from the capacity needs assessments being conducted under Activity 2.1.1. Specifically, this will be achieved through implementing targeted adaptation interventions in Haiti's TR watershed to optimise the agricultural value chain and promote the use of climate-resilient agricultural techniques across the sector. Such interventions will include enhancing the transportation of agricultural	2.2.1. Strengthen the technical capacity of farmers and communities in the seven target communes to implement climate-resilient agricultural practices and SLM. 2.2.2. Provide climate-resilient agricultural inputs (e.g. climate-resilient seeds and infrastructure upgrades) to households in the TR watershed.	• Trainings for farmers and communities in the seven target communes • Strengthened agricultural finance mechanism • Climate-resilient agricultural inputs/inputs and/or infrastructure provided to smallholder farmers

	produce (e.g. by developing agricultural tracks), as well as improving water supply systems (e.g. irrigation and water pumps) to address decreasing water security caused by the impacts of climate change-induced flooding. Infrastructural upgrades will also be undertaken in the form of increasing the storage capacity and quality of post-harvest facilities, as well as establishing local processing facilities for agricultural produce to optimise agricultural production in the watershed. These interventions will be supported by the provision of climate-resilient agricultural inputs (e.g. climate-resilient seed varieties) to vulnerable households that depend on subsistence agriculture for their livelihoods. This will be supplemented by delivering technical skills training on climate-resilient agricultural techniques and SLM to farmers. Specifically, this will involve: i) enhancing their ability to effectively manage and organise irrigation and other associations; ii) facilitating participatory access to agricultural inputs, farming equipment, maintenance facilities; and iii) upskilling of maintenance staff. Strengthening these key elements of the agricultural value chain in Haiti's TR watershed will not only enhance the climate resilience of agricultural production in the country, but also contribute to increased food security in the watershed by improving market access for farmers and those people who depend on agriculture for their livelihoods. This increased market access will be supported by the establishment of an agricultural finance mechanism under the Agricultural	2.2.3. Enhance access to finance for smallholder farmers through the VSCA.	
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	Village Savings and Credit Associations (VSCA). Financing will be made available to smallholder farmers under this mechanism to facilitate their adoption and implementation of sustainable, climate-resilient agricultural practices. These farmers will be able to access financing from this mechanism during and immediately after the impacts of a climate change-related event, such as flooding. This will promote the adoption of more sustainable land-use practices that are resilient to climate change.		
Activity 2.3. Implement a social protection system to support vulnerable households at risk to food insecurity because of climate change	While Activity 3.1.2 provides a long-term solution to addressing these negative impacts, this activity will provide the required immediate relief to vulnerable people affected by these events, concomitantly reducing further pressure on the environment by providing an alternative option to resorting to unsustainable charcoal production. Specifically, Activity 2.1.3 will facilitate the development of a social safety net whereby food coupons will be provided to vulnerable households affected by food insecurity. To determine eligible households for these coupons, vulnerable households will be identified through a site survey and/or assessment, with their details captured in a database for future reference. Recipients of these coupons will be able to redeem the coupons for local produce at selected primary and secondary markets. These coupons will also serve as a form of currency (or e- coupon), which will provide recipients with purchasing power to exchange the coupons for specific goods. A total of 934 farmers will be selected to participate in the operation of the food	2.3.1. Conduct a site survey/assessment to determine eligible households to receive food coupons to be developed under Sub-activity 2.3.2 2.3.2. Establish and operationalise a food coupons mechanism in the TR watershed. 2.3.3. Engage and train 934 farmers on the implementation of the food coupons mechanism established under Sub-activity 2.3.2. 2.3.4. Create 150 new VSCAs in the TR watershed to enhance the food security of vulnerable households. 2.3.5. Establish a network of village agents to monitor existing VSCAs and establish new ones, as well as to monitor market performance contributing to food coupons mechanisms.	- Operational food coupons mechanism in the TR watershed. - Site survey/assessment - Training for 934 farmers on the implementation of the food coupons mechanism 150 new VSCAs created in the TR watershed - A network of village agents established for the monitoring and establishment of VSCAs

	coupons mechanism. Participating farmers will receive training on <i>inter alia</i> hygiene, modalities of exchange and personal security. To support the efficient and effective implementation of the food coupons mechanism, existing VSCAs will receive operational improvements under this activity, supplemented by the creation of 150 new VSCAs in the TR watershed. The VSCAs will provide additional financial support to the coupons mechanism, which will secure its effectiveness as a food relief mechanism. To ensure that this system remains operational throughout the lifetime of the project, a network of village agents will be established, with their primary responsibilities being the monitoring of existing VSCAs and creation of new ones.		
Activity 3.1. Strengthen national capacities for the implementation of the Water Act	Under Activity 3.1, provisions will be drafted to the Water Act to ensure that it promotes an integrated approach to water resources management, including considerations for the coordinated development of water, land and related resources. This will contribute towards: i) maximising economic and social welfare in an equitable manner without compromising the sustainability of vital ecosystems under future climate change conditions; and ii) facilitating efficient, integrated adaptation responses to the impacts of climate change in Haiti. Provisions to the Water Act will also recommend the establishment of Catchment IWRM Committees (<i>Committees de GIRE des Bassins Versants, CGBVs</i>) within each of the 31 catchments in the country, and the establishment of Sub-	3.1.1. Draft provisions to Haiti's Water Act for the adoption of an integrated approach to water resources management and implement the act in the TR watershed. 3.1.2. Establish Catchment IWRM Committees (<i>Committees de GIRE des Bassins Versants, CGBVs</i>) within each of the 31 catchments in Haiti, as per the provisions drafted in Sub-activity 3.1.1. 3.1.3. Establish Sub-Catchment Water Resource Users Associations (<i>Associations d'Usagers des Ressources en Eau du Sous-bassin versant; AssURES</i>) within the sub-catchments of the 31 catchments in	• Provisions drafted to the Water Act • 31 Catchment IWRM Committees established

	Catchment Water Resource Users Associations (Associations d'Usagers des Ressources en Eau du Sous-bassin versant, or AssURES).	Haiti, as per the provisions drafted in Sub-activity 3.1.1	
Activity 3.2. Develop an integrated, climate-resilient water management governance framework targeting the catchment and sub-catchment levels in the Trois-Rivières watershed	Under this activity, an inclusive and equitable water management governance framework will be developed for the TR catchment, namely the Catchment Integrated Water Resources Management Committee (CIWRMC). This committee will work with representatives from productive sectors (e.g. agriculture), land users at sub-catchment level and Sub-Catchment Water Resources User Associations (SCWRUAs) to achieve two main goals. First, these groups will work together to develop climate-resilient IWRM plans for end users at the sub-catchment and catchment levels that promote sustainable water resources management and an equitable distribution of the available resources among all users. The IWRM plans will define roles and responsibilities of institutions and civil society stakeholders for water resources management (complementary to the decentralised management of water resources to be established in the Water Act under Activity 3.1). It will also promote a landscape management approach that will be linked to the implementation of community land-use plans (Activity 1.1), considering gender dynamics. Second, the CIWRMC, together with the relevant land users and SCWRUAs, will ensure that the management of water resources in the catchment considers climate change impacts, particularly the expected impacts at landscape level as well as on upstream and downstream flow dynamics. Training	3.2.1. Establish the Catchment Integrated Water Resources Management Committee (CIWRMC) to manage water resources in the TR watershed. 3.2.2. Develop climate-resilient IWRM plans for end users at the sub-catchment and catchment levels in a participatory, gender-responsive way with representatives from CIWRMC and productive sectors (e.g. agriculture), land users and Sub-Catchment Water Resources User Associations (SCWRUAs). 3.2.3. Promote a climate-resilient, integrated landscape management approach in the TR watershed that considers gender dynamics, using the community land-use plans (Activity 1.1). 3.2.4. Train the CIWRMC (established under Sub-activity 3.2.1) and representatives from productive sectors (e.g. agriculture), land users and Sub-Catchment Water Resources User Associations (SCWRUAs) on the application of climate-resilient IWRM plans in decision making.	• Catchment Integrated Water Resources Management Committee (CIWRMC) established • Climate-resilient IWRM plans • Promotion of climate-resilient, integrated landscape management approach in the TR watershed • Trainings delivered to the CIWRMC (established under Sub-activity 3.2.1) and representatives from productive sectors (e.g. agriculture), land users and Sub-Catchment Water Resources User Associations (SCWRUAs)

	on the climate-resilient IWRM plans will also be delivered focusing on the application of these plans in decision-making. The combined result of these actions will be the establishment and operation of an integrated governance framework that will contribute to preserving climate-resilient, sustainable land and water resources management beyond the lifetime of the project. During the implementation of Activity 3.2, considered efforts will be made to securing equitable representation, participation and leadership of women.		
Activity 3.3. Implement regular monitoring and evaluation of water resources at the catchment and sub-catchment levels to support the implementation of integrated water resources management (IWRM) plans	From project inception (Year 1), continuous health monitoring and impact evaluations will be undertaken and will continue throughout the implementation phase. By monitoring the evolution, frequency and location of flood-related illnesses from the start of the project, the impact of the project will be assessed once a sufficient time series of observations is completed. UNDP will commission an expert to undertake the health monitoring and impact evaluations of the EbA flood management solutions. The World Health Organisation (WHO) will provide technical support to this expert, as they are currently monitoring cases of water- and vector-borne diseases in Haiti and therefore have a pre-defined methodology that would allow comparison studies to be undertaken across watersheds. The contracted expert will undertake a baseline assessment of the prevalence of water- and vector-borne diseases within the TR watershed and analyse the impact of climate change-induced flooding on the spread of these	3.3.1. Commission an expert to undertake the health monitoring and impact evaluations of the EbA flood management solutions (Activity 1.2), supported by WHO. 3.3.2. Undertake a baseline assessment of the prevalence of water- and vector-borne diseases — as well as monitor dry season baseflow — within the TR watershed, assisted by local representatives from SCWRUAs. 3.3.3. Assess the effectiveness of EbA interventions (Activity 1.2) in reducing the spread of water- and vector-borne diseases. 3.3.4. Generate lessons through the implementation of all project interventions and disseminate these lessons through the committees and councils established under Output 2	• Expert contracted to undertake the health monitoring and impact evaluations of the EbA flood management solutions • Baseline assessment of the prevalence of water- and vector-borne diseases within the TR watershed • Assessment of the effectiveness of EbA interventions (Activity 1.2) in reducing the spread of water- and vector-borne diseases. • Workshops for the generation and dissemination of lessons learned

	diseases. Following this assessment, the expert will report on her/his findings and provide the required health impact data to assess the health benefits of the EbA interventions implemented under the project. Simultaneously, this expert, assisted by representatives from SCWRUAs, will monitor dry season baseflow at four locations along the main stem of the Trois-Rivières (TR) river. This will be done to provide a stable indicator of the increased infiltration capacity of the soil because of forest rehabilitation and restoration activities to be implemented under Activity 1.2 of the project. The monitoring will facilitate an evidence-based assessment of these EbA solutions as effective measures for reducing the impacts of climate change-induced flooding in the TR watershed and for improving the health of the watershed's population. Finally, lessons generated through the implementation of all project interventions will be shared through the relevant committees and associations at the catchment and sub-catchment levels to inform adaptative management of these interventions under future climate change conditions.		
E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)			
200. Project-level monitoring and evaluation will be undertaken in compliance with the UNDP POPP and the UNDP Evaluation Policy. The Project Manager and Monitoring and Gender Officer, under the oversight of UNDP as the AE, will be responsible for monitoring the progress of all project activities against the outcome-based indicators presented in the Logical Framework (Section E of this Funding Proposal). Any delays or challenges experienced during the implementation phase will be communicated timeously to the Project Board and the UNDP Country Office by the Project Manager, so that any appropriate support and corrective measures required can be actioned efficiently. The Project Manager will also ensure that all project staff maintain transparency, responsibility and accountability in monitoring and reporting on project results. 201. Monitoring and evaluation of project activities will be undertaken on an annual basis. The type of monitoring activities to be conducted during the implementation phase, as well as the responsible parties for undertaking these activities, are detailed in Annex 11: Monitoring and Evaluation (M&E) Plan. Such			

activities include field observations, key informant interviews and focus group sessions. The costs associated with undertaking the necessary monitoring, assessments and engagements required to inform the project evaluations are presented in Annex 4: Detailed budget.

202. A project inception workshop will be held once the proposed project has been approved to: i) re-orient project stakeholders to the project strategy and discuss any changes in the overall context that may influence project implementation; ii) discuss the roles and responsibilities of the project team, including reporting and communication lines and conflict resolution mechanisms; iii) review the project Logical Framework; iv) discuss reporting, monitoring and evaluation roles and responsibilities and finalise the M&E plan (Annex 11); v) review the financial reporting procedures and mandatory requirements, and agree on the arrangements for the annual audit; and vi) plan and schedule Project Board meetings and finalise the first year annual work plan for the project. The final inception report will be shared with the UNDP Country Office, UNDP Regional Technical Advisor and Project Board for clearance. Once the report has been cleared by the relevant parties, it will be submitted to the GCF for approval. This will happen within six months after the Funded Activity Agreement (FAA) becomes effective.
203. An annual progress report (APR) will be prepared by the Project Manager, UNDP Country Office, and UNDP Regional Technical Advisor for submission to the GCF, Project Board and relevant project stakeholders. The Project Manager, supported by the Monitoring and Gender Officer, will ensure that the indicators included in the project Logical Framework are monitored annually and that the results from regular M&E activities are presented in the GCF APR. Any challenges and delays experienced during the implementation phase will be monitored by the Project Manager and reported in the GCF APR.
204. During the implementation phase, specific means of verification will be used to assess the: i) progress of project interventions towards achieving GCF Fund-level impacts and Fund-level outcomes; and ii) performance of interventions against the project performance indicators presented in Section E.5 of this Funding Proposal. A full-time Monitoring and Gender Officer will be employed to conduct and coordinate on-the-ground monitoring and evaluation (M&E) of the project and ensure that gender targets are met. The Monitoring and Gender Officer will design and operationalise a performance monitoring framework to track the project's progress towards achieving its targets. This will be achieved by: i) measuring the performance against the project indicators (Sections E.3–5) to evaluate the progress of the project; ii) reporting the project's performance to the National Steering Committee (NSC) and Project Management Unit (PMU); and iii) providing technical support to the Project Manager. Additionally, the Monitoring and Gender Officer will be responsible for overseeing and monitoring the application of gender-disaggregated indicators¹⁴⁰. At particular milestones (e.g. baseline, mid-term and project closure), the Monitoring and Gender Officer will undertake evidence-gathering exercises to verify the project's performance against the outcomes-based indicators presented in the Logical Framework, M&E Plan and Gender Action Plan. The results from this process will be presented in periodic reports developed by the Monitoring and Gender Officer, supported by the Project Manager. These reports will be submitted to the NSC and validated during the Independent Interim and Final Evaluations for the project, as described below.
205. UNDP will be responsible for managing the Independent Interim (mid-term) and Final Evaluations of the project, including the process for contracting these positions. For the Interim Evaluation, an independent consultant will be contracted to provide an assessment of project performance against its targets at the project's mid-term (end of Year 3). This will be an outcomes-based assessment and will include analysing whether the project is on track, what problems and challenges the project is encountering, and what corrective actions are required, so that the project can achieve its intended outcomes by project completion in the most efficient and sustainable way. Following this evaluation, the NSC will develop a management response to the evaluation recommendations along with an implementation plan for undertaking the required actions. It is the responsibility of UNDP to assess whether the agreed recommendations are being

¹⁴⁰ The project's gender targets as well as the associated gender-disaggregated indicators are presented in Annex 8: Gender Assessment and Action Plan.

implemented during the remainder of the project period. The completed Interim Evaluation report will be cleared by the UNDP Country Office as well as the UNDP Regional Technical Advisor and submitted to the Project Board for approval.

206. An independent consultant will be contracted by UNDP to conduct the Final Evaluation Report at the end of project implementation. UNDP will be responsible for overseeing the Final Evaluation Report, which is a summative evaluation. This evaluation will include an independent assessment of the project's overall performance against standard evaluation criteria (e.g. strategic relevance, effectiveness, efficiency, likelihood of impact and sustainability), as well as against the project indicators presented in the Logical Framework. This evaluation will be based on inter alia documentaries of evidence, field observation visits and key informant interviews. Following completion of the evaluation, the Final Evaluation report will be developed and shared with the UNDP Country Office as well as the UNDP Regional Technical Advisor for clearance. Once the report has been cleared, it will be submitted to the Project Board for approval.

207. The UNDP Country Office will include the planned project evaluations in the UNDP Country Office evaluation plan. All evaluation reports, including responses to feedback received from the public, will be uploaded in English to the UNDP Evaluation Resource Centre (ERC)¹⁴¹. The UNDP Country Office will retain all M&E records for this project for up to eight years after project financial closure to support ex-post evaluations.

¹⁴¹ available at: www.erc.undp.org.

RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures. Insert additional rows if necessary.

For probability: High has significant probability, Medium has moderate probability, Low has negligible probability

For impact: High has significant impact, Medium has moderate impact, Low has negligible impact

Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistleblowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices

Selected Risk Factor 1

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>High</u>

Description

Low survival rates of crops and trees planted during the project makes stabilisation of slopes ineffective, hampering improvement in soil conservation measures.

Mitigation Measure(s)

Risk mitigation measures will include:

- Specifically selecting and using robust (ideally indigenous) species that are well adapted to local conditions, with high levels of fire resistance and good re-sprouting capacity;
- Providing technical assistance to households implementing agroforestry;
- Involving the community in planning at the inception stage to foster ownership and promote long-term monitoring and protection of planted sites;
- Practicing participatory planning for reforestation activities; and
- mobilising civil society to raise awareness of the importance of monitoring and protection of ground cover for ecosystem service provision.

Selected Risk Factor 2

Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>High</u>

Description

Weak land-tenure, -use and -management systems prevent or delay implementation if agreements with landowners or communities cannot be reached.

Mitigation Measure(s)

Ownership of project interventions will be supported by the development of Community Land Use Plans (under Activity 1.1), including assessing land-tenure and prioritising areas for interventions based on community recommendations and agreements.

Since modifying the legislative framework for non-climate related aspects of land management is not within the scope of the project, other risk mitigation measures include:

- ensuring the involvement and participation of target communities in the TR watershed at the inception and planning stage to establish trust and support in the validation of land ownership;
- using the consultation process during the assessment phase as a platform to address concerns and increase buy-in of communities;
- encouraging community involvement in IWRM at the sub-catchment level, helping to work towards more sustainable agricultural and sub-catchment management practices; and
- capitalising upon top-down (e.g. establishment of the Water Act) and bottom-up (e.g. establishment and capacitating of sub-catchment Water Resource Users' Associations (WRUA)) approaches to land and water resources management to ensure that local needs and experiences are considered during project implementation.

Selected Risk Factor 3		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>High</u>
Description		
Limited uptake of project interventions by target communities because of a preference towards fulfilling the immediate need for sustenance through conventional agricultural techniques over long-term climate action.		
Mitigation Measure(s)		
Risk mitigation measures will include: • building community interest by engaging Haitian civil society organisations (CSOs) and working with community governance systems and associations to raise awareness of the challenges and impacts of climate change on environmental degradation and food security; • training and knowledge sharing initiatives to facilitate capacity building and ownership; and • emphasising the introduction of more sustainable farming systems (i.e. agroforestry), sustainable (sub-) catchment management and IWRM, as well as food security and nutrition interventions, which are designed to be inclusive. • Working with community members in monitoring the project's indicators to establish a link between project action and benefits.		
Selected Risk Factor 4		
Category	Probability	Impact
<u>Governance</u>	<u>Low</u>	<u>Medium</u>
Description		
Inadequate national focus on adaptation measures because of other development pressures affecting the sustainability of the proposed project beyond GCF involvement.		
Mitigation Measure(s)		
Risk mitigation will consider: • building resilience into watershed management is increasingly becoming a national priority and is expected to develop into a sustained focus on adaptation in Haiti; and • a participatory and inclusive processes to help institutionalise climate change adaptation and ensure the sustainability of the integrated management approach. • Embedding the project within national policy efforts including water management and the National Adaptation Plan process		
Selected Risk Factor 5		
Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>High</u>
Description		
Continued overexploitation and unregulated land-use and -tenure renders soil conservation measures redundant, causing increasing degradation and permanent losses in soil fertility.		
Mitigation Measure(s)		
• Through an integrated flood management planning approach, the project is intended to address environmental degradation partially and indirectly by framing regulations pertaining to land-use in areas directly impacted by climate change. • Additionally, improved institutional capacity will also contribute to reducing degradative practices through better planning, policy decision-making and natural resource-management capacities. • The risk will further be mitigated by shifting practices and garnering buy-in of the local community through consultations to promote active involvement in sub-catchment IWRM, participation in crop/tree planting activities and building awareness of climate change impacts by mobilising the CSOs.		
Selected Risk Factor 6		

Category	Probability	Impact
<u>Governance</u>	<u>Low</u>	<u>Low</u>
Description		
Weak coordination between stakeholders, climate initiatives and implementation partners reduce the effectiveness of IWRM approaches in Haiti.		
Mitigation Measure(s)		
The risk will be mitigated by promoting coordination through the strategic identification of established multi-stakeholder groups, including the WRUAs at sub-catchment level and the CIWRMC at catchment level. Coordination between these groups will be strengthened through workshops and knowledge-sharing initiatives aimed at: i) building awareness of activities being undertaken; ii) developing best practices to draw from; iii) synchronising efforts; and iv) replicating successes from each project. Risk will further be reduced by involving multiple actors in supporting catchment IWRM, including NGOs, private development organisations and community-based organisations. The participation of these entities will complement the relevant stakeholders within MoE and other relevant sector ministries, contributing towards coordination and integration of interests, needs and perspectives. The project will report regularly to the Technical Secretariat of the National Committee on Climate Change to ensure alignment with the GoH own Climate Strategies including the country's National Adaptation Plan. The project will also shares result with development partners and agencies through the the various donor groups for the environmental sector and for disaster risk reduction.		
Selected Risk Factor 7		
Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>High</u>
Description		
Extreme weather event and/or natural disaster during project implementation and/or operations inhibits or delays progress.		
Mitigation Measure(s)		
Mitigation measures include: • land-use plans taking these risks into account; • accelerating project execution during non-peak wet weather periods; and • choosing tree varieties and timing of planting to favour fast-growing species during the first planting cycle to shield crops and newly planted seedlings from extreme weather events.		
Selected Risk Factor 8		
Category	Probability	Impact
<u>Governance</u>	<u>Medium</u>	<u>High</u>
Description		
Socio-political instability undermines envisaged project outputs because of disruptions and changes in national priorities and government operations.		
Mitigation Measure(s)		
This risk will be mitigated by ensuring a continuity of process through: • Ensuring outreach to new officials/political administration at both local and national levels based on technical results to allow for an easy transition. • mainstreaming integrated flood governance into the institutional framework at multiple levels and amongst diverse jurisdictions and mandates; • defining roles and responsibilities and developing clear standard operating procedures (SOPs) for implementing stakeholders, this includes working through NGOs, and Responsible Parties such as Heifer with local presence and with clear knowledge on how to address instability at a territorial level when working with communities;		

- undertaking capacity building initiatives at various levels looking to empower local government and mid level technical officials that are less vulnerable to political disruptions. These actions allow technical work to continue despite social and political upheavals;
- focusing on ensuring buy-in and participation of vulnerable communities by mobilising local CSOs to raise awareness on climate change, climate impacts and adaptation strategies and creating local trainers. Working at community level as well as mobilizing community associations has proven effective to ensure transformative action on the ground despite economic and social disruptions.
- Enhancing community and local governance structures around common objectives that result in clear value adds and hence improve buy in from local actors for the development of project outputs.
- Enabling local structures to enhance local capacities and resilience.
- Ensuring outreach to new officials/political administration at both local and national levels based on technical results to allow for an easy transition.

Selected Risk Factor 9

Category	Probability	Impact
Other	Medium	Medium

Description

Insecurity within the country may seep into project sites particularly in Bassin Bleu and Gros Morne and cause delays in the implementation of project activities.

Mitigation Measure(s)

An initial conflict analysis has been prepared by the project and included within the project's ESMF. The analysis and the project's ESFM provided guidelines for the development of Security Plans and Conflict Assessments that will be developed as part of the activity level EISAs/ESMFs during the project's first year and will result in activity specific management plans per grouped areas. To reduce the potential of project beneficiaries or staff being targeted by illegal actors, the project has avoided the provision of direct cash benefits focusing more on community based benefits (capacity building, agroforestry inputs, enhanced associative capacity, etc.). In terms of mobility, UNDSS rules will be used and strictly adhered to by the project particularly during internal travel and security issues will be continuously monitored to allow for adaptive management in case it is required. Procurement strategies will favor local purchases to avoid any potential danger in the transport of project equipment and its delivery to project sites. In addition, a logistics and operations officer has been foreseen by the project to ensure that logistics take into account insecurity risks when planning the delivery of project activities and on the ground missions. The project's community based approach will allow for the project to ensure strong community buy in and enhance social capital, this approach will help develop community informed practices in the implementation of project activities to avoid potential security risks while creating an opportunity to maximise the peacebuilding potential of the project.

GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

208. This project has completed the UNDP social and environmental screening procedure undertaken to ensure it complies with UNDP's Social and Environmental Standards. The social and environmental risk category for this project is: Moderate (Category B). A detailed risk assessment undertaken for the project Social and Environmental Risk Procedure (SESP), as well as the Environmental and Social Management Framework (ESMF), are presented in Annex 6.
209. UNDP will be responsible for the ESMF, ensuring it is adequate and followed. The PMU will be responsible for the revision or updates of the ESMF document during project implementation and will ensure timely corrective actions are taken, as identified. Any non-conformances to the ESMF are to be noted in environmental inspections and logged into the register. Depending on the severity of the non-conformance, the Site Supervisor may specify a corrective action on the site inspection report. The ESMF and its procedures are to be reviewed at least every year by UNDP staff and the MoE. The objective of the review is to update the document to reflect knowledge gained during project delivery/construction and to reflect new knowledge and changed community standards (values).
210. The progress of all corrective actions will be tracked using the register. Any non-conformances and the issue of corrective actions are to be advised to UNDP. UNDP and the MoE will also release updates on the project on a regular basis to provide interested stakeholders with information on project status. Updates may be via a range of media, for example print, radio, social media or formal reports, and all material must be published in English and Haitian Creole (Kreyòl) as appropriate. A publicised telephone number will be maintained throughout the project to serve as a point of contact for enquiries, concerns, complaints and/or grievances. All enquiries, concerns, complaints and/or grievances will be recorded on a register and the appropriate manager will be informed.
211. The project has the potential to cause low to moderate environmental and social impacts. The Project's design has avoided all high-risk environmental and social impacts and favoured a wholly climate resilient infrastructure approach to flood management, eschewing high-risk grey infrastructure for flood control. Furthermore, the project does not require any land acquisition and/or resettlement, does not promote the use of pesticides and fertilisers, prohibits the use of alien invasive species in planting, and promotes diverse agroforestry systems and reforestation that will not only build climate resilience and food security, and decrease the impact of floods, but also provide significant environmental and social co-benefits.
212. When undertaking the risk assessment, all activities were assessed, with an emphasis on those with the potential for some adverse risk, specifically Output 1, Activity 1.2 which involves the project's physical interventions (agroforestry and reforestation) financed by GCF. Potential risks identified include: i) the possibility that activities reproduce discriminations against women based on gender, due to existing structural inequalities and roles in agriculture and decision-making; ii) the possibility that activities lead to the unintentional increased use of pesticides and fertilisers in agroforestry systems because of improved income; and iii) a low risk that there is a possibility of child labour use in the promoted agroforestry systems. Specific measures for each environmental and social risk are discussed along with mitigation measures in the ESMF.
213. The preparation of the ESMF included public consultation as part of the Stakeholder Engagement Plan. The full Stakeholder Engagement Plan is appended as part of the proposal in Annex 7: Stakeholder Engagement Plan, which includes the full list of stakeholders consulted, and a summary of the consultations. The project was discussed with a wide range of stakeholders including relevant government departments, industry groups, NGOs, and individual community members (including marginalised groups) and was approved by the Government of Haiti. These extensive on-the-ground consultations undertaken during the design of the project (as well as during the earlier interventions that this project complements and up-scales) closely informed overall project design and the ESMF, and it is expected that consultations with affected communities and beneficiaries will continue. It is anticipated that based on the communities' needs, the projects will be fully accepted.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

214. A Gender Assessment and Action Plan (GAAP) was developed for this project in order to provide an overview of the issue of gender in Haiti, with a specific focus on the gender differentiated impacts of climate change and how this information can be used to design a gender-sensitive adaptation intervention. The GAAP identifies gender issues relevant to the project interventions and situated in the Haitian context, across elements such as use and access to resources, gender-differentiated roles in employment and livelihoods, and roles in political decision

making. The Gender Action Plan specifically describes opportunities for gender mainstreaming in the project design so that the proposed project interventions address gendered climate resilience across the household, community and institutional levels.

215. Haiti ranks third globally on the Long-Term Climate Risk Index¹⁴², demonstrating its extreme vulnerability to climate change — to which women are especially vulnerable. Widespread deforestation and soil degradation have so severely impacted the island's environment that it can no longer provide critical ecosystem functions, such as mitigating the effects of heavy rain and flooding. This has progressively increased human vulnerability to climate change. Extreme climate events, such as tropical storms and hurricanes, are expected to increase in frequency and intensity, while rainfall is expected to become more erratic across seasons¹⁴³. In an already denuded and degraded landscape, this increased rainfall engenders severe erosion and landslides, flash flooding, and sedimentation of river channels¹⁴⁴.
216. Climate change impacts are not gender-neutral and many of the consequences of climate-induced impacts are more severe for women, and other socio-economically marginalised groups, given their specific livelihood circumstances. In relative terms, women lack access to productive resources, and this has impacts on their health, food security and safety in the context of flooding and other extreme climate events. The additional socio-cultural barriers limiting participation and movement outside the household sphere is worsened by phenomena such as flooding, drought and erratic rainfall. These constraints cause women to have to work much harder to secure food and water, and generate additional income through livelihoods which, in turn, diminishes their ability to advance out of poverty, particularly when they have, for example, lost their land due to the impacts of flooding during cyclone events.
217. Gender equality and women's empowerment are not addressed in most of Haiti's strategy or action plan documents for climate change and disaster risk management^{145, 146}. Flooding has cumulative livelihood effects that unfairly impact women and exacerbate gender inequality. In Haiti, women are disproportionately represented among the poor and there is a significant portion of women-headed households, so impacts such as crop loss, animal drowning, destroyed homes, health risks or displacement and loss of life resulting from floods ultimately affect women to a higher extent. Yet, during these extreme flooding events, the importance of grassroots organisation and mobilisation of women's groups during disaster relief efforts has been unparalleled and critical, demonstrating the resilience of Haitian women in the face of extreme adversity.
218. Over 40% of households in Haiti are headed by women, and this percentage has been on the rise since 2007^{147, 148}. Today, there are more women living under extreme poverty than men, with a large proportion residing in rural areas¹⁴⁹. Most of Haiti's population is rural, and two-thirds of this population is dependent on agriculture; with women comprising a significant portion of the labour force involved across all aspects of agricultural processes¹⁵⁰.
219. To address gender dimensions within the project design and implementation, the Gender Action Plan will account for:
- specific strategies to include/target female-headed households;
 - differing conservation incentives faced by women;
 - identification of gaps in gender equality through the use of sex-disaggregated data enabling the gender action plan to close those gaps, devoting resources and expertise for implementing such strategies, monitoring the results of implementation, and holding individuals and institutions accountable for outputs that promote gender equality;
 - advocacy and awareness that is adjusted to most effectively reflect gender-specific differences; and
 - the inclusion of a Gender Specialist position/provision of advice within the project to implement gender-related activities.

¹⁴² German Watch, Global Climate Risk Index 2020. https://germanwatch.org/sites/germanwatch.org/files/20-2-01e%20Global%20Climate%20Risk%20Index%202020_14.pdf.

¹⁴³ Ibid.

¹⁴⁴ IUNC 2011 gender action plan.

¹⁴⁵ USAID (2016).

¹⁴⁶ Shadow Report (2011).

¹⁴⁷ IHSI, enquête, emploi, économie informelle, 2010.

¹⁴⁸ MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir.

¹⁴⁹ MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir.

¹⁵⁰ USAID (2016).

220. Gender benefits through this project have been optimised by promoting the full involvement of women in decision-making regarding integrated flood management at the community and institutional levels, as well as promoting women as the stewards of agroforestry systems, playing key roles in the selection of species and transformation of products. In addition, as during the design of the project, strong efforts will continue to be made during implementation to ensure the equitable participation of women. Additionally, going forward beyond the implementation phase, equal participation of women will be pursued and achieved in ongoing management and water governance especially on the Water Governance Committees.
221. During project implementation, qualitative assessments will be conducted on the gender-specific benefits that can be directly attributed to the project. This will be incorporated in the annual Project Implementation Report, Mid-Term Report, and Terminal Evaluation. Indicators to quantify the achievement of project objectives in relation to gender equality will include men and women who had access to affordable solutions, number of men and women employed from the jobs created by the project, training opportunities, knowledge management and information dissemination.
222. The stakeholder consultations and engagement of women's organisations promote gender equality at the local as well as the national level. The involvement of women's organisations in the project design aided in identifying relevant gender issues within the country's social context, as well as in implementing and monitoring the gender aspects of the project.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

223. The financial management and procurement of this project will be guided by UNDP financial rules and regulations available [here](#)¹⁵¹. Further guidance is outlined in the financial resources management section of the UNDP Program and Operations Policies and Procedures (POPP) available [here](#). UNDP has comprehensive procurement policies in place as outlined in the 'Contracts and Procurement' section of UNDP's POPP. See [here](#).
224. The project will be implemented following the full-support National Implementation Modality (NIM) following the UNDP's POPP guidelines available [here](#)¹⁵². UNDP will ascertain the national capacities of the implementing partner by undertaking an evaluation of capacity following the Framework for Cash Transfers to Implementing Partners (part of the Harmonized Approach to Cash Transfers - Harmonized Approach to Cash Transfer ([HACT](#))). All projects will be audited following the UNDP financial rules and regulations noted above and applicable audit guidelines and policies.
225. The Full Support to NIM guidelines are a formal part of UNDP's policies and procedures, as set out in the UNDP POPP. The NIM guidelines were corporately developed and adopted by UNDP and are fully compliant with UNDP's procurement and financial management rules and regulations.
226. The national executing entity MoE (also referred to as the national 'Implementing Partner' in UNDP terminology), is required to implement the project in compliance with UNDP rules and regulations, policies, and procedures (including the NIM guidelines). In legal terms, this is ensured through the national Government's signature of the UNDP Standard Basic Assistance Agreement (SBAA), together with a UNDP project document which will be signed by the Implementing Partner to govern the use of the funds. Both of these documents require compliance. During implementation, UNDP will provide oversight and quality assurance in accordance with its policies and procedures, and any specific requirements in the Accreditation Master Agreement (AMA) and project confirmation to be agreed with the GCF. This may include, but is not limited to, monitoring missions, spot checks, facilitation and participation in project board meetings, quarterly progress and annual implementation reviews, and audits at project level or at implementing partner level on the resources received from UNDP.

¹⁵¹ In line with UNDP National Implementation (NIM) Guidelines, the Government is responsible for the management and procurement of the Project to achieve project outputs. Government regulations, rules and procedures apply to project implementation to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. UNDP Financial Regulations section 16.05 state:

- "The administration by executing entities or, under the harmonized operational modalities, implementing partners, of resources obtained from or through UNDP shall be carried out under their respective financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP.

- Where the financial governance of an executing entity or, under the harmonized operational modalities, implementing partner, does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition that of UNDP shall apply."

The project will be audited in accordance with UNDP policies and procedures on audits, informed by and together with any specific requirements agreed in the AMA. According to the current audit policies, UNDP will be appointing the auditors. In UNDP, scheduled audits are performed during the project cycle as per UNDP assurance/audit plans, based on the implementing partner's risk rating and UNDP's guidelines. A scheduled audit is used to determine whether the funds transferred to the implementing partner were used for the appropriate purpose and in accordance with the work plan. A scheduled audit can consist of a financial audit and/or an internal control audit.

G.4. Disclosure of funding proposal

Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.

Indicate below whether or not the funding proposal includes confidential information.

☒ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☒ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☒ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☐ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)¹⁵³
 - ☐ Eligibility Criteria
 - ☐ Beneficiary Calculation Worksheet
 - ☐ Results Framework sheet

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

¹⁵³ Annex 22 is mandatory for mitigation and cross-cutting projects.

No-objection letter issued by the national designated authority(ies) or focal point(s)



MINISTÈRE DE L'ENVIRONNEMENT

AUTORITÉ NATIONALE DESIGNÉE POUR LE FONDS VERT CLIMAT

MDE/GCF/023

Port-au-Prince, le22 avril 2019.....

Mr. Yannick GLEMAREC
Executive Director
Green Climate Fund
Incheon City, Songdo/Korea

Re: *Funding proposal for the GCF by United Nations Development Program (UNDP) regarding the project "Enhanced climate resilience in the Trois-Rivières region of Haiti through Integrated Flood Management"*

Dear Sir,

We refer to the project "Enhanced climate resilience in the Trois-Rivières region of Haiti through Integrated Flood Management" in Haiti as included in the funding proposal submitted by United Nations Development Program (UNDP) to us on 27 March 2019.

The undersigned is the duly authorized representative of the Ministry of Environment, the National Designated Authority of Haiti.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Haiti has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with Haiti's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely,

Approved by :

Joseph JOUTHE
Minister of Environment




Moïse JEAN-PIERRE
GCF Focal Point

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Enhanced climate resilience in the Trois-Rivières region of Haiti through Integrated Flood Management
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	United Nations Development Programme (UNDP)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Trois Rivières Watershed, Republic of Haiti (within the communes of Marmelade, Plaisance, Pilate, Gros-Morne, Bassin Bleu, Chansolme, and Port-de-Paix.)
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Monday, June 5, 2023
Language(s) of disclosure	English and French
Explanation on language	French is an official language of Haiti.
Link to disclosure	English: https://www.undp.org/sites/g/files/zskgke326/files/2023-06/Annex%2006%20-%20Annex VI-A - FP-UNDP-0562023-5996-Haiti-ESMF%20highlight 4.pdf French: https://www.undp.org/sites/g/files/zskgke326/files/2023-06/Annex%2006%20-GCF PIMS5996-Haiti-ESMF June2023.pdf
Other link(s)	<u>The Ministry of Environment of the Government of Haiti</u> English: https://mde.gouv.ht/index.php/fr/nos-publications/documents-officiels# French: https://mde.gouv.ht/phocadownload/Annex%2006%20-GCF PIMS5996-Haiti-ESMF June2023.pdf <u>Heifer International</u> English: https://heiferinternationalhaiti.org/wp-content/uploads/2023/06/Annex-06-UNDP-5996-GCF-Haiti-ESMF English.pdf French: https://heiferinternationalhaiti.org/wp-content/uploads/2023/06/Annex-06-GCF PIMS5996-Haiti-ESMF June2023.pdf

Remarks	An ESIA consistent with the requirements for category B project is contained in the “Environmental and Social Management Framework (ESMF)”.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity’s website	Monday, June 5, 2023
Language(s) of disclosure	English and French
Explanation on language	French is an official language of Haiti.
Link to disclosure	English: https://www.undp.org/sites/g/files/zskgke326/files/2023-06/Annex%2006%20-%20Annex VI-A - FP-UNDP-0562023-5996-Haiti-ESMF%20highlight 4.pdf French: https://www.undp.org/sites/g/files/zskgke326/files/2023-06/Annex%2006%20-GCF PIMS5996-Haiti-ESMF June2023.pdf
Other link(s)	<u>The Ministry of Environment of the Government of Haiti</u> English: https://mde.gouv.ht/index.php/fr/nos-publications/documents-officiels# French: https://mde.gouv.ht/phocadownload/Annex%2006%20-GCF PIMS5996-Haiti-ESMF June2023.pdf <u>Heifer International</u> English: https://heiferinternationalhaiti.org/wp-content/uploads/2023/06/Annex-06-UNDP-5996-GCF-Haiti-ESMF English.pdf French: https://heiferinternationalhaiti.org/wp-content/uploads/2023/06/Annex-06-GCF PIMS5996-Haiti-ESMF June2023.pdf
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the “Environmental and Social Management Framework (ESMF)”.
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity’s website	Gender Analysis and Action Plan, Social and Environmental Safeguards Procedure (SESP)/ Monday, June 5, 2023

Language(s) of disclosure	English and French
Explanation on language	French is an official language of Haiti.
Link to disclosure	English: https://www.undp.org/sites/g/files/zskgke326/files/2023-06/Annex%2006A-UNDP-5996-%20GCF-Haiti-SESP_cleared%207june2023_0.pdf https://www.undp.org/sites/g/files/zskgke326/files/2023-06/Annex%2008-UNDP-5996-GCF%20Haiti-GAAP_cleared%207june2023.pdf French: https://www.undp.org/sites/g/files/zskgke326/files/2023-06/Annex%2006-A%20-%20Sauvegarde%20Environnementale%20et%20sociale%20Haiti_FP-UNDP5996_GCF%5B60141%5D.pdf https://www.undp.org/sites/g/files/zskgke326/files/2023-06/Annex%2008%20-Plan%20d%27action%20genre%20Haiti_FP-UNDP5996_GCF.pdf
Other link(s)	<u>The Ministry of Environment of the Government of Haiti</u> English: https://mde.gouv.ht/index.php/fr/nos-publications/documents-officiels# French: https://mde.gouv.ht/phocadownload/Annex%2006-A%20-%20Sauvegarde%20Environnementale%20et%20sociale%20Haiti_FP-UNDP5996_GCF60141.pdf https://mde.gouv.ht/phocadownload/Annex%2008%20-Plan%20daction%20genre%20Haiti_FP-UNDP5996_GCF60142.pdf <u>Heifer International</u> English: https://heiferinternationalhaiti.org/wp-content/uploads/2023/06/Annex-06A-UNDP-5996-GCF-Haiti-SESP_English.pdf https://heiferinternationalhaiti.org/wp-content/uploads/2023/06/Annex-08-UNDP-5996-GCF-Haiti-GAAP_English.pdf French: https://heiferinternationalhaiti.org/wp-content/uploads/2023/06/Annex-06-A-Sauvegarde-Environnementale-et-sociale-Haiti_FP-UNDP5996_GCF60141.pdf https://heiferinternationalhaiti.org/wp-content/uploads/2023/06/Annex-08-Plan-daction-genre-Haiti_FP-UNDP5996_GCF60142.pdf

Remarks	The project's SESP as well as its Gender Analysis and Action Plan is also disclosed by UNDP Haiti's website, the Ministry of Environment's website and Heifer international's website.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, June 9, 2023
Place	A physical copy of the documents has been made available as within the mayor's office of each commune • Marmelade, Mayors Office, 22 Rue Geuerrier, Marmelade • Plaisance Mayor's Office, Poudriere, Rue, Sylla, Ducheine, Plaisance Nord Haiti • Pilate Mayor's Office Complexe Administratif de Pilate, Route Frosiine • Gros-Morne, 34 Rue Simone Complex Administratid de Gros Morne • Bassin Bleu, Rue de la Independance 1er La Plate Palais Municipal de BAssin Bleu • Chansolme Mayor's Office, 5 Route Nationale, Chansolde • Port de Paix Mayor's Office Rule du Commisariat, Port de Paix Information meetings have also been planned to occur later in June in each of the Project target Communes (Marmelade, Plaisance, Pilate, Gros-Morne, Bassin Bleu, Chansolme, and Port-de-Paix) through Public Hearing Meetings.
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, July 10, 2023

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP208

Proposal name:	Enhanced climate resilience in the Trois-Rivières region of Haiti through Integrated Flood Management
Accredited entity:	United Nations Development Programme (UNDP)
Country/(ies):	Republic of Haiti
Project/programme size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
Direct benefits to 292,600 people (2.7 per cent of the total population), targeting the most vulnerable population, and 40 per cent reduction in peak run-off with the increase in tree cover (17,740 ha agroforestry and 7,700 ha restoration) and reduce floodplain extents 26 in 1 per cent AEP.	Ongoing crises including gang-related insecurity could pose a risk that needs to be carefully managed.
Shifting away from unsustainable land-use practices to an integrated and climate-resilient approach to land and water resources management, including community-led governance.	
Increasing household incomes, community health outcomes, improved water and soil quality and greater biodiversity, bridging sustainable land use and guaranteeing income.	
Economically viable (economic internal rate of return: 22 per cent; net present value: USD 38.5 million). Highly efficient with USD 32/beneficiary (direct).	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XVII, titled "List of proposed conditions and recommendations", respectively.

II. Summary of the Secretariat's assessment

2.1 Project background

3. Haiti is a small island developing State (SIDS) and the only least developed country (LDC) in the Caribbean region. The geographical location, heavy dependence on climate-sensitive livelihoods, and high level of longstanding macroeconomic challenges make Haiti highly vulnerable to hydrometeorological disasters. In particular, projected climate change impacts including temperature rise, greater annual rainfall variability, and more frequent and intense hurricanes will lead to increased and more severe flooding events. Between 1975 and 2012, floods, tropical storms and hurricanes resulted in economic damage and losses equivalent to 2 per cent of the country's gross domestic product on average per year. As such, the Government of Haiti has prioritized watershed management actions in its national adaptation plan in the Northwest and the Aribonite regions where the Trois-Rivières (TR) watershed is located.
4. Although the TR watershed is not the largest watershed in the country, its population is particularly vulnerable to the impacts of climate change-induced flooding, with limited resources to adequately adapt or respond to such events. The rural areas of the TR catchment have an extreme poverty rate of 40 per cent, which is almost twice the national average of 23.9 per cent and much higher than any other area of Haiti. This project will provide a suite of interventions to enhance the resilience of the population and ecosystems of the TR region and promote a transformative approach to the flood management system in the TR watershed.
5. The objective of this project is to enhance the climate resilience of the TR region by initiating a shift away from uncoordinated decision-making and unsustainable land-use practices towards an integrated, climate-resilient approach to land and water resources management. This will be achieved through the following integrated components:
 - (a) Ecosystem-based flood management solutions implemented in 25,440 ha of the TR watershed;
 - (b) Climate-resilient agricultural practices, optimized value chains and social safety nets established to promote sustainable landscape management and reduce degradation in the TR watershed; and
 - (c) Strengthened governance and capacity for climate-resilient integrated water resources management (IWRM).
6. The project will directly benefit at least 292,600 people, of which 52 per cent are women, in the seven target communes across the TR region of Haiti, which accounts for about 3 per cent of the country's total population. These are the most vulnerable populations of Haiti who have limited adaptive capacity to cope with the increased frequency of flooding and will benefit from the project's combination of interventions detailed below. The project will also indirectly benefit an additional 441,272 people, of which 52 per cent are women (approximately 4 per cent of the entire population of Haiti).
7. The total project financing is USD 31,290,561, with a request for GCF grants of USD 22,427,206. Heifer International Haiti will provide co-finance of USD 8,303,355 in the form of grants, and United National Development Programme (UNDP) will also provide grant co-financing of USD 560,000. The project is submitted under environmental and social safeguards (ESS) category B.

2.2 Component-by-component analysis

Component 1: Ecosystem-based flood management solutions implemented in 25,440 hectares of the Trois-Rivières watershed (total cost: USD 18.40 million; GCF cost: USD 18.20 million)

8. Component 1 aims to directly enhance the resilience of the TR catchment by implementing agroforestry systems and restoration of arboreal land cover on degraded mountainous slopes in the TR watershed. An increase in concentrated and heavy rainfall,

combined with the increasing frequency and intensity of hurricanes and tropical storms, has resulted in the rapid erosion of fertile soils throughout the TR watershed. This is exacerbated by the steep topography and highly deforested landscape of the watershed, which intensifies flow rates because of reduced ground cover and further increases erosion and river sedimentation. The combination of these factors often results in the destruction of crops, farmlands and agricultural infrastructure, as well as a loss of livestock and human lives during the flood events. The implementation of the proposed ecosystem-based adaptation (EbA) flood management solutions is expected to address these issues by significantly reducing peak run-off and flooding extent in the area by increasing vegetative cover in the watershed and the infiltration capacity of the soil. More specifically, the agroforestry systems will be implemented in 17,740 ha of degraded land, and reforestation and restoration will be implemented in 7,700 ha of forest areas across the target areas.

9. The proposed activities include establishing nurseries, planting seedlings (e.g. for fruit production and reforestation) and providing ongoing technical assistance to end users including on establishing and monitoring of nurseries, scheduling seedling planting, using technical packages for agroforestry and monitoring the agroforestry systems to account for the growing times of many agroforestry species. The spaces between the agroforestry species will be filled with perennial food crops (e.g. plantains, sugar cane, cassava, pigeon peas and pineapple), in combination with annual crops (e.g. sweet potato, yams and other annuals).

10. This intervention, which is an example of EbA, will be supplemented by developing the capacity of the target community groups, including farmers' and women's associations regarding climate-resilient land-use planning. Thirty-three communities within the seven target communes in the TR watershed will receive this training, which will then result in the development of 33 community land-use plans, through community engagement in a consultative process. These plans will focus on identifying the impacts of climate change, particularly flooding, on the TR watershed and designing appropriate actions for addressing these impacts. Such actions include the implementation of EbA solutions to reduce flood risk in the watershed as well as the promotion of sustainable, climate-resilient land-use practices and techniques to reduce the risk of increased land degradation under future climate change conditions. The farmers in the target areas will receive training on developing sustainable agroforestry business models that will contribute to both improving livelihood security and enhancing the use of sustainable land-use practices. This participatory approach will ensure that the proposed interventions will have buy-in from beneficiary groups.

Component 2: Climate-resilient agricultural practices, optimized value chains and social safety nets established to promote sustainable landscape management and reduce degradation in the Trois-Rivières watershed (total cost: USD 8.30 million; GCF cost: USD 0)

11. Component 2 will facilitate a transformative shift in Haiti's agricultural sector from unsustainable practices (e.g. the cultivation of annual crop types and clearing of forests for charcoal production) towards sustainable options that are resilient to the impacts of climate change-induced flooding. This shift will result in sustainable livelihoods for the populations of the TR region, while reducing deforestation and land degradation and maintaining the integrity of its land and soil. In turn, this will contribute to decreasing flood damage, and a virtuous cycle will be created in the integrated flood management practices in this region. This component will be entirely funded and implemented by Heifer International for this project with its co-finance funding. A suite of complementary activities will be implemented to achieve this component including: (1) strengthening the institutional capacity of the people who are involved in the technical aspects of agriculture at the target commune, inter-commune and department levels for implementing sustainable and climate-resilient agricultural practices; (2) enhancing access to financial resources for smallholder farmers in order to facilitate the uptake and implementation of climate-resilient agricultural practices over the long term; and (3) establishing a social protection system for households that face chronic food insecurity. The co-

finance funding for this component will be catalysed by the approval of the GCF amount, and the sequence of activities will be aligned in order to maximize the impacts of these activities.

Component 3: Strengthened governance and capacity for climate-resilient integrated water resources management (total cost: USD 3.27 million; GCF cost: USD 3.27 million)

12. Component 3 will strengthen the capacity of the Government of Haiti to develop and implement the Water Act, which will be an indispensable legal foundation for enabling participatory IWRM in the TR watershed. The current water governance system in Haiti is organized at both the departmental and the communal level, which creates an overlap between different bodies, particularly at the catchment and sub-catchment levels. This poses challenges in coordination and planning among stakeholders at different levels and leads to ineffective and inefficient implementation of an integrated approach to water resources management in the TR watershed. The activities envisaged under this component include: (1) drafting provisions to Haiti's Water Act for the adoption of an integrated approach to water resources management; (2) establishing catchment IWRM committees within 31 catchments in Haiti; and (3) establishing sub-catchment water resource users' associations within the sub-catchments of the 31 catchments. These committees will work with different stakeholders in the 31 catchments and sub-catchments to manage the efficient use of land and water resources and will be responsible for fostering community empowerment and ownership over these critical resources under future climate change conditions. These activities will ensure that the Water Act is developed and implemented to respond not only to national level priorities, but also to the adaptation needs of vulnerable communities living in the TR watershed.

Project management (total cost: USD 1.31 million; GCF cost: USD 0.95 million)

13. The GCF portion of this project management cost is less than 5 per cent of the total requested GCF funding and is compliant with the GCF policy on fees.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: N/A

14. The proposed project will directly benefit 292,600 people (2.7 per cent of the total population), targeting the most vulnerable population.

15. The project net present value (NPV) of interventions is estimated to be USD 207 million with an internal rate of return (IRR) of 37.9 per cent, and therefore represents a highly cost-effective spend for the Government of Haiti. The estimated health benefits under the economic and financial analysis list water-borne diseases (such as diarrhoea and cholera) that would, without the project interventions, likely increase in prevalence because of increased flooding and related impacts.

16. The increase in tree cover (17,740 ha agroforestry and 7,700 ha restoration) is expected to result in a 40 per cent reduction in peak run-off and reduce floodplain extents by 26 per cent in the areas with a 1 per cent annual exceedance probability (AEP) of floods.¹

3.2 Paradigm shift potential

Scale: N/A

17. The proposed project will catalyse a paradigm shift in Haiti's approach to flood management in the TR watershed by initiating a shift away from uncoordinated decision-

¹ A 1 per cent AEP implies a 1 in 100 chance of flood in any one year in the given area.

making and unsustainable land-use practices towards an integrated, climate-resilient approach to land and water resources management.

18. Firstly, agroforestry systems will be established and degraded forests restored at priority intervention sites across the watershed to: (1) reduce peak surface run-off and flood extent; and (2) provide alternative livelihood options to the unsustainable land-use practices on which most vulnerable communities in the TR watershed depend. Secondly, complementary interventions focused on enhancing technical and institutional capacity at the commune, inter-commune and department levels will be implemented to promote productive climate-resilient agricultural practices and sustainable land management. Finally, the necessary governance framework for IWRM will be established in Haiti to support the uptake of climate-resilient land management systems and to facilitate sustainable use and management of water resources over the long term.

3.3 Sustainable development potential

Scale: N/A

19. Under the project, the technical and institutional capacity of relevant stakeholders at the national, department and commune levels will be strengthened regarding implementation of an integrated approach to land and water resources management that is centred around sustainability and climate resilience. By building the capacity of the targeted institutions, the transfer of knowledge and skills related to sustainable watershed and flood management will extend beyond the project.

20. Strengthening the Water Act will establish a legal foundation for optimizing the use of water resources without compromising the sustainability of vital ecosystems under future climate change conditions, consequently securing equitable economic and social welfare. The Water Act will also provide a framework for the institutionalization of integrated watershed management to empower regional administrations across Haiti – through establishing catchment water resources management committees and sub-catchment water resource users associations.

21. This country-driven project has been developed using a participatory and consultative process, enabling all relevant stakeholders to contribute to the design of interventions, ensuring buy-in and commitment from national- and local-level stakeholders. More importantly to communities, the social protection system provided through the project, as well as the EbA interventions that will be installed, will provide tangible benefits to communities linking the IWRM approach to reduced vulnerability.

3.4 Needs of the recipient

Scale: N/A

22. Haiti is a SIDS and an LDC hit by cascading and compounding climate hazards (floods, hurricanes, tropical storms, erosions, etc). The target beneficiaries endure double the national average poverty rate. The watershed is particularly vulnerable to the impacts of climate change-induced flooding. Given the near-complete absence of embankments and levees in the watershed, such flooding causes severe damage to crops and farmland, as well as agricultural infrastructure and equipment. Flooding in the region also has severe consequences for both human and livestock health, resulting in an increased risk of loss of human life and livestock. Addressing the impact of climate change-induced flooding on vulnerable communities in the region requires the adoption of an integrated approach to flood management that responds to the impacts of climate change.

23. At present, the Government of Haiti does not have the capacity to fully finance integrated climate-resilient flood management interventions through domestic finance in both

the private and public sector. Without GCF support, the Government of Haiti will continue to operate under the existing paradigm of reactive recovery from the impacts of climate change.

3.5 Country ownership

Scale: N/A

24. The project is a direct contribution to Haiti's adaptation targets in relation to watershed management, coastal and marine protection, and reforestation, as reflected in the updated nationally determined contribution submitted to the United Nations Framework Convention on Climate Change in 2022 and responds to the institutional capacity needs identified in Haiti's national adaptation plan submitted in early 2023. The project is also a prioritized intervention with GCF and is included in the advanced draft country programme which, at the time of writing, was in the last stages of preparation. In this regard the no-objection letter was issued and cleared in April 2022.

25. Extensive stakeholder consultations were held during the project elaboration and feedback was incorporated into the development of the proposal. Active stakeholder engagement, notably in the intervention areas, is planned to ensure optimal execution of the project and as part of the adaptive management strategy of the project.

3.6 Efficiency and effectiveness

Scale: N/A

26. The funding proposal package presents economic and financial analyses with reasonable assumptions as well as a clear definition of scope and the counterfactual scenarios. The economic analysis contains estimations of all the relevant economic values from the project activities that are consistent with the information provided in the funding proposal. The model uses a social discount rate of 10 per cent in line with all the other UNDP projects. In summary, the economic model results show the NPV ranging between USD 375 million and USD 412 million with an IRR of 41 per cent and 43 per cent depending on the different flood reduction scenarios. The model shows that the proposed project is economically viable.

27. The financial analysis has been conducted at the individual farmer's perspective for the agroforestry activity, assessing whether the farmers would be able to invest in the agroforestry activities described in the funding proposal through their own resources or other existing sources of finance (microfinance, bank loans, etc.) The weighted average cost of capital (WACC) used is 3.7 per cent as the mean of the lowest and highest nominal monthly rates charged by microfinance institutions in peer group and village banks. The financial IRR is above the WACC, although grants are still requested, on the basis of the sensitivity analysis conducted with loan scenarios.

28. Financial adequacy and appropriateness of concessionality is assessed to be high. The accredited entity (AE) has conducted a sensitivity analysis for the agroforestry model and calculated financial profitability indicators for the agroforestry model (NPV, financial IRR and payback period) in four cases: (1) when activities are fully financed by farmers on their own ("without project"); (2) when activities are financed with GCF grants ("with project"); (3) when activities are financed with bank loans at the lowest interest rates (2.75 per cent); and (4) when activities are financed with bank loans at the highest interest rates (4.8 per cent). The results show that the agroforestry activities cannot be financed with full bank loans (negative cash flows), and the payback period would be too long for low interest rate loans. An analysis of the mix of loans and grants scenario has not been conducted.

29. The need for grants in this project is also justified by the public nature of the project activities. The funding proposal describes the absence of financing for the proposed activities for farmers as one of the major barriers. Alternative sources of funding and credit are not available or are difficult to access, leading farmers to go for the option of short-term relief

activities, such as the production of charcoal at unsustainable rates to meet pressing short-term financial needs.

30. The project's adaptation cost is USD 78/direct beneficiary and USD 31/total beneficiary. Cost per direct beneficiary is in line with the portfolio average for Latin America and the Caribbean. The co-financing ratio is 1:0.39, which is lower than the target set in the updated Strategic Plan for the GCF 2020–2023. The proposal does not present a clear roadmap or prospect about the potential for leveraging or mobilizing additional financial resources.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

31. **Project background.** The project aims to take a sustainable land and water resource management (IWRM) approach as adaptive measures to build resilience in the Trois-Rivières (TR) watershed in Haiti. The main interventions proposed by the project include: i) implementing green infrastructure (agroforestry and restoration activities) to strengthen flood resilience, supported by capacity-building programmes for land-use management; ii) establishing and promoting climate-resilient agricultural and land-use practices to enhance food security in the watershed; and iii) improving national and local governance frameworks for the implementation of integrated water resources management at national, catchment and sub-catchment levels to contribute to creating an enabling environment for climate-resilient flood management in Haiti. The project will be implemented through community-based governance and planning mechanisms to be developed in seven communes targeted by the project in the TR watershed, namely Port-de-Paix, Chansolme, Bassin Bleu, Plaisance, Pilate, Marmelade, and Gros Morne. The project will improve climate resilience and food security, decrease the impact of floods, and provide significant environmental and social co-benefits, through ecosystem-based flood management and improved flood management governance.

32. **Environmental and social risk category and safeguards instruments.** The AE has classified the project substantial environmental and social (E&S) risk against AE's own screening criteria. The AE considers the project as equivalent to Category B per GCF's E&S risk categorization. The project is expected to have potential limited adverse environmental and social risks and/or impacts which are site-specific, largely reversible, and can be mitigated with appropriate environmental and social management measures. Any high-risk activities will be excluded by the project, favouring a climate resilient and green infrastructure approach to flood management, and eschewing high-risk grey infrastructure for flood control. The Secretariat confirms that the risk category is aligned with GCF's E&S risk categorization and is within the E&S risk accreditation level of the AE. The AE has prepared a screening report against its own social and environment screening procedure (SESP) and an environmental and social management framework (ESMF) to guide the E&S screening, assessment and management to be conducted during the project implementation phase, where the preparation of a strategic environmental and social assessment (SESA), site-specific environmental and social impacts assessments (ESIAs) and environmental and social management plans (ESMPs) will be required.

33. **Compliance with GCF's ESS standards.** The paragraphs below describe the programme's compliance with GCF's ESS standards.

34. **ESS 1: Assessment and management of environmental and social risks and impacts.** The ESMF provides the general information about the project activities, the baseline socio-economic and environmental conditions of the project, and an overview of the laws and regulations as well as AE's E&S standards applicable to the project. Using the SESP screening, the ESMF identifies and briefly describes a total of 18 potential social and environmental risks associated with this project. Among which, three substantial risks identified relate to the

project's potential negative impacts on economic displacement, critical habitats and KBA, and security concerns. The rest are considered moderate risks.

35. In response to the security concerns, the ESMF includes a conflict and fragility assessment and identifies that the project will unlikely increase or exacerbate the national-level political or security crisis. At the local level, the AE considers the security issue a destabilizing factor that makes the project's operating context more prone to conflict and increases its risk on conflict and security. The ESMF states that the project may inadvertently trigger new and/or existing factors of conflict in certain project sites where social tensions around land tenure have been observed. Such risk is currently not considered by the AE as flashpoints for wider violence or conflict because such tensions are not to be related to inter-community grievances or to issues of identity, ethnicity, or religion. As the project will promote an inclusive community approach and continuous consultations during the implementation phase, it will mitigate the social tensions by working with stakeholders to ensure that all land and water users in the project areas will be involved in the project and will have the space to share their views and needs.

36. The AE will adopt a proactive approach to conflict sensitivity. This will begin with a detailed conflict analysis to be conducted at each project site as part of the site-specific ESIA. The ESMF provides a methodological guide for conducting a conflict and development analysis (CDA). The CDA will include management measures related to the security situation and the human rights and conflict related risks specific to the project site. These management measures identified through the CDA will be incorporated into the site-level ESMPs and/or stakeholder engagement plan (SEP). The CDA will also provide recommendations to tailor the delivery of the project interventions and support. It will also inform the design of community land use plans. The project will also take an environmental peacebuilding approach by bringing communities together with a common objective on natural resources management and livelihoods, which will enable the project to integrate conflict resolution methodologies into the participatory process for designing community land use plans. In addition, the AE will develop a security plan, including an initial assessment of potential security risks in target site areas as well as the relevant protocols and structures to manage the identified risk in alignment with UN policy for work in the field. The detailed procedures will be integrated within the project's site specific ESIA and ESMPs and will be updated as needed.

37. **ESS 2: Labour and working conditions.** The AE has triggered its standard on this as the project activities will involve the use of labour. Agroforestry activities could potentially involve child labour that can expose children to disrupted physical, mental, moral, and social development, extreme weather conditions, injuries, infections, and diseases. Activities to be supported by the project on agriculture and agroforestry supply chain, land restoration, and erosion control could potentially involve practices that do not comply with national and/or international labour standards or safety standards. The Project will avoid hiring workers below the age of 18 and will exclude activities or production of activities that involve child labour through the project's exclusionary criteria. The project and contractor's staff and workers will potentially be exposed to occupational health and safety hazards. The project will exclude activities involving purchase, application or storage of harmful pesticides and hazardous materials. Furthermore, the ESMF includes an indicative outline for labour management procedures, that will be developed as part of the site-specific ESIA and ESMPs. The procedures set out the conditions in which project workers will be employed, engaged, and managed, in accordance with the requirements of AE's standard and applicable labour laws, rules and regulations. The procedures will be appropriate to the size, locations, and workforce of project activities.

38. **ESS 3: Resource efficiency and pollution preventions.** The ESMF identifies that to a limited extent the project activities interventions on agroforestry and reforestation may generate waste and litter, such as human waste, food waste, plastic, soiled material, and scrap

equipment around the project areas. Poor management of those waste product and/or its poor handling during storage and transport could cause pollution or physical hazards and pose a threat to the health of the environment as well as to the workers and community. The AE has triggered its standard on pollution prevention and resource efficiency and will require that any material threat related to pollution and discharges of waste be detailed as part of the site-specific ESIA. This will be applied to activities that aim to (i) improve existing waste management practices; (ii) generate or cause generation of solid, liquid, or gaseous waste; (iii) use, cause use of, or manage the use, storage or disposal of hazardous materials and chemicals, including pesticides; and (iv) significantly consume or cause consumption of water, energy, or other resources. AE is suggested to capacitate local communities to identify the related risks and incorporate mitigation measures as part of the community land use plan to ensure long-term sustainability of the project interventions.

39. **ESS 4: Community health, safety and security.** The AE has broadly identified that the project activities may generate risks to the health and safety of communities and considered the health impacts related to the flood risk and associated threat to the communities as well as the impact of the COVID-19 pandemic. As the small infrastructure and/or equipment to be used at each project site is yet to be identified, the ESMF requires further assessment to be conducted as part of the site-specific ESIA. The project will also incorporate community health monitoring as part of the project's wider efforts of implementing regular monitoring and evaluation of water resources at the catchment and sub-catchment levels (under Activity 3.3 to support the implementation of IWRM plans). As it is likely that some of the small infrastructure projects may create noise, dust, or traffic risks for communities, and that if water supply systems are not designed or maintained properly, they may create risk of water- or vector-borne diseases. The AE is suggested to provide basic guidance on and include a list of standard mitigation measures for the main types of small-scale infrastructures and equipment likely to be deployed during project implementation.

40. **ESS 5: Land acquisition and resettlement.** The AE has triggered its own standard on displacement and resettlement, noting that the project's agroforestry or reforestation activities (under Activity 1.2, establishing agroforestry systems areas over 17,740 hectares and restoring forestation to 7,700 hectares) can potentially lead to the spatial or temporal restrictions on lands previously used by local communities and on their livelihoods. This could then create economic displacement and pose hardship on local communities. The project will exclude activities that will require any physical displacement of people or changes of land tenure. The ESMF describes the possibility that land acquisition for small infrastructure investments may emerge during the project implementation, such as those activities that will involve deploying agricultural tracks, developing water infrastructure, and providing upgrades to water infrastructure to address decreasing water security caused by the impacts of climate change-induced flooding. The ESMF also notes that in such case the project will only work with those who are willing participants and seeking infrastructural upgrades, and in areas that are dedicated as public lands. The project will ensure that all legislation is followed. The ESMF requires that a livelihood action plan (LAP) to be prepared if project activities will cause economic displacement, namely where the livelihoods of individuals or communities are restricted, partially or fully, and either directly or indirectly, in their access to land or resources to support their economic well-being. To be developed as part of the ESMP, the LAP will address beneficiary participation in economic displacement decision-making, adequate (full and fair) compensation and assistance, and risk management to ensure that livelihoods are at least as good as prior to the program implementation and that the livelihoods of poor and marginalized are improved. The LAP will include independent monitoring procedures, clarification of land rights consistent with applicable law, and outline capacity, training, and development actions targeting beneficiaries' livelihoods. Following the indicative outline for the LAP as annexed in the ESMF, project will be required to justify why the land and/or resources to which access will be restricted, the necessity of economic displacement to achieve the project objectives, and the

linkage between resettlement implementation and initiation of civil works for each of the project components. The AE will ensure that its own safeguards requirements, best practice standards, and the mitigation measures are being met, such that project activities involving economic displacement cannot proceed until completion of the ESIA and the related LAP.

41. **ESS 6: Biodiversity conservation and sustainable management of living natural resources.** As listed in the project's exclusionary criteria, the project will exclude project activities that will be implemented in areas of critical habitats, unless all of the following are demonstrated. That is, (i) there are no measurable adverse impacts on the criteria or biodiversity values for which the critical habitat was designated, and on the ecological processes supporting those biodiversity values (determined on an ecologically-relevant scale); (ii) there is no reduction of any recognized Endangered, Vulnerable or Critically Endangered species, (iii) any lesser impacts are mitigated, and (iv) a robust, appropriately designed, and long-term Biodiversity Action Plan is in place to achieve net gains of those biodiversity values for which the critical habitat was designated. The project will exclude activities that involve the application of chemical fertilizers or pesticides as a result of the agroforestry and reforestation interventions of the project, as well as activities that risk the introduction of alien and potentially invasive alien species. Despite such exclusion, the ESMF identifies that the project could expose substantial risks on biodiversity due to that Port-de-Paix, one of the targeted project sites, is a KBA and an Alliance for Zero Extinction (AZE) site. This KBA falls within the project's AREA 2 encompassing the semiarid areas across Gros-morne, Chansolme, Port-de-Paix, and Bassin Bleu. The AE expects that the community land use plan for this area (under Sub activity 1.1.1) may be developed potentially within or adjacent to this KBA. The ESMF requires that the decision as to whether project activities will take place within or adjacent to this KBA and other critical habitats must follow the mitigation hierarchy as described in the ESMF. If project activities take place within or adjacent to a KBA, those activities must be guided by an appropriately scoped and detailed Biodiversity Action Plan (BAP) to create positive impacts on biodiversity conservation and in line with the requirements of AE's relevant and guidance. The project will assess its potential impacts on biodiversity and natural resources more clearly at the start of the project implementation, through conducting a SESA and site-specific ESIA's. Depending on the scope of the project activities to be designed at the site-level, mitigation measures will be formulated as part of the site-specific ESMP or as a standalone Biodiversity Management Plan (BMP). The ESMF provides detailed guidance for the development of context specific BAP or BMP to ensure that the project will achieve net gains to the biodiversity values and underpinning ecosystem services for which the critical habitat was designated.

42. **GCF Indigenous Peoples Policy and ESS 7: Indigenous Peoples.** Based on a review of UNDPs' and other agencies' programmes in Haiti and field consultations carried out by the AE, the AE concluded that there are no sub-group in the project area and area of influence that meet the criteria for indigenous peoples (e.g. has pursued its own way or developed distinct group identity, languages or no distinct ethnicity nor group self-identifying as indigenous peoples in Haiti or no special connection between Haitian people and their territory other than the national one). Accordingly, the AE concluded that the GCF Indigenous Peoples Policy and its own Social and Environmental Standard 6 were not triggered.

43. **ESS 8: Cultural heritage.** The ESMF identifies moderate risk to cultural heritage that may arise from undertaking project activities that will involve the movement of earth over large areas. This may lead to the accidental discovery of or damage to cultural heritage. The ESMF provides chance finds procedures to be followed at site- and activity-level in all plans and contracts. The procedures will be applied to project-related construction works, including excavations, demolitions, movement of earth, flooding, or other changes in the physical environment. The procedures establish how chance finds of tangible cultural heritage shall be managed, including notification of relevant authorities and stakeholders, avoidance of further disturbance or damage, protection, documentation, and assessment of found objects by relevant experts. Per AE's own standard, the procedures will be applied in the case of archaeological sites

and materials, built heritage, landscapes and natural features with cultural significance, and movable cultural heritage. This is no baseline information available in the ESMF. The project will ensure that through the site-specific ESMPs a chance finds procedure will be implemented during all works of the project. The ESMF requires that the community participation, stakeholder consultations and use of experts will be exercised fully and effectively to deal with cultural heritage properties, and that all discoveries and any archaeological sites discovered by the project will be immediately reported to local authorities.

44. **Sexual exploitation, sexual abuse and sexual harassment (SEAH) safeguarding.**

The ESMF has identified the project's moderate risks on gender-based violence (GBV) and SEAH. The project plans to provide significant support to women through enhancing their access to natural resources and increasing their ability to participate and derive equal benefits from project-related outcomes, including the outcomes targeting livelihoods. While the project expects to increase women's income through project-sponsored activities, the traditional power dynamics in the household and in a broader societal context could be changed and that the gender norms and an existing culture of discrimination can be challenged. Such shifts can lead to and exacerbate the existing GBV and SEAH. The AE has conducted an initial SEAH risk assessment and identified the risk factors, indicators, and mitigation measures related to preventing SEAH (PSEAH). Based on the PSEAH safeguarding approach taken at the project-level, the ESMF requires that in-depth SEAH risk assessment to be undertaken as part of the site-specific ESIs. Subsequently, the corresponding PSEAH measures to prevent, mitigate and respond to the SEAH risks for each site will be specified in the ESMPs. The project will establish a survivor-centred grievance redress mechanism (GRM) for SEAH-related grievances and complaints in line with good practices identified in the ESMF. The project will hire a gender and safeguards specialist to guide project safeguards compliance on SEAH.

45. **Implementation arrangements.** The Ministry of Environment of Haiti will act as the executing entity (EE) for this project and will establish a Project Management Unit (PMU). The PMU will comprise of a national project manager, who will be supported by specialists in environment, social development, and gender and social inclusion, as well as technical specialists. The PMU will be responsible for supervising and managing the implementation of measures defined in the ESMF, assign specific responsibilities within the implementing partner for the ESMF implementation, report to the AE and the Project Steering Committee on the implementation of the ESMF, and ensure that all service providers are informed of their responsibilities for the day-to-day compliance with the ESMF. The AE will provide quality assurance for the project, ensure the project's compliance with the AE's policies and procedures, including its own E&S standards, and ensure the implementation of the requirements set forth in the ESMF.

46. **Stakeholder engagement and information disclosure.** The ESMF outlines the overall requirements on stakeholder engagement to be followed during project implementation. The details of the existing stakeholder consultations are summarized in a stakeholder consultation and engagement plan provided separately by the AE, that includes a stakeholder analysis and provides an SEP to be followed at the site-level during project implementation. The SEP outlines processes and mechanisms for continued consultations with all the above stakeholders during project implementation, with targeted communication methods for different stakeholders. The SEP also includes specific outreach mechanisms for women and disadvantaged/ vulnerable groups in the engagement plan. The stakeholders will be assured access to relevant project information to understand the potential project-related opportunities and risks and to engage in the detailed project design and implementation. The project will disclose information on the potential project-related E&S impacts and mitigation/management measures in advance of decision-making whenever possible. As the project risk has been assessed as substantial, the project will follow the disclosure framework specified in the ESMF to ensure that all the E&S safeguards documentation will be disclosed and consulted on prior to implementation of activities that may give rise to potential adverse E&S impacts.

47. **Grievance redress mechanism (GRM).** The ESMF provides the principles for the development of a project-level GRM and a compliant register, which will handle community-related issues during construction works. Any complaint and/or grievance will be advised to the AE and EE within 24 hours of receipt. The ESMF requires a summary list of complaints and/or grievances received, and their disposition to be published in a report produced every six months. The project-level GRM will be operationalized at the start of the project implementation to address all complaints and/or grievances in the project. The stakeholders will be informed of the various means for making complaints and/or grievances, including in addition to the project-level and national grievance redress mechanisms, the option to access AE's own accountability mechanism that has both the compliance and grievance functions. A social and environmental compliance unit hosted by the AE's office of audit and investigations will investigate allegations regarding the AE's standards, screening procedure, or other E&S commitments that are not being implemented adequately, and that harm may result to people or the environment. A compliance review is available to any community or individual with concerns about the impacts of the AE's programmes or projects. The unit is mandated by the AE to independently and impartially investigate valid requests from locally impacted people, and to report its findings and recommendations publicly. The ESMF also include the GCF's Independent Redress Mechanism as an integral of the project's GRM.

4.2 Gender policy

48. The AE provided a gender assessment and action plan with the funding proposal and therefore complies with the requirements of the GCF Gender Policy. The assessment was informed by secondary data and outcomes from the consultations during mission visits by the project development team.

49. The climate change and disaster risk management legislative frameworks in Haiti do not adequately address the issues of gender equality and the empowerment of women. However, flooding and other natural disasters have cumulative livelihood effects that unfairly impact women and exacerbate gender inequality. Evidence and impacts were drawn from the 2010 earthquake and 2015 Hurricane Matthew. In Haiti, women are disproportionately represented among the poor and there is a significant proportion of women-headed households, so impacts such as crop loss, livestock drowning, destroyed homes or displacement, and health risks and loss of life due to floods ultimately affect women first and foremost. Gender-based violence is another issue that normally goes unnoticed and unreported, including police violence targeting women and marginalized groups. Of the cases of violence reported between 2009 and 2011, the majority were physical in nature (59.74 per cent), followed by sexual violence (29.80 per cent), while cases of psychological and economic violence constituted a minority (5.88 per cent and 4.58 per cent, respectively).

50. From this gender assessment, various gender-responsive approaches have been proposed, including building a strong collaboration with public water resources departments in ensuring that the design and installation of water infrastructure facilities are based on gender assessments; increasing the capacity of local government on women's leadership and participation in water management; supporting local municipal authorities and communities to develop and maintain gender-equitable distribution of water resources; and ensuring that partnerships with the private sector include a commitment to gender equality and that women receive equal access to technical or management training, peer-to-peer learning and other resources. Other gender recommendations have been made in the areas of agricultural and climate change resilience, reforestation and agroforestry systems, including promoting land ownership and other productive resources in forest management and climate mitigation strategies.

51. A gender and safeguards expert (at programme management unit level) will be responsible for the implementation of the gender action plan (GAP) in collaboration with Heifer International (the EE for outcome 2). Gender technical expertise and assistance will also be sought on an ad hoc basis to supplement the available skill set and experience within the programme management unit. The expected outcome of the GAP is improved capacities for an estimated 87,780–146,300 women (30–50 per cent of project direct beneficiaries) on sustainable water and land management as a means to enhance their resilience to climate change. Some of the activities outlined in the GAP include (1) strengthening the capacity of community groups, farmer and women’s associations in climate-resilient and land-use management; (2) including gender modules during training sessions for communities and farmer associations in the seven targeted communes on the application of community land-use plans and implementation of climate resilient land-use practices (50 per cent of trainers will be women); (3) establishing commune-level development councils and multi-sectoral management structures to oversee and manage the implementation of project interventions, including strongly encouraging women’s leadership in these institutions (with a target of 50 per cent leadership and participation by women); and (4) drafting the provisions of Haiti’s Water Act for the adoption and implementation of an integrated approach to water resources management in the TR watershed, including provisions on gender responsiveness to ensure inclusive water governance and other inclusive clauses at the policy level.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

52. The GCF is requested to provide a grant of USD 22.9 million, which will be invested to address climate change-induced flood impacts in the TR watershed and the impacts on population residing nearby. Heifer International Haiti and the AE will provide co-financing of USD 8.3 million and USD 0.5 million, respectively. The AE is requesting full grant support for this project. As an LDC and SIDS, Haiti has been experiencing increases in public debt which is placing the country at a high risk of debt distress.

4.3.2. Accredited entity/executing entity capability to execute the current programme (medium risk)

53. UNDP is the AE for the project. UNDP Haiti country office has been working in Haiti since 1973 and has experiences implementing water and land use-related climate change projects through policy formulation and strategic planning. The country office has a track record of managing projects with an EbA approach, and these projects have had a budget range of between USD 2 million and USD 15 million in recent years.

54. The Ministry of Environment (MoE) is the EE of this project. UNDP has carried out a capacity assessment of the EE, which concluded an overall moderate risk with a significant risk regarding monitoring and evaluation (M&E), and procurement and financial management. Requested by the Government of Haiti, UNDP will provide direct project services by assisting procurement processes and making payments to the suppliers and responsible parties at the instruction of MoE.

4.3.3. Programme-specific execution risks (medium risk)

55. The AE has identified the weak land tenure and management system as one of the risks that could delay or prevent the implementation of the project. The feasibility study stated that there are many constraints in terms of obtaining land titles, fragmented land holdings and legal provisions limiting the use of land as collateral, and so on. The current mitigation measure

suggested by the AE is through the community buy-in and implementation of the Water Act. In case an agreement is not reached, the project will work with other communities and participants nearby who are willing to participate in the project activities. The non-participants may continue the unsustainable land use practices and the project does not impose enforcement mechanism. Therefore, the AE/EE's ability to engage communities and establish the participatory community land-use plans will be key to dealing with the land tenure issue.

- (a) Financial sustainability beyond the project implementation period may be achieved through the Village Saving and Credit Association (VSCA) and the budgetary contribution from the national government. However, the VSCA under component 2 is fully financed by co-financing, which is subject to the completion of the integrated approach to climate-resilient flood management in component 1. The Government of Haiti is committed to undertaking the operation and management of the project and VSCA is expected to increase access to finance for smallholder farmers. Given the budgetary constraints of the Haitian Government, its financial commitment is uncertain and there is no co-financing provided by the Government. As a soft measure, it is anticipated that the strengthened technical and institutional capacity of government departments and communes in terms of implementing the land and water resource management will help to enhance the sustainability of the project.
- (b) The AE carried out an economic analysis that identified an economic IRR of 41 per cent and 43 per cent, depending on the choice of flood reduction scenarios, over 25 years. The analysis considered the benefits from the reduction of water-borne diseases (health benefits), reduction of infrastructure damage, and increase in the livelihood income. With regards to the livelihood benefits, a financial analysis has been conducted at an individual farmer's perspective for the agroforestry activity (see paragraph 28 above). The analysis shows that the agroforestry activity cannot be financed with loans and farmers will have difficulty in accessing additional financing such as credit. Therefore, the full grant support is justified. As noted above, the project viability will also depend on the capacity of the national government to take over the operation and management of the project, and the analysis assumed that there would be no additional operation and management costs beyond project implementation.

4.3.4. Compliance risk (medium risk)

- 56. The UNDP, as the AE, confirmed that no individual or entity that is listed on any United Nations Security Council sanctions list, including the United Nations Consolidated List, will be involved in any manner with the project or its activities, either as a counterparty, implementer or beneficiary.
- 57. UNDP confirmed that the risks for the project have been assessed and are mitigated in accordance with the rules, policies and risk management processes of UNDP. This includes risks linked to engagement with implementing partners and responsible parties, where specific capacity assessments have been performed and mitigation measures are put in place. These measures have been incorporated into the project.
- 58. UNDP confirmed that it conducts partner capacity assessments as part of project preparation. This is in line with UNDP's policy on anti-money laundering and countering the financing of terrorism (AML/CFT) and its fraud and corrupt practices (FCP) policy included within the UNDP Programme and Operations Policies and Procedures (POPP).
- 59. In the case of both MoE and Heifer, the assessment using the UNDP Partner Capacity Assessment Tool included a question regarding a history of fraud, corruption, money laundering, financing terrorism and the like. For both partners, the assessment did not find

evidence for such past practices. In the case of Heifer, the assessment identified specific policies for the prevention of these practices².

60. The UNDP Harmonized Approach to Cash Transfers (HACT) assessments for both partners also evaluated the existence of policies against fraud and corruption, policies for reporting wrongdoing (including the misuse of funds) and policies on fair procurement and capacity for reporting and monitoring the use of donor and partner funds.

61. Heifer was evaluated to have an overall low risk in all those factors and a moderate risk in the procurement and contract of awards due to the fact that its procurement system is currently being computerized under the AGRESSO system (<https://agresso.heifer.org>). Nonetheless, the evaluation identified Heifer's buying process as adequate and responsible, in that it follows specific written policies and procedures for procurement. The evaluation highlighted the existence of a supplier database that is updated annually by Heifer and where background checks are done systematically.

62. The MoE was evaluated with a moderate risk score in the case of governance and leadership, operations and management, finance management, programme management, accountability and coordination. Its total overall risk rating was assessed as moderate to high. The assessment of MoE identified the lack of a specific code of conduct for the ministry; however, MoE did mention that it is accountable under all public decrees and laws for governance and public rules, including the Law of Public Function.

63. In response to the HACT evaluation, the MoE has developed its *Manual of Financial, Accounting and Administrative Procedures* in the last year, which includes a specific mention of anti-fraud and corruption policies.

64. UNDP confirmed that the Government of Haiti has put in place a set of institutions to fight corruption. These include:

- (a) Central Unit of Financial Information, which provides information on the laundering of assets from drug trafficking and other serious crimes;
- (b) Unit for the Fight against Corruption, which has the mission of working to combat corruption and its manifestations in all its forms within the public administration;
- (c) National Commission for Public Contracts;
- (d) National Commission for the Fight against Drugs; and
- (e) Superior Court of Accounts and Administrative Litigation, which is a financial and administrative jurisdiction responsible for judging the accounts of the authorizing officers and accountants of the State and the territorial collectives and for ruling on disputes between the State and the territorial collectives, the administration and public officials, the public services and the citizens.

65. UNDP informed the Secretariat that the HACT evaluation did identify challenges in the MoE's procurement processes, including in the integration of a list of suppliers. However, the MoE also demonstrated a positive history of working with partners for project implementation. With this in mind, and at the request of the Government of Haiti, the project will work under support in the form of a national implementation modality in which UNDP will provide project

² These policies include: 1) Code of conduct policy ([https://media.heifer.org/About Us/inside-heifer/Procurement/Code-of-Conduct-Policy-Global.pdf](https://media.heifer.org/About%20Us/inside-heifer/Procurement/Code-of-Conduct-Policy-Global.pdf)); 2) Conflict of interest policy ([https://media.heifer.org/About Us/financial-information/conflict-of-interest.pdf](https://media.heifer.org/About%20Us/financial-information/conflict-of-interest.pdf)); 3) Whistle-blower policy ([https://media.heifer.org/About Us/financial-information/whistleblower-policy.pdf](https://media.heifer.org/About%20Us/financial-information/whistleblower-policy.pdf)); and 4) Procurement and contracting services policy ([https://media.heifer.org/About Us/inside-heifer/Procurement/Procurement-Policy-Global.pdf](https://media.heifer.org/About%20Us/inside-heifer/Procurement/Procurement-Policy-Global.pdf)).

support services for procurement as well as in the payment of vendors on behalf of the MoE (as per the approved workplan prepared by the MoE and approved by the project board).

66. UNDP confirmed that the processes and requirements for the UNDP Partner Capacity Assessment Tool and HACT require UNDP to carry out thorough due diligence of the risks in relation to the programme/project it intends to engage in, as well as in relation to the potential partners who will be involved in the implementation of such programmes/projects. Such capacity assessments include, among other things, a screening of all entities being considered to be contracted by or partner with UNDP against the United Nations Security Council Consolidated List.

67. On the basis of the HACT assessment of the MoE, UNDP will process payments to vendors and third parties through the “direct payment modality”. Payments will be made on the basis of a prior approved annual workplan that is prepared by the MoE and is presented and approved by the project board. The project will also work with Heifer as a responsible party for output 1 and implement all activities under output 2 (fully funded by co-finance funds from Heifer). Heifer has been evaluated as a low-risk partner under the HACT assessment. It should be noted that during the HACT assessment a particular strength was noted; namely that project and beneficiary monitoring uses a quick response (QR) code method.

68. UNDP informed the Secretariat that it applies a zero-tolerance policy in relation to fraud and corruption. The guiding principles of UNDP’s commitment to prevent, identify and address all acts of fraud and corruption have been laid out in the UNDP FCP policy, which applies to all activities and operations of UNDP, including projects and programmes.

69. The fundamental principles of the FCP policy serve as the basis for and are integrated in the UNDP policy frameworks in relation to procurement, financial management, internal control and accountability, and staff rules and regulations. Under the terms of the FCP policy and related policies, UNDP has a commitment, when developing a new programme or project, to ensure that risks related to fraud and corrupt practices are fully identified and considered in the programme/project design and processes, and that adequate and effective measures to mitigate such risks are put in place.

70. UNDP confirmed that the UNDP policy for AML/CFT will be made applicable to the project during its implementation. The policy includes clear risk-based due diligence processes that take into account screening procedures against international sanctions lists, and also includes contractual arrangements with termination clauses for AML/CFT; and monitoring, reporting and investigation protocols, including protection against retaliation.³

71. UNDP confirmed that, during project implementation, UNDP will also provide supervisory missions to project areas to ensure that project equipment/technology is not misused.

72. UNDP also referenced the procurement plan and how this is vetted and approved, specifically indicating how all procurement activities are linked to the implementation of specific activities under the project.

73. These stipulations are included within the programme document that will be signed between the MoE and UNDP as well as in the Standard Basic Assistance Agreement entered into by UNDP and the Government of Haiti,⁴ which states (in section III, paragraph 7) that “Technical and other equipment, materials, supplies and other property financed or provided by the UNDP shall belong to the UNDP unless and until such time as ownership thereof is transferred, on

³ Available at https://popp.undp.org/layouts/15/WopiFrame.aspx?sourcedoc=/UNDP_POPP_DOCUMENT_LIBRARY/Public/AC_Accountability%20AML%20CFT.docx&action=default&DefaultItemOpen=1.

⁴ Available at http://www.ilo.int/dyn/legprot/en/f?p=2200:11002:614574251622::NO:11002:P11002_AGREEMENT_ID:101072:NO.

terms and conditions mutually agreed upon between the Government and the UNDP, to the Government or to an entity nominated by it.” As such, ownership of project equipment remains with UNDP until it is officially transferred at the end of the project.

74. UNDP confirmed that the project will be providing food vouchers as part of output 2. This will be fully financed through co-financing funds provided by Heifer. The HACT assessment noted Heifer’s capacity to provide direct benefits to beneficiaries. This capacity is high enough to warrant a low risk assessment, due to Heifer’s monitoring system that provides QR codes to individual beneficiaries. The beneficiaries will include the farming population residing within the project’s target area, prioritized on the basis of criteria on social and economic vulnerability. This prioritization will respond to an on-site survey, including a wealth assessment. This assessment will also undergo a community-based screening process. UNDP confirmed that Heifer is familiar with the intervention area, having worked through various initiatives, and because it has a field office in the TR region.

75. According to the UNDP *Legal Framework for Addressing Non-Compliance with UN Standards of Conduct*, the Office of Audit and Investigation is the principal channel to receive allegations. UNDP informed the Secretariat that anyone with information regarding fraud against UNDP programmes or involving UNDP staff is strongly encouraged to report this information through the Investigations Hotline.

76. The guiding principles of UNDP’s commitment to prevent, identify and address all acts of fraud and corruption provided in the UNDP *Policy against Fraud and Corrupt Practices*, which applies to all activities and operations of UNDP, including projects and programmes. The fundamental principles of the policy serve as the basis for and are integrated into UNDP’s policy frameworks in relation to procurement, financial management, internal control and accountability and staff rules and regulations. The policy also refers to UNDP’s *Policy for Protection against Retaliation* or “whistle-blower protection” policy. People reporting wrongdoing to the Investigations Hotline have the option to leave relevant contact information or to remain anonymous.

77. UNDP confirmed that the progress of all corrective actions will be tracked, and that UNDP and the MoE will also release updates on the project on a regular basis to provide interested stakeholders with information on project status. Updates may be via a range of media, for example print, radio, social media or formal reports, and all material must be published in English and Haitian Creole (Kreyòl) as appropriate. A publicized telephone number will be maintained throughout the project to serve as a point of contact for enquiries, concerns, complaints and/or grievances. All enquiries, concerns, complaints and/or grievances will be recorded on a register and the appropriate manager will be informed.

78. The project will also have its own GRM, which is described in greater detail within the project’s environmental and social management framework. During project implementation, information will also be provided regarding the GCF Independent Redress Mechanism to project stakeholders, contractors and consultants.

79. **Risk rating:** The Office of Risk Management and Compliance (ORMC)/Compliance Team has conducted a review of the project in accordance with relevant GCF Board approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the ORMC/Compliance Team have determined a risk rating of Medium.

4.3.5. GCF portfolio concentration risk (low risk)

80. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of result area and single proposal is not material.

4.3.6. Recommendation

81. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall programme	Medium
Accredited entity (AE)/executing entity (EE) capability	Medium
Project-specific execution	Medium
GCF portfolio concentration	Low
Compliance	Medium

4.4 Fiduciary

82. The UNDP will be the AE of the project. It will provide oversight and quality assurance in accordance with its policies and procedures, and any specific requirements in the accreditation master agreement (AMA) and project confirmation to be agreed with the GCF.

83. The MoE will be the EE for the project (also known as “implementing partner” in UNDP terminology). The MoE will be accountable to UNDP for managing the project, including M&E of project interventions achieving project outcomes and for the effective use of UNDP resources. It will implement the project in compliance with UNDP rules and regulations, and its POPP guidelines, including the guidelines on the national implementation modality. These include relevant requirements on fiduciary, procurement, ESS and other performance standards. A subsidiary agreement (project document) will be established between UNDP and the MoE as per article 3 of the UNDP Standard Basic Assistance Agreement, and all references in that agreement to “executing entity” shall be deemed to refer to “implementing partner”.

84. At the request of the Government of Haiti, UNDP will provide direct project services for this project, including procurement of goods and services through UNDP processes, providing direct payments to suppliers and responsible parties, hiring of project staff, and delivery of training services. Under these terms UNDP will only make payments and provide related services at the instruction of the MoE, which will be responsible for the preparation of terms of references and selection criteria.

85. UNDP will also provide implementation support in the delivery of activities included within the project’s output 3. A firewall will be implemented to separate the work being provided for the implementation of output 3 and UNDP’s oversight responsibilities in its role as an AE. A Letter of Agreement will be signed between the Government of Haiti and UNDP specifying UNDP’s role in output 3.

86. Heifer International Haiti will act as a responsible party for output 1. It will execute all activities under output 2 through its own co-finance. Once the funded activity agreement is signed between UNDP and the GCF and upon indication by the MoE, a responsible party agreement will be signed between the MoE and Heifer International Haiti for the provision of services for the execution of these outputs.

87. Heifer International Haiti will be overseen by and report to the MoE in its role as EE. As such, Heifer International will enter into written agreements with the MoE specifying its role in project implementation including deliverables, budgets and workplan. Heifer International will report on the usage of project funds and delivery of the workplan to the MoE.

88. GCF funding will be received by the administrative agent (UNDP) as the AE based on a standard letter of agreement (the funded activity agreement) signed between the GCF and UNDP. As funds will be channelled through UNDP, a “direct payments” option will be utilized, whereby transfers will be made directly from UNDP to the goods and service providers based on annual workplans submitted by the MoE. Heifer’s co-financing resources, which are

conditional on the approval of this project by the GCF, will be used for implementing the project interventions outlined in output 2.

89. A project board will ensure that activities under all three outputs of the project comply with the strategies, principles and overall approach outlined in this proposal. The project board will be responsible for making decisions by consensus. Its responsibilities include recommending to UNDP and EE/implementing partner approval of project plans, substantive revisions and acceptance of the project progress reports.

90. To ensure UNDP's ultimate accountability, decisions taken by the project board will be made in accordance with standards that shall ensure management for development results, the best value for money, fairness, integrity, transparency and effective international competition. In the case where a consensus cannot be reached, the final decision shall rest with UNDP. Project reviews by the project board will take place at least twice a year, or as necessary when raised by the project manager or one of the members of the project board.

91. The project manager will provide strategic orientation to the proposed project and ensure technical and operational coordination of the project management unit, with UNDP providing training to the unit's staff. Specifically, the project manager will undertake day-to-day management and decision-making for the project. In line with this mandate, the project manager will be primarily responsible for guiding the implementation of adaptation interventions to achieve the targets specified in the logical framework of the funding proposal and ensuring that these interventions are implemented to the required standards, within the specified time periods and according to the project budget.

92. The financial management and procurement of this project will be guided by UNDP financial rules and regulations. The project will be implemented following the national implementation modality and in accordance with the UNDP's POPP guidelines. UNDP will ascertain the national capacities of the implementing partner by undertaking an evaluation of capacity following its Framework for Cash Transfers to Implementing Partners (part of HACT). All projects will be audited following the UNDP financial rules and regulations noted above and applicable audit guidelines and policies.

93. UNDP will provide oversight and quality assurance in accordance with its policies and procedures, and any specific requirements in the AMA and project confirmation to be agreed with the GCF. This may include, but is not limited to, monitoring missions, spot checks, facilitation and participation in project board meetings, quarterly progress and annual implementation reviews, and audits at project level or at implementing partner level on the resources received from UNDP.

94. The project will be audited in accordance with UNDP policies and procedures on audits, informed by and together with any specific requirements agreed in the AMA. According to the current audit policies, UNDP will be appointing the auditors. In UNDP, scheduled audits are performed during the project cycle as per UNDP assurance/audit plans, based on the implementing partner's risk rating and UNDP's guidelines. A scheduled audit is used to determine whether the funds transferred to the implementing partner were used for the appropriate purpose and in accordance with the workplan. A scheduled audit can consist of a financial audit and/or an internal control audit.

4.5 Results monitoring and reporting

95. As an adaptation initiative, the project aims at generating various adaptation results across three GCF result areas. It is expected that 292,600 direct beneficiaries and 441,272 indirect beneficiaries will receive adaptation benefits in Haiti, representing 3 per cent and 4 per cent of its total population, respectively. The AE has provided a methodology for estimating numbers of direct and indirect adaptation beneficiaries based on expected adaptation benefits.

The Secretariat provided further guidance of how to disaggregate the beneficiaries against GCF results areas, which can feed into the target setting of GCF core indicator 2 and its supplementary indicators in the logical framework.

96. The theory of change diagram adequately captures various levels of expected changes and logical linkages between them from the goal statement to the proposed activities. There was an issue on the description of outcomes; however, it was discussed well with the AE and it was agreed to have three project outcomes. The three outcomes are (1) increased resilience in ecosystems, (2) improvements to the livelihoods of beneficiaries and (3) strengthened governance as an enabling environment. Co-benefits have also been identified, including carbon sequestration. The barriers and risks that may hinder the achievement of these changes are identified clearly and linked to pertinent activities.

97. The logical framework is structured in alignment with the GCF integrated results management framework. It shows how the project can contribute to three GCF adaptation results areas with project outcomes and outputs. Relevant GCF core and supplementary indicators have been chosen for monitoring and reporting the key performance of the project. The project-specific indicators also seem to be adequate to measure the progress of project implementation along with project deliverables.

98. There have been discussions on the project M&E plan (annex 11) and project implementation timetable (annex 5). The Secretariat recommended that the AE ensure that all M&E-related budgets, including project-specific indicators, are identified in the M&E plan and these should be identical to the M&E budget in the project budget plan. The AE has been responsive and annexes 11 and 5 have been revised accordingly.

4.6 Legal assessment

99. The original AMA was signed with the AE on 5 August 2016, and it became effective on 23 November 2016. The AE's first term of accreditation expired on 22 November 2021. The AE was re-accredited by the Board pursuant to decision B.30/05. The first amendment and restatement agreement amending and restating the AMA ("AMA Amendment") was signed on 16 March 2023 and it became effective on 3 May 2023.

100. The AE has provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

101. The proposed project will be implemented in Haiti, a country in which GCF is not provided with privileges and immunities. This means that, among other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. The GCF provided an updated draft agreement on privileges and immunities with a background note to the Government of Haiti on 25 May 2018.

102. The Heads of the Independent Redress Mechanism and Independent Integrity Unit have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where the GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by the GCF are made only after the GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities.

4.7 List of proposed conditions (including legal)

103. In order to mitigate risk, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval;
- (b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP208

Proposal name:	Enhanced climate resilience in the Trois-Rivières region of Haiti through Integrated Flood Management
Accredited entity:	United Nations Development Programme (UNDP)
Country/(ies):	Haiti
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel

1. This funding proposal for a small-sized public sector adaptation project in environmental and social safeguards category B, to be implemented over an 8-year period and with benefits accruing over a 25-year project lifespan, is submitted by the Government of Haiti (GoH) through the United Nations Development Programme (UNDP) as accredited entity (AE), with the Ministry of Environment of Haiti (MoE) as the executing entity (EE). The total cost of the funding proposal (FP) is USD 31.29 million, of which USD 22.43 million is requested from GCF, with USD 8.86 million in co-finance from Heifer International Haiti (Heifer) and UNDP.

2. The proposed project will contribute to GCF's paradigm shift objective of increased climate-resilient sustainable development by enhancing the resilience of Haiti's Trois-Rivières (TR) watershed and its communities to the impacts of climate change intensified flooding. This shift will be achieved by implementing a transformative approach to the way in which the GoH and farming communities address flood impacts, incorporating ecosystem-based adaptation (EBA), and integrated land and water resource planning and management that is informed by climate risk. Specifically, direct investments from GCF combined with co-financing from Heifer will be used for:

- **output 1: Ecosystem-based flood management solutions implemented in 25,440 hectares of the Trois- Rivières watershed:** activities to implement agroforestry systems and rehabilitate 'water towers' through reforestation of degraded landscapes at priority intervention sites; also including climate change awareness with local government and communities, and participatory land-use planning;
- **output 2: Climate-resilient agricultural practices, optimized value chains and social safety nets established to promote sustainable land management and reduce degradation in the TR watershed** – activities to enhance technical and institutional capacity for productive climate-resilient land management at the national and local levels, including support to tree nurseries, agri-processing cooperatives and village savings and credit associations (VSCAs);
- **output 3: Strengthened governance and capacity for climate-resilient integrated water resources management (IWRM)** – activities to establish the required governance framework, to support the climate-resilient land and water management, including amendments to the Water Act, and establishment of catchment committees and sub-catchment water user associations across the country.

3. The third component involves capacity-building at the national level and at catchment level throughout the country, and the other two components will focus only on the TR

watershed in the country's north-west, which is highly vulnerable to floods, that are expected to become more frequent and severe with the changing climate. This work will be carried out in 33 communal sections with 118 communities. These communities are based in seven target communes, falling into three departments – Port-de-Paix, Chansolme and Bassin-Bleu (Northwest); Plaisance and Pilate (North); and Marmelade and Gros-Morne (Artibonite).

1.1 Impact potential

Scale: N/A

4. **Climate change basis:** The FP highlights Haiti's extreme vulnerability to climate change impacts, as well as earthquakes (which are generally considered natural disasters). In the past two decades, Haiti has experienced 26 floods, 12 of which were catastrophic, 8 hurricanes, and numerous other tropical storms and earthquakes, with severe damage and loss caused by climate change intensified flooding in particular. For example, in 2004, Hurricane Jeanne caused severe flooding in the northern part of Haiti, resulting in more than 2,800 deaths and approximately USD 41 million in damage. More recently, Hurricane Matthew struck Haiti in 2016, causing 546 deaths and the loss of up to 90 per cent of crops and livestock in some areas.

5. The FP and feasibility study (FS) use historical data and future projections to demonstrate convincingly that climate change is causing an increase in annual average temperatures and creating the conditions for increased frequency and intensity of hurricanes in the Caribbean region. Climate change is causing a decrease in total annual rainfall in Haiti, but also a shortening of the wet season, with a resulting increase in concentration and intensity of rainfall events. These changing climatic conditions are contributing to an increase in the frequency and intensity of flooding in the country, particularly in the north-western TR watershed of the Massif du Nord, with its steep topography.

6. The FP also shows how climate change impacts are compounded by direct human pressures on the environment, with degraded landscapes caused by unsustainable agricultural practices, and historical deforestation, that involved felling of 99 per cent of the primary forest cover of the country, much of this for charcoal production.¹ During heavy rainfall events, degraded areas experience increased surface run-off, worsened by decreased infiltration of water into soil because of reduced vegetation cover. These factors, combined with steep slopes and the lack of embankments along the river, increase the flow of water into the low-lying plains, causing flooding of downstream villages. Increased water flow intensifies erosion and causes sediment deposits in the riverbed and drainage infrastructure, further exacerbating flooding, including in the downstream city of Port de Paix. Drinking water contaminated with sewage and pollutants, and pools of standing floodwater increase the spread of water- and vector-borne diseases.

7. **Adaptation impact:** The project aims to deliver impacts against three GCF adaptation results areas in the target communes of the TR watershed:

- (a) Increased resilience of most vulnerable people and communities;
- (b) Increased resilience of health and well-being, and food and water security; and
- (c) Increased resilience of ecosystems and ecosystem services.

¹ Although 99 per cent of Haiti's original forest cover and its high biodiversity was lost, significant greening has occurred in recent years through smallholder production of fast-growing tree species for use as fuelwood and sale into charcoal value chains – with up to 36 per cent of the country now 'reforested' in this manner, and harvested on a sustainable basis using coppicing techniques. See https://www.climate-links.org/sites/default/files/asset/document/2021-06/2021_USAID-ProLand-Haiti-Woodfuel-Case-Study-Final.pdf

8. Overall, the project will confer direct and indirect adaptation benefits to approximately 292,600 and 441,272 vulnerable people in Haiti, respectively. By implementing 17,740 ha of agroforestry systems and rehabilitating 7,700 ha of forests in priority areas in the TR watershed to address the impacts of climate change-induced flooding, it is anticipated that the interventions will reduce peak surface run-off and floodplain extents, resulting in a 35 per cent reduction in the number of households at risk from 20-year flood events, and a more than 50 per cent reduction in the number of households affected by 100-year flood events. This will be achieved through revegetation which i) restores the ecosystems' capacity for water infiltration and reduces flash floods; ii) reduces the soil erosion that worsens downstream flooding through siltation of drainage structures; and, although this is less emphasized in the FP, iii) improves groundwater availability and baseflow² in the dry season. Agroforestry interventions will enhance 15,000 vulnerable households' resilience and improve food security, by diversifying and increasing agricultural production and adding value through agri-processing support.

9. Interventions to reduce the extent of flooding are expected to reduce the spread of water- and vector-borne diseases, including, *inter alia*, cholera, diarrhoea and typhoid, with an expert hired to monitor disease incidence across the target communes, assisted by water user associations. The agroforestry and reforestation interventions under this project are also expected to result in an important co-benefit of reducing greenhouse gas emissions. The combined project investments will transform Haiti's approach to land and water resources management and create an enabling environment for the uptake of climate-resilient, sustainable flood management practices in the public sector and in farming communities. The project aims to create pathways to scale up and replicate these practices in catchments across Haiti through activating catchment management structures.

10. **Adaptation options and impacts:** The project proponents conducted an options analysis to choose between 'grey' engineered solutions to flooding, such as embankments, check dams, canals and discharge channels on the one hand, and 'green' ecosystems-based solutions, such as tree planting, natural regeneration, slope stabilization, riverbank vegetation, and contour line farming, on the other. Green solutions were selected because they are less costly and require less complex and expensive maintenance, and also because of the wide range of co-benefits they provide – for food security and human health, ecosystem health and biodiversity.

11. The justification for the selection of an EBA approach is set out in the FS, though it is noted that less technical detail is provided on the green solutions in section 4 in the options analysis. This is true also of section 7, which provides much less detail on the means of catchment reforestation, in comparison with the technical guidance on agroforestry. In answer to a question from the independent Technical Advisory Panel (iTAP) about how detailed protocols would be developed for reforestation, as well as monitoring and community engagement, the AE responded that an operations manual would be developed in year 2 of the project, in collaboration with participating communities. The iTAP welcomes the planned annual monitoring of dry season baseflow at four locations along the main stem of the TR river – to provide an indicator of improvements in soil infiltration capacity as a result of upstream reforestation. Technical monitoring protocols will be developed as part of the manual, using a low-tech "[infiltration rate tool](#)" and "[water flow metering](#)" techniques to "increase community engagement and limit technical dependence/maintenance without compromising accuracy of data".

12. In response to a question about how the project's actual results in mitigating flood risk will be tracked, the AE highlighted the significant budget allocation (through Heifer and UNDP co-finance) for impact analysis, evaluation and studies to monitor results – including, if/when a flood does occur, conducting post-flood community surveys in the project intervention sites, and integrating this with official data gathered by government and United Nations disaster risk

² Water that seeps into the stream from the groundwater, as opposed to from surface run-off.

teams. The hydrographic modelling outlined in the FS will be updated, both with geographical information system information to be collected in year 1 as a baseline, and with a full hydrographic analysis in year 8, to be presented to government for future planning and replication in other areas of the country.

13. **Conflict sensitivity analysis:** In response to a query about the potential for project impacts to be limited by the country's fragile security situation, with lawlessness in and around Port-au-Prince, and gangs now controlling much of Haiti's territory, UNDP indicated that the activities are located in relatively secure parts of Haiti, with the partial exception of Gros-Morne and Bassin-Bleu. Nonetheless, it is acknowledged that "the security issue should be considered a destabilizing factor that makes the operating context more prone to conflict and increases risks. To mitigate the impact of the security situation and maximise the peacebuilding potential of the project, a proactive approach to conflict sensitivity will be adopted and will be promoted by UNDP. This will begin with a detailed conflict analysis of project sites³" as outlined in the environmental and social management framework (section 6.1.6). A security plan and a conflict assessment will be developed for each commune, as part of the community land-use planning process, that "will map issues and dynamics that bring groups together and divide them, particularly with regard to natural resources, and assess more deeply how the project may interact with local conflict dynamics". Procurement logistics are planned to circumvent the capital city, and project benefits are not provided in cash, in order to reduce the risk that beneficiaries and staff are targeted by criminal gangs in the delivery of project activities.

14. The iTAP appreciates the conflict sensitivity analysis conducted, and the process to continue conflict assessment and mitigation during project implementation. We note that work on the ground in the north-west (i.e. components 1 and 2, and outputs 3.2 and 3.3), will be supported with supplies from the region or airlifted in by the United Nations Humanitarian Air Service, with no supplies coming by road from Port-au-Prince, which is worst affected by gang violence and roadblocks. Having a project management office in the local directorate of the MoE in Port-de-Paix will also mean that staff do not need to travel from the capital in the south, and that they will help to enable effective delivery. It is noted that similar arrangements are in place for a number of other rural projects in the MoE-UNDP environment portfolio, which currently has a 90 per cent delivery rate.

15. We remain concerned, however, about the ability of the project to carry out the national level work of output 3.1 effectively, strengthening the national governance framework for IWRM and scaling up the approach by establishing catchment and sub-catchment management structures in 31 catchments across the country. This aspect, important for maximizing adaptation impact, would appear very difficult to achieve under the current conditions, with travel by MoE officials from the capital restricted. Even without such restrictions, it would be challenging to establish new structures in other catchments, without the kind of in-depth capacity development happening in the TR catchment. Under the current security situation, it would seem near impossible.

16. Overall, the project's adaptation impact in reducing flood risk is dependent on the success of the reforestation and agroforestry efforts, as well as the development of the governance framework for climate-risk informed land and water management, all of which are taking place in a challenging socio-economic and security context. The work on the ground in the TR watershed may have a medium to high impact potential, while the impact potential of output 3.1 may be low, given the anticipated challenges in implementation. The iTAP assesses the impact potential of the project as medium overall.

1.2 Paradigm shift potential

Scale: N/A

³ Written responses to the iTAP questions received from UNDP on 17 May 2023.

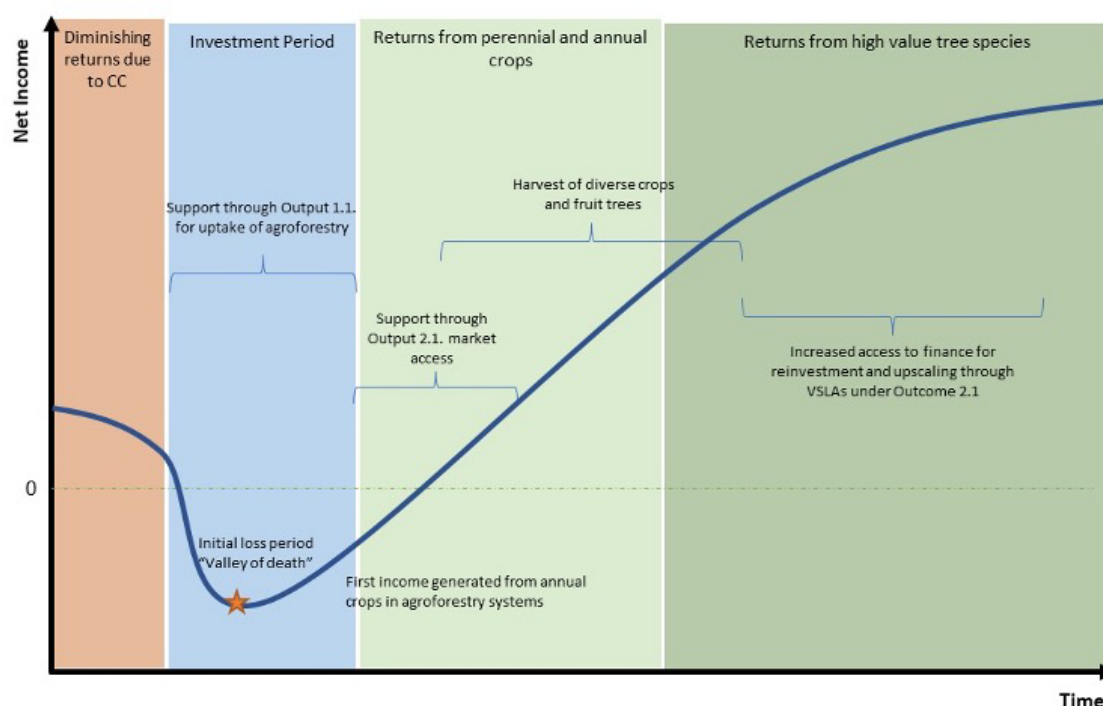
17. **Overall paradigm shift:** The proposed project intends to catalyse a paradigm shift in Haiti's approach to managing climate change intensified flooding in the TR watershed and beyond – by initiating a shift away from the present fragmented, uncoordinated decision-making and unsustainable land-use practices towards an integrated approach to land and water resource management that takes into account the current and projected impacts of climate change. Without GCF support, the GoH will continue to operate under the existing paradigm of reactive recovery from the impacts of climate change intensified disasters, including severe flooding, and failure to prepare for slow-onset events like drought. The project aims to incentivize this shift and make it sustainable through an integrated approach – restoring strategic areas of the TR catchment (C1), while promoting sustainable incomes based on regenerative agroforestry (C2) and building a governance framework that supports rural committees to plan land use and manage catchments effectively for climate risk (C3).

18. The lead role of MoE in executing the project, and the strong emphasis on governance and capacity development for local government, national government and community structures, help to embed the integrated land and water management approach to climate change adaptation for the long term. A pathway to scaling up is built into the project through output 3.1, although The iTAP notes the likely challenges that will be faced in carrying out national government level work, and work across all catchments nationwide, given the security context. The intended scale-up involves amendments to Haiti's draft Water Act to include, *inter alia*, provisions for the adoption of IWRM structures across the country, and the subsequent establishment of 31 new catchment IWRM committees, each with sub-catchment water resource users associations. In terms of sustainability, the iTAP notes the commitment of MoE to develop and sustain IWRM structures, and its letter to GCF, committing USD 7.5 million for post-project operations and maintenance (O&M), supporting the "maintenance of the cost of the green infrastructure set up by the project" (annex 22).

19. **Challenges of reforestation:** The iTAP posed a question to the AE about whether the intended paradigm shift might be overambitious, particularly the attempts to reforest the TR catchment on a sustainable basis. In the context of Haiti's extreme development challenges, many past reforestation efforts have failed to sustain tree growth, and stakeholder structures set up to support these efforts have often collapsed once donor funds dry up. The AE acknowledged the depth of the challenges, but pointed to the systematic analysis conducted by UNDP and MoE of 22 past and current related initiatives (as shown in the FS) to understand the elements of success and failure, and to design the proposed project accordingly. Based on the written responses and verbal interactions with the AE, the iTAP has identified three key elements built into the project design that improve the chances of achieving the desired shift, despite the difficult circumstances:

- (a) **Agroforestry business as an incentive:** Despite decades of support to agroforestry, many farming communities maintain a focus on annual crops, because of the short-term returns, and a lack of access to finance. In the proposed project's output 1.2, in addition to GCF investment in reforesting 7,700 water towers (USD 5 million), the largest single portion of the GCF investment (USD 11 million) will go into best-practice agroforestry support on 17,740 ha of farmers' own land (including women-headed households). Technical assistance will focus on maximizing the use of native and exotic tree species with value for fruit, medicine and fodder, in combination with food crops, for which they provide shade and soil enrichment, as well as support for beekeeping and small livestock. The Heifer co-finance in outputs 2.1 and 2.2 will sustain this investment by enabling commercial agroforestry as a long-term incentive for keeping trees standing – through technical support to farmer field schools, promotion of cooperative nurseries and agri-processing facilities, and access to appropriate micro-loans through VSCAs.
- (b) A transition model underpinning the paradigm shift for the agroforestry interventions is depicted in the figure below taken from the FP, demonstrating how the proposed eight-

year project will allow farmers to bridge the ‘valley of death’ of the first years of investment in agroforestry, when costs are higher than revenues. By providing initial inputs to farmers, the costs of implementing agroforestry are lowered, enabling the farmer to shift to this more sustainable land use and guarantee income from year 1 (through harvests of annual crops intercropped with trees). Over time, with support on access to markets and finance, returns are generated from faster-maturing fruit trees, and eventually higher returns from high-value tree species, enabling reinvestment and upscaling of operations. The diagram below is from paragraph 148 of the FP:



- (c) **Social capital approach to community engagement:** While many past reforestation efforts have worked hard to engage communities in tree planting, and sometimes also in monitoring tree survival, the approach to community engagement has not necessarily been a bottom-up approach, driven primarily by communities’ own needs and knowledge. As explained by the AE, “[p]ast projects have mainly been guided by top-down directives that, while taking into account socialization with the communities, have not made them co-participants in the design of interventions that will be made within their own lands. This in turn has resulted in lack of ownership and sustainability as well as success of the forestry investment themselves”⁴.
- (d) By contrast, the proposed project has been designed around a social capital approach,⁵ in which community engagement is focused explicitly on building social cohesion and shared value, as a sound basis for sustaining project interventions. Both reforestation and agroforestry interventions will be designed to take place on and in walking distance of farming households’ own lands, and will be guided by community land-use plans developed through a participatory process, with community-based agreements to be signed governing reforestation plantings and their long-term security. A diverse range of agroforestry species will be used, to be selected by farmers themselves based on traditional usage and preferences. Intensive capacity development will focus on understanding the tangible benefits of the interventions to the communities – in terms

⁴ Written responses to the iTAP questions received from UNDP on 17 May 2023.

⁵ The AE highlighted [a resource published in Community Development in 2021](#), which guided the project proponents, outlining a social capital approach to agroforestry.

of soil fertility, crop yields, nutrition, water availability, health and sanitation, and slope stability, in addition to the less direct benefit of downstream flood prevention. The principles of the farmer field school approach will be built into the reforestation work (as well as the agroforestry), with farmer's association and women's groups, guided by locally based agronomists and field technicians, learning surveying and monitoring skills, and taking long-term responsibility for the areas of land designated by themselves through the land-use plans as having value through reforestation.

- (e) In answer to the iTAP question about how restoration work outside people's own farms will be incentivized, the AE explained that the project will apply the traditional 'Kombit' system to mobilize communal labour for tree planting, irrigation and agri-processing, which, according to the AE "has proven to be effective not only in delivering project impacts but also in enhancing social cohesion and ownership"⁶. This is reinforced by Heifer's proven '[Passing of the Gift](#) (*pase kado*)' approach, in which beneficiaries are required to "pass on the gift" to their community through capacity-building and support for replication. "This system has been used by Heifer successfully in various type of projects and is a key part of the project's sustainability strategy, as it will create a sense of local ownership, community and appropriation and allowing for the implementation of community best practices that are locally appropriate"⁷. Stakeholder consultations demonstrated that this approach would be welcomed and would be seen as less of a top-down approach than that taken in many past projects. Additional incentive measures that have proven to be successful include competitions around environmental protection and monitoring, and special campaigns with students and women's associations that are involved in the protection of seedlings in public areas.
- (f) **Funding enabling conditions, not running costs:** A number of past initiatives in Haiti have set up their own project-specific multi-stakeholder structures and funded their operations, only to find that these structures have not survived long-term (see useful analysis on this in a [May 2021 report](#) on the USAID Haiti Reforestation Project). The proposed project utilizes structures that already exist, or for which legal provision is already made by government. This includes Heifer's support in output 2.1 for existing communal development councils in the seven target communes to produce community land-use plans, and GCF support in outputs 3.1 and 3.2 for a new (but required by law) Trois Rivières Catchment Water Resources Management Committee and sub-catchment water resource user associations. Most importantly, the project will not provide for the running costs of these organizations, which will largely be expected to be supplied in kind by local government and by community organizations (e.g. producer associations, women's groups). Instead, the project invests in creating the necessary enabling conditions for these structures to operate and bring value to communities, such that their continued existence is enabled.
- (g) As indicated by the AE to the iTAP: "As evident in the project's budget, a key part of the project's sustainability approach is in fact not covering the full cost of the community structures that will be supported by the project, rather creating the enabling conditions (feasibility studies, supporting the creation of common bylaws, networking, expert support, covering costs related to training workshops of their members) to allow these to operate and link their work towards sustainable water and land management as evidence[d] by the various benefits that will be created (improved health, food security, water supply, access to financial capital, diversified livelihoods)... The project further will follow MoE rules on the ground that are applied when there is no project, and avoid

⁶ Written responses to the iTAP questions received from UNDP on 17 May 2023.

⁷ Written responses to the iTAP questions received from UNDP on 17 May 2023.

paying travel and per diem cost for committee members that the MoE will never be able to sustain”⁸.

- (h) This approach also applies to the enabling conditions for VSCAs, selected by the project proponents as the best way to support producers (through Heifer co-finance) to invest continuously in agroforestry, rather than outside microcredit provision, which the AE convincingly demonstrates is not affordable in this context. Output 2.2 is focused on creating the enabling conditions for VSCAs and de-risking their formation – including feasibility studies, market assessment and financial mapping that will inform the establishment and registration of the 150 new structures, and help them shape their own rules. According to the AE, “[p]utting in place the VSCAs will create a culture of financial management within the communities as well as creating resources for the access to collateral (by providing financial inputs through the pooling of savings that are provided as peer-to-peer loans that are sufficient to cover the purchase of capital goods), while providing capital under terms that are considered to be socially appropriate and sustainable by communities. In time this will also enhance the capacity for saving and create a community-validated financial history that may come in use when these same producers look towards more formal financial institutions”⁹.

20. **Food coupon strategy:** An interesting element of the project’s theory of change in bringing about the paradigm shift relates to the provision of food coupons (exchangeable for produce in local markets) to 934 highly food-insecure families in the project sites, using Heifer co-finance in output 2.2 for a temporary (eight-year) intervention. Following a query from the iTAP, the AE indicated that this is planned as a transition measure: “The logic behind the food coupons responds to a key driver of deforestation, which is the need to address short term food insecurity as a result of flood-related losses, through the selling of charcoal as a livelihood option... providing access to a basket of produce to address immediate food insecurity needs, while allowing the space required for the agroforestry interventions to take hold”¹⁰. In response to concerns about sustainability and potential market distortions, the AE explained that these are addressed by gradually phasing out the coupons from year 6, and by designing the mechanism to work in association with local food producers, helping to create a market for the newly developed products being supported by the project.

21. **Excluding charcoal value chains:** the iTAP notes that the proposed project provides agroforestry support for a wide range of benefits and income generation opportunities. Tables 3 and 4 in the FP list useful indigenous and non-invasive exotic tree species that can be used to generate income and benefits, including food, shade, soil fertility, fodder, medicines, hedges and erosion control. Timber (poles, planks, fuelwood) for household use is also envisaged.

22. The iTAP notes that inefficient charcoal production and consumption contributes to greenhouse gas emissions and air pollution, and suggests that it is preferable that GCF funds are not utilized for planting trees to supply charcoal value chains. It is noted that the environmental awareness and capacity-building in activity 1.1 includes training in the 118 target communities on “the role of the community in sustainable charcoal production and reforestation, as a strategy for increasing climate resilience in the watershed”, which is welcomed. This could include awareness of the negative impacts of traditional charcoal production, and the extent to which these can be mitigated through the use of improved low-emission charcoal kilns (for production), as well as fuel-efficient and/or cleaner cookstoves (for consumption).

23. The iTAP proposes inclusion of a covenant in the funded activity agreement, that the AE shall ensure that GCF Proceeds under Activity 1.2 of the project are not used for producing wood for sale into charcoal value chains. The project’s interventions in this activity should focus

⁸ Written responses to the iTAP questions received from UNDP on 17 May 2023.

⁹ Written responses to the iTAP questions received from UNDP on 17 May 2023.

¹⁰ Written responses to the iTAP questions received from UNDP on 17 May 2023.

on supporting agroforestry with other benefits, such as sale of fruits, provision of shade for intercropping with food crops, harvesting leaves with medicinal benefits, or slope stabilization.

24. Overall, the project's paradigm shift potential is significant, and mechanisms have been put in place to improve the chances of sustainable outcomes on the ground in the project sites. Replication and scale-up may be difficult, however, given the challenges noted in section 1.1 above, around work in other catchments. The paradigm shift potential of the project is assessed as medium.

1.3 Sustainable development potential

Scale: N/A

25. The proposed project will contribute towards the achievement of at least 5 of the 17 United Nations Sustainable Development Goals (SDGs), including: i) SDG 2 – Zero hunger; ii) SDG 5 – Gender equality; iii) SDG 6 – Clean water and sanitation; iv) SDG 11 – Sustainable cities and communities; and v) SDG 15 – Life on land. In addition, project interventions will generate several environmental, social, economic and gender-responsive co-benefits, as described below.

26. **Environmental co-benefits:** The restoration of degraded forests through targeted reforestation interventions and expanded agroforestry practices, as well as the implementation of IWRM practices, will generate several environmental co-benefits. Reduced surface run-off and soil erosion will contribute to soil conservation, with increased water infiltration and storage capacity of the soil. Reduced erosion and flooding will also mean less contamination of food and water by heavy chemicals and/or pesticides transported by flooding. Water quality will improve because of reduced sedimentation, including in near-shore marine environments. Implementing agroforestry and reforestation activities will contribute to enhancing the carbon sequestration potential of the restored landscapes. This expected co-benefit is quantified at 3,639,950 tonnes of carbon dioxide equivalent over the 25-year project lifespan, and monitoring is planned and budgeted for – see the monitoring and evaluation plan, annex 11. Lower-emission processes for charcoal production and cooking will also be promoted through the project's capacity-building interventions.

27. **Social co-benefits:** Implementing targeted landscape restoration and IWRM, as well as enhancing the technical capacity of smallholder farmers and communities in the TR watershed to utilize climate-resilient agricultural practices will lead to improved food security, social empowerment, and greater energy security, as well as health benefits. Training communities and individual farmers in climate-resilient agroforestry practices, as well as expanding access to agricultural markets and credit facilities will improve household nutrition and increase income. Project interventions will also generate positive social co-benefits for beneficiary communities in the form of greater social empowerment, with revived communal efforts for irrigation, and cooperation on sustainable use of non-timber forest products. This will be reflected in the development of community land-use plans and the establishment of communal development councils, catchment IWRM committees and sub-catchment resource user associations that will improve the decision-making power of traditionally marginalized groups, including women and local communities, on natural resource management.

28. **Economic co-benefits:** Flood management interventions are expected to reduce peak surface run-off and floodplain extents in Haiti's TR watershed, reducing economic damages to existing infrastructure, generating public savings and indirectly increasing the fiscal space available for other social and developmental investments. Additional employment opportunities will be created through the project interventions in agroforestry and reforestation, such as the establishment of nurseries, and operation of agri-processing cooperatives. Encouraging farmers to participate in intercropping of cash crops, timber trees as well as a variety of food crops and fruit trees will provide them with an important economic safety net to reduce the economic impacts of climate change-induced flooding on their production. These impacts will be further reduced by expanding agricultural markets, which will increase the resilience of the agricultural

sector to external climate change shocks, such as flood and hurricane impacts. Increasing access to agricultural finance through VSCAs will allow beneficiaries to purchase necessary agricultural inputs and improve their yields.

29. **Gender co-benefits:** Approximately 52 per cent of direct beneficiaries of the proposed project are women. Haiti has a high percentage of female-headed rural households (approximately 39 per cent), the majority of whom are smallholder farmers. These women will benefit from training on climate-resilient agricultural practices and sustainable land management, as well as the extension of agricultural finance and provision of climate-resilient agricultural inputs. In addition, implementing an integrated approach to flood management in the TR watershed will reduce the rate of displacement, and in turn reduce the amount of violence, including sexual violence, experienced by women. Reduced flooding will result in a decline in the spread of waterborne diseases, as the contamination of water supplies will be reduced, with a disproportionately positive impact on women, who are most exposed to contaminated water because of gender roles in the home.
30. The iTAP assesses the project's sustainable development potential as high.

1.4 Needs of the recipient

Scale: N/A

31. **Vulnerability of Haiti:** Ranked 163rd out of 191 countries on the 2021 United Nations Human Development Index,¹¹ Haiti is both a small island developing State (SIDS) and a least developed country (LDC), and faces multiple development challenges related to food and economic security, health, education, political violence and environmental degradation. Poverty is endemic, with approximately 59 per cent of the population living below the World Bank poverty line, and 74 per cent below USD 5 a day¹². Poverty has been exacerbated by a series of natural and climate change intensified disasters over the past two decades, including severe storms, flooding, landslides, drought, the devastating earthquake that rocked the country in 2010, the category 4 Hurricane Matthew (which left 806,000 people in need of urgent food assistance in 2016), and an earthquake which devastated the country's southern regions in August 2021. On the 2021 Climate Risk Index, Haiti is third among countries that were most affected by extreme weather events from 2000 to 2019¹³.
32. The country is currently locked into a reactive cycle of costly disaster response and rebuilding that is preventing the adoption and implementation of sustained climate-resilient approaches to managing the cross-sectoral impacts of flooding in the country. These impacts are further exacerbated by the country's fiscal challenges and dependency on neighbouring countries for importing critical supplies. For example, approximately 50 per cent of the national budget comes from foreign aid, and 60 per cent of food requirements are imported, despite agriculture being the primary economic activity (40 per cent of the population is dependent on the agricultural sector for its livelihood, mostly through small-scale and subsistence farming). The FP quotes a statistic from the World Food Programme that 35 per cent of the population faced food insecurity in 2016¹⁴ – this has subsequently risen to 43 per cent in 2022¹⁵. The already alarming situation worsened in 2022 due to global conflict and the resulting economic and supply crisis, which has caused prices to rise and sharply decreased the purchasing power of Haitians, to the point that an estimated 4.5 million people are currently in need of food assistance¹⁶. The project addresses these challenges by employing cost-effective EBA and IWRM

¹¹ <https://hdr.undp.org/data-center/specific-country-data#/countries/HTI>

¹² World Bank Poverty and Inequality Platform, <https://pip.worldbank.org/home>

¹³ <https://www.germanwatch.org/en/19777>

¹⁴ Funding proposal, footnote 99

¹⁵ <https://www.wfp.org>

¹⁶ <https://www.wfp.org>

approaches to reducing flood risk and aiming to strengthen the governance framework of the whole country.

33. **Vulnerability of the Trois Rivières watershed:** Poverty is prevalent in the TR watershed, including parts of the Artibonite, North and Northwest departments. The Northwest department has the joint highest percentage of poor people in the country, with an extreme poverty rate exceeding 40 per cent, and including the city of Port-de-Paix, which is the third-most food-insecure area in Haiti. The TR region is particularly vulnerable to the impacts of climate change-induced flooding because of its steep topography and exposure to the impacts of hurricanes and tropical storms coming from the north-east. The cumulative negative impacts of climate change-induced flooding and extreme events are compounded by land degradation in the TR region, resulting from deforestation and unsustainable crop farming on the surrounding steep slopes, including annual crop cultivation which leaves the soil bare during most of the year, contributing to decreased infiltration and increased surface run-off in the catchment. This, in turn, has resulted in increased erosion and soil degradation within the catchment area, further exacerbating flood impacts downstream in the catchment. Outcomes 1 and 2, and outputs 3.2 and 3.3, focused on the TR watershed, address these issues.

34. **Current socio-political and economic situation:** The FP quotes 2017 figures on Haiti's high level of public debt. The iTAP notes that the country's public debt has nearly doubled since then and stood at USD 5.1 billion in 2022 – representing 25 per cent of gross domestic product and having risen rapidly from zero after the 2010 earthquake saw the cancellation of all outstanding external debt. With an external debt to export ratio of 93 per cent, projected to rise rapidly in the next decade, the country faces a high risk of external debt distress. The government is in no position to take loans for climate change adaptation and even allocating funding from the national budget for this purpose is challenging.

35. The country also faces severe socio-political instability, worsened since the assassination of President Jovenel Moïse in 2021, followed by a violent 7.2 magnitude earthquake a few weeks later, resulting in more than 2,200 deaths and USD 2 billion worth of damage in southern Haiti. Haiti has been hit hard by the recent global food price shock, with food inflation reaching 44 per cent in September 2022. This shock compounds the hardships of an already highly fragile country, also suffering an ongoing public health emergency (cholera) and serious security risks. With the country's small tax and market base as well as its high public debt, limited opportunities exist for the public and private sector to access financial resources for addressing climate change impacts, which is estimated to have cost the country approximately USD 20 million per annum in loss and damage in recent years, according to Germanwatch in 2020¹⁷. The grant funding by GCF addresses the country's need for support in the midst of all these challenges, whilst the Heifer co-finance helps to put agroforestry and agri-processing on a sustainable business footing, including through access to finance through establishing and empowering VSCAs through outcome 2 (activity 2.2).

36. **Institutional capacity needs:** Facing myriad environment, climate change, security and development challenges, the Government of Haiti is nonetheless committed to implementing rural development policies that improve the quality of life of its citizens, while more effectively managing climate change risks. The proposed project will help to build the capacity of a number of government entities, including the Ministry of Agriculture, Natural Resources and Rural Development (MARNDR) and the MoE to ensure the sustainability of the project's investments. The MoE has committed USD 7.5 million in kind to cover the O&M costs of the reforestation investments, *ex situ* conservation of genetic resources, and the continued work of the watershed and sub-watershed committees.

37. The AE has expressed high confidence in the materialization of the O&M support, on account of UNDP's own experiences, where various project investments that have been made

¹⁷ <https://www.germanwatch.org/en/19777>

through Global Environment Facility projects continue to be operational 10 years after the project's end (for example, the Centre de Propagation des Végétaux à Dosmond). Strengthening of the governance and technical capacities of institutions, and the establishment of appropriate legal frameworks for integrated land and water resource management have been built into the project in outcomes 1 (activity 1.1), 2 (activities 2.1 and 2.2), and 3 (activities 3.1–3.3) to address these needs. The AE acknowledges the need to build the capacity of the MoE as EE through project implementation, and the iTAP believes that this is crucial and will need to be prioritized, with resources committed to it.

38. Overall, based on the above, the needs of the recipient are rated as very high.

1.5 Country ownership

Scale: N/A

39. **Policy alignment:** The proposed project is strongly aligned with Haiti's national priorities, including the revised national adaptation programme of action (NAPA) of 2017, focused on soils, agriculture, coastal zones and water resources. The NAPA itself was designed to align with the country's earlier national environmental action plan, Strategic Development Plan for Haiti, interim cooperation framework and poverty reduction strategy paper, including their goals to ramp up agricultural production in a sustainable way, preventing land degradation, improving access to clean water, and reducing disaster risk. The proposed project aims to build on recent results of the national adaptation plan (NAP) process, supported by UNDP through a GCF Readiness and Preparatory Support project, by strengthening institutional and technical capacities and existing frameworks and systems. In particular, the project responds to the identified need to ensure that the NAP is implemented with a strong evidence base for adaptation planning and that priorities are subsequently integrated into the strategic development plan and disaster risk reduction plan and strategy.

40. The FP was not updated to reflect Haiti's revised nationally determined contribution (NDC) of 2022. It reflects on the project's consistency with the 2015 NDC and its adaptation priorities, including enhancing the resilience and sustainable management of 15 watersheds; and undertaking afforestation and/or reforestation activities of 137,500 ha of land. The iTAP notes that the new NDC includes in its non-exhaustive list of costed priority actions the "Afforestation of strategic hydrogeological places". Haiti's second national communication on climate change to the United Nations Framework Convention on Climate Change highlighted the importance of managing watersheds to address the impacts of climate change and emphasized the need to: i) implement climate-resilient solutions that will address the impacts of climate change-induced flooding on vulnerable communities and the main economic sectors (e.g. agriculture); ii) promote the adoption of an integrated approach to land and water resource management in the country; and iii) strengthen the cohesion between existing and planned initiatives focused on landscape and watershed management, informed by best practices generated from past projects and/or programmes across the country and Caribbean region. It is noted that the project's IWRM interventions build well on this approach, as well as the framework first set out in the 1999 Action Plan for Water Resources Management in Haiti.

41. **Stakeholder engagement:** The proposed project has been designed following extensive consultations with members of the Haitian government, public authorities, civil society and the private sector. Haiti's national designated authority, the MoE, has been fully involved in the development of this project since inception, through its contributions to the consultation process and inputs during field missions. UNDP has facilitated dialogue between the MoE, the MARNDR, and other relevant institutions, to confirm that the project responds to the adaptation priorities of the country. Following technical assessments of flooding risk, stakeholder consultations were held in the North-West, North and Artibonite departments of Haiti. Discussions were held with the regional branches of the MoE, Civil Protection Agency, Ministry of Economy and Finance, the National Directorate for Water Supply and Sanitation in the

Ministry of Public Works, Transport and Communications, the Regional Water Supply and Sanitation Office and irrigation associations. These discussions underlined the need for a coordinated effort toward improved flood risk management for increasing resilience to climate change.

42. **Capacity of entities:** As the EE, the MoE will be part of the project board and will lead the Project Management Unit through the designation of a National Director. The project will also report to the National Technical Committee for Climate Change. As the AE, UNDP will build on its substantial portfolio of IWRM projects in the Latin America and Caribbean region, in providing supervision and monitoring of the project. Based on a capacity assessment that identified gaps in the ministry's capacity, the UNDP Haiti country office, at the request of the GoH, will also provide some direct execution support to government in its EE role, using UNDP's financial management and procurement processes. A strict firewall will be maintained between UNDP's AE role and this execution support role. In practical terms, this means that different teams within the AE will be responsible for the two roles, with no overlap allowed. It is noted that the MoE will retain responsibility for drafting and approving all terms of reference, and approving payments to vendors and transfer of funds. The iTAP also notes the AE's intention to use its execution support role consciously to develop ministry capacity. The iTAP believes this is essential and recommends that a capacity development plan be developed by the project manager to spell out how this will be undertaken.

43. UNDP's support to Haiti in adaptation is framed within the NAP process, integrating diverse initiatives to enhance Haiti's national adaptive capacity, through the implementation of adaptation solutions and improved capacity for early disaster alerts. These efforts are supported by various projects within UNDP's portfolio that reinforce each other through a wider portfolio approach that is guided by the GoH. The project will coordinate with ongoing donor groups chaired by UNDP, including the Technical and Financial Partner Group for the Environment Sector and the International Community Support Group for Disaster Risk Management. In addition, the project will provide regular briefings to the technical secretariat of the National Committee on Climate Change and ensure alignment with the NAP implementation process. During project implementation, all target beneficiary groups will remain engaged to ensure that the benefits conferred by project interventions respond directly to the needs of these groups.

44. Country ownership of the project is thus seen to be high.

1.6 Efficiency and effectiveness

Scale: N/A

45. Of the project's total proposed budget of USD 31.29 million, 72 per cent of funds are requested from GCF, with 26 per cent in co-finance from Heifer and 2 per cent from UNDP. Although there is no government co-finance for the project, the provision of a letter of government commitment to financing long-term maintenance of green infrastructure is seen as positive. The proposed financial structure is adequate and reasonable in order to achieve the proposal's objectives, and the iTAP agrees that a 100 per cent grant is the most appropriate level of concessionality for the GCF funding, given: i) the public good of the project's activities and lack of financial reflows; ii) Haiti's status as a SIDS and LDC; iii) the level of debt of the country, with its limited capacity to repay loans; and iv) the fact that the project beneficiaries are poor smallholder farmers who need grants to make these initial investments.

46. An economic analysis was conducted, focused on determining costs and benefits associated with: i) implementing EBA interventions, including agroforestry and reforestation activities, to enhance the resilience of the TR watershed to climate change-induced flooding; ii) facilitating sustainable catchment management practices with the strengthening of agricultural value chains; and iii) supporting the GoH in the development of a national-level policy framework to facilitate IWRM in Haiti. The economic analysis estimated the net present value of

interventions to be USD 376 million, with an internal rate of return of 40.7 per cent, representing a highly cost-effective spend. Project benefits included in the economic analysis were estimated on an annual basis, comparing with- and without-project scenarios in the presence of climate change. This included health benefits (avoidance of water-borne disease), rural livelihoods benefits (agroforestry income) and infrastructure benefits (avoided flood damage to homes and infrastructure). It is noted that the first two sets of benefits would not have accrued, had a 'grey' infrastructure approach to flood mitigation been chosen over a 'green' ecosystems-based solution.

47. Although the proposed project is not a climate information/early warning system project, its efficiency is enhanced by the fact that its design complements ongoing initiatives to build Haiti's meteorological capacity and strengthen early warning of disasters. In particular, the project will build on the past decade of support from the Government of Canada to Haiti to rebuild and modernize the national meteorological and hydrological service, [implemented by the World Meteorological Organization](#) (WMO). UNDP is also working with the WMO to improve multi-hazard alerts, including for flooding, through: (i) the Climate Risk and Early Warning Systems (CREWS) project that will develop an environmental information system; and (ii) a Least Developed Countries Fund project that will enhance capacity for the delivery of climate information analysis to guide water management planning. As part of output 3, where national government capacity for climate risk-informed IWRM is being built, hydrologists from the Hydrometeorological Unit of Haiti will be consulted and trained through a learning-by-doing approach in the recreation of the flood modelling that will be undertaken by the project to gauge its impacts, equipping them to recreate the analysis in other areas.

48. The project's overall efficiency and effectiveness is rated as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

49. The iTAP believes that this climate change adaptation project from the Government of Haiti, supported by UNDP and with significant co-finance from Heifer, has a well thought through and integrated project design, and represents an important initiative to build societal and community resilience in a challenging context.

50. The iTAP endorses this funding proposal, subject to inclusion of the following covenant in the funded activity agreement:

- (a) the Accredited Entity shall ensure that GCF Proceeds under Activity 1.2 of the project are not used for producing wood for sale into charcoal value chains.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP208)

Proposal name:	Enhanced climate resilience in the Trois-Rivières region of Haiti through Integrated Flood Management
Accredited entity:	United Nations Development Programme (UNDP)
Country/(ies):	Haiti
Project/programme size:	Small

Impact potential

We appreciate the detailed analysis and considerations of the ITAP in regards to the project's impact regarding its interventions on the ground which have been assessed to have a medium to high impact potential. We also recognize ITAP's concerns regarding the replicability of the project's governance component arising from the current national context.

To this effect the project will ensure active coordination with the GoH to ensure that project actions are streamlined within existing national and local structures and create a basis of replication through local and national frameworks and by enhancing capacities. The GoH has informed UNDP that the current project will be a cornerstone to its own National Adaptation Framework and will serve as a demonstrative experience for adaptation action. While we acknowledge the challenging conditions on the ground to deliver results in the project, particularly Activity 3.1, the strong commitment from the GoH and the above described measures will help deliver the adaptation benefits expected in the project.

Paradigm shift potential

The level of analysis undertaken by the ITAP in its assessment of the project's approach is highly appreciated. As has been highlighted in the assessment, the project has built in lessons learned from both failed and positive experiences around reforestation and agroforestry in Haiti and will look to put these in action to enable the project's theory of change focusing on sustainable integrated water and land management as an adaptation solution to climate induced flooding.

We recognize ITAP's concern in regard to the challenging implementation context. However, as it is stated within the assessment, measures have been put in place to ensure a paradigm shift as a result of a community oriented approach as well as by integrating a strong sense of national ownership around the project interventions.

We also recognize ITAP's concern in regard to the charcoal value chains and agree to the inclusion of the covenant as suggested.

Sustainable development potential

We appreciate and are in full agreement of ITAP's assessment regarding the HIGH sustainable development potential of the project that is in turn related to the project's ecosystem based approach.

Needs of the recipient

We appreciate ITAP's assessment and are in agreement of the VERY HIGH needs of the project's recipients that was evidenced during the project's preparation and has been reiterated at various moments by the GoH. An initial baseline study will be conducted to ensure that the recipient needs are comprehensively assessed.

Country ownership

We appreciate ITAP's assessment as it pertains to the project's HIGH degree of country ownership. We confirm that throughout project implementation UNDP will work with the MoE and other ministries to ensure that national capacities are enhanced as a result of this project.

The MoE is fully engaging with its local structures to already put in place an operational guidebook that will be finalized based on additional consultations as part of the initial baseline study. This operational guidebook will fix roles and responsibilities of each local counterparts under the MoE leadership to ensure proper monitoring and sustainability of the project activities on the ground.

Efficiency and effectiveness

We appreciate ITAP's assessment in regards to MEDIUM to HIGH efficiency and effectiveness of project intervention. As has been stated by the project's economic and financial analysis and is highlighted by the ITAP, the project's internal rate of return is calculated at 40.7% taking into account the multiple benefits that will arise through its ecosystem based adaptation strategy including a reduction of flood losses, improved livelihoods and health.

Overall remarks from the independent Technical Advisory Panel:

ITAP's full endorsement of the project is appreciated as well as its comprehensive analysis of the project's strategy, its benefits and potential impact. We also believe that the project will create an important foundation and model for community based resilience in Haiti and look forward to its implementation.

UNDP recognizes ITAP's concern regarding charcoal value chains and agrees on the inclusion of the suggested covenant.



Enhanced climate resilience in the Trois-Rivières region of Haiti through Integrated Flood Management

Annex VIII – Gender Assessment and Action Plan

Table of Contents

1. Objective of the Gender Assessment and Action Plan	2
2. Overview of the Proposed Project	2
3. Overview of Gender Issues in Haiti	4
4. Facets of gender inequality in Haiti relevant for the Project design:	5
5. Gender-based Violence (GBV) and Sexual Exploitation, Abuse and Harassment (SEAH)	18
6. Legal, Policy and Administrative Frameworks for Protecting Women and Gender Equality	20
The Ministry of Women and Women's Rights (MCFDF)	21
The Office for the Fight against Violence Against Women	21
Criminal Code	21
Climate change governance and gender in Haiti	22
7. Project-specific design consideration resulting from the Gender Assessment	23
8. Recommendations from Haiti's existing on Climate Resilience and Gender Projects and Programmes.....	24
9. Gender Action Plan (GAP)	27
Activity-level Gender-Responsive Budgeting envisioned for the Project (in USD\$)	45
Proposed SEAH Mitigation Measures	45
Please refer to Section 6.1.4. within Annex 6b (ESMF) for proposed actions to be implemented by the project mitigate SEAH.	45
Appendix 1: Glossary and List of Stakeholders	46
Appendix 2: List of stakeholders consulted	48

1. Objective of the Gender Assessment and Action Plan

This Gender Assessment and Action Plan (GAAP) – is a supporting document for the design of the following proposed Green Climate Fund (GCF) project: *Enhanced climate resilience in the Trois-Rivières region of Haiti through integrated flood management*. It is the Annex 8 of the Funding Proposal, which is being proposed as a full-scale project by the Accredited Entity (AE) – United Nations Development Programme (UNDP) and has been developed in close coordination with the Government of Haiti.

The main objective of this Gender Assessment is to screen the gender context which will inform where the GCF project will be implemented, and to subsequently strengthen the gender-responsive actions through the project, to ensure that the benefits of the project accrue equitably to women, men, boys and girls in the targeted projects sites selected with country partners and stakeholders in Haiti.

In particular, this gender assessment aims to provide an analysis of gender equality and women empowerment in Haiti, with a specific focus on the gender differentiated impacts of climate change and how the findings can be used to drive the design of an adaptation intervention, that is gender-responsive and (where possible) gender-transformative. The objectives of the GAAP are as follows:

- To align the proposed project design with Haiti's national priorities on gender as well as GCF's Gender Policy and Action Plan¹ and UNDP's Gender Equality Strategy 2022 - 2025²
- To incorporate information and lessons learned from the gender assessment conducted for this project, and the previous national gender studies and evaluations in Haiti to both understand the context, and ensure that project activities are gender-responsive and tailored
- To present gender-related results from stakeholder consultations and involving women, men, girls and boys residing in the areas where the Project will be developed from an early stage in project design
- To integrate gender considerations into project indicators, goals and activities, and identify ways in which women can act as leaders and decision-makers in community-based adaptation interventions.

2. Overview of the Proposed Project

The Government of Haiti (GoH) is requesting US\$25 million in GCF grant finance to implement climate-resilient integrated water resources management (IWRM) to respond to the impacts of flooding on the Trois Rivières (TR) watershed. This GCF investment will be supported by the Heifer International, which has committed US\$8.3 million in co-financing for the implementation of project interventions.

The proposed project will seek to contribute to GCF's paradigm shift objective of increased climate-resilient sustainable development by enhancing the resilience of Haiti's Trois-Rivières (TR) watershed to the impacts of climate change-induced flooding.

¹ The GCF adopted a revised version of its 2014 Gender Policy and Action Plan on June 2018 in Korea. The revised Policy addresses pertinent issues on gender and climate change: the expansion of gender mainstreaming beyond the preserve of 'women's issues'; and the identification of synergies with the in-house Indigenous People (IP) Policy as well as the United Nations Framework Convention on Climate Change (UNFCCC)'s Gender Action Plan (GAP), Sustainable Development Goals (SDGs) and Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW).

² UNDP Gender Equality Strategy: <https://www.undp.org/publications/gender-equality-strategy-2022-2025>

Haiti is particularly affected by the climate change impacts of increasing frequency and intensity of floods, tropical storms and hurricanes. Between 1975 and 2012, such climate change-related events resulted in damages and losses equivalent to 2% of the country's GDP on average per year.

In 2004, one storm — Hurricane Jeanne — caused severe flooding in northern Haiti. The flooding caused by this storm resulted in approximately US\$41 million in damages, more than 2,800 deaths and the need to remove approximately 592,000 m³ of mud that clogged the drainage networks and infrastructure across the country. Since 2012, the impacts of climate change have continued to negatively affect the population of Haiti. For example, the Category 5 storm — Hurricane Matthew — that struck Haiti in 2016 caused widespread damages and economic losses and negatively affected more than 2 million people (20% of the country's population) primarily located in the poorest regions of the country. The storm also inflicted economic damages and losses estimated at more than 22% of the country's GDP in that year (2016), including the losses of up to 90% of crops and livestock in some areas of the country. In addition, it caused a sharp increase in the number of cholera cases experienced throughout Haiti.

The intense rainfall associated with extreme climate events such as Hurricane Matthew also leads to severe flooding, which exacerbates these associated negative impacts. Such high-impact flooding has been a regular occurrence in Haiti in recent decades. During the period 1993–2012, for example, Haiti experienced 31 floods, causing the deaths of approximately 2,870 people and adversely affecting approximately 9 million people through livelihood losses and personal damages. Of these flood events, 10 were directly attributed to hurricanes and a further 16 attributed to extreme rainfall events.

Implementing a transformative approach by making adaptation investments is crucial to ensure that the GoH can improve flood response, and improve watershed management — thus ushering a shift from climate impacted-communities towards climate-resilience communities. Specifically, the funds will be used to:

- implement agroforestry systems and rehabilitate 'water towers' through reforestation of degraded landscapes at priority intervention sites;
- enhance technical and institutional capacity for productive climate-resilient land management at the national and local levels; and
- establish the required governance framework for integrated water resources management (IWRM) to support the climate-resilient land management systems and facilitate sustainable use and management of water resources over the long term.

The combined effect of project interventions will result in the adoption and implementation of a climate-resilient, integrated approach to flood management that can be readily scaled up and replicated nationally and across the Caribbean region. The Haitian government's prioritization of flood resilience and watershed management, and food security as key elements of building resilience to climate change threats is clearly demonstrated by several policies and assessment reports, including the National Adaptation Programme of Action (NAPA) highlight the importance of managing watersheds to combat the extreme climate risks. The Risks Disaster Management National Framework (PNGRD), which addresses vulnerability in Haiti, highlights the importance of harmonizing the efforts and applying better practices for watershed management. National Watershed Policy, an initiative by the Ministry of Agriculture, Natural Resources and Rural Development (MARNDR), focuses on building resilience through participatory planning.

Both UNDP and the Haitian government recognize that social issues – which include gender dimensions, sociocultural norms and factors affecting the marginalization of particular groups within communities – form an integral part of the climate change driven flood management process. In addition to extreme weather events linked to climate change, factors of vulnerability, exposure and capacity to anticipate, adapt to and cope with flood risks heavily determine impacts of flooding. Therefore, effective integrated flood management requires appropriate options for managing flood risks for different social groups.

Social norms and values also determine how well the negative impacts of flooding can be overcome and how well the positive effects of flood management use can be utilized and shared equitably. Therefore, the proposed project will give special attention to the gender and social dynamics that influence access to flood management decision-making, with emphasis on promoting the empowerment of women and other disadvantaged groups.

3. Overview of Gender Issues in Haiti

Haiti's Human Development Index – HDI – value for 2021 is 0.535, which puts the country in the Low human development category, positioning it at 163 out of 191 countries and territories.³

The Gender Development Index – GDI – measures gender gaps in achievements in three basic dimensions of human development: health (measured by female and male life expectancy at birth), knowledge (measured by female and male expected years of schooling for children and mean years of schooling for adults aged 25 years and older) and living standards (measured by female and male estimated GNI per capita). It is the ratio of female to male HDI. The 2021 female HDI value for Haiti is 0.506 in contrast with 0.564 for males, resulting in a GDI value of 0.898, placing it in Group 5.

The Gender Inequality Index – GII – measures gender inequalities (the loss in human development due to inequality between female and male achievements) in three key dimensions – reproductive health, empowerment, and labour market. Haiti has a GII value of 0.635, ranking it 163 out of 170 countries in 2021. Reproductive health is measured by maternal mortality ratio and adolescent birth rates; empowerment is measured by the shares of parliamentary seats held and population with at least some secondary education by each gender; and labour market participation is measured by the labour force participation rates for women and men.

Similarly, the Social Institutions and Gender Index – SIGI – of the Organization for Economic Cooperation and Development (OECD) calculated a value for Haiti of 0.1466 in 2014, suggesting that discrimination against women is medium.⁴

This implies that the country faces consistent challenges in the area of health and education, political participation, and employment that by and large unevenly affect women and girls. Persistent inequalities across the social, economic and political spheres of life are intricately related to climate change vulnerability. Projects that understand and address these through gender mainstreaming will enhance the climate resilience of women and ultimately increase the chances for more effective risk prevention and/or mitigation⁵. The assessment and analysis herein focus on and highlight the

³ <https://hdr.undp.org/data-center/specific-country-data#/countries/HTI> (2021)

⁴ OECD. Social Institutions and Gender Index 2014. Country Profile <http://www.genderindex.org/ranking>

⁵ The Haiti Gender Shadow Report (A Coalition Gender Shadow Report of the 2010 Haiti Post-Disaster Needs Assessment (PDNA)): <https://reliefweb.int/report/haiti/haiti-gender-shadow-report> (2010).

differentiated impacts that climate change and flooding have on women, men, girls, and boys, reflecting their differential capacity to adapt to climate change and respond to natural disasters.

Methodology

This document is based upon available data and information retrieved from a review of literature and key documentation from the Government of Haiti, reports from multi-lateral institutions and development agencies, and interviews and field visits with relevant stakeholders during several missions by the project development team in Haiti.

It includes insight from national gender experts, agronomists, and focus group consultations with community stakeholders, and has also ensured the engagement of women in the design phase to understand potential impacts of the project, and builds on lessons learned from past studies and assessments on gender.

To summarize, the gender analysis and stakeholder participation and consultation, enabled the following:

- A strong participative process and accordingly commitment by both local government officials and community members in fulfilling the objectives of the project.
- Demonstration of the need for gender-disaggregated data and indicators to establish a baseline in which to measure improvements and identify areas of interest; and
- Establishment of recommendations to be incorporated into the Gender Action Plan.

Accordingly, the Gender Action Plan presented in this Annex has been designed by considering findings both the comprehensive gender assessment presented, as well as the results of the consultations. It has been built to tie with the logical framework of the Project, and identifies mainstreaming actions for the activities planned, with indicative timeline, means of verification and indicators.

4. Facets of Gender Inequality in Haiti relevant for the Project:

4.1 Climate change

Haiti ranks third globally on the Long-Term Climate Risk Index⁶, demonstrating its extreme vulnerability to climate change – to which women are especially vulnerable. Widespread deforestation and soil degradation have so severely impacted the island's environment that it can no longer provide critical ecosystem functions, such as mitigating the effects of heavy rain and flooding. This has progressively increased human vulnerability to climate change. Extreme weather events such as cyclones and hurricanes, are expected to increase in frequency and intensity, while rainfall is expected to become more erratic across seasons⁷. In an already denuded and degraded landscape, this engenders severe erosion and landslides, flash flooding, and sedimentation of river channels. The agricultural sector is especially vulnerable to climate change, and both women and men are central actors in this sector⁸.

Climate change impacts are not gender-neutral and many of the consequences of climate-induced impacts are more severe for women, and other socio-economically marginalized groups, given their specific livelihood circumstances. In relative terms, women lack access to productive resources, and

6 German Watch, Global Climate Risk Index 2016. <https://germanwatch.org/fr/download/13503.pdf>

7 Ibid.

8 IUCN (2011) – ccGAP: https://www.climatelinks.org/sites/default/files/asset/document/2011_IUCN_Climate-Change-Gender-Action-Plan-Haiti.pdf

this has impacts on their health, food security and safety in the context of flooding and other extreme climate events. The additional socio-cultural barriers limiting participation and movement outside the household sphere, is worsened by phenomena such as flooding, drought and erratic rainfall. These constraints cause women to have to work much harder to secure food and water, and generate additional income through livelihoods which, in turn, diminishes their ability to advance out of poverty, particularly when they have, for example, lost their land due to the impacts of flooding during cyclone events.

Gender equality and women's empowerment are not addressed in most of Haiti's strategy or action plan documents for climate change and disaster risk management.^{9,10} Flooding has cumulative livelihood effects that unfairly impact women and exacerbate gender inequality. In Haiti, women are disproportionately represented among the poor and there is a significant portion of women-headed households, so impacts such as crop loss, animal drowning, destroyed homes, health risks or displacement and loss of life due to floods ultimately affect women first and foremost. And the evidence is clear, recent natural disasters; such as the 2010 earthquake and 2015 Hurricane Matthew have shown surges in GBV, and food insecurity. Yet, during these extreme events, the importance of grassroots organizing and mobilization women's groups during disaster relief efforts was unparalleled and critical, demonstrating the resilience of Haitian women in the face of extreme adversity.

4.2 Sociocultural norms and economic sectors

Women have played a crucial role in Haitian society since the beginning of the nation's history, commanding armies during the battle for independence, contributing to the 1805 constitution, and as activists mobilizing for the right to vote in the 1950s. Today they continue to play central although largely invisibilized roles, especially with respect to home life and family care, agricultural production, and grassroots disaster relief. While gender equality was translated into Haiti's current constitution in 1987, decreeing a 30-percent quota for women in all levels of public office, gender gaps are recurrent and found in all aspects of Haitian life.

Over 40% of households in Haiti are headed by women, and this has been on the rise since 2007.^{11,12} Today, there are more women living under extreme poverty than men, with a large portion residing in rural areas.¹³ In fact most of Haiti's population is rural, and two-thirds of this population is dependent on agriculture; among them, women make up the majority of labor force involved in all aspects of agricultural processes.¹⁴

While Haiti has indeed made advances to close gender gaps, persistent challenges remain. Sexist attitudes and beliefs that permeate Haiti's social fabric continue to limit women's job opportunities and meaningful participation in public political life. Women are also socialized to be primary caretakers of the household and the burden of this unpaid, invisibilized labor is an additional obstacle. Specific legislative reforms are needed to redress the gender imbalances and eliminate discriminatory provisions against women within public institutions, notably regarding domestic and child labor laws, penal codes regarding sexual harassment, intimate partner violence, and rape¹⁵. Additionally, legal gaps regarding abortion and common-law partnerships hinder women's ability to exercise and enjoy their

9 USAID (2016)

10 Shadow Report (2010)

11 IHSI, enquête, emploi, économie informelle, 2010

12 MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir

13 MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir

14 USAID (2016)

15 IUNC 2011

rights¹⁶.

Although Haitian women are active in every economic sector and gender-disaggregated statistics show that girls have near equal education opportunities as boys, women are nonetheless underrepresented among the higher-paying jobs and there are very few women in public positions of power^{17,18}. This ultimately reinforces the gendered poverty gap and women's disproportionate vulnerability to climate change. Indeed, post-disaster inquiries have consistently shown that, in times of climate crisis, women and girls suffer the most adverse impacts, as victims of violence, facing higher health and sanitation risks, and significant material losses¹⁹. In response, women's groups have been among the first to organize and support disaster relief efforts²⁰.

Intensifying floods and other climate changes threaten agricultural production and food security. Women are key players in agriculture, especially the areas of commercialization, transformation and processing, so they will be impacted, but this also represents an opportunity for building resilience to climate change. From UNFPA, 2017: The UN supports Haiti in reforms aimed at giving women the same rights to economic resources, as well as access to property and control of land and other forms of property, financial services, inheritance and natural resources.

Overall, this paints a portrait of Haiti, whereby efforts towards gender equality are overshadowed by the threat of floods and other natural disasters linked to climate change that not only impact the country's institutional capacities and resources, but also directly impact the lives of Haiti's most vulnerable populations, including low-income families, women, and other marginalized groups. The experiences and impacts of climate change will be different among Haitian men and women. Addressing the underlying drivers of poverty and gender inequality within the broader challenge of building climate change resilience is complex, yet critical. The following subsections further explore the gendered dimensions of flood and environmental management and gendered impacts in the context of climate change.

4.3 Gender roles, power, and decision-making

In the household, women generally have decision-making independence, taking on the majority of household decisions related to budgeting and the allocation of financial resources (e.g., small expenses, food purchasing and preparation etc.)²¹. In fact, some 78% of married women have sole or joint decision-making power for major household purchases²². Nonetheless, prevailing sexist beliefs and social norms regarding men and women in Haitian society inform the "acceptable" gender roles at school, in the workforce and at home, such that overall, women do not uphold the social status as men, nor the same economic and political independence^{23,24}. Manifested along geographic and class divides, the position and expectations of women in society are important barriers that reinforce existing gender inequalities in the country.

Traditionally, men hold social positions of importance and influence, namely as politicians, spiritual

16 THE MCFBF, 2015 (politique égalité 2015)

17 MCFDF 2014 – politique

18 MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir

19 Post Disaster Needs Assessment, Gender Shadow Report, 2010

20 Post Disaster Needs Assessment, Gender Shadow Report, 2010

21 2012 EMMUS/DHS survey (République d'Haïti Ministère de La Santé Publique et de La Population (MSPP) (2012). Enquête Mortalité et Utilisation des Services EMMUS-V Haïti 2012 verify)

22 2012 EMMUS/DHS survey (République d'Haïti Ministère de La Santé Publique et de La Population (MSPP) (2012). Enquête Mortalité et Utilisation des Services EMMUS-V Haïti 2012 verify)

23 USAID (2016)

24 MCFDF (2014) politique d'égalité

leaders, and school headmasters²⁵, while women and girls assume the responsibility of primary caretakers of the household and family well-being^{26,27}. Women spend twice as much time on domestic tasks compared to men (15.4 h/wk, vs to 7h/wk)²⁸, and this difference is more pronounced in rural areas²⁹. Cooking, cleaning, child rearing are burdensome tasks, and women may additionally spend up to five hours per day collecting firewood and numerous hours per day collecting water³⁰. In fact, over a third of households must travel 30 minutes or more to access potable water supplies³¹. Moreover, not only are women expected manage social relations and conflict in the home, but they are also called upon to do so at the community-level³². Although a main thread in the social fabric and primordial to the daily functioning of Haitian life, this work is typically undervalued, invisibilized and non-remunerated.

With climate change and the threat natural disasters, women inevitably devote more time and effort on these invisibilized yet critical social and family tasks on a daily basis³³. This can affect their psychological and physical health, and further limit their capacity to adapt and recover from these events. For example, girls (and sometimes boys) are pulled from school to help with domestic tasks in climate crises.

Women also play a key role in the rural economy, being the main actors in the commercialization, processing, and transformation of agricultural products, while men engage in the heavier labor activities and animal husbandry. In rural areas, where small-scale agricultural production is a livelihood pillar, Both men and women engage in agricultural wage labor, but frequently at different stages of the value chain – Men engage in heavy agricultural labor, such as land clearing and plowing, production of export crops (coffee, mangoes, etc.), and large livestock husbandry; they also have wage jobs. Women, on the other hand, produce food for local markets, plant and weed crops, and ensure household food security, including the supply of drinking water. In times of recent climate crises and natural disasters, women have been invaluable to relief efforts, mobilizing resources and organizing activities for the security and health of women, girls and other vulnerable groups. This demonstrates the power and potential of grassroots women groups in the fight for gender equity, despite the aforementioned barriers.

4.4 Gender and cross-cutting issues:

Poverty

Haiti is one of the poorest countries in the world, with 59% of the population living on less than 2.5 USD per day (equivalent to about 162.5 gourds)³⁴. Over the last thirty years, there has been a “feminization” of poverty³⁵ and today are more women in extreme poverty than men, living on under USD 1.23 per day³⁶. Haitian women earn, on average, 32% less than men. About a third of this earnings gap can be explained by factors such as age, number of children, education, and industry of employment, while the remaining two-thirds is unexplained, suggesting some gender discrimination in the labor market

25 USAID (2016)

26 Bellanca (2007)

27 USAID (2016)

28 USAID 2016 gender assessment

29 MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir

30 Gardella, 2006 (primary not found, from Feed the Future)

31 EMMUS (2012)

32 Shadow report, 2011

33 Shadow report, 2011

34 Haiti Overview: World Bank (<http://www.worldbank.org/en/country/haiti/overview>)

35 MCFDF 2014 politique d'égalité

36 Haiti Overview: World Bank (<http://www.worldbank.org/en/country/haiti/overview>)

(World Bank, 2014c). The gender earnings gap is largest in the urban areas and is greater than in any other Latin American or African country. It is important to note that most women are employed in the informal sector, not only making them vulnerable to labor exploitation.

The incidence of poverty is much higher among the rural population (75% compared to 41% in urban areas), especially with a high rate of unemployment and precariousness of production. In fact, the highest poverty headcount index is highest in the Northeast (84%) and Northwest (72%) departments³⁷, where project activities will take place. Poverty affects all aspects of life, including housing, nutrition, education, and human and environmental health. In fact, these spatial disparities closely parallel the geographic distribution of health outcomes and access to basic services³⁸.

Almost half of the households in Haiti are women-led, yet the per capita income in these households is 7% less than those headed by men³⁹. In rural areas, female-headed households (of which 62 percent are poor) are much more likely to fall below the indigence poverty line than are male-headed households (54 percent are poor).⁴⁰ With natural disasters and extreme climate events, however, these numbers may have shifted with displacement, migration, and casualties⁴¹.

Health and Sanitation

In Haiti, access to health care services is precarious, especially for remote rural households, due to cost, distance, and quality of facilities. This was already an outstanding issue for the health of women, young children, and other vulnerable groups (Persons with disabilities, the elderly), but recent natural disasters have further debilitated the healthcare system and its infrastructure⁴². The impact on women is evident: maternal mortality rates in Haiti are among the highest in Latin America and Caribbean region^{43,44}, while limited reproductive agency and access to contraceptives⁴⁵ means that unplanned and adolescent births have long-term impacts on women's lives.

The widespread occurrence of contaminated water sources, inadequate sanitation facilities, and low access to clean water in rural areas affects the health of many Haitians, and women are especially vulnerable to health risks as they are the primary providers and handlers of water in households⁴⁶. Waterborne diseases are therefore a serious concern; 20% of children under the age of five years are affected by diarrhea and, to-date, cholera has killed thousands of people - with severe outbreaks being triggered by recent natural disasters. This is particularly relevant to young and pregnant women's health and well-being, as cholera can lead to fetal death, especially in the third trimester.⁴⁷

Severe climate events and flooding are associated with an increased risk of infection of waterborne diseases, including typhoid fever, leptospirosis, hepatitis A, as well as cholera. Moreover, increased sedimentation and pesticide run-off due to flooding further contribute to unsafe water. Risks are especially high when there is significant population displacement and/or contamination of drinking-water sources (i.e., flooding of sanitation infrastructure, sewage treatment plants, latrines, etc.)⁴⁷.

37 Gardella, 2006 (primary not found, from Feed the Future)

38 Feed the Future, 2016

39 DSCRNP, 2007 (primary not found, citing from MCFDF 2014 politique d'égalité)

40 IMF (2020). Haiti Selected Issues. Available at <https://www.elibrary.imf.org/view/journals/002/2020/122/article-A001-en.xml>. Accessed May 2022.

41 Shadow report (2010)

42 Ibid.

43 UNFPA (2017) Faits et chiffres clés sur la situation des femmes en Haïti, Novembre 2017. Available at : <http://www.unfpa.org> . Accessed May 2018

44 World Bank, 2015 from USAID 2016

45 UNFPA Faits et chiffres clés sur la situation des femmes en Haïti, Novembre 2017. Available at : <http://www.unfpa.org> . Accessed May 2018

46 USAID (2016) gender assessment

47 WHO flooding and communicable diseases fact sheet. Accessed May, 2018. Available at: http://www.who.int/hac/techguidance/ems/flood_cds/en/

Above and beyond the primary risks of injury or drowning, floods can also increase risks of vector-borne diseases, such as malaria, dengue, yellow fever, and West Nile Fever, as they may expansion the number and range of vector habitats⁴⁸.

When cholera, or other illnesses, affect households and communities, women and girls disproportionately carry the burden of the work of caring for ill family members and uphold the responsibility of restoring household and collective well-being⁴⁹. Consequently, health behaviors promoted by education campaigns to reduce the risk of cholera and other communicable diseases often end up placing more work on the shoulders of women and girls.

Previous reports on the gendered effects of natural disasters and disaster relief ^{50,51} highlight the importance of safety and sanitation planning that considers and consults with vulnerable groups, including women and girls as well as LGBTI and PWD. The increased incidence of gender-based violence during climate crises and population displacement is significant, and there are security risks of utilizing public sanitation facilities in relief camps⁵². Coping with climate uncertainty and extreme events such as flooding can therefore engender significant psychological burdens and potentially important psycho-social impacts on all Haitians, but especially women and girls. Yet, these effects are often minimized or understood as secondary rather than central in projects and relief efforts⁵³.

Education

Nearly half of all Haitian children do not attend school⁵¹, highlighting the myriad challenges both girls and boys face in accessing free, quality education - especially those from poor and rural households. The public education system faces shortages of qualified teachers, limited supplies and a lack of infrastructure – which is has only been further weakened by climate change and recent disasters. For example, hurricane Matthew (2015) damaged 1,633 out of 1,991 schools in the most impacted regions⁵⁴, while losses from the 2010 earthquake meant that low-income families had fewer resources to send their children to school⁵⁵. The bulk of functioning schools (85%) are run by private entities, many of the elite operated by Canada, France or the United States, with school fees that are prohibitively unaffordable for low-income families⁵⁶. Consequently, more than 500,000 children and youth remain out of primary and secondary school⁵⁷, which has created and reinforces an educational divide and continual poverty gap that exists in Haiti.

Although the available data suggests that both boys and girls have equal access to education - with a near gender parity in terms of education levels and similar illiteracy rates among men and women⁵⁸ - sexist attitudes and gender stereotypes are prevalent. Consequently, education tends to favor the advancement of boys, while young girls face numerous obstacles⁵⁹. Indeed, in post-secondary institutions, women students are underrepresented compared to men⁶⁰, and in the job market they

48 WHO flooding and communicable diseases fact sheet. Accessed May, 2018. Available at: http://www.who.int/hac/techguidance/ems/flood_cds/en/

49 USAID (2016) gender assessment

50 USAID (2016) gender assessment

51 Shadow report (2011)

52 Shadow report (2011)

53 Shadow report (2011)

54 HRW (2017). Accessed May 2018. available at: <https://www.hrw.org/world-report/2018/country-chapters/haiti>

55 Shadow report (2011)

56 HRW (2017). Accessed May 2018. available at: <https://www.hrw.org/world-report/2018/country-chapters/haiti>

57 Shadow report (2011)

58 MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir

59 MCFDF 2013. Le rapport de diagnostique des inégalité des genre

60 MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir

tend to get lower paying jobs in more “feminized” niches^{61,62}. Gaps among rural and urban areas also highlight the geographic and class distribution of education, where only about 64% of women in rural areas are literate, compared to 84% in urban areas, and this is most acute among poor households⁶³.

To counteract the gender stereotypes that shape women’s life options, the MCFDF signed *protocole d'accord* in 2007 with the Ministry of Education, focusing on the issue of the exclusion of girls in the education system. Nonetheless, gender roles and the unequal division of unpaid labor continue to limit the opportunities of girls receiving a quality education, by confining them to the household and burdening them with domestic chores.

Enforced child labor is a prevailing risk in Haiti in specific sectors, such as domestic work for girls and construction for boys, and occurs especially among uneducated girls from low-income families⁶⁴ who may be sold to more affluent families to work as “restavèks” (domestic workers) under unregulated and often abject conditions⁶⁵. Recurrent climate threats further exacerbate this gendered risk, as impoverished and highly impacted families may feel they have no other choice but to take children out of school to cope with livelihood stress and, in extreme cases, sell children for labor as a survival strategy⁶⁶.

Political participation

While women face no legal obstacles to vote and stand for election, in the 2015 presidential and legislative elections, over 90% of the candidates were men⁶⁷. While a constitutional amendment in 1987, decreed a 30-percent quota for women’s participation in all levels of public office, gender inequalities continue to exist, and tis especially evident among the positions of power that are dominated by men⁶⁸. Furthermore, representation is regressing, with currently only 3% of women in the Haitian parliament (2017), while in 2003 there was 9% of women⁶⁹.

This gender gap is evident at all levels and branches of the government. At the local level, only 3% of CASEC (*Conseil d'Administration de la Section Communale*) representatives are women, and 11% of mayors are women (data from 2012)⁷⁰. Each municipality is governed by a board of 3 members. There is a limited number of cities that have a woman at the board level and these women often do not occupy leadership roles.

In judiciary and prosecutorial positions, only 32 out of 583 judges are women (5.2%), while in the superior court, there are only 2 out of 9 judges⁷¹. Of note, there are no women deputies and senators among the current 92 Deputies and 24 Senators recently elected into office⁷². In Haitian Senate, on a staff of 30 (29 active) senators, there is only one woman. For the Chamber of Deputies, there are 3 women out of 118 (according to information available on the parliament's website), so women represent less than 3% at the Legislative Power. Moreover, throughout the history of certain Ministries (e.g., Ministry of Agriculture), almost all the important posts have been and are still held by men.

Rural women are often insufficiently informed, consulted, or included in the design of public policies

61 MCFDF 2013. Le rapport de diagnostique des inégalité des genre

62 MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir

63 MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir

64 HRW (2017). Accessed May 2018. available at: <https://www.hrw.org/world-report/2018/country-chapters/haiti>

65 Shadow report (2011)

66 Shadow report (2011)

67 MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir

68 MCFDF 2013. Le rapport de diagnostique des inégalité des genre

69 UNFPA Faits et chiffres clés sur la situation des femmes en Haïti, Novembre 2017. Available at : <http://www.unfpa.org> . Accessed May, 2018

70 MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir

71 MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir

72 USAID (2016) gender assessment

and, at the national level, few actions have been taken to create a space where women can fully and meaningfully participate in environmental management⁷³. This is even more pronounced in areas outside the capital and in the target regions of this project⁷⁴.

Watershed management is led by the Ministry of Agriculture and the Ministry of Environment – which are primarily composed of men – while groups and associations engaged in management activities (e.g., *association d'irrigants*) typically include very few women⁷⁵. With respect to climate change and disaster risk reduction and preparedness, women are not explicitly included in policy making processes, despite recognition of the gendered dimensions of mitigation and adaptation⁷⁶.

MCFDF's recent inquiry into women's low rates of political participation⁷⁷ indicates that primary barrier to wider engagement is the burden of family responsibilities that mostly (sometimes exclusively) concern women. Lack of financial resources and sexist stereotypes are additional obstacles that are especially relevant for poor rural women. While women have historically been highly involved in social and political demonstrations⁷⁸, the danger linked to these activities is also noted as a limitation, notably the fear of losing one's life and leaving children behind⁷⁹. These obstacles are only compounded by climate-related livelihood stress, such as periodic and increasingly intensified flooding, which amplifies the burdens of work and care demands on women.

Recognizing that women have been historically excluded from political debates and decisions on the direction of society, the Government has introduced gender aspects to their projects "to reinforce the capacity of women and their participation in development projects"⁸⁰, and four priority areas have been identified in relation to gender mainstreaming in climate change (Agriculture and Food Security, Water Resources Management, Disaster and Risk Management, and Health). However, policy decisions and practices are still being implemented, and women are still largely excluded from policymaking and decisions, especially with respect to disaster risk reduction and agriculture⁸¹. Wide-spread corruption with the added impact of climate related natural disasters impede the effective implementation. For example, the 2015 elections were postponed due to Hurricane Matthew and election fraud allegations⁸².

Labor force participation

The work force is segregated by gender, and women have considerably fewer employment opportunities than men across all age groups⁸³. The most recent available data shows that women make up only 38.4% of the active population (or 47.7% of the working population, although women earn less than half of what men earn⁸⁴. While equal salary policies have been adopted in the public administrative positions, women are unrepresented in the higher paying, more influential positions in this sector⁸⁵. In fact, this is echoed across all sectors, where women tend to have the lowest paying jobs, with

73 USAID (2016) gender assessment

74 PFS gender assessment

75 PFS gender assessment

76 USAID (2016) – CHECK REF 193: Republic of Haiti (2016). Submission for the U.N. Universal Periodic Review 26th Session of the Working Group on the UPR Human Rights Council [31 Oct, – 11 Nov., 2016].

77 MCFDF (2014) politique d'égalité femme hommes 2014 - 2034

78 Shadow report (2011)

79 Myriam Merlet. La participation politique des femmes en Haïti, quelques éléments d'analyse, 2002 (primary not found verify)

80 REF 193 From USAID: Republic of Haiti (2016). Submission for the U.N. Universal Periodic Review 26th Session of the Working Group on the UPR Human Rights Council [31 Oct, – 11 Nov., 2016]. Climate Change and the Right to Food). (primary not found verify)

81 USAID (2016) gender assessment

82 HRW (2016) Accessed May 2018, Available at: From <https://www.hrw.org/world-report/2018/country-chapters/haiti>

83 MCFDF, 2013. Le rapport de diagnostic des inégalités de genre.

84 Shadow report, 2011 – verify source within report.

85 MCFDF, 2013. Le rapport de diagnostic des inégalités de genre.

particularly high disparities in private sector salary⁸⁶, and they are underrepresented among professional jobs such as engineering, agronomy, and law. In the project's region of application, the level of employment is especially low, with an unemployment rate of more than 42%. This, among other influences, promotes out-migration, particularly of young people looking for opportunities to other countries, namely in Dominican Republic or North America – many working as undocumented immigrants.

Women predominate in the informal sector (75.1%), where work conditions are unregulated and undocumented, so exploitation and income stability are relevant gender concerns⁸⁷. This also means that, in socio-economic crises, women will be the primary victims and it will be largely undetectable or unreportable as this work is invisibilized⁸⁸. The prevalence of child domestic workers is also common; in fact, pre-earthquake data indicates that almost all Haitian girls between the ages of 5 and 9 work in the informal market⁸⁹. Often, this is to help low-income families with domestic tasks, however, the prevalence of “restavèks” – children sent/sold to wealthier families to perform chores in exchange for room and board – is a major concern, especially for the rights of young girls⁹⁰. According to some estimates, there may be as many as 300,000 children working as restavèks in Haiti⁹¹. Although the minimum age is set to 15 for work in industrial, agricultural, and commercial enterprises, there is currently no labor code that sets a minimum age domestic work⁹².

Women also hold a majority (70%) of jobs in trade in the primary sector, especially in agriculture where they are the primary actors working in commercialization as well as processing and transformation⁹⁷. The work of women farmworkers is often undervalued, and they tend to suffer wage discrimination. This means that they have little access to stable income sources, that are dependent on the changing environmental conditions, contributing to gender-specific vulnerability to climate change. This also indicates that there is a lot of potential for projects to support women's role in agriculture and to benefit women as part of climate change resilience-building initiatives.

Agriculture and food security

Agriculture has been the backbone of Haiti's economy and an architect of Haitian society since the colonial area. Today, a majority of agricultural production occurs on small farms, primarily oriented towards subsistence production in a variety of micro-climates and altitudes (USG, 2011, from Feed the Future). Both women and men participate in the cultivation of subsistence crops (namely of tubers and beans) as well as staple crop production that is marketed nationally (such as maize and rice)^{93,94}. Although these crops (beans, rice, corn) are most vulnerable to crop loss from climate change⁹⁵, demonstrating the disproportionate impacts incurred by women and rural, low-income households.

Women make up a majority of the labor force in agriculture, yet income generating activities in this sector are skewed by gender. While women primarily work with domestic trade and commerce⁹⁶, the lucrative export commodities, such as essential oils, mango, cacao and coffee, tend to be controlled by men⁹⁷. Women face additional income generating obstacles in this sector: with limited access to private

86 MCFDF, 2013. Le rapport de diagnostic des inégalités de genre.

87 MCFDF 2015. Statistiques de Genre: Comprendre pour mieux agir

88 MCFDF 2014. Politique d'égalité femmes hommes 2014-2034

89 Shadow report (2011)

90 HRW (2018) Accessed May 2018, Available at: From <https://www.hrw.org/world-report/2018/country-chapters/haiti>

91 HRW (2018) Accessed May 2018, Available at: From <https://www.hrw.org/world-report/2018/country-chapters/haiti>

92 MCFDF 2014. Politique d'égalité femmes hommes 2014-2034 (CHECK, also HRW website)

93 FAOSTAT

94 Feed the future (2016)

95 USAID 2016

96 MCFDF – which one?

97 FAOSTAT

transportation they are more likely to use public services to undergo transactions⁹⁸. This not only contributes to the time poverty Haitian women already face, but also highlights their livelihood vulnerability to extreme flooding, which decreases transportation options and limits their ability to actively engage in the commercial transactions that many depend on.

Not only do domestically consumed crops generate less income, but they also receive less attention in agrarian development agendas compared to cash crops of coffee, cacao and mango⁹⁹. Given the gender distribution of agricultural production and sales, a prioritization of cash crops risks directly excluding women and their access to development program benefits. Indeed, export commodity value chains are often promoted a favorable pathway to generating more income in the country, however, over-investment in export crops may ultimately widen the economic gap between men and women¹⁰⁰.

Furthermore, there has been minimal support and regulation from the state for women in agricultural value chains, nonetheless women have innovated by establishing expansive information and trade networks¹⁰¹. In the fields, men are often responsible for large livestock (cattle, horses, etc.) and heavier, machine labor, while women assist with tasks such as harvesting and weeding or tending to smaller livestock (chickens, goats, etc.)¹⁰².

Over the last 50 years, agricultural production, processing and marketing have decreased, and today it accounts for about 25% of gross national product¹⁰³; given women's central role in agricultural this has implications for their well-being and livelihoods. With climate change and continual land degradation, productivity is often low on farms; compounded by population growth, Haiti's agricultural sector is struggling to meet the national food demand (WFP, 2008 USG, 2011 from FTF).

For rural farming families, production does not often generate enough income to meet basic dietary and household needs. In fact, only a quarter of their cash income originates from farming, so livelihoods must be supplemented by additional income sources¹⁰⁴. At the same time, malnutrition affects over half the population, while Haiti imports as much as 60% of its food, making it especially vulnerable to global economic and food crisis¹⁰⁵. Trade liberalization has exposed farmers to competition from subsidized U.S. rice exports and made consumers vulnerable to volatile global food prices. Food aid has also disrupted local markets and agriculture (USG, 2011 from USAID 2016). As such, access to nutritious and stable food supplies depends more on household purchasing power than on household farm production, and this is only exacerbated by climate change. In fact, among the communes most acutely affected by recent droughts, food insecurity affected between 300,000 and 560,000 people¹⁰⁶. Food security is a thus central, cross-cutting component that is relevant to any climate resilience intervention for both men and women. Given women's role in the production, distribution, and provision of food to their communities and families, it is an opportunity ripe for gender-transformative actions.

Land access

Notwithstanding their central role in agriculture and food security, women's access to and rights over productive land remain limiting factors that contribute to gender gaps and poverty¹⁰⁷. Women often

98 Shadow Report (2011)

99 Shadow Report (2011)

100 Shadow Report (2011)

101 Oxfam. (2012). PLANTING NOW: Revitalizing Agriculture for Reconstruction and Development in Haiti. Briefing Paper. Port-au-Prince, Haiti. Oxfam. Retrieved from <https://www.oxfam.org/en/research/planting-now-2nd-edition>

102 USAID (2016) – citing gardella, 2006

103 Feed the future (2016) – check original reference, ALSO ref 54 from Shadow report – seems to be from World Bank (2010) data.worldbank.org

104 shadow report (2011) – see if they refer to original data?

105 Shadow report (2011) – see if they refer to original data?

106 Republic of Haiti (2016) – UN report to HR Council

107 MCFDF 2014. Politique d'égalité femmes hommes 2014-2034

face difficulties obtaining legal titles and do not often gain equal land inheritance; as such, they rarely own land or tend farm on land that belongs to their male relatives^{108,109}. Exacerbated by climate extremes and population growth, low-income families and women-led households are thus often relegated to farming on the most unproductive, infertile lands for subsistence production, namely denuded, steep sloping terrains¹¹⁰.

Article 36 of Haiti's Constitution recognizes and guarantees the right to private property. In practice, however, customary law governs rural land rights and use. Customary rights are less evident in urban areas (Smucker et al. 2000; Howard 1998).

Rural land ownership is generally governed by customary law and traditional practice as opposed to formal law. Land is largely individualized; rural communities in Haiti do not have a tradition of communal land management although land held by families may remain undivided (Smucker et al. 2000).

Haiti's formal laws pronounce principles of nondiscrimination. The 1987 Constitution (Article 18) mandates the equality of all Haitians. Women achieved equality under the law in 1982, when a decree was passed that established the equality of the sexes, particularly within marriage (GOH Constitution 1987; De Ferranti 2002).

In practice, the formal law often does not impact the reality faced by Haitian women. A majority of couples in Haiti have common law marriages that are not registered with the state. Common law marriage and cohabitation create insecurity for women because the law does not recognize their inheritance rights to their husbands' land (De Ferranti 2002).

Under both formal and customary law, all recognized children have equal rights to the land owned by their deceased parents, although female heirs tend to receive smaller shares of land. Customary law favors the possessor of land. Heirs who remain on the land assume control of absentee shares and consolidate adjoining shares, which can deprive daughters who have married outside of the community of their access to inherited land (Smucker et al. 2000; Gardella 2006).

Married-out daughters usually lose their shares of land to relatives because unregulated customary practice restricts selling land to outsiders. Thus, they will not sell their inherited share and will not personally farm it. Instead, they give the land to a relative (Smucker et al. 2000).

In general, women's property rights derive from their status as wives, mothers, or wards, and thus women often lose their land rights when their status within the household changes. The customary laws that are prevalent in rural areas exclude women from land ownership. Anecdotal evidence suggests that the land rights of women who are household heads are more secure than the rights of women in households headed by men (De Ferranti et al. 2002).

Even though a woman doesn't hold the title to property – it's rare to find one who has a title. Women usually work the land that her husband owns. They live with, supports, and collaborates with the [husband or father's] family to make sure that they and their children can eat, that the children can go to school, that they can pay for their health needs, etc.

Community and peasant groups need to make a strategic plan for what will be the priorities and objectives for merging women's issues and land rights, and then launch a program that all the women's organizations can participate in. There should be more training for women, and more support to augment their technical, political, social, and economic capacity. What's most important, consciousness-raising, training and action. It is necessary for community and peasant groups to raise

108 USAID (2016)

109 Oxfam (2012)

110 Feed the future (2016)

women's consciousness of these issues. Unfortunately, rural Haitian women face challenges in organizing around gender at several levels. First, women suffer from high levels of illiteracy and lack of education, so even when they're in peasant organizations, there are a lot of decisions in the women's movement that they can't make. Second, women don't have economic means in Haiti. They don't have the resources to advocate for their rights or to address problems related to land.

Forest resources and charcoal production

Haiti is infamously known as one of the most deforested areas in the world, with population growth, and growing demands for energy and arable land being commonly cited as direct drivers of extreme forest loss and degradation over the last century. While the FAO statistics of 2% - 4% forest cover are often used as a reference^{111,112}, more rigorous, local land cover classifications show that forested area is between 19.5% - 32.4%, depending on the data and methods used¹¹³. This incites a re-evaluation of the predominant narrative about environmental degradation and the use of forest resources in the country. Of the forested areas, coffee and cacao plantations represent a large portion of this cover (50% in the case of coffee)¹¹⁴.

The majority (over 90%) of Haitian households meet their energy needs by charcoal and firewood, as only about a third have access to electricity – primarily in urban areas¹¹⁵. In fact, about 10,000 sacs of charcoal are used a day¹¹⁶. Both men and women are actively involved in the production of charcoal, while women are the main traders and users of this wood-based energy source (for cooking and cleaning)¹¹⁷. As such, this is an issue of relevance to both gender and environmental management.

With a stable national demand for charcoal, it is often a supplementary or primary source of revenue for poor rural families¹¹⁸. While charcoal production has long been painted as unsustainable, practices involving the cutting of fast growing trees such as cassia and acacia (that coppice and regrow within 4 – 6 years), are bringing into question this assertion that prevails across green development narratives¹¹⁹. As such, interest is surging on how to integrate, rather than eliminate, charcoal production and use into programs and projects.

With differentiated gender roles and responsibilities in the household and the economy, the intended use and preferences of forest resources are often distinct between men and women. For example, recent reforestation initiatives in Haiti lead by the CCAf found that men tended to favor tree species for charcoal production and construction, while women expressed preferences for fruit trees and vegetable crops to integrate into home gardens¹²⁰. In some regions of Haiti, there may also be taboos around the use, planting or cutting of certain tree species¹²¹. With the prevalence of men in spiritual leadership positions¹²², their power and control over belief systems that shape these practices may be particularly relevant, yet it is a largely unexplored and overlooked aspect of climate resilience building endeavors and gender-mainstreaming that involve reforestation efforts. Indeed, evaluation of previous reforestation attempts in Haiti indicate that interventions are most successful when there is a strong

111 FAO (2010) – FRA

112 FAO (2015) – FRA

113 Churches et al. (2014)

114 CIAT (2011) – coffee value chains

115 EMMUS (2012) – from USAID (2016)

116 FAO (check référence from <http://www.haitilibre.com/article-17656-haiti-environnement-l-onu-appelle-haiti-aux-bonnes-pratiques-environnementales-et-agricoles.html>)

117 UNEP (2016) Charcoal report

118 UNEP (2016) – charcoal report

119 See: <https://haitiliberte.com/charcoal-is-not-the-cause-of-haitis-deforestation/>

120 Canada-UNDP (2017)

121 See for example: Tarter (2015)

122 USAID (2016) – gender assessment

incentive for local populations to ensure the survival of seedlings and maintain tree-cover ¹²³.

Access to productive land

In Haiti, women make up a majority of the labor force in agriculture, contributing to all aspects of production and especially commercialization and trade¹²⁴, yet ownership and rights over productive land resources is highly skewed by gender. The national statistics are a testament to this, with only 9% property owners in the country being women, while 20% of women have joint-ownership of a property¹²⁵. Land tenure is regulated by a civil code (enacted in 1962), that requires tenure registration for all land, yet a third of parcels do not have a legally recognized title, while 19% of land ownership is *de facto* declared by a sales receipt¹²⁶, indicating that the land tenure situation is uncertain and fragile across many Haitian households.

Most rural households (80%) have access to at least one hectare of land¹²⁷, with climate change and soil degradation however, the productivity of land is declining. Fertile land may have decreased by as much as 70% since the 1970s¹²⁸; in fact, estimates put losses of 10,000 to 15,000 fertile hectares per year due to erosion linked to a combination of land use practices and climate change¹²⁹.

Haiti has a particular and highly informal land ownership scheme rooted in traditional family relations that are structured around the family patriarch, or “lakou”^{130, 131}. Land is typically passed through inheritance and distributed among children, which has contributed to the fragmentation of land into smaller and smaller parcels over time. With each generation, the inheritance situation is re-evaluated, such that relationships to home and land may be impermanent and unstable for families from one generation to the next¹³².

For women and girls, these tenure customs are a serious obstacle regarding their access to and securing their rights over productive land. Although inheritance is bilateral, with both daughters and sons being included in the division of land and homes, daughters traditionally receive smaller inheritances^{133, 134}. Haitian marriage law recognizes that both men and women have equal inheritance rights, yet women have little legal and intuitional recourse to affirm these rights, as common law relationships and unofficial unions are extremely common¹³⁵. In the case of adopted children, those born of wedlock and “restaveks” (child domestic workers, who are often young girls), they are not recognized in informal hereditary arrangements; moreover, they retain their original family names and consequently have no legal rights to inherited land¹³⁶.

For landless households, sharecropping is common, particularly under informal agreements of the “mèt té” system, whereby up to half of the agricultural production is collected by the property owner¹³⁷, most of which are men. Moreover, these parcels are often the most remote and degraded lands^{138, 139}.

¹²³ UNEP (2016) Charcoal report

¹²⁴ MCFDF

¹²⁵ UNFPA (2017)

¹²⁶ FAO (n.d.) – land rights database

¹²⁷ FAO (n.d.) – landrights database

¹²⁸ USAID (2015) – ICTP report

¹²⁹ Arias, Brearly and Damais (2006)

¹³⁰ CIRAD (2016) – prob fonicer

¹³¹ FAO (n.d.) – land rights database

¹³² CIRAD (2016) – prob fonicer

¹³³ Gardella (2006) – from USAID (2016)

¹³⁴ Feed the future (2016) – landscape evaluation of USAID not oxfam.

¹³⁵ Feed the future (2016) – landscape evaluation of USAID not oxfam.

¹³⁶ FAO (n.d.) – land rights database

¹³⁷ FAO (n.d.) – land rights database

¹³⁸ FAO (n.d.) – land rights database

¹³⁹ Feed the future (2016)

Within Haitian polygamist practices – which account for up to one-third of unions¹⁴⁰ – secondary wives do not have the same family status, yet men may still collect yields and thus revenue from their farms or gardens¹⁴¹.

Overall, the situation of tenure rights and land access is especially precarious for women-led and poor, landless farming households, and this deeply interconnected to climate vulnerability. Being constrained to farming on small, unfertile plots and steep slopes, the livelihood security and well-being of these marginalized families is especially at risk with the intensification of prolonged droughts, extreme tropical storms, and natural disasters¹⁴². In fact, conflict over increasingly scarce productive resources has multiplied in recent years^{143,144}. Further land conflicts may emerge from natural disasters, as post-earthquake inquiries found that urban populations moved back to rural areas to farm or claim land¹⁴⁵.

Access to extension services and finances

With the prevalence of rural aid projects from donors, NGOs and the private sector, agricultural extension services are readily available in Haiti – although the portion of public extension services has diminished in the last decades¹⁴⁶. According to a World Bank (2013) study, both men and women are equally likely to receive agricultural extension services in certain regions of Haiti, although this is probably due to women's higher demand and active seeking out these services, rather than an outcome of gender equal access¹⁴⁷. Moreover, the report found that there are fewer positive outcomes of agricultural training among women-led households.

In terms of credit, the MCFDF highlights that eligibility criteria of microfinancing institutions create barriers for women to access sufficient funds to effectively implement agrarian projects or transition to other economic sectors^{148,149}. With women as key actors in agricultural value chains, particularly the commercialization and sale of produce, low access to credit is a missed opportunity to build their climate resilience and increase livelihood security.

5. Gender-based Violence (GBV) and Sexual Exploitation, Abuse and Harassment (SEAH)

In Haiti, gender-based violence (GBV) is a pervasive problem, and current legal and judiciary structures do not adequately address this vulnerability. It is a major obstacle for the empowerment of women and girls, as it reduces their ability to fully participate in the public and private spheres of life. One in three Haitian women aged 15-49 has experienced physical and/or sexual violence, and the rate of intimate partner violence has risen since 2005¹⁵⁰.

Climate and natural disasters only increase the incidence, as violence and sexual assault in displaced persons and relief camps has been reported to be widespread^{151,152}. The World Bank summarizes that research indicates that women and girls are up to 14 times more likely to be harmed during a

140 <http://www.rebuildingrespectforwomen.org/haiti.html>

141 FAO (n.d.) – land rights database

142 USAID (2016)

143 USAID (2016)

144 Shadow report (2011)

145 Shadow report (2011)

146 Feed the future (2016)

147 The World Bank (2013)

148 MCFDF 2014. Politique d'égalité femmes hommes 2014-2034

149 USAID (2016)

150 EMMUS (2012)

151 Shadow report (2011)

152 HRW (2018) - <https://www.hrw.org/world-report/2018/country-chapters/haiti>

disaster. For those who survive, climate-induced disasters can amplify gender inequalities, making them more vulnerable to GBV, even when life for others may go back to normal.

Solidarite Fanm Ayisyèn (Haitian Women's Solidarity – SOFA) – Haitian feminist organization of approximately 10,000 members nationwide, 80% of whom are peasant women – along with the Global Justice Clinic and the Institute for Justice and Democracy in Haiti have also presented a compelling case on how land grabbing is a source of violence against women and other human rights violations in the country.¹⁵³

A majority of GBV cases go unnoticed and unreported – as the most recent surveys indicate that 43.1% of women never told anyone, while 65% percent of women did not seek help of any kind¹⁵⁴. When they do seek help, it is most commonly among family members and neighbors¹⁵⁵, as official reporting accompanies a fear of revenge violence from abusers and the perception that official avenues are ineffective in terms¹⁵⁶. Indeed, the prevalence of police violence targeting women and other marginalized groups is inevitably deters the denouncement of aggressors or making formal complaints regarding conflicts of any kind¹⁵⁷. Among cases of violence reported in 2009 – 2011, the majority were physical in nature (59.74%), followed by sexual violence (29.80%), while cases of psychological and economic violence constituted a minority (5.88% and 4.58% respectively)¹⁵⁸. Although very few of these cases are investigated or prosecuted¹⁵⁹. See data below from UN Women's Global Database on Violence against Women:¹⁶⁰

Prevalence Data on Different Forms of Violence against Women:

Lifetime Physical and/or Sexual Intimate Partner Violence : 26 % ⁽¹⁾
Physical and/or Sexual Intimate Partner Violence in the last 12 months : 13.9 % ⁽²⁾
Lifetime Non-Partner Sexual Violence : Official National Statistics Not Available
Child Marriage : 14.9 % ⁽³⁾

Reports submitted by UN Human Rights Bodies:

Report of the Special Rapporteur on Violence against Women, its Causes and Consequences
Concluding observations of the Committee on the Elimination of Discrimination against Women (CEDAW)
Report of the Working Group on the Universal Periodic Review (UPR)

Gender Equality Indexes:

Gender Inequality Index Rank : 152 ⁽⁴⁾



Even when cases that are reported, justice for victims of GBV is hindered by the significant gender imbalance in Haiti's legal system. In 2016, there were only 32 women judges out of a total of 615¹⁶¹ and today the percentages remain about the same. Moreover, with women disproportionately represented among the population with low education and living in poverty, they generally have lower access to legal aid and justice systems¹⁶².

¹⁵³ [https://chrgj.org/wp-content/uploads/2022/04/220331_SOFA_GJC_IJDH_Submission-to-Special-Rapporteur - English.pdf](https://chrgj.org/wp-content/uploads/2022/04/220331_SOFA_GJC_IJDH_Submission-to-Special-Rapporteur-English.pdf)

¹⁵⁴ EMMUS (2012)

¹⁵⁵ EMMUS (2012)

¹⁵⁶ USAID (2016)

¹⁵⁷ Republic of Haiti (2016) – UN report on HR commission

¹⁵⁸ FROM PFS GENDER ASSESSMENT: Kay Fanm and SOFA - which mainly receive abused women and girls - Gheskio and Médecins sans Frontières / MSF France

¹⁵⁹ Shadow Report (2011)

¹⁶⁰ <https://evaw-global-database.unwomen.org/en/countries/americas/haiti>

¹⁶¹ FROM THE PFS: the Ministry of Justice and Public Security, Legal Service, 2013, cited in "Diagnosis of gender inequalities" 2016 by UN Women

¹⁶² Shadow Report (2011)

Currently, there is no specific legislation to deal with domestic violence or sexual harassment, and while rape was criminalized just over a decade ago, alarmingly the law still does not recognize spousal rape as a crime. Although the Government has made efforts to combat violence against women and girls, with the support of the UN Committee on the Elimination of Discrimination Against Women (CEDAW) and other agencies, recent political crises and natural disasters have obstructed the effective reform and adoption of gender protective legislations^{163,164}.

Considering this context of GBV in Haiti, there is a risk that any shifts in power balances that the project could cause and might exacerbate GBV, including sexual exploitation, abuse, and harassment (SEAH), and therefore, the Project has identified this as a moderate risk (Risk 08 – see Annex 6[a] – SESP; see also Annex 6[b] – ESMF, which includes a preliminary SEAH risk assessment completed during the design phase and an initial list of mitigation measures).

It should be noted that during the ESIA, the Project will carry out in-depth SEAH risk assessment. The Project will then update the mitigation measures and include them in the ESMP, in order to prevent, mitigate and respond to these SEAH risks. The Project will adhere to UNDP's as well as GCF's policies for protection against Harassment, Sexual Harassment, Discrimination and abuse of Authority, as well as special measures for protection from sexual exploitation and sexual abuse. UNDP will request that contractors, suppliers, and partners adhere to zero tolerance for SEAH conduct and commit to taking adequate action if faced with SEAH allegations, in the absence of which, contractual arrangements can be terminated. Other mitigation measures will include, but not limited to:

- Conducting training and sensitization sessions for project staff and the target population on SEAH issues in collaboration with the Ministry for Women's Affairs and Women's Rights.
- Establish a separate Grievance Redress Mechanism that is survivor-centered to adequately respond to SEAH incidents.
- Integrate the Ministry for Women's Affairs and Women's Rights into the project's technical implementation committee
- Encourage competent authorities to legally address SEAH-related issues.

6. Legal, Policy and Administrative Frameworks for Protecting Women and Gender Equality

Several constitutions have succeeded each other and the first to establish a turning point for the situation of women was the 1950 Constitution, guaranteeing them the right to vote. The Constitution in force, adopted on 10 March 1987 by a Constituent Assembly, which was approved in a referendum on 29 March 1987, as amended, unequivocally enshrines the principle of equality of men and women before the law and equal rights in the protection and enjoyment of human rights (articles 17, 18, 19, 28, 32.1 and 32.6) .

Thus, at the level of national legislation, the numerous laws published between 1825 and 1826 contained discriminatory provisions that were maintained until the 20th century: the Civil Code (27 March 1825) and the Criminal Code (19 May 1826), inspired by the French Napoleonic Code, were amended much later, in 1982 and 2005, largely eliminating discriminatory aspects against women; on the other hand, the Code of Civil Procedure of 3 May 1825 and the Code of Criminal Instruction of 12 April 1826 did not recognize the right of a married woman to take legal action. For the Commercial Code of 28 March 1826; (art. 4) a woman cannot have the status of trader without her husband's consent. Only the Labour Code of 1961 affirms the equality of the sexes in terms of access to

¹⁶³ HRW (2018). Accessed May, 2018; updated in 2018; available at: <https://www.hrw.org/world-report/2018/country-chapters/haiti>

¹⁶⁴ USAID (2016)

employment, salary and wages³. The principle of a 30% quota for women in all decision-making positions in national life, particularly in the public services, was recognized in 2012 in the so-called amended version, in article 17-1, of the 1987 Constitution.

The Ministry of Women and Women's Rights (MCFDF) is tasked with ensuring that signed agreements and conventions (domestic and international) are implemented, as part of its broader mission to guide the formulation, implementation, and enforcement of equitable public policies. Recently, the adoption of the Gender Equality Bill as well as the implementation of the National Plan for Gender Equality and the National Plan for the Fight Against Gender Equality (2017), attests to Haiti's continued efforts to protect the rights of women¹⁶⁵. The ministry's operations are divided across four areas: Women's Rights Promotion Directorate (DPDDF), Gender Analysis Directorate (DPAG), the Directorate of Administrative Affairs (DAF), and Direction of coordination of the departmental offices.

MCFDF representatives (or Gender Focal Points/ *Point Focal genre*) are in place across ministries and state structures to coordinate and collaborate on women's rights and gender equality in the public sphere. Notably, a key function of this department is the production of national gender analyses and awareness-raising and training activities. However, the MCFDF faces insufficient budget allocation, as well as limited human resources and technical capacities to carry out its intended mission, while a lack of regulation for the functioning of this strategic institution means that its level of authority and influence in orienting governance activities is weak¹⁶⁶.

The Office for the Fight against Violence Against Women (*Bureau de Lutte contre la Violence Faite aux Femmes*) created in 2014, brings together three ministries (Women's Rights, Health and Justice) was created. Its management is formalized according to a Protocol signed by the three ministries. The Office is a public service office offering support and quality care to women and girls who are victims of specific violence and ensuring access to information and quick response to cases.

Criminal Code – Under Haitian law, rape is classified as a crime and therefore it is subject to a 10-year statute of limitations. Rape is not considered as being a "*crime de sang*", meaning a crime which resulted in the death of a person or a political crime. The maximum penalty for rape in Haiti is a ten years prison sentence if the victim is an adult and 15 years maximum if the victim is a minor under the age of 15. The Decree of 6 July 2005 modified article 278 of the Criminal Code regarding rape. The decree changed the categorization of rape from a crime against morals to a crime against the person. Whilst bringing some positive change, the decree did not properly address the issue of rape and sexual violence as it failed to accurately define the crime of rape and to list its elements. Judges and prosecutors are therefore required to resort to definitions of rape found outside of the code¹⁵ thus leading to an inconsistent application of the decree and a generally low level of predictability in how cases will be treated.

Incorporating gender and gender-responsive policy and planning is an essential part of effectively and strategically dealing with climate change impacts. Overall, Haiti's mechanisms and reform plans for gender equality are based on a number of international agreements and conventions for women's rights. Several such instruments have been ratified in Haiti, including:

- The Convention for the Suppression of Human Trafficking and Exploitation of the Prostitution of Others September 2, 1952

¹⁶⁵ Shadow report (2011)

¹⁶⁶ UNDP (2016) - *Projet de Renforcement des capacités adaptatives des communautés côtières aux changements climatiques en Haïti* (ACC - PNUD/FEM/MDE)

- the Convention on the political Rights of Women, July 31, 1957 (Please note that the Haitian woman has the right to vote since 1950)
- the Convention on the Elimination of all forms of Discrimination with regard to Women (CEDAW), 7 April 1981
- the Convention on the fight against discrimination in the field of Education, October 15, 1984
- the International Covenant on Civil and Political rights, 23 November 1990
- the Convention on Children's rights, December 23, 1994
- the Inter American Convention on the prevention, punishment and eradication of violence against Women, 3 April 1996

Climate change governance and gender in Haiti – There are few links between Haiti's climate change governance structures and the established frameworks for gender equality and women's empowerment (i.e., MCFDF, National Gender Equality Policy)¹⁶⁷. The National Adaptation Programme of Action (NAPA, 2006, revised 2017) and the National Climate change policy (PNCC, 2017) – both aligned with the UNFCCC Paris Agreement – promise gender integration in all aspects of its implementation, however details on specific measures to address gender inequalities are lacking¹⁶⁸. In the Second National Communication on Climate Change (DCNCC), gender considerations were taken into account in four priority areas linked to climate change adaptation: agriculture and food security, water resources management, disaster and risk management, and health. The *Direction des Changements Climatiques* in the Ministry of the Environment (MDE) is primarily responsible for national climate change strategies and ensuring the inclusion of gender its initiatives. This department participates in a Climate Change Consortium along with the Ministry of Health, Ministry of Education, and Ministry of Agriculture.

A **National Disaster Risk Reduction System** is in place, with coordination across 10 ministries, and a Disaster Risk Reduction Plan has been developed. While change adaptation plans do note disaster risks as a priority area, there is limited coordination and information sharing among the two¹⁶⁹. Day-to-day disaster risk reduction activities are overseen by the Department of Civil Protection (DPC), which is managed by the Ministry of the Interior and Territorial Communities. Women are highly vulnerable to the livelihood impacts of extreme climate events and natural disasters, while the high incidence of GBV in emergency shelters is a serious concern¹⁷⁰. The National Contingency Plan (2015) for emergency planning does give some consideration to the specific vulnerabilities of women (and other vulnerable groups such as people with disabilities), however the inclusion of women in planning committees is inconsistent and weak across all administrative levels¹⁷¹. Donor agencies are supporting the development of gender protection practices at temporary emergency shelters and there are on-going capacity building and training activities to support gender-responsive disaster and emergency responses in the country¹⁷².

¹⁶⁷ USAID (2016)

¹⁶⁸ USAID 2016

¹⁶⁹ UNDP (2018) – NAP lessons

¹⁷⁰ Shadow report (2011)

¹⁷¹ USAID (2016) -(check refs: 189, 190)

¹⁷² USAID (2016)

7. Project-specific design consideration resulting from the Gender Assessment

Although Haiti faces challenges with respect to its capacities and resources for watershed governance and climate change adaptation and mitigation^{173,174}, the country is attempting to make advances in improved and integrated water management despite significant setbacks related to recent natural disasters and political corruption. There is a wide variety of support from donor agencies, foreign governments, international NGOs, and researchers that aim to strengthen capacities and support Haiti in its commitment to building climate resilience and prioritizing gender equality within its policies and planning.

The gender assessment carried out highlights the complexities of gender relations in Haiti and identifies barriers to gender equality. The analysis is a foundation, supporting the identification of opportunities to improve gender-responsive approaches to management activities and risk reduction in the context of climate change adaptation.

Building on this analysis, the following challenges and opportunities for gender-responsive integrated flood management and development of green and grey infrastructure in the context of climate change adaptation in Haiti are highlighted:

- I. Women are disproportionately vulnerable to flooding and climate change, with impacts on their livelihoods, health and safety, and of unpaid labor burdens:
 - A majority of women and women-led households depend on agriculture for their livelihoods making them vulnerable to climate change and flooding. They tend to produce some of the most vulnerable crops and are more likely to face obstacles of unproductive land and insecure land tenure.
 - Health risks associated to flooding and climate change are higher for women (especially pregnant and lactating) as well as other vulnerable populations (e.g., children and elderly). Women also undertake the primary care labour within the household, therefore, increased health burdens impact women's existing time poverty.
 - Women face high risks of violence during and in the aftermath of disasters: in [rapid analyses](#) post the 2021 Earthquake conducted by humanitarian organizations, for example – it was revealed that gender-based violence became a post-disaster crisis. 70% of women and men surveyed said their fear of sexual violence had increased since the earthquake. 43% of community leaders and 75% of youth say GBV has increased since the earthquake. 70% of organizations say women and girls are most at risk of GBV.
 - Climate changes such as flooding or drought, place an additional burden of unpaid labor on women, as they are primarily responsible for family/community water use/provisioning.

Through the Activity 1.1, that aims to capacitate farmer's groups as well as women's cooperatives and associations, specific gender-transformative sub-activities have been designed to ensure the above challenges are addressed. Further, the food coupon distribution (Activity 2.3) will also target vulnerable households, with a specific focus on women-headed, female-led, widow-headed households.

¹⁷³ Stoa (2017)

¹⁷⁴ UNDP (2018) – NAP : lessons from Haiti

- II. There is an institutional and policy-making gender gap. Gender equality is not fully addressed in Haiti's climate change, disaster risk, and water resource management strategies or action plans; as such, measures to promote gender-responsive adaptations and mitigate gender specific impacts are lacking. Women and women's associations do not fully participate in policymaking and decisions related to water resource management, disaster risks, and climate change across all levels.

Through the Activities 3.1 and 3.2, this challenge will be addressed. These activities aims to improve Haiti's Water Act, build national capacities, as well as catchment-level governance framework, with a focus on improved gender and social inclusions.

8. Recommendations from Haiti's existing on Climate Resilience and Gender Projects and Programmes

Lessons Learnt

Lessons learnt from previous programmes and projects in the country and elsewhere can be drawn upon to improve the transformative potential of integrated flood management and effective flood control through soil conservation in the Trois Rivières region and ensure equal participation and benefit sharing among men and women.

A recent cross-review of policies, institutions, and interventions in water resource governance across several countries conducted by UNDP (2017)¹⁷⁵ found that gender-responsive approaches should:

- Build strong collaboration with public water resources departments, ensuring that designing and installing water infrastructure and facilities are based on gender-responsive assessments
- Increase the capacity of local government representatives to ensure the effective participation and leadership of women in water management mechanisms
- Support communities in upholding principles of gender equity when establishing water-user groups
- Support local municipal authorities and communities to develop and maintain social mechanisms for gender-equitable distribution of water resources
- Ensure that partnerships with the private sector include a commitment to gender equality and women receive equal access to technical or management training, peer-to-peer learning, and other resources.

With respect to the agricultural sector and climate change resilience, UNDP (2016) recommends that gender-responsive initiatives¹⁷⁶:

- Close the gap in access to land: Eliminate discrimination under the law; Recognize the importance and power of customary land rights; Educate officials and evaluate them on gender targets; Educate women regarding land rights; Ensure that women's voices are heard; Adjust bureaucratic procedures; Gather sex-disaggregated data for policy design and monitoring
- Closing the gap in rural labor markets: Target women's multiple trade-offs; Reduce gender inequalities in human capital; Capitalize on public work programmes; Strengthen women's

¹⁷⁵ Canada-UNDP (2017)

¹⁷⁶ UNDP (2016) – gender, climate change and food security training module 3

rights and voice

- Closing the financial services gap; Promote financial literacy; Design products that meet the needs of women; Promote a women-friendly and empowering culture Close the gap in social capital through women's groups; Use technology and innovative delivery channels
- Closing the technology gap; Develop technologies and environments that address women's needs Improve extension services; Scale up farmer field schools

With respect to the reforestation and agro-forestry systems in the context of climate change, IUNC (2008) recommended that gender-responsive initiatives¹⁷⁷:

- From the onset, ensure full participation and integration of women, from local communities and promote systematic attention to the participation of women in forestry development in policies, strategies and capacity-building efforts related to the conservation and sustainable development of forests and trees and their use.
- Programmes should promote equal access of women to land ownership and other resources necessary for effective socio-economic participation in forest management and climate mitigation strategies (e.g., land, capital, technical assistance, technology, tools, equipment, markets and time).
- Both women and men must be trained in methods and forestry technologies, including nursery techniques, site selection, and selection of species, land preparation, planting, weeding, and maintenance.

Project Design and Implementation

Gender benefits through this Project have been optimized by promoting the full involvement of women in decision-making regarding integrated flood management at the community and institutional levels, as well as the stewards of agroforestry systems, playing key roles in the selection of species and transformation of products. In addition, as during the design of the project, during implementation, strong efforts will continue to be made, in order to ensure the equitable participation of women. Additionally, going forward beyond the implementation phase, equal participation of women will be pursued and achieved in ongoing management and water governance especially on the Water Governance Committees.

Addressing gender dimensions within the project design and implementation, this proposal works to identify and integrate interventions to provide gender responsive and transformative results.

The project design and implementation will take into consideration the following gender implications:

- Specific strategies to include / target female-headed households;
- Differing conservation incentives faced by women;
- Identification of gaps in gender equality through the use of sex-disaggregated data enabling development of a gender action plan to close those gaps, devoting resources and expertise for implementing such strategies, monitoring the results of implementation, and holding individuals and institutions accountable for outcomes that promote gender equality.
- Advocacy and awareness is adjusted to most effectively reflect gender-specific differences. Strategies used in the project are then tailored, taking into account such differences;
- Inclusion of a Gender Specialist position / provision of advice within the project to implement gender related activities.

¹⁷⁷ IUNC (2008) – Reforestation, afforestation, deforestation, climate change and gender

During project implementation, qualitative assessments will be conducted on the gender-specific benefits that can be directly associated to the project. This will be incorporated in the annual Project Implementation Report, Mid-Term Report, and Terminal Evaluation. Indicators to quantify the achievement of project objectives in relation to gender equality will include men and women who had access to affordable solutions, number of men and women employed from the jobs created by the project, training opportunities, knowledge management and information dissemination.

Stakeholder engagement

The stakeholder consultations and engagement of women's organizations promote gender equality at the local as well as the national level. The involvement of women's organizations in the project design, aided in identifying relevant gender issues within the country's social context, and implementing and monitoring the gender aspects of the project.

Monitoring and evaluation

Through onset analysis, data will be collected and collated in the project's inception phase to establish a baseline. This data shall be monitored against throughout implementation and evaluation. In order to monitor and evaluate progress of the project, the following indicators can be measured:

Quantitative outcomes:

- Female-headed households as beneficiaries;
- Improvements in health and well-being;
- Improved livelihoods;
- Business development services component targeting rural women entrepreneur groups.

Qualitative outcomes:

- Opportunities to generate additional income. Women are more likely to respond to incentives that address their family's basic needs, such as better health and nutrition
- Time-saving for women as a result of lower hours in labour required for agricultural and water management practices prior to the implementation of the project;
- Contribution to improved self-esteem and empowerment of women in the community;
- Expanded involvement in public and project decision-making as a result of initiation of women into active participation in income generating activities;
- Support for training and educational activities which may include activities related to climate change, agriculture, leadership, business, finance, entrepreneurship and decision-making, thereby enabling empowerment and involvement (or increased involvement) of women to participate with confidence in community meetings;
- Effectiveness of awareness raising.

9. Gender Action Plan (GAP)

This Gender Action plan provides suggested entry points for gender-responsive actions to be taken under each of the Activity areas of the project, based on the Logical Framework of the Project (Section E – full Funding Proposal). In addition, specific indicators are also proposed to measure and track progress on these actions at the activity level. This can be incorporated into the detailed M&E plan which will be developed at the start of implementation, and provides concrete recommendations on how to ensure gender (including disaggregated data) continues to be collected and measured throughout implementation.

UNDP – as the Accredited Entity – will be responsible for the implementation of this GAP, and the responsible parties column (see below) lays down the architecture at the activity-level. A Gender and Safeguards Expert will be part of the Project Management Unit (PMU) with the remit to deliver the GAP, in collaboration with Heifer International, which is the Executing Entity for Outcome 2. Gender technical expertise and assistance may be sought on an ad-hoc basis to supplement the available skill set and experience on the PMU – this has been mentioned (indicative) where necessary in the Action Plan below.

The expected impact of this GAP is: **Increased resilience to climate change of vulnerable communities, including women and girls; gender-equitable participation in territorial planning and household food and water security for increased climate change resilience.**

The expected outcome of this GAP is: **Improved capacities for an estimated 87,780-146,300 women/ 30-50% of project direct beneficiaries for sustainable water and land management as a means to enhance their resilience to climate change.**

OUTPUT 1: Ecosystem-based flood management solutions implemented in 25,440 hectares of the Trois-Rivières watershed

ACTIVITY 1.1: Strengthen the capacity of community groups, including farmer and women's associations, for climate-resilient land-use planning in seven target communes in the Trois-Rivières watershed solutions

SUB-ACTIVITIES	GENDER MAINSTREAMING OPPORTUNITY	INDICATORS	BASELINE	TARGET	M&E TIMELINE	RESPONSIBLE PARTIES AND MEANS OF VERIFICATION	GENDER TECHNICAL ASSISTANCE AND BUDGET (in USD)
There is a mention of additional/follow-up gender assessments in the previous section, where is this reflected in this GAP?							
1.1.1 Develop 33 community land-use plans.	Under this activity, community members and farmer and women's associations in 33	Gender-inclusive stipulations in the	0	33 community land-use plans	Annual document review (33	PMU (Project Manager and Gender Expert)	

Annex VIII – Gender Assessment and Action Plan

	communal sections within the 7 target communes in the TR watershed (Port-de-Paix, Chansolme, Bassin Bleu, Plaisance, Pilate, Marmelade, Gros Morne) will be trained on climate-resilient land-use management. Through this activity, the Project will develop land-use plans, co-developed with communities, and include clear delineation of roles and responsibilities, and ensure women's concerns and needs are mainstreamed into these plans.	land-use plans ¹⁷⁸ (qualitative indicator)			land use plans expected by Y5)	Community land use plans developed through the project	Total: 220,050 ¹⁷⁹
1.1.2	Train communities, farmer and women's associations in the seven target communes on the application of the community land-use plans and implementation of climate-	Through this Activity, communities within the 33 sub communes will be reached by these plans and the associated training. The training will be in a ToT format to ensure continuity and sustainability of these investments. The Project will include gender modules / sensitization modules (where relevant) in this Activity to ensure that groundwork for	# of women and men trainers with improved understanding of climate resilience and the role of communities in reforestation	0	50% trainers trained (ToT) are women, 14,153 farmers trained are women (accounting for all associated farmers that are women identified within target area per 2008 Agricultural Census)	Annual review	PMU Heifer Training participation logs/records and training reports

¹⁷⁸ UNDP has extensive experience and expertise in delivering projects and programmes around LUPs, as well as developed guidance on integrated spatial planning. Please review: <file:///Users/deby1605/Downloads/UNDP-GEF-Integrated-Spatial-Planning-Workbook.pdf>

¹⁷⁹ Accounts for 1 community mobilizer per BN 1B (84,000), 30% of the time of the KM Officer in the design of the workshops (25,200) per BN 1B, 30% of the internal travel costs for the delivery of the campaigns per BN 1C, 20% of stationary costs for the delivery of the campaigns per BN 1D (15,400), 40,000 for the workshops dedicated to ensuring gender participation in the awareness per BN 1I, 25,200 for the awareness campaign with a gender focus BN 1H, 30% of the time of the ME officer who will monitor the effectiveness of these campaigns (25,200) per BN 1ME1

	resilient land-use practices.	increased women's participation is accepted and driven forward by communities.						
1.1.3	Conduct one awareness campaign in each of the seven target communes in Haiti.	To support these capacity-building initiatives, awareness-raising campaigns will be designed and implemented in the seven target communes through city hall meetings and community radio broadcasts. The Project will encourage equitable participation of men and women in these campaigns, as well as target men and women equitably through these campaigns.	# of female and male participants, per awareness campaign	0	7 communes, with equitable participation (50% male and 50% female) in each	Annual review	PMU Heifer Minutes (Official Summaries) of City Hall Meetings Radio campaigns Participation lists	
ACTIVITY 1.2: Implement ecosystem-based flood management solutions in the Trois-Rivières watershed								

Annex VIII – Gender Assessment and Action Plan

1.2.1	Establish 17,740 ha of agroforestry systems in priority areas the TR watershed, including: i) agrisilvicultural systems — which are a combination of crops and trees, such as alley cropping; ii) silvopastoral systems — which combine forestry and grazing of domesticated animals; and iii) agrosilvopastoral systems that combine trees, crops and animals.	This sub-activity will focus on reducing the impacts of climate change-induced flooding in the TR catchment and Port-de-Paix. This will be achieved by implementing 17,740 ha of agroforestry systems. This is the ecosystem-based investment, which form the continuum of actions that the Project will invest in, and therefore, gender-sensitive.	# of direct beneficiaries, disaggregated by male and female	0	At most 70% male and at least 30% female participation, prioritizing the participation of women farmers that account for 14,153 of the 54,714 total associated farmers in the region	Annually	PMU Heifer Agroforestry logs and project participation logs	2,731,840 ¹⁸⁰
1.2.2	Restore 7,700 ha of forests in priority areas the TR watershed through planting indigenous tree species in	This sub-activity will focus on rehabilitating 7,700 ha of forests in priority areas in the watershed. This is the ecosystem-based investment, which form the continuum of actions that the	# of participants in conservation/restoration activities, disaggregated by gender	0	At most 70% male and at least 30% female participation, prioritizing the participation of women farmers that account for 14,153 of	Annually	PMU Heifer Project logs and participation lists	

¹⁸⁰ Accounts for the 15% work dedicated to ensure that women are well benefitted within the EBA interventions, including targeting, support in the establishing of these plots per BN 10 and 30% of the time of the ME specialist per 1ME2 to ensure that the gender target is well monitored and the strategy is adjusted accordingly

Annex VIII – Gender Assessment and Action Plan

	degraded areas within the seven target communes.	Project will invest in, and therefore, gender-sensitive.			the 54,714 total associated farmers in the region			
1.2.3	Hold 8 engagement workshops with community leaders, farmer and women's associations to facilitate the implementation of Sub-activities 1.1.2 and 1.1.3.	This sub-activity will be conducted in tandem with 1.1.2 and 1.1.3. These engagement workshops will complement the awareness and training components of the previous activity, and ensure that women's associations, community groups and other farmer associations are adequately engaged and their stakeholder buy-in is ensured for the proposed interventions of the Project.	# of female and male participants, per engagement campaign	0	7 communes, with equitable participation (50% male and 50% female) in each	Annually	PMU Participation lists	

OUTPUT 2: Climate-resilient agricultural practices, optimised value chains and social safety nets established to promote SLM and reduce degradation in the Trois-Rivières watershed

ACTIVITY 2.1: Strengthen institutional capacity at the commune, inter-commune and department levels in Haiti's Trois-Rivières watershed for productive and sustainable land-use management

SUB-ACTIVITIES	GENDER MAINSTREAMING OPPORTUNITY	INDICATORS	BASELINE	TARGET	M&E TIMELINE	RESPONSIBLE PARTIES AND MEANS OF VERIFICATION	GENDER TECHNICAL ASSISTANCE AND BUDGET (in USD)
2.1.1 Conduct capacity gap/needs	Under this broader activity and through the sub-activity, a	Inclusion of gender questions/concerns	0	Capacity gap/needs assessments are	Year 2 of the project	Heifer	159,865 ¹⁸¹

¹⁸¹ 30% of the time by the project staff to ensure gender monitoring and targeting per BN CoF2, 50% of the training and workshop for farmers and audiovisual materials that will include targeting for women farmers

Annex VIII – Gender Assessment and Action Plan

assessments for the agricultural sector in seven priority communes in the TR region.	capacity needs assessment will be conducted for the agricultural sector to identify the technical and institutional requirements for transitioning the sector from unsustainable land-use practices towards a sustainable landscape management (SLM) approach to land and water resources management that is climate resilient. Gender considerations and further analysis will be mainstreamed into the assessment structure.	in assessment structure (qualitative indicator)		conducted with gender assessments mainstreamed into the overall structure.	and annually verified to ensure mainstreaming within the implementation	Capacity gaps assessment (document review) Meeting minutes and summary of socialization of the capacity assessment as well as of the meetings held within its development Meeting/ workshop participation list	
2.1.2 Establish communal development councils and multi-sectoral management structures to oversee and manage the implementation of interventions designed to address priority areas identified in Sub-activity.	Through this sub-activity, the project will also establish multi-sectoral governance structures will be established to oversee and manage the implementation of interventions designed to address priority areas identified under the capacity needs assessment conducted in the previous sub-activity. Women's leadership will be strongly encouraged in these institutions. The target will be 50% both in leadership, and also in participation.	# of women (in ratio to men) participating in communal development councils # of women (in ratio to men) participating at CDC and FSN meetings	0	50% of councils led by women, 50% of all participants / trainees are women (prioritizing women-headed households)	Annually	Heifer Statutes of the Communal Development councils or similar origination documents that specify leadership positions Participation lists of FSN meetings	

	to reflect realities of Haitian society, where women may be too time poor or may not always be able to ascent traditional and faith-based norms to participate fully.						
2.1.3 Train representatives from appropriate government entities, including MARNDR, NCFS and MSAL, on the implementation of climate-resilient agricultural techniques and SLM.	Through this sub-activity, representatives will be trained from across different governance levels to ensure implementation of climate-resilient agricultural techniques and SLM are done successfully, with the requisite institutional and capacity-building. Modules on gender will be made part of the curricula to ensure there is buy-in on gender-responsive climate-resilient agricultural techniques and SLM.	# of representatives trained, gender disaggregated	0	50% of all participants / trainees are women (prioritizing women-headed households)	Annually	Heifer Participation lists of trainings	
ACTIVITY 2.2: Enhance the technical capacity and access to finance of national and local representatives involved in agriculture in the Trois-Rivières watershed for adopting climate-resilient sustainable land-use practices							
2.2.1. Strengthen the technical capacity of farmers and communities in the seven target communes to	This broader Activity will increase the technical capacity of farmers to implement climate-resilient agricultural techniques and land-use practices, using the findings from the capacity needs	# of trainer of trainers trained, disaggregated by gender	0	50% trainers trained (ToT) are women thus allowing to empower the existing under representation of women in farming		Heifer Participation records and logs	585,000 ¹⁸²

¹⁸² 10% of the work regarding the capacity and social capital development to ensure gender is well reflected within the capacity gaps, 315,00 for gender and capacity building initiatives, 20,000 for trainings for women related to the strengthening of the finance mechanisms, 20,00 for trainings on community awareness per CoF3

implement climate-resilient agricultural practices and SLM.	assessments being conducted under Activity 2.1.1. Trainings conducted under this sub-activity will be gender-responsive and target vulnerable farming communities – as well as marginalised groups within them such as women agricultural workers. In general, in many of these communes, women are heavily involved in each part of the agricultural processes and value chains.			associations (currently identified at 30%, from the stakeholder consultations)			
2.2.2. Provide climate-resilient agricultural inputs (e.g. climate-resilient seeds and infrastructure upgrades) to households in the TR watershed.	These interventions will be supported by the provision of climate-resilient agricultural inputs (e.g. climate-resilient seed varieties) to vulnerable households that depend on subsistence agriculture for their livelihoods. Inputs provided through this sub-activity will be equitably distributed, in line with the findings of the gap assessments conducted in the previous Activity, and in proportion to the farmers identified in the region.	# of direct beneficiaries, disaggregated by gender	0	At most 70% male and at least 30% female, considering that identified associated women farmers currently only account for 14,153 of the 54,714 total associated farmers in the region	Annually	Heifer Project logs regarding the usage and access of the inputs provided Site visits	

Annex VIII – Gender Assessment and Action Plan

2.2.3. Enhance access to finance for smallholder farmers through the VSCA.	Actions (feasibility and risk analysis) to promote the creation of community finance associations to enhance access and mobilize savings to be used for investment in the capital goods required for the and implementation of sustainable, climate-resilient agricultural practices. The Project will aim to include women in the project-supported savings groups, by ensuring requisite mechanisms and conducive environments to ensure favourable lending conditions, in light of the assessment findings that women may lack access to collateral/with limited financial literacy.	Number of feasibility and risk analysis developed to enable VSCAs that look to promote the access of women into the project supported saving groups	0	100% of risk/feasibility/scoping analysis developed by the project take into account how to promote women's access and participation within savings groups 60% of village agents will be women	Annually	Heifer	Document review of the risk/ feasibility/s coping analysis developed under this activity	
ACTIVITY 2.3: Implement a social protection system to support vulnerable households at risk to food insecurity because of climate change								
2.3.1. Conduct a site survey/assessment to determine eligible households to receive food coupons to be developed under Sub-activity 2.3.2	This sub-activity will provide the required immediate relief to vulnerable people affected by climate induced flooding events, concomitantly reducing further pressure on the environment by providing an alternative option to resorting to unsustainable charcoal production.	Inclusion of gender questions/concerns in assessment structure (qualitative indicator)	0	Capacity gap/needs assessments are conducted with gender assessments mainstreamed into the overall structure.	Y2 and Y3 of the project (when the assessment is expected to take place and	Heifer	Document review of the Capacity gap/needs assessments	200,000

	This sub-activity will conduct the requisite site surveys and assessments to pinpoint the households. To determine eligible households for these coupons, vulnerable households will be identified through a site survey and/or assessment, with their details captured in a database for future reference. Gender considerations will be mainstreamed into the assessment structure. Increased spotlight will be on female-headed or widow-headed households, which may not be performing at the level of male-headed households – the scoping exercise for the food coupons will isolate and identify the most vulnerable and in-need households.				is completed)		
2.3.2. Establish and operationalise a food coupons mechanism in the TR watershed.	Food coupons will be distributed through this sub-activity, targeting vulnerable and marginalized households prioritizing those that are headed by women (female-headed households, widow-headed households).	# of households, disaggregated by household head gender (ie. # of MHHs and # of FHHs) # of direct beneficiaries, gender disaggregated	0	At least 60% of households prioritized are headed by women	Annually starting Y3	Heifer Beneficiary monitoring system	

Annex VIII – Gender Assessment and Action Plan

2.3.3. Engage and train 934 households on the implementation of the food coupons mechanism established under Sub-activity 2.3.2.	A total of 934 households will be selected to participate in the operation of the food coupons mechanism. Participating households will receive training on <i>inter alia</i> hygiene, modalities of exchange and personal security. There will be focus on capacitating women-headed households.	# of households in the food coupon mechanism, disaggregated by gender		At least 561 women headed households	Annually starting Y3	Heifer Participation longs	
2.3.4. Create 150 new VSCAs in the TR watershed to enhance the food security of vulnerable households.	To support the efficient and effective implementation of the food coupons mechanism, existing VSCAs will receive operational improvements under this activity, supplemented by the creation of 150 new VSCAs in the TR watershed.	- No gender-specific indicator required however participation of women in the VSCAs will ensured through a favorable enabling environment and will also be monitored by the project					
2.3.5. Establish a network of village agents to monitor existing VSCAs and establish new ones, as well as to monitor market performance contributing to food coupons mechanisms.	To ensure that this system remains operational throughout the lifetime of the project, a network of village agents will be established, with their primary responsibilities being the monitoring of existing VSCAs and creation of new ones	- No gender-specific indicator required however participation of women in the VSCAs will ensured through a favorable enabling environment and will also be monitored by the project					

OUTPUT 3: Strengthened governance and capacity for climate-resilient integrated water resources management (IWRM)							
ACTIVITY 3.1: Strengthen national capacities for the implementation of the Water Act							
SUB-ACTIVITIES	GENDER MAINSTREAMING OPPORTUNITY	INDICATORS	BASELINE	TARGET	M&E TIMELINE	REONSIBLE PARTIES AND MEANS OF VERIFICATION	GENDER TECHNICAL ASSISTANCE AND BUDGET (in USD)
3.1.1. Draft provisions to Haiti's Water Act for the adoption of an integrated approach to water resources management and implement the act in the TR watershed.	Under Activity 3.1, provisions will be drafted to the Water Act to ensure that it promotes an integrated approach to water resources management, including considerations for the coordinated development of water, land and related resources. This will contribute towards: i) maximising economic and social welfare in an equitable manner without compromising the sustainability of vital ecosystems under future climate change conditions; and ii) facilitating efficient, integrated adaptation responses to the impacts of climate change in Haiti. The provisions will include gender-transformative and gender-responsive stipulations, to ensure water governance in Haiti incorporates gender and social inclusion-related clauses at the policy level. As evinced in the	Inclusion of gender questions/concerns in assessment structure (qualitative indicator)	0	1 Water Act prepared for adoption	Annually	PMU Text of the Draft Water Act presented	135,200 ¹⁸³

¹⁸³ 10% of effort dedicated to including gender considerations within ESMP and ESIA per BN 3A, 30% of the efforts of the policy specialist to ensure inclusion of women within the draft water act provisions and IWRM per 3C

Annex VIII – Gender Assessment and Action Plan

	assessment, climate and environmental governance in Haiti does not adequately address gender concerns yet.						
3.1.2. Establish Catchment IWRM Committees (<i>Comittes de GIRE des Bassins Versants, CGBVs</i>) within each of the 31 catchments in Haiti, as per the provisions drafted in Sub-activity 3.1.1.	IWRM Committees will be designed to increase women's leadership, and ensure women's equitable participation. The target is being kept at 30% to reflect realities of Haitian society, where women may be too time-poor or may not always be able to ascent traditional and faith-based norms to participate fully.	# of women participating in communal development councils # of women participating at CDC and FSN meetings	0	30% of committees led by women, 50% of all participants / trainees are women (prioritizing women-headed households)	Annually	PMU UNDP (Project Support) Statutes of Creation of the Committees Published Committee Agreements Participation log included within meeting minutes	
3.1.3. Establish Sub-Catchment Water Resource Users Associations (<i>Associations d'Usagers des Ressources en Eau du Sous-bassin versant; AssURES</i>) within the sub-catchments of the 31 catchments in Haiti, as per the provisions drafted in Sub-activity 3.1.1.	Sub-Catchment Water Resource Users Associations will be designed to increase women's leadership, and ensure women's equitable participation. The target is being kept at 30% to reflect realities of Haitian society, where women may be too time-poor or may not always be able to ascent traditional and faith-based norms to participate fully.	# of representatives trained, gender disaggregated	0	30% of associations led by women, 50% of all participants / trainees are women (prioritizing women-headed households)	Annually	PMU UNDP (Project Support) Statutes of Creation of the Committees Published Committee Agreements Participation log included within meeting minutes	

Activity 3.2: Develop an integrated, climate-resilient water management governance framework targeting the catchment and sub-catchment levels in the Trois-Rivières watershed

3.2.1. Establish the Catchment Water Resources Management Committee (CWRMC) to manage water resources in the TR watershed.	Under this activity, an inclusive and equitable water management governance framework will be developed for the TR catchment, namely the Catchment Water Resources Management Committee (CWRMC). This committee will work with representatives from productive sectors (e.g. agriculture), land users at sub-catchment level and Sub-Catchment Water Resources User Associations (SCWRUAs) to achieve two main goals. Increased women's leadership will be encouraged in the Committee, as well as gender parity in participation of members.	# of participants, disaggregated by gender	0	50% of all participants / trainees are women	Annually	PMU UNDP (Project Support) Statutes of Creation of the Committees Published Committee Agreements Participation log included within meeting minutes	443,124 ¹⁸⁴
3.2.2. Develop climate-resilient IWRM plans for end users at the sub-catchment and catchment levels in a participatory, gender-responsive way with	The IWRM plans will define roles and responsibilities of institutions and civil society stakeholders for water resources management (complementary to the decentralised management of water resources to be established in the Water Act under Activity 3.1).	Inclusion of gender questions/concerns in IWRM plans (qualitative indicator)	0	31 IWRM plans	Annually	PMU UNDP (Project Support) IWRM Plans (document review)	

¹⁸⁴ 30% of the time allocated to the IWRM adviser to provide gender considerations in the IWRM plans and ensuring participation within the committees (87,477) per 3I, salary of the gender specialist per 3I (185,247), 82,200 travel costs for gender monitoring travel per 3J, 72,200 per workshops focused on women per 3M, 30% of stationary and audio visual costs per 3K and 3L for use in the delivery of the workshops towards women (16,000).

representatives from CWRMC and productive sectors (e.g. agriculture), land users and Sub-Catchment Water Resources User Associations (SCWRUAs).	This will also include stipulations on increased women's participation and leadership, with the aim to making a business case of how gender-responsive IWRM yields improved results. Lessons learnt and best practices for other contexts (https://cap-net.org/genderiwrn/; https://www.fsnnetwork.org/resource/why-gender-matters-iwrn-tutorial-water-managers) may be brought in and tailored.						
3.2.3. Promote a climate-resilient, integrated landscape management approach in the TR watershed that considers gender dynamics, using the community land-use plans (Activity 1.1).	- As above.						
3.2.4. Train the CWRMC (established under Sub-activity 3.2.1) and representatives from productive sectors (e.g. agriculture), land users and Sub-Catchment Water	Training on the climate-resilient IWRM plans will also be delivered focusing on the application of these plans in decision-making. The combined result of these actions will be the establishment and operation of an integrated governance framework that will contribute to preserving climate-	Inclusion of gender modules (qualitative indicator)	0	Gender module included in the training provided to all of the 31 Sub-Catchment Water Resource Users Associations	Annually	PMU Gender modules developed Training plans Training reports	

Resources User Associations (SCWRUAs) on the application of climate-resilient IWRM plans in decision making.	resilient, sustainable land and water resources management beyond the lifetime of the project. During the implementation of Activity 3.2, considerable efforts will be made to securing equitable representation, participation and leadership of women. To that end, training will also contain modules / informally highlight (in the context of traditional communities/stakeholders) the importance of women's increased leadership roles and participation in water governance bodies at local and sub-local levels.						
Activity 3.3: Implement regular monitoring and evaluation of water resources at the catchment and sub-catchment levels to support the implementation of integrated water resources management (IWRM) plans							
3.3.1. Commission an expert to undertake the health monitoring and impact evaluations of the EbA flood management solutions (Activity 1.2), supported by WHO.	From project inception (Year 1), continuous health monitoring and impact evaluations will be undertaken and will continue throughout the implementation phase. By monitoring the evolution, frequency and location of flood-related illnesses from the start of the project, the impact of the project will be assessed once a sufficient time series of observations is completed. UNDP will commission an expert to	Inclusion of gender considerations	0	1 overall study	Annually	PMU UNDP (Project Management Support) ToRs developed for the study Study itself	156,800

	undertake the health monitoring and impact evaluations of the EbA flood management solutions. The World Health Organisation (WHO) will provide technical support to this expert, as they are currently monitoring cases of water- and vector-borne diseases in Haiti and therefore have a pre-defined methodology that would allow comparison studies to be undertaken across watersheds. As with all assessments that will be undertaken during the project life cycle, gender considerations will be made a part of the evaluation structure to ensure data and information (on which there is a paucity currently) on differential impacts on men and women, regional disparities, between different households can be captured.						
3.3.2. Undertake a baseline assessment of the prevalence of water- and vector-borne diseases — as well as monitor dry season baseflow — within the TR watershed,	The contracted expert will undertake a baseline assessment of the prevalence of water- and vector-borne diseases within the TR watershed and analyse the impact of climate change-induced flooding on the spread of these diseases.	- As above					

assisted by local representatives from SCWRUAs.						
3.3.3. Assess the effectiveness of EbA interventions (Activity 1.2) in reducing the spread of water- and vector-borne diseases.	- As above.					
3.3.4. Generate lessons through the implementation of all project interventions and disseminate these lessons through the committees and councils established under Output 2	Lessons generated through the implementation of all project interventions will be shared through the relevant committees and associations at the catchment and sub-catchment levels to inform adaptative management of these interventions under future climate change conditions. Gender and social inclusion learnings will be captured and mainstreamed to ensure gender-responsive design, planning and implementation of future programming.	Inclusion of gender-responsive case studies/lessons learnt/best practices (qualitative indicator)	0	1 overall study	- As above	



Activity-level Gender-Responsive Budgeting envisioned for the Project (in USD\$)

Activity 1.1	220,050
Activity 1.2	2,731,840
Activity 2.1	159,865
Activity 2.2	585,00
Activity 2.3	200,00
Activity 3.1	135,200
Activity 3.2	443,124
Activity 3.3	156,800

Total: USD 4,846,879

Proposed SEAH Mitigation Measures

Please refer to Section 6.1.4. within Annex 6b (ESMF) as well as its appendices for proposed actions to be implemented by the project mitigate SEAH.

[Please see next pages for Appendix 1 and 2.](#)



Appendix 1: Glossary and List of Stakeholders

To ensure a common understanding, the following UNDP definitions apply to key terminology used throughout the document.¹⁸⁵ It should be noted that UNDP operates in line with the quadrennial comprehensive policy review of operational activities for development call on all United Nations entities to improve interagency coordination and maximize impact. At the country level, UNDP supports the integration of gender equality in all common country analyses and United Nations Sustainable Development Cooperation Frameworks.

Gender	“Refers to the social attributes and opportunities associated with being male and female and the relationships between women and men and girls and boys, as well as the relations between women and those between men. These attributes, opportunities and relationships are socially constructed and are learned through socialization processes. They are context/time-specific and changeable. Gender determines what is expected, allowed and valued in a woman or a man in a given context. In most societies there are differences and inequalities between women and men in responsibilities assigned, activities undertaken, access to and control over resources, as well as decision-making opportunities. Gender is part of the broader socio-cultural context. Other important criteria for socio-cultural analysis include class, race, poverty level, ethnic group and age.”
Gender Equality	“Refers to the equal rights, responsibilities and opportunities of women and men and girls and boys. Equality does not mean that women and men will become the same but that women’s and men’s rights, responsibilities and opportunities will not depend on whether they are born male or female. Gender equality implies that the interests, needs and priorities of both women and men are taken into consideration —recognizing the diversity of different groups of women and men. Gender equality is not a ‘women’s issue’ but should concern and fully engage men as well as women. Equality between women and men is seen both as a human rights issue and as a precondition for, and indicator of, sustainable people-centred development.”
Gender Mainstreaming	“Mainstreaming a gender perspective is the process of assessing the implication for women and men of any planned action, including legislation, policies or programmes, in all areas and at all levels. It is a strategy for making women’s as well as men’s concerns and experiences an integral dimension of the design, implementation, monitoring and evaluation of policies and programmes in all political, economic and societal spheres so that women and men benefit equally and inequality is not perpetuated. The ultimate goal is to achieve gender equality.”
Women’s Rights	“The human rights of women and of the girl child are an inalienable, integral and indivisible part of universal human rights. The full and equal participation of women in political, civil, economic, social and cultural life, at the national, regional and international levels, and the eradication of all forms of discrimination on grounds of sex are priority objectives of the international community.”

¹⁸⁵ Annexes to the UNDP Gender Equality Strategy 2014-2017: The Future We Want: Rights and Empowerment:
<https://www.undp.org/sites/g/files/zskgke326/files/publications/Annex%201%20Terminology.pdf>



	“As defined in Article 1, ‘discrimination against women’ shall mean any distinction, exclusion or restriction made on the basis of sex which has the effect or purpose of impairing or nullifying the recognition, enjoyment or exercise by women, irrespective of their marital status, on a basis of equality of men and women, of human rights and fundamental freedoms in the political, economic, social, cultural, civil or any other field.”
Women’s Empowerment	“Women’s empowerment has five components: Women’s sense of self-worth; their right to have and to determine choices; their right to have access to opportunities and resources; their right to have the power to control their own lives, both within and outside the home; and their ability to influence the direction of social change to create a more just social and economic order, nationally and internationally.” “The concept of empowerment is related to gender equality but distinct from it. The core of empowerment lies in the ability of a woman to control her own destiny. This implies that to be empowered women must not only have equal capabilities (such as education and health) and equal access to resources and opportunities (such as land and employment), they must also have the agency to use those rights, capabilities, resources and opportunities to make strategic choices and decisions (such as are provided through leadership opportunities and participation in political institutions. And to exercise agency, women must live without the fear of coercion and violence.”
Gender Parity	“...equal numbers of men and women at all levels of the organization. It must include significant participation of both men and women, particularly at senior levels. Gender parity is one of several integrated mechanisms for improving organizational effectiveness.”
Gender-based Violence	“Gender-based violence is a form of discrimination that seriously inhibits women’s ability to enjoy rights and freedoms on a basis of equality with men...Gender-based violence, which impairs or nullifies the enjoyment by women of human rights and fundamental freedoms under general international law or under human rights conventions, is discrimination within the meaning of Article 1 of (CEDAW).” “...any act of violence that results in, or is likely to result in, physical, sexual or psychological harm or suffering to women, including threats of such acts, coercion or arbitrary deprivation of liberty, whether occurring in public or in private life.”¹⁰ “...any harmful act that is perpetrated against a person’s will and that is based on socially associated differences between males and females. As such violence is based on socially ascribed differences. Gender-based violence includes, but it is not limited to sexual violence. While women and girls of all ages make up the majority of the victims, men and boys are also both direct and indirect victims. It is clear that the effects of such violence are both physical and psychological, and have long term detrimental consequences for both the survivors and their communities.”
Sexual Abuse	Actual or threatened physical intrusion of a sexual nature, whether by force or under unequal or coercive conditions.

Sexual Exploitation	Any actual or attempted abuse of position of vulnerability, differential power or trust, for sexual purposes, including, but not limited to, profiting monetarily, socially or politically from the sexual exploitation of another.
Sexual Harassment	Sexual harassment is not SEA. Sexual harassment refers to prohibited conduct in the work context and can be committed against UN staff and related personnel. In context of the United Nations, sexual harassment primarily describes prohibited behaviour against another UN staff or related personnel, which may also include nationals of the host state. It is defined for UN staff by ST/SGB/2008/5 and similar directives for uniformed personnel and involves any unwelcome sexual advance, request for sexual favour, verbal or physical conduct or gesture of a sexual nature, or any other behaviour of a sexual nature that might reasonably be expected or be perceived to cause offence or humiliation to another, when such conduct interferes with work, is made a condition of employment or creates an intimidating, hostile or offensive work environment. Conduct or behaviour of a sexual nature is considered SEA when this conduct or behaviour amounts to either sexual exploitation or sexual abuse as defined in ST/SGB/2003/13.

Appendix 2: List of stakeholders consulted

Department	Commune	Organisation/Stakeholder	Date of Consultation
Artibonite	Gonaïves	MDE (Government)	January 2018
Artibonite	Gonaïves	Delegation (Government)	August 2018
Artibonite	Gonaïves	MPCE (Government)	August 2018
Artibonite	Gonaïves	MARNDR (Government)	August 2018
Artibonite	Gros-Morne	CASEC (Government)	August 2018
Artibonite	Gros-Morne	Mayor's Office	August 2018
Artibonite	Gros-Morne	BAC (Government)	August 2018
Artibonite	Gros-Morne	Farmers	August 2019
Artibonite	Gros-Morne	Women	August 2020
Artibonite	Marmelade	Mayor's Office	August 2018
Artibonite	Marmelade	BAC (Government)	September-18
North	Cap	MDE (Government)	August 2018
North	Cap	MCDFD (Government)	August 2018
North	Cap	MPCE (Government)	August 2018
North	Cap	MARNDR (Government)	August 2018

North	Pilate	CASEC (Government)	January 2018
North	Pilate	BAC (Government)	January 2018
North	Pilate	Women	August 2018
North	Plaisance	CASEC (Government)	January 2018
North	Plaisance	BAC (Government)	January 2018
North	Plaisance	Women	August 2018
North West	Port-de-Paix	MDE (Government)	January 2018
North West	Port-de-Paix	MCFDF (Government)	July 2018
North West	Port-de-Paix	Delegation (Government)	July 2018
North West	Port-de-Paix	MPCE (Government)	July 2018
North West	Port-de-Paix	TPTC (Government)	July 2018
North West	Port-de-Paix	MARNDR (Government)	July 2018
North West	Chansolme	CASEC (Government)	January 2018
North West	Chansolme	Mayor's Office	July 2018
North West	Chansolme	BAC (Government)	July 2018
North West	Chansolme	Farmers	July 2018
North West	Chansolme	Women	July 2018
North West	Bassin Bleu	CASEC (Government)	January 2018
North West	Bassin Bleu	Mayor's Office	December-18
North West	Bassin Bleu	BAC (Government)	January 2018
North West	Port-de-Paix	Mayor's Office	August 2018
North West	Port-de-Paix	NGO	July 2018
National	Port-au-Prince	NGO	July 2018
National	Port-au-Prince	Bilateral donor	July 2018
North	Cap	Bilateral donor	July 2018
National	Port-au-Prince	Bank	December-18